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ASSIGNMENT

Circles

Concept – Definition – Formulae

Circle : In a plane the locus of points which are at a constant distance from a fixed point is called a circle.

(a) Types of circles in circles

The equation of circle with centre $c(h, k)$ and radius 'r' is $(x - h)^2 + (y - k)^2 = r^2$

If $r = 0$ then the circle is point circle

$r = 1$ then the circle is unit circle

(b) Standard equation of a circle in circles

1) The equation of circle in general form is

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (g^2 + f^2 - c \geq 0)$$

Centre $C(-g, -f)$, radius $r = \sqrt{g^2 + f^2 - c}$

2) The conditions that the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a circle are (i) $a = b$ (ii) $h = 0$ (iii) $g^2 + f^2 - ac \geq 0$

3) The equation of circle passing through (0,0) is $x^2 + y^2 + 2gx + 2fy = 0$
($C = 0$)

4) The parametric equations of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ are

$$x = -g + r \cos \theta, \quad y = -f + r \sin \theta$$

Where ' θ ' is parameter and $0 \leq \theta < 2\pi$

5) Equation of any circle concentric with the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is
 $x^2 + y^2 + 2gx + 2fy + k = 0$

6) Four or more points lie on a same circle called concyclic points

7) There exist only one circle passing through 3 non collinear points

(c) Lengths of tangents and chords in circles

1) Length of tangent drawn from an external point $P(x_1, y_1)$ to the circle $S = 0$ is $\sqrt{S_{11}}$

2) The length of the chord intercepted by the line $lx + my + n = 0$ on the circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \text{ is } 2\sqrt{r^2 - d^2}$$

(d) Intercepts made by a circle on axes in circle

The condition for a circle $x^2 + y^2 + 2gx + 2fy + c = 0$

(i) touching x-axis is $g^2 = c$

(ii) intersecting x-axis is $g^2 > c$

(iii) neither touching nor intersecting x-axis is $g^2 < c$

(iv) touching y-axis is $f^2 = c$

(v) Intersecting y-axis is $f^2 > c$

(vi) neither touching nor intersecting y-axis is $f^2 < c$

The length of the intercept made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on

(i) x-axis is $2\sqrt{g^2 - c}$

(ii) y-axis is $2\sqrt{f^2 - c}$

(e) Equation of chord joining $P(x_1, y_1)$ and $Q(x_2, y_2)$ in circles

1) Equation of chord joining $P(x_1, y_1)Q(x_2, y_2)$ on the circle $S = 0$ is $S_1 + S_2 = S_{12}$

2) Equation of chord joining two points $P(\theta_1), Q(\theta_2)$ on the circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \text{ is } (x+g)\cos\left(\frac{\theta_1+\theta_2}{2}\right) + (y+f)\sin\left(\frac{\theta_1+\theta_2}{2}\right) = r\cos\left(\frac{\theta_1-\theta_2}{2}\right)$$

(f) Position of a point w.r.t. circle

1) The power of a point $P(x_1, y_1)$ w.r.t. to the circle $S = 0$ is $S_{11} = cp^2 - r^2$

2) The maximum distance from the point $P(x_1, y_1)$ to the circle $S = 0$ with centre 'C' and radius 'r' is $CP + r$

3) Minimum distance is $|CP - r|$

(g) Tangent line equation in circles

1) The line $L = 0$ cuts the circle $S = 0$ at two distinct points $\Rightarrow r > d$

2) The line $L = 0$ does not meet the circle $S = 0 \Rightarrow r < d$

3) The line $L = 0$ touches the circle (tangent) $S = 0 \Rightarrow r = d$

4) If the circle touches the line then radius

$r =$ perpendicular distance from centre of circle to the line

5) Equation of tangent to the circle $S = 0$ at $P(\theta)$ is $(x+g)\cos\theta + (y+f)\sin\theta = r$

6) Equation of tangent to the circle $S = 0$ at (x_1, y_1) is $S_1 = 0$

7) The point of contact of the line $lx + my + n = 0$ be a tangent to the circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \text{ is } \left(-g + \frac{lr^2}{lg + mf - n}, -f + \frac{mr^2}{lg - mf - n} \right)$$

8) There exist two tangents drawn from an external point $P(x_1, y_1)$ to the circle $S = 0$

9) If m_1, m_2 are the slopes of tangents drawn from an external point $P(x_1, y_1)$ to the circle $x^2 + y^2 = a^2$ then m_1, m_2 are roots of the equation $(x_1^2 - a^2)m^2 - 2x_1y_1m + y_1^2 - a^2 = 0$

$$m_1m_2 = \frac{y_1^2 - a^2}{x_1^2 - a^2}, \quad m_1 + m_2 = \frac{2x_1y_1}{x_1^2 - a^2}$$

10) The equation of pair of tangents drawn from $P(x_1, y_1)$ to the circle $S = 0$ is $S_1^2 = SS_{11}$

11) Let θ be the angle between the tangents drawn from $P(x_1, y_1)$ to $S = 0$ then

$$\tan\left(\frac{\theta}{2}\right) = \frac{r}{\sqrt{S_{11}}}$$

12) The equation of director circle of $x^2 + y^2 = a^2$ is $x^2 + y^2 = 2a^2$

(h) Normal to a circle in circles

1) The equation of normal to the circle $S = 0$ at (x_1, y_1) is

$$(y_1 + f)(x - x_1) - (x_1 + g)(y - y_1) = 0$$

2) The line $lx + my + n = 0$ be a normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ then

$$lg + mf - n = 0$$

3) The equation of normal to the circle $S = 0$ at $P(\theta)$ is $(x + g)\sin\theta - (y + f)\cos\theta = 0$

(i) Chord of contact of a circle in circles

1) The equation of chord of contact of $P(x_1, y_1)$ to the circle $S = 0$ is $S_1 = 0$

2) The equation of chord of circle $S = 0$ having $P(x_1, y_1)$ as its midpoint is $S_1 = S_{11}$

3) The length of chord of the circle $S = 0$ having $P(x_1, y_1)$ as its midpoint is $2\sqrt{|S_{11}|}$

(j) Pole and polar of a circle

1) The equation of polar of $P(x_1, y_1)$ w.r.t. to the circle $S = 0$ is $S_1 = 0$

2) The pole of the line $lx + my + n = 0$ w.r.t. to $S = x^2 + y^2 + 2gx + 2fy + C = 0$ is

$$\left(-g + \frac{lr^2}{lg + mf - n}, -f + \frac{mr^2}{lg + mf - n}\right)$$

3) The condition that the points $P(x_1, y_1), Q(x_2, y_2)$ are conjugate points w.r.t. to $S = 0$ is

$$S_{12} = 0$$

4) The condition that the lines $l_1x + m_1y + n_1 = 0, l_2x + m_2y + n_2 = 0$ are conjugate lines w.r.t.

to $x^2 + y^2 + 2gx + 2fy + c = 0$ is $r^2(l_1l_2 + m_1m_2) = (l_1g + m_1f - n_1)(l_2g + m_2f - n_2)$

Analysis of two circles in circles

1) Let θ be the angle between two circles then $\cos\theta = \frac{d^2 - r_1^2 - r_2^2}{2r_1r_2}$

Where $d = C_1C_2$

If two circles intersect orthogonally then (i) $d^2 = r_1^2 + r_2^2$ (or) $2gg' + 2ff' = C + C'$

(a). Relative position of two circles

1) $C_1C_2 > r_1 + r_2$ circles do not intersect, four common tangent can be drawn two direct and two transverse

2) $C_1C_2 = r_1 + r_2$ circles touch each other externally, three common tangents can be drawn

3) $|r_1 - r_2| < C_1C_2 < r_1 + r_2$

Circles intersect in two points

Two common tangents can be drawn

4) $C_1C_2 = |r_1 - r_2|$

Circles touch each other internally

Only one common tangent can be drawn

5) $0 \leq C_1C_2 < |r_1 - r_2|$

One circle lie completely inside other

No common tangent can be drawn

6) If two circles with centres $C_1(x_1, y_1)$ and $C_2(x_2, y_2)$ and radii r_1 and r_2 respectively then

(i) Internal centre of similitude is $\left(\frac{r_2x_1 + r_1x_2}{r_1 + r_2}, \frac{r_1y_2 + r_2y_1}{r_1 + r_2} \right)$

ii) External centre of similitude is $\left(\frac{r_1x_2 - r_2x_1}{r_1 - r_2}, \frac{r_1y_2 - r_2y_1}{r_1 - r_2} \right)$

If C_1, C_2 are centres of two circles and r_1, r_2 are radii then

(i) length of direct common tangent is $\sqrt{(C_1C_2)^2 - (r_1 - r_2)^2}$

(ii) length of transverse common tangent is $\sqrt{(C_1C_2)^2 - (r_1 + r_2)^2}$

If two circles touch each other at P then the equation of common tangent at P is $S - S' = 0$

(b) System of circles in circles

The equation of any circle passing through point of intersection of circle $S = 0$ and line $L = 0$ is $S + \lambda L = 0$

The equation of any circle passing through points of intersection of circle $S = 0$ and $S' = 0$ is of the form $S + \lambda S' = 0$ (or) $\lambda_1 S + \lambda_2 S' = 0$

(c) Radical axis

The equation of radical axis of two circles $S = 0$ and $S' = 0$ is $S - S' = 0$

The radical axis is perpendicular to the line joining the centres of the given circles

The radical axis bisects the common tangents of two circles

If two circles cut a third circle orthogonally then the radical axis of two circles will pass through centre of the third circle

The equation of common chord of two circles $S = 0$ and $S' = 0$ is $S - S' = 0$

A system of circles, every pair of circles have the same radical axis, is called coaxial system of circles

Point of intersection of any two radical axes of three circles is called radical centre

The point circles in the coaxial system are called limiting points.

Exercise : I

(Straight Objective Including PYQ's)

1. Circles**1) Standard equation of a circle, Lengths of tangents and chords;**

- 1) If the length of the chord of the circle, $x^2 + y^2 = r^2$ ($r > 0$) along the line, $y - 2x = 3$ is r then r^2 is equal to : (JEE MAIN 2020 SHIFT 1)

- a) $\frac{9}{5}$ b) $\frac{24}{5}$ c) $\frac{12}{5}$ d) 12

Ans : C

Sol : Let chord AB = r

 $\therefore \Delta AOM$ is right angled triangle

$$\therefore OM = \frac{r\sqrt{3}}{2} = \text{perpendicular distance of line}$$

AB from (0,0)

$$\frac{r\sqrt{3}}{2} = \left| \frac{3}{\sqrt{5}} \right|$$

$$r^2 = \frac{12}{5}$$

- 1a) The length of the chord cut by $4y = 3x + 1$ on the circle $x^2 + y^2 - 6x = 0$ is

(JEE MAIN 2020 SHIFT 1)

- a) $2\sqrt{5}$ b) -8 c) 4.5 d) -4

Ans : A

Sol : Given circle $x^2 + y^2 - 6x = 0$

Centre = (3,0)

radius $r = \sqrt{9} = 3$ d = distance from (3,0) to $4y = 3x + 1$

$$= \frac{|9 + 1|}{\sqrt{9 + 16}} = 2$$

$$\therefore \text{length of chord} = 2\sqrt{r^2 - d^2}$$

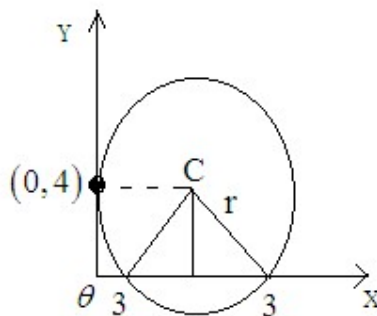
$$= 2\sqrt{9 - 4}$$

$$= 2\sqrt{5}$$

- 1b) A circle touches the y-axis at the point (0,4) and cuts the x-axis in a chord of length 6 units. Then find the radius of circle.

a) 5 b) $\sqrt{5}$ c) 25 d) 15

Ans : A



Sol :

From figure $r = \sqrt{4^2 + 3^2} = 5$

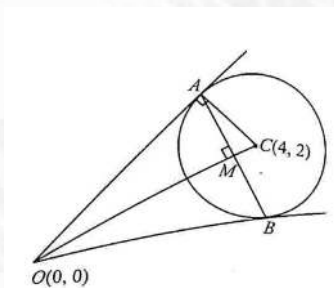
- 2) Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the point A and B. Then $(AB)^2$ is equal to:

(JEE MAIN 2020 SHIFT 2)

a) $\frac{64}{5}$ b) $\frac{24}{5}$ c) $\frac{8}{5}$ d) $\frac{8}{13}$

Ans : A

Sol :



Centre of the circle is $C(4, 2)$

$$OC = \sqrt{16 + 4} = 2\sqrt{5}$$

$AC = \text{Radius of the circle} = 2$

$$OA = \sqrt{OC^2 - AC^2} = 4$$

For area of triangle OAC we have $\frac{1}{2} \times AC \times OA = \frac{1}{2} \times OC \times AM$

$$\Rightarrow 2 \times 4 = 2\sqrt{5} \times AM$$

$$\Rightarrow AM = \frac{4}{\sqrt{5}}$$

$$\Rightarrow AB = 2AM = \frac{8}{\sqrt{5}}$$

$$\Rightarrow AB^2 = \frac{64}{5}$$

2a) Let the tangents drawn from the point $(1, 2)$ to the circle $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the point A and B then AB^2 is equal to (JEE MAIN 2020 SHIFT 1)

- a) $\frac{80}{9}$ b) 4 c) $\frac{16}{9}$ d) $\frac{20}{9}$

Ans : A

Sol : Chord of contact of $(1, 2)$ is $S_1 = 0$

$$x + 2y - 4(x + 1) - 2(y + 2) + 16 = 0$$

$$-3x + 8 = 0$$

Centre $(4, 2)$ radius $r = 2$

$$d = \frac{|-12 + 8|}{3} = \frac{4}{3}$$

$$\text{Length of chord} = 2\sqrt{r^2 - d^2} = 2\sqrt{4 - \frac{16}{9}} = 2\sqrt{\frac{20}{9}}$$

$$\therefore AB^2 = \frac{80}{9}$$

2b) Let the tangents drawn from the origin to the circle $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the point A and B then area of ΔOAB is (JEE MAIN 2020 SHIFT 1)

- a) $\frac{64}{5\sqrt{5}}$ b) $\frac{8}{\sqrt{5}}$ c) $\frac{32}{5}$ d) $\frac{128}{5\sqrt{5}}$

Ans : C

Sol : Centre $(4, 2)$ radius $r = \sqrt{16 + 4 - 16} = 2$

Chord of contact of $(2, 0)$ is $S_1 = 0 \Rightarrow -4x - 2y + 16 = 0$

$$2x + y - 8 = 0 \quad \dots(1)$$

$$\perp^r \text{ distance from } (0, 0) \text{ to the line (1)} = \frac{|-8|}{\sqrt{4+1}} = \frac{8}{\sqrt{5}}$$

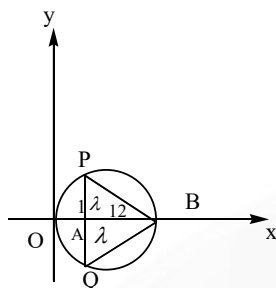
$$d = \frac{|8 + 2 - 8|}{\sqrt{4+1}} = \frac{2}{\sqrt{5}}$$

$$\begin{aligned} \text{Length of chord} &= 2\sqrt{r^2 - d^2} \\ &= 2\sqrt{4 - 4/5} = \frac{3}{\sqrt{5}} \end{aligned}$$

$$\text{Area of } \Delta OAB = \frac{1}{2} \times \frac{3}{\sqrt{5}} \cdot \frac{8}{\sqrt{5}} = \frac{32}{5} \quad (\text{or}) \text{ formula} = \frac{r(S_{11})^{3/2}}{S_{11} + r^2}$$

- 3) In the circle given below, let $OA = 1 \text{ unit}$ $OB = 13 \text{ unit}$ and $PQ \perp OB$. Then, the area of the triangle PQB (in square units) is:

(JEE MAIN 2021 SHIFT 1)



Ans: (b)

- a) $24\sqrt{2}$ b) $24\sqrt{3}$ c) $26\sqrt{3}$ d) $26\sqrt{2}$

Sol: let $PA = AQ = \lambda$ NOW, $OA \cdot AB = AP \cdot AQ$

$$\Rightarrow 1 \times 12 = \lambda^2$$

$$\Rightarrow \lambda = 2\sqrt{3}$$

so, area of triangle

$$PQB = \frac{1}{2} \times 2\lambda \times AB$$

$$= \frac{1}{2} \times 4\sqrt{3} \times 12 = 24\sqrt{3}$$

- 3A) AB is a chord of the circle $x^2 + y^2 = 9$. The tangents at A and B intersect at C. If $M(1,2)$ is midpoint of AB then area of triangle ABC is

(JEE MAIN 2021 SHIFT 1)

Ans: 4

- a) 5 b) 8 c) $2\sqrt{\frac{2}{5}}$ d) $\frac{8}{\sqrt{5}}$

Ans : (d)

Sol:

Equation of chord $x + 2y - 9 = 1 + 4 - 9$

$$= x + 2y - 5 = 0$$

$$AB = \text{length of chord} = 2\sqrt{|S_{11}|} = 2\sqrt{4} = 4$$

$$C = \left(-\frac{lr^2}{n}, -\frac{mr^2}{n} \right)$$

$$C = \left(\frac{9}{5}, \frac{18}{5} \right)$$

$$\text{Height} = \frac{\left| \frac{9}{5} + \frac{36}{5} - 5 \right|}{\sqrt{5}} = \frac{4}{\sqrt{5}}$$

$$\text{Area} = \frac{1}{2} \times b \times h = \frac{1}{2} \times 4 \times \frac{4}{\sqrt{5}}$$

$$= \frac{8}{\sqrt{5}}$$

3b) The equation of the circle which passes through the point $A(2,0)$ and whose centre is the limit of the point of intersection of the lines $3x + 5y = 1$ and $(2+c)x + 5c^2y = 1$ as $c \rightarrow 1$ is _____

a) $25x^2 + 25y^2 - 20x + 2y - 60 = 0$

b) $25x^2 + 25y^2 - 20x - 2y - 60 = 0$

c) $25x^2 + 25y^2 + 20x - 2y - 60 = 0$

d) $25x^2 + 25y^2 + 20x - 2y - 60 = 0$

Ans: (a)

Sol: Given lines

$$3x + 5y = 1 \quad (1)$$

$$(2+c)x + 5c^2y = 1 \quad (2)$$

if $c = 1$ two lines are identical $c \neq 1$

$(1) - (2)$ gives

$$(1-c)x + 5y(1-c^2) = 0$$

$$x + 5y(1+c) = 0$$

as $c \rightarrow 1 \Rightarrow x + 10y = 0$ and

$3x + 5y = 1$, by solving we get

$$C = \left(\frac{2}{5}, -\frac{1}{25} \right)$$

$\therefore$ equation of circle is

$$\left(x - \frac{2}{5} \right)^2 + \left(y + \frac{1}{25} \right)^2 = CA^2$$

$$= \left(2 - \frac{2}{5} \right)^2 + \left(0 + \frac{1}{25} \right)^2$$

$$\Rightarrow 25x^2 + 25y^2 - 20x + 2y - 60 = 0$$

4) If the locus of the mid-point of the line segment from the point $(3,2)$ to a point on the circle, $x^2 + y^2 = 1$ is a circle of radius r , then r is equal to: **(JEE MAIN 2021 SHIFT 2)**

a) 1

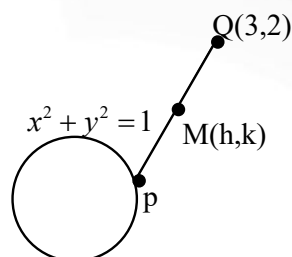
b) $\frac{1}{2}$

c) $\frac{1}{3}$

d) $\frac{1}{4}$

Key: (a)

Sol: let $Q = (3,2)$



Point P is variable point on the circle.

Let $M(h,k)$ be mid-point of PQ .

So, $P \equiv (2h-3, 2k-2)$

P lies on the circle.

$$\text{So, } (2h-3)^2 + (2k-2)^2 = 1.$$

Hence, locus of point M is $(x-3/2)^2 + (y-1)^2 = 1/4$

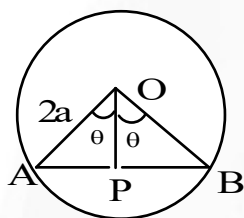
$$\text{So, radius is } r = \frac{1}{2}.$$

- 4a) The locus of the middle point of the chords of the circle $x^2 + y^2 = 4a^2$ which subtend a right angle at the centre of the circle is **(JEE MAIN 2021 SHIFT 1)**

a) $x + y = 2a$ b) $x^2 + y^2 = a^2$ c) $x^2 + y^2 = 2a^2$ d) $x^2 + y^2 = x + y$

Ans: 3

Sol: Let $P = (x_1, y_1)$ be the midpoint



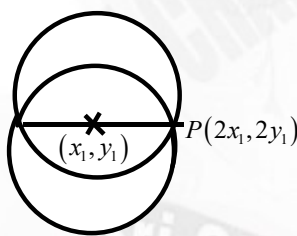
$$\cos 45 = \frac{OP}{2a} \Rightarrow x^2 + y^2 = 2a^2$$

- 4b) A variable chord of the circle $x^2 + y^2 - 2ax = 0$ is drawn through the origin. Locus of the centre of the circle drawn on this chord as diameter is **(JEE MAIN 2021 SHIFT 1)**

a) $x^2 + y^2 + ax = 0$ b) $x^2 + y^2 + ay = 0$ c) $x^2 + y^2 - ax = 0$ d) $x^2 + y^2 - ay = 0$

Ans: (c)

SOL:



$\therefore P(2x_1, 2y_1)$ lies on the given circle

$$4x_1^2 + 4y_1^2 - 4ax_1 = 0$$

$$\Rightarrow \text{Locus in } x^2 + y^2 - ax = 0$$

5. Let $A(1,4)$ and $B(1,-5)$ be two points. Let P be a point on the circle $(x-1)^2 + (y-1)^2 = 1$ such that $(PA)^2 + (PB)^2$ have maximum value, then the points, P, A and B lie on (2021)

a) a straight line b) a hyperbola c) an ellipse d) a parabola

Ans: (a)

Sol: P is a point on $(x-1)^2 + (y-1)^2 = 1$.

$$\text{Let } P = (1 + \cos \theta, 1 + \sin \theta)$$

$$A = (1, 4), B = (1, -5)$$

$$\therefore (PA)^2 + (PB)^2$$

$$= (\cos \theta)^2 + (\sin \theta - 3)^2 + (\cos \theta)^2 + (\sin \theta + 6)^2$$

$$= 47 + 6 \sin \theta$$

$$(PA)^2 + (PB)^2 \text{ is maximum when } \sin \theta = 1$$

$$\Rightarrow \cos \theta = 0 \text{ and } P = (1, 2)$$

Clearly, points $P(1, 2), A(1, 4), B(1, -5)$ lie on line $x = 1$.

- 5a) For the circle $x^2 + y^2 = r^2$, find the value of r for which area enclosed by the tangents drawn from the point $P(6, 8)$ to the circle and chord of contact is maximum is

(JEE MAIN 2021 SHIFT 1)

- a) 5 b) 10 c) 3 d) 4

Ans: (a)

$$\text{Sol: } \Delta = \text{area} = \frac{r s_{11}^{3/2}}{s_{11} + r^2} = \frac{r(100 - r^2)^{3/2}}{100 - r^2 + r^2}$$

$$\Delta = \frac{r(100 - r^2)^{3/2}}{100}$$

$$\frac{d\Delta}{dr} = 0 \Rightarrow r = 5$$

- 5b. A variable chord of circle $x^2 + y^2 = 4$ is drawn from the point $P(3, 5)$ meeting the circle at the point A and B. A point Q is taken on their chord such that $2PQ = PA + PB$ locus of Q is (2021)

- a) $x^2 + y^2 + 3x + 4y = 0$ b) $x^2 + y^2 = 36$ c) $x^2 + y^2 = 16$ d) $x^2 + y^2 - 3x - 5y = 0$

Ans: (d)

Sol:



$$2PQ = PA + PB$$

$$PQ + PQ = PA + PA + AB$$

$$PA + AQ + PA + AQ = PA + PA + AB$$

$$AQ + AQ = AB$$

$$AQ + AQ = AQ + QB$$

$$\Rightarrow AQ = QB$$

$\Rightarrow Q$ in mid point

$$\text{Let } Q = (x_1, y_1)$$

$$\text{equation of chord } xx_1 + yy_1 = x_1^2 + y_1^2$$

passing through $P(3,5)$

$$= 3x_1 + 5y_1 = x_1^2 + y_1^2$$

$$= \text{locus is } x^2 + y^2 - 3x - 5y = 0$$

- 6) The points of intersection of the line $ax + by = 0$ and the circle $x^2 + y^2 - 2x = 0$ are $A(\alpha, 0)$ and $B(1, \beta)$. The image of the circle with AB as diameter in the line $x + y + 2 = 0$ is

(JAN 2023 SESSION - I)

{Circle general equation or Equation with given diameter ends}

a) $x^2 + y^2 + 5x + 5y + 12 = 0$

b) $x^2 + y^2 + 3x + 5y + 8 = 0$

c) $x^2 + y^2 + 3x + 3y + 4 = 0$

d) $x^2 + y^2 - 5x - 5y + 12 = 0$

Key : (a)

Solution: $A(\alpha, 0) = (0, 0)$ and $B(1, \beta) = (1, 1)$ Hence centre is Centre $(1/2, 1/2)$ and radius $= 1/\sqrt{2}$

Image of centre is $(-5/2, -5/2)$

- 6A) The circle $x^2 + y^2 - 10x - 6y + 9 = 0$ and X-axis intersect at A and B. The image circle of the circle with AB as diameter, in the line $x - y = 0$, is

a) $x^2 + y^2 - 10x + 9 = 0$

b) $x^2 + y^2 + 10x + 6y + 9 = 0$

c) $x^2 + y^2 - 6y + 9 = 0$

d) $x^2 + y^2 - 10y + 9 = 0$

Key : (d)

Solution: $x^2 - 10x + 9 = 0 \Rightarrow A(1, 0), B(9, 0) \Rightarrow \text{Centre} = (5, 0)$ and $r = 4$

Image of centre is $(0, 5)$

- 7) Consider a triangle whose two sides lie on the x-axis and the line $x + y + 1 = 0$. If the orthocenter of Δ is $(1, 1)$, then the equation of the circle passing through the vertices of the triangle Δ is (IIT 2021)

a) $x^2 + y^2 - 3x + y = 0$

b) $x^2 + y^2 + x + 3y = 0$

c) $x^2 + y^2 + 2y - 1 = 0$

d) $x^2 + y^2 + x + y = 0$

KEY: (b)

Solution: Two sides of triangle are x-axis and $x + y + 1 = 0$. $\rightarrow$ their intersection point is vertex $A(-1, 0)$

Remaining vertices are $B(a, 0)$ on x axis and $C(x, -x - 1)$ on $x + y + 1 = 0$. Given Ortho centre is $H(1, 1)$

Using AH is perpendicular to BC , $m_1 m_2 = \frac{1}{2} \left(\frac{-x-1}{x-a} \right) = -1 \rightarrow x+1 = 2(x-a) \rightarrow x = 1+2a$

Similarly CH is perpendicular to AB (i.e. x axis) Hence slope of CH is $1/0 \rightarrow x = 1$, $a = 0$

Triangle vertices now are $A(-1, 0), B(0, 0), C(1, -2)$

- 7A) A, B are two points in xy plane which are $2\sqrt{2}$ units apart such that AB subtend 90° at the point C(1,2). The point C, on the line $x - y + 1 = 0$, is having maximum angle formed by AB than the remaining points of the line. The equation of one of the circles through the points A, B, C is -----

a) $x^2 + y^2 + 6y + 7 = 0$

b) $x^2 + y^2 + 6y - 17 = 0$

c) $x^2 + y^2 + 4x + 2y + 3 = 0$

d) $x^2 + y^2 - 4x - 2y + 3 = 0$

KEY: (d)

Solution: Angle $ACB = \frac{\pi}{2}$, is max possible angle for any point C on $x - y + 1 = 0$

Circle has diameter AB, tangent at C is $x - y + 1 = 0$

$$r = \sqrt{2} \Rightarrow \text{centre} \left(1 \pm \sqrt{2} \left(\frac{-1}{\sqrt{2}} \right), 2 \pm \sqrt{2} \left(\frac{1}{\sqrt{2}} \right) \right) = (0, 3), (2, 1)$$

CONCEPT: End point of diameter of a circle

FORMULA: Equation of circle is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

- 8). Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $(a + b - c)$ is (2022_26 July - Shift 2)
- a) 12 b) 13 c) 14 d) 16

Key: (a)

Sol: $P = (x_1, y_1)$ $Q = (x_2, y_2)$

$$x^2 - 4x - 6 = (x - x_1)(x - x_2)$$

$$y^2 + 2y - 7 = (y - y_1)(y - y_2)$$

Eq to having $\overline{PQ}$ as diameter $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

$$x^2 - 4x - 6 + y^2 + 2y - 7 = 0$$

$$\Rightarrow x^2 + y^2 - 4x + 2y + 13 = 0$$

$$x^2 + y^2 + 2ax + 2by + C = 0$$

$$a = -2, \quad b = 1, \quad c = -13$$

$$a + b - c = -2 + 1 - (-13) = 12$$

- 8a) Let α, β be the roots of $x^2 + 5x + 6 = 0$ and γ, δ be the roots of $y^2 + 6y + 7 = 0$. Then the equation of the circle with (α, γ) and (β, δ) as the extremities of a diameter is

a) $x^2 + y^2 + 5x + 6y + 10 = 0$

b) $x^2 + y^2 + 5x + 6y + 11 = 0$

c) $x^2 + y^2 + 5x + 6y + 13 = 0$

d) $x^2 + y^2 + 5x + 6y + 12 = 0$

Key: (c)

Sol: Let (α, β) are the roots of $x^2 + 5x + 6 = 0$

$$\Rightarrow (x - \alpha)(x - \beta) = x^2 + 5x + 6$$

$$(\gamma, \delta) \text{ be the roots of } y^2 + 6y + 7 = 0 \Rightarrow (y - \gamma)(y - \delta) = y^2 + 6y + 7$$

Eq: of the circle with (α, γ) and (β, δ) as ends of diameter is

$$(x - \alpha)(x - \beta) + (y - \gamma)(y - \delta) = 0$$

$$\Rightarrow x^2 + y^2 + 5x + 6y + 13 = 0.$$

- 8b) The abscissa of two points P, Q are the roots of the equation $2x^2 + 4x - 7 = 0$ and their ordinates are the roots of the equation $3x^2 - 12x - 1 = 0$. Then the centre of the circle with PQ as a diameter is
- a) $(-1, 2)$ b) $(-2, 6)$ c) $(1, -2)$ d) $(2, -6)$

Key: (a)

Sol: Let $P(x_1, y_1)$ $Q(x_2, y_2)$

$$(x - x_1)(x - x_2) = x^2 + 2x - \frac{7}{2}$$

$$(y - y_1)(y - y_2) = y^2 - 4y - 1/3$$

Eq. Of circle with PQ as diameter is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

$$x^2 + y^2 + 2x - 4y - 23/6 = 0$$

$$= (-1, 2)$$

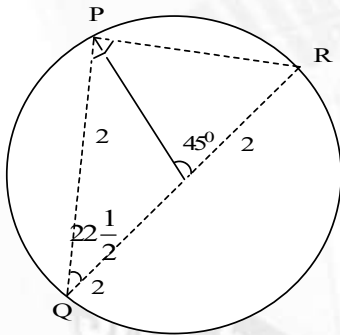
- 9) Let C be the centre of the circle $x^2 + y^2 - x + 2y = \frac{11}{4}$ and P be a point on the circle. A line passes through the point C, makes an angle of $\frac{\pi}{4}$ with the line CP and intersects the circle at Q and R then the area of the triangle PQR (in unit²) is:

[2022_28 July - Shift 1]

- a) 2 b) $2\sqrt{2}$ c) $8 \sin\left(\frac{\pi}{8}\right)$ d) $8 \cos\left(\frac{\pi}{8}\right)$

Key: (b)

Sol:



$$x^2 + y^2 - x + 2y = \frac{11}{4}$$

$$\left(x - \frac{1}{2}\right)^2 + (y + 1)^2 = (2)^2$$

From triangle PQR

$$PR = QR \sin 22\frac{1}{2}^\circ$$

$$= 4 \sin \frac{\pi}{8}$$

$$PQ = QR \cos 22\frac{1}{2}^\circ = 4 \cos \frac{\pi}{8}$$

$$\text{As } \Delta PQR = \frac{1}{2} PR \times PQ = \frac{1}{2} \left(4 \sin \frac{\pi}{8} \right) \left(4 \cos \frac{\pi}{8} \right) = 4 \sin \frac{\pi}{4} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

9a) Let B be the Centre of the circle $x^2 + y^2 - 2x + 4y + 1 = 0$

let the tangents at two points P and Q on the circle intersect at the point A(3,1). Then

$$8 \left(\frac{\text{Area } \Delta APQ}{\text{Area } \Delta BPQ} \right) \text{ is } \underline{\hspace{2cm}}$$

a) 17

b) 18

c) 20

d) 21

Ans: (b)

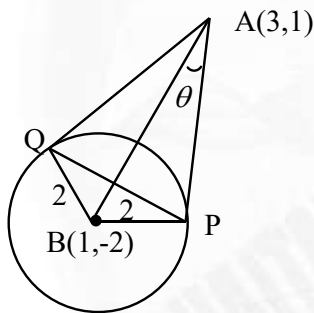
Sol: B(1,-2) is the centre, $r = 2$

$$Ap = \sqrt{s_{11}} = 3$$

$$\tan \theta = \frac{BP}{Ap} = 2/3$$

$$\text{Area } \Delta APQ = \frac{1}{2} (PQ)(AR) = \frac{1}{2} (PQ)(3 \cos \theta)$$

$$= \frac{3}{2} (PQ) \cos \theta.$$



$$\text{Area of } \Delta BPQ = \frac{1}{2} (PQ)(BR) = \frac{1}{2} PQ \left(2 \cos \left(\frac{\pi}{2} - \theta \right) \right) = PQ \sin \theta$$

$$8 \left(\frac{\text{Area } \Delta APQ}{\text{Area } \Delta BPQ} \right) = 8 \left(\frac{3PQ \cos \theta}{2PQ \sin \theta} \right) = 12 \cot \theta$$

$$= 12(3/2)$$

$$= 18$$

9b) The area of triangle formed with Co-ordinate axes and the tangent at the point (x_1, y_1) as the circle $x^2 + y^2 = a^2$ is _____

a) $\frac{a^4}{|x_1 y_1|}$

b) $\frac{a^4}{2|x_1 y_1|}$

c) $\frac{2a^4}{|x_1 y_1|}$

d) $\frac{a^2}{2|x_1 y_1|}$

Key: (b)

Sol: Equation of tangent is $S_1 = 0$

$$\text{i.e. } xx_1 + yy_1 = a^2$$

$$\text{Area of triangle} = \frac{c^2}{2|ab|} = \frac{(-a^2)^2}{2|x_1 y_1|} = \frac{a^4}{2|x_1 y_1|}$$

10) Concept: Area of the triangle

Formula: $\frac{1}{2}(\text{Base}) \times \text{height}$

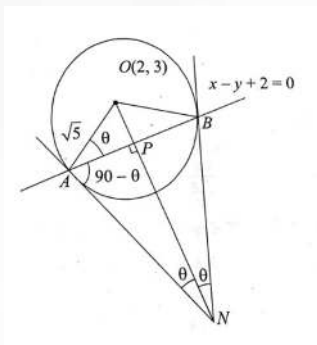
Let the tangent to the circle $C_1 : x^2 + y^2 = 2$ at the point $M(-1,1)$ intersect the circle

$C_2 : (x-3)^2 + (y-2)^2 = 5$ at two distinct points A and B. If the tangents to C_2 at the points A and B intersect at N, then Area of triangle ANB is equal to (29Jun 2022 shift -I)

- a) $\frac{1}{2}$ b) $\frac{2}{3}$ c) $\frac{1}{6}$ d) $\frac{5}{3}$

Ans : (c)

Sol :



$$OP = \left| \frac{2-3+2}{\sqrt{2}} \right| \Rightarrow OP = \frac{3}{\sqrt{2}}$$

$$\text{Now, } AP = \sqrt{OA^2 - OP^2} = \frac{1}{\sqrt{2}}$$

$$\tan \theta = 3 \therefore \sin \theta = \frac{3}{\sqrt{5}} = \frac{AP}{AN} \Rightarrow AN = \frac{\sqrt{5}}{3} = BN$$

$$\text{Area of } \triangle ANB = \frac{1}{2} \cdot (AN^2) \sin 2\theta = \frac{1}{6}$$

10a) If tangent at (1,2) to the circle $x^2 + y^2 = 5$ intersect circle $x^2 + y^2 = 9$ at 'A' and 'B' and tangents at 'A' and 'B' to the second circle intersect at 'C' then area of $\triangle ABC = ?$

- a) $\frac{8\sqrt{5}}{5}$ b) $\frac{4\sqrt{5}}{5}$ c) $\frac{6\sqrt{5}}{5}$ d) $\frac{2\sqrt{5}}{5}$

Ans: (a)

$$\text{Sol: } x^2 + y^2 - 5 = 0 \dots\dots (1)$$

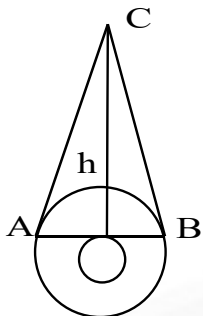
$$x^2 + y^2 - 9 = 0 \dots\dots (2)$$

Tangent at (1,2) w.r.t (1) is $S_1 = 0$

$$\Rightarrow x + 2y - 5 = 0 \dots\dots (3)$$

$$c = \left(\frac{-r^2 l}{n}, \frac{-r^2 m}{n} \right) = \left(\frac{-9 \times 1}{-5}, \frac{-9 \times 2}{-5} \right)$$

$$d = \frac{|0+0-5|}{\sqrt{1+4}} = \sqrt{5}$$



$$AB = 2\sqrt{r^2 - d^2} = 2\sqrt{9 - 5} = 4$$

$$h = \frac{\left| \frac{9}{5} + \frac{36}{5} - 5 \right|}{\sqrt{1+4}} = \frac{|9+36-25|}{5\sqrt{5}} = \frac{20}{5\sqrt{5}} = \frac{4}{\sqrt{5}}$$

$$\text{Area of } \triangle ABC = \frac{1}{2}(AB)(h) = \frac{1}{2} \times 4 \times \frac{4}{\sqrt{5}} = \frac{8}{\sqrt{5}} = \frac{8\sqrt{5}}{5} \text{ sq units}$$

10b) If 'c' be the Point of Intersection of the tangents drawn to the circle $x^2 + y^2 = 12$ at the points where the circle intersect the circle $x^2 + y^2 - 5x + 3y - 2 = 0$ at 'A,B' then area of $\triangle ABC = ?$

a) $\frac{154 \times \sqrt{77}}{85}$

b) $\frac{157 \times \sqrt{77}}{85}$

c) $\frac{97 \times \sqrt{77}}{84}$

d) $\frac{158 \times \sqrt{77}}{65}$

Key: (a)

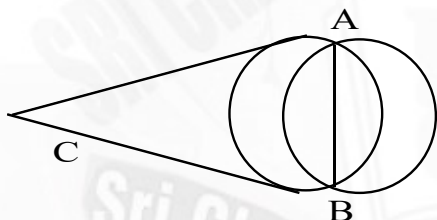
Sol: $x^2 + y^2 - 12 = 0 \dots (1)$

$c = (0,0) \quad r^2 = 12$

$x^2 + y^2 - 5x + 3y - 2 = 0 \dots (2)$

AB is R.A of (1)&(2)

AB is $5x - 3y - 10 = 0 \dots (3)$



$$c = \left(\frac{-r^2 l}{n}, \frac{-r^2 m}{n} \right)$$

$$= \left(\frac{-12 \times 5}{-10}, \frac{-12 \times -3}{-10} \right)$$

$$= \left(\frac{-60}{-10}, \frac{-36}{-10} \right) = \left(6, \frac{-18}{5} \right)$$

$$d = \frac{|0 - 0 - 10|}{\sqrt{25 + 9}} = \frac{10}{\sqrt{34}}$$

$$AB = 2\sqrt{r^2 - d^2} = 2\sqrt{12 - \frac{100}{34}}$$

$$= 2\sqrt{\frac{408-100}{34}} = 2\sqrt{\frac{308}{34}}$$

$$h = \frac{\left| 5 \times 6 - 3 \times \frac{-18}{5} - 10 \right|}{\sqrt{25+9}} = \frac{\left| 30 + \frac{54}{5} - 10 \right|}{\sqrt{34}} = \frac{\left| 20 + \frac{54}{5} \right|}{\sqrt{34}}$$

$$= \frac{\left| \frac{100+54}{5} \right|}{\sqrt{34}} = \frac{154}{5\sqrt{34}}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times (AB) \times h$$

$$= \frac{1}{2} \times 2\sqrt{\frac{308}{34}} \times \frac{154}{5\sqrt{34}}$$

$$= \frac{2\sqrt{77}}{\sqrt{34}} \times \frac{154}{5\sqrt{34}} = \frac{2 \times 154 \times \sqrt{77}}{5 \times 34}$$

$$\frac{154 \times \sqrt{77}}{85} \text{ sq units}$$

Concept: Area of the triangle inscribed in a circle

Formula: $\frac{1}{2}(\text{Base}) \times \text{height}$

- 11) Let a triangle ABC be inscribed in a circle $x^2 - \sqrt{2}(x+y) + y^2 = 0$, such that $\angle BAC = \frac{\pi}{2}$. If the length of the side AB is $\sqrt{2}$ then the area of triangle ABC is equal to

(29 Jun 2022 Shift-II)

- a) 1 B) 2 C) 3 D) 4

Ans : (a)

Sol: $C = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \quad r = 1$

$$BC = \text{diameter} = 2$$

$$AB = \sqrt{2}$$

$$AC = \sqrt{BC^2 - AB^2} = \sqrt{2}$$

$$\text{Area} = \frac{1}{2}(AB)(AC) = 1$$

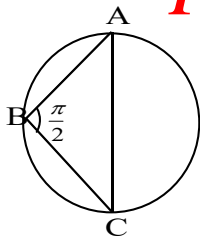
- 11 a) Let a triangle ABC be inscribed in the circle $x^2 - \sqrt{2}y + y^2 = 0$ where $\angle ABC = \frac{\pi}{2}$ and $AB = 1$ then area of $\triangle ABC = ?$

- a) 1 b) $\frac{1}{2}$ c) $\frac{1}{4}$ d) 2

Key: B

Sol: $x^2 + y^2 - \sqrt{2}y = 0$

Centre = $\left(0, \frac{1}{\sqrt{2}} \right)$



$$\text{Radius } r = \sqrt{0 + \frac{1}{2}} = \frac{1}{\sqrt{2}}$$

$$AC = 2r = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$AB = 1$$

$$AB^2 + BC^2 = AC^2$$

$$1 + BC^2 = 2 \Rightarrow BC^2 = 1$$

$$BC = 1$$

$$\text{Area} = \frac{1}{2}(AB)(BC) = \frac{1}{2} \times 1 \times 1$$

$$= \frac{1}{2} \text{ sq unit}$$

11 b) Let a triangle ABC be inscribed in the circle $x^2 + y^2 - 4x - 2y - 4 = 0$ where $\angle BAC = \frac{\pi}{2}$ and

AB=4 then area of $\triangle ABC = ?$

a) $5\sqrt{5}$

b) $3\sqrt{5}$

c) $4\sqrt{5}$

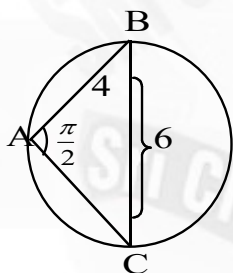
d) $6\sqrt{5}$

Key: (c)

Sol: $x^2 + y^2 - 4x - 2y - 4 = 0$

Centre = (2, 1)

$$r = \sqrt{4 + 1 + 4} = 3$$



$$BC = 2r = 6, AB = 4$$

$$AB^2 + AC^2 = BC^2$$

$$16 + AC^2 = 36$$

$$AC^2 = 20$$

$$AC = \sqrt{20} = 2\sqrt{5}$$

$$\text{Area} = \frac{1}{2}(AB)(AC)$$

$$= \frac{1}{2} \times 4 \times 2\sqrt{5} = 4\sqrt{5} \text{ sq unit}$$

12) Let C be a circle passing through the points $A(2, -1)$ and $B(3, 4)$. The line segment AB is not a diameter of C. If r is the radius of C and its centre lies on the circles

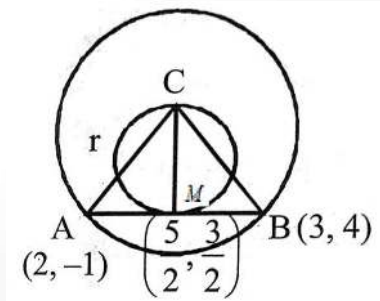
$(x-5)^2 + (y-1)^2 = \frac{13}{2}$, then r^2 is equal to :

- a) 32 b) $\frac{65}{2}$ c) $\frac{61}{2}$ d) 30

Ans: (b)

Solution: Since $AM = CM$

So, the given circle will touch the chord AB



$$AB = \sqrt{26}$$

$$r^2 = CM^2 + AM^2$$

$$= \left(2 \times \sqrt{\frac{13}{2}} \right)^2 + \left(\sqrt{\frac{13}{2}} \right)^2 \Rightarrow r^2 = \frac{65}{2}$$

12a. A circle passes through the points $(2, 3)$ and $(4, 5)$. If its centre lies on the line $y - 4x + 3 = 0$ then its radius is equal to

- a) $r = 1$ b) $r = 2$ c) $r = 3$ d) $r = 5$

Ans: b

Sol: equation of circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \text{ -----(1)}$$

Equation (1) passes through $(2, 3)$ & $(4, 5)$

$$(I, e) S_{11} = 0 \quad 13 + 4g + 6f + c = 0 \text{ -----(2)}$$

$$\& \quad 41 + 8g + 10f + c = 0 \text{ -----(3)}$$

$$(3) - (2)$$

$$4g + 4f + 28 = 0 \text{ -----(4)}$$

Centre $(-g, -f)$ lies

On $y - 4x + 3 = 0$

$$4g - f + 3 = 0 \text{ -----(5)}$$

$$(4) - (5)$$

$$\rightarrow 5f + 25 = 0$$

$$f = -5$$

From (5)

$$4f + 8 = 0$$

$$g = -2$$

From (2)

$$-8 - 30 + 13 + c = 0$$

$$c = 25$$

$$r = \sqrt{g^2 + f^2 - c}$$

$$= \sqrt{4 + 25 - 25}$$

$$= 2$$

12b. A circle passes through the pts (0,0) and (0,1) and also touches the circle $x^2 + y^2 = 16$. The radius of the circle is

a) 1

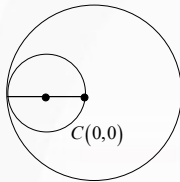
b) 2

c) 3

d) 4

Ans: (b)

Sol:



Given circle $x^2 + y^2 = 16$

$$c(0,0)$$

$$r = 4$$

(0,0) & (0,1) are inside the given circle so the diameter of required circle is the radius of given circle

So $r = 2$

2) Intercepts made by a circle on axes, Equation of chord, Position of a point w.r.t circle

1) Let the length of intercepts on x-axis and y-axis made by the circle

$x^2 + y^2 + ax + 2ay + c = 0, (a < 0)$ be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is equal to

(2021)

a) $\sqrt{11}$

b) $\sqrt{7}$

c) $\sqrt{6}$

d) $\sqrt{10}$

Ans: (c)

For circle $x^2 + y^2 + ax + 2ay + c = 0$

$$x - \text{intercept} = 2\sqrt{g^2 - c}$$

$$\Rightarrow 2\sqrt{\frac{a^2}{4} - c} = 2\sqrt{2}$$

$$\Rightarrow \frac{a^2}{4} - c = 2 \dots \dots \dots (1)$$

$$y - \text{intercept} = 2\sqrt{f^2 - c}$$

$$\Rightarrow 2\sqrt{a^2 - c} = 2\sqrt{5}$$

$$\Rightarrow a^2 - c = 5 \dots \dots \dots (2)$$

Solving (1) and (2), we get

$$a = -2 (a < 0) \text{ and } c = -1$$

So, equation of circle becomes

$$x^2 + y^2 - 2x - 4y - 1 = 0$$

$$\text{or } (x-1)^2 + (y-2)^2 = 6$$

Tangent perpendicular to $x + 2y = 0$ is $2x - y + \lambda = 0$.

$$\text{So, } \frac{|2 - 2 + \lambda|}{\sqrt{5}} = \sqrt{6}$$

$$\Rightarrow \lambda = \pm\sqrt{30}$$

So, equation of tangents are $2x - y \pm \sqrt{30} = 0$

$$\text{Their distance from origin is } \left| \frac{\pm\sqrt{30}}{\sqrt{4+1}} \right| = \sqrt{6}$$

- 1a) The sum of the squares of the length of chord intercepted by the line $x + y = n$, $n \in N$ on the circle $x^2 + y^2 = 4$ is

- a) 11 b) 22 c) 33 d) 44

Ans: (b)

$$\text{Sol: } d = \frac{n}{\sqrt{2}}$$

$$\text{length of chord} = 2\sqrt{4 - \frac{n^2}{2}}$$

$$(Lc)^2 = 4\left(4 - \frac{n^2}{2}\right)$$

$$= 16 - 2n^2$$

$$\text{sum} = 14 + 8 = 22$$

- 1b) The circle passing through origin and making intercepts 6 and +4 on x and Y-axes respectively has centre

(JEE MAIN 2021 SHIFT 1)

- a) (3, +2) b) (-2, 4) c) (8, -4) d) Both 1 & 2

Ans: (a)

$$\text{Sol: Let the circle be } x^2 + y^2 + 2gx + 2fy = 0$$

$$2\sqrt{g^2 - 0} = 6 \Rightarrow |g| = 3, g = \pm 3$$

$$2\sqrt{f^2 - 0} = 4 \Rightarrow f = \pm 2$$

$$C = (-3, -2) \text{ or } (3, 2)$$

- 2) If the line $3x + 4y - 24 = 0$ Intersects the x-axis at A and 4-axis at B then the in centre of triangle OAB, where o is origin is

(JEE MAIN 2019 SHIFT 2)

a) (3, 4)

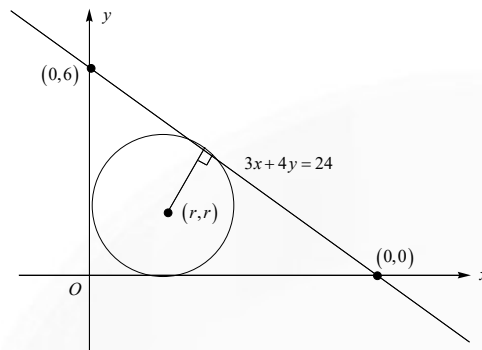
b) (2, 2)

c) (3, 4)

d) (4, 3)

Ans : (b)

Sol:



From the figure,

$$\left| \frac{3r + 4r - 24}{5} \right| = r$$

$$\Rightarrow 7r - 24 = \pm 5r$$

$$\Rightarrow 2r = 24 \text{ or } 12r = 24$$

$$\Rightarrow r = 14, r = 2$$

So, incentre is (2, 2).

2a) If the line $3x + 4y = 48$ intersects the x-axis at A, Y-axis at B. then incentre of triangle OAB is

(JEE MAIN 2019 SHIFT 2)

a) (2, 2)

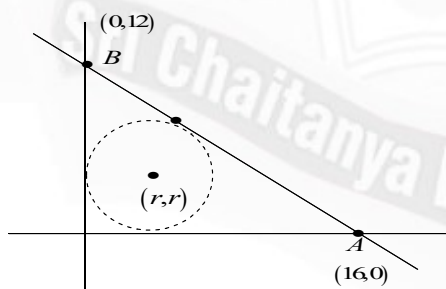
b) $\left(\frac{2}{3}, \frac{2}{3}\right)$

c) $\left(\frac{4}{3}, \frac{4}{3}\right)$

d) (4, 4)

Ans: (d)

Sol:

Let $c(r, r)$ be centre r be radius

Perpendicular distance = radius

$$\left| \frac{3r + 4r - 48}{5} \right| = r$$

$$|7r - 48| = 5r$$

$$7r - 48 = 5r$$

$$\text{Or } 7r - 48 = -5r$$

$$2r \neq 48 \quad 12r = 48$$

$$r \neq 24 \quad r = 4$$

$$\text{In centre} = (4, 4)$$

2b) If $p = (0, 0)$ $Q = (1, 0)$ $R = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ then centre of circle for which the lines PQ, QR, and

RP are Tangents is

(JEE MAIN 2019 SHIFT 1)

a) $\left(\frac{1}{2}, \frac{1}{4}\right)$

b) $\left(\frac{3}{2}, \frac{\sqrt{3}}{4}\right)$

c) $\left(\frac{1}{2}, \frac{1}{2\sqrt{3}}\right)$

d) $\left(\frac{1}{2}, \frac{-1}{\sqrt{3}}\right)$

Ans: (a)

Sol: $PQ = QR = RP = 1$

Δ is equilateral centroid is incentre

$$\text{Incentre} = \left(\frac{0+1+\frac{1}{2}}{3}, \frac{0+0+\frac{\sqrt{3}}{2}}{3} \right)$$

$$= \left(\frac{1}{2}, \frac{1}{2\sqrt{3}} \right)$$

3) The sum of the squares of the lengths of the chord intercepted on the circle $x^2 + y^2 = 16$ by the lines $x + y = n$, $n \in N$ where N is set of all natural numbers is

a) 105

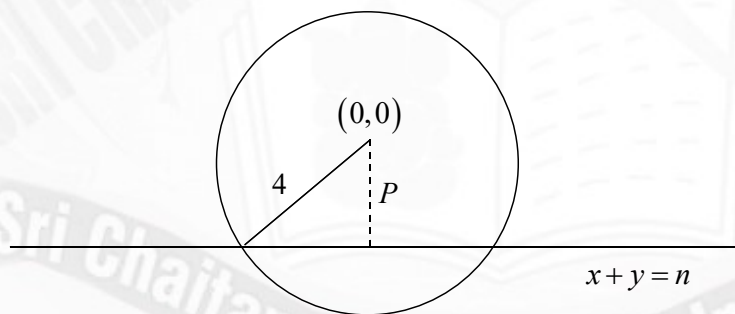
b) 210

c) 320

d) 160

Ans : 2

Sol :



$$p = \frac{n}{\sqrt{2}}$$

Now, $\frac{n}{\sqrt{2}} < 4$

$$\Rightarrow n < 4\sqrt{2}$$

$$\text{Length of intercepts} = 2\sqrt{16 - n^2/2}$$

$$\therefore \text{square of intercepts} = 4 \times \left(16 - \frac{n^2}{2} \right), n \in N$$

$$\begin{aligned}\text{Sum of square of intercepts} &= 4 \times \left(\left(16 - \frac{1}{2} \right) + \left(16 - \frac{4}{2} \right) + \left(16 - \frac{9}{2} \right) + \left(16 - \frac{16}{2} \right) + \left(16 - \frac{25}{2} \right) \right) \\ &= 4 \left(80 - \frac{1}{2} \times 55 \right) = 210\end{aligned}$$

3a) If the circles $x^2 + y^2 - 2x - 2y - 7 = 0$ and $x^2 + y^2 + 4x + 2y + k = 0$ cut orthogonally. If the length of common chord of the circles is $\frac{m}{\sqrt{n}}$ then $m+n$ is **(JEE MAIN 2019 SHIFT 2)**

- a) 22 b) 24 c) 25 d) 26

Ans: (c)

Sol: here $c_1 = (1, 1)$, $c_2 = (-2, -1)$ $r_1 = 3$

Given circles touch each other orthogonally.

$$2gg^1 + 2ff^1 = c + c^1$$

$$\Rightarrow k = 1$$

Equation of common chord is $s - s^1 = 0$

$$\Rightarrow 3x + 2y + 4 = 0$$

$$\text{Now } d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{|3(1) + 2(1) + 4|}{\sqrt{9 + 4}} = \frac{9}{\sqrt{13}}$$

$$\text{Length of common chord} = 2\sqrt{r^2 - d^2}$$

$$= 2\sqrt{9 - \frac{81}{13}}$$

$$\frac{m}{\sqrt{n}} = \frac{12}{\sqrt{13}}$$

$$\therefore m + n = 12 + 13 = 25$$

4) If a variable line $3x + 4y - \lambda = 0$ is such that the two circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ are on its opposite sides then the set of all values of λ is the interval **(JEE MAIN 2019 SHIFT 1)**

- a) [12, 21] b) [2, 17] c) [23, 31] d) [13, 23]

Ans: (a)

Sol: Centre $(c_1) = (1, 1)$

Radius $(r_1) = 1$

Centre $(c_2) = (9, 1)$

Radius $(r_2) = \sqrt{81 + 1 - 78} = 2$

Centres lie on the opposite sides of the line $3x + 4y - \lambda = 0$

$$\Rightarrow (3 + 4 - \lambda)(27 + 4 - \lambda) < 0$$

$$\Rightarrow (\lambda - 7)(\lambda - 31) < 0$$

$$\Rightarrow \lambda \in (7, 31) \text{ ----- (1)}$$

$$d_1 \geq r_1 \text{ and } d_2 \geq r_2$$

$$\left| \frac{3+4-\lambda}{5} \right| \geq 1 \text{ and } \left| \frac{27+4-\lambda}{5} \right| \geq 2$$

$$|\lambda - 7| \geq 5 \text{ and } |\lambda - 31| \geq 10$$

$$\lambda \in (-\infty, 2] \cup [12, \infty) \text{ -----(2)}$$

$$\lambda \in (-\infty, 21] \cup [41, \infty) \text{ -----(3)}$$

From (1), (2) and (3)

$$\lambda \in [12, 21]$$

4a) If the variable line $3x + 4y = \alpha$ lies between the two circles $(x-1)^2 + (y-1)^2 = 1$ and $(x-9)^2 + (y-1)^2 = 4$, without intercepting a chord on either circle, then the sum of all integral values of α is **(JEE MAIN 2019 SHIFT 1)**

a) 145

b) 165

c) 155

d) 75

Ans: (b)

Sol: Let $L \equiv 3x + 4y - \alpha = 0$

$$L(1,1).L(9,1) < 0$$

$$\Rightarrow (3+4-\alpha)(27+4-\alpha) < 0$$

$$\Rightarrow \alpha \in (7, 31) \text{ -----(1)}$$

$$d_1 \geq r_1 \Rightarrow \frac{|3+4-\alpha|}{\sqrt{3^2+4^2}} \geq 1$$

$$\Rightarrow |7-\alpha| \geq 5$$

$$\Rightarrow 7-\alpha \geq 5, 7-\alpha \leq -5$$

$$\Rightarrow \alpha \leq 2, \alpha \geq 12 \text{ -----(2)}$$

$$d_2 \geq r_2 \Rightarrow \alpha \leq 21, \alpha \geq 41 \text{ -----(3)}$$

$$(1) \cap (2) \cap (3) \Rightarrow \alpha \in [12, 21]$$

Sum of all integral values

$$= 12 + 13 + 14 + 15 + 16 + 17 + 18 + 19 + 20 + 21 = 165$$

4b) The values of α for which the point $(\alpha-1, \alpha+1)$ lies in the larger segment of the circle

$x^2 + y^2 - x - y - 6 = 0$ made by the chord whose equation is $x + y - 2 = 0$ are

a) $-1 < \alpha < 1$

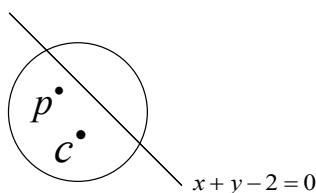
b) $1 < \alpha < \infty$

c) $-\infty < \alpha < -1$

d) $\alpha \leq 1$

Ans: (a)

Sol:



$$S = x^2 + y^2 - x - y - 6 = 0$$

$$\text{centre } (c) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

$$p(\alpha-1, \alpha+1) \text{ lies inside the circle} \Rightarrow s_{11} < 0$$

$$\Rightarrow (\alpha-1)^2 + (\alpha+1)^2 - (\alpha-1) - (\alpha+1) - 6 < 0$$

$$\Rightarrow \alpha^2 + 1 - 2\alpha + \alpha^2 + 1 + 2\alpha - \alpha + 1 - \alpha - 1 - 6 < 0$$

$$\Rightarrow 2\alpha^2 - 2\alpha - 4 < 0$$

$$\Rightarrow \alpha^2 - \alpha - 2 < 0$$

$$(\alpha+1)(\alpha-2) < 0$$

$$-1 < \alpha < 2 \text{ -----(1)}$$

C and P lie on the same side

$$\Rightarrow L\left(\frac{1}{2}, \frac{1}{2}\right) \cdot L(\alpha-1, \alpha+1) > 0$$

$$\Rightarrow \left(\frac{1}{2} + \frac{1}{2} - 2\right)(\alpha-1 + \alpha+1 - 2) > 0$$

$$(-1)(2\alpha-2) > 0$$

$$-2(\alpha-1) > 0$$

$$\alpha-1 < 0$$

$$\alpha < 1 \text{ -----(2)}$$

From (1) & (2)

$$-1 < \alpha < 1$$

- 5) The circle passing through the point $(-1, 0)$ and touching the y-axis at $(0, 2)$ also passes through the point [IIT 2021]

a) $(-3/2, 0)$ b) $(-5/2, 2)$ c) $(-3/2, -5/2)$ d) $(-4, 0)$

KEY: (d)

Solution: Required circle has centre at $(-a, 2)$ with radius 'a'.

$$(x+a)^2 + (y-2)^2 = a^2 \text{ is } x^2 + y^2 + 2ax - 4y + 4 = 0. \text{ It is through } (-1, 0) \rightarrow$$

$$a = 5/2$$

$$\text{So circle equation is } x^2 + y^2 + 5x - 4y + 4 = 0.$$

It is satisfied by the point $(-4, 0)$

- 5A) The circle which touches both the coordinate axes and the line $x + 2y + 2 = 0$ and having minimum radius, has centre at

a) $\left(\frac{3-\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}\right)$ b) $\left(\frac{-3+\sqrt{5}}{2}, \frac{-3+\sqrt{5}}{2}\right)$ c) $\left(\frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}\right)$ d) $\left(\frac{7-\sqrt{5}}{5}, \frac{7+\sqrt{5}}{5}\right)$

KEY: (b)

Solution: $x + 2y = -2 \Rightarrow \frac{x}{-2} + \frac{y}{-1} = 1$, So circle in Q3, has centre $(-a, -a)$ where radius is 'a'

Using tangent condition $r = d$, for the line $x + 2y + 2 = 0$ we get $a^2 - 3a + 1 = 0$

$$\text{Hence } a = \frac{3-\sqrt{5}}{2}$$

CONCEPT: Length of the intercept made by circle on x-axis**FORMULA:** $2\sqrt{g^2 - c}$

- 6) If the circle $x^2 + y^2 - 2gx + 6y - 19c = 0, g, c \in \mathbb{R}$ passes through the point (6,1) and its centre lies on the line $x - 2cy = 8$, then the length of intercept made by the circle on

x-axis is

[2022_27 July - Shift 1]

- a) $\sqrt{11}$ b) 4 c) 3 d) $2\sqrt{23}$

Key: (d)

Sol: Circle is $x^2 + y^2 - 2gx + 6y - 19c = 0$ It passes through (6,1) $\therefore 36 + 1 - 12g + 6 - 19c = 0$

$$12g + 19c = 43 \dots\dots(1)$$

Line $x - 2cy = 8$ passes through Centre $S(g, -3)$

$$\therefore g + 6c = 8$$

By (1) & (2) $g = 2, c = 1$

$$\therefore \text{Circle is } x^2 + y^2 - 4x + 6y - 19 = 0$$

$$\therefore \text{x-intercept} = 2\sqrt{g^2 - c} = 2\sqrt{4 + 19} = 2\sqrt{23}$$

- 6b) If a circle passing through the point (-1,0) and touches the y-axis at (0,2), then the length of the chord of the circle along the x-axis is

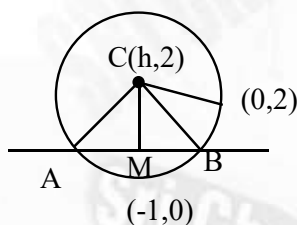
- a) $3/2$ b) 3 c) $5/2$ d) 5

Key: (b)

Sol: Perpendicular drawn from Centre bisect the chord.

Let the Co-ordinates of centre of circle be (h,2)

$$\Rightarrow BC^2 = h^2$$



$$\Rightarrow (h+1)^2 + 2^2 = h^2 \Rightarrow 2h + 5 = 0$$

$$h = -5/2$$

$$AB = 2(AM) = 2\sqrt{AC^2 - CM^2}$$

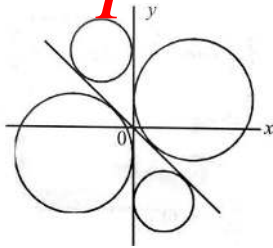
$$= 2\sqrt{\frac{25}{4} - 4} = 2 \cdot \frac{3}{2} = 3$$

- 7) A circle touches both the y-axis and the line $x + y = 0$. Then the locus of its centre is

- a) $y = \sqrt{2}x$ b) $x = \sqrt{2}y$ c) $y^2 - x^2 = 2xy$ d) $x^2 - y^2 = 2xy$

Ans: (d)

Solution:



Let (h,k) is centre of circle

$$\left| \frac{h-k}{\sqrt{2}} \right| = |h|, k^2 - h^2 - 2hk = 0$$

7a. A circle touches x-axis and cuts off constant length $2p$ from y-axis then the locus of its centre is

- a) $x^2 + y^2 = p^2$ b) $x^2 - y^2 = p^2$ c) $y^2 - x^2 = p^2$ d) $y^2 + x^2 = 4p^2$

Ans: (c)

Sol:

Let the equation of circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

Circle touches x-axis

$$(i.e) g^2 = c$$

Length of the intercept made by the circle

On y-axis is $2p$

$$(i.e) 2\sqrt{f^2 - c} = 2p$$

$$\Rightarrow f^2 - c = p^2$$

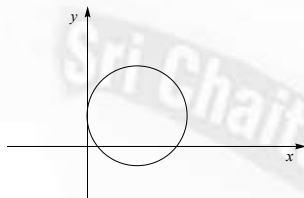
$$\Rightarrow f^2 - g^2 = p^2$$

7b. A circle is drawn to cut a chord of length $2a$ unit along x-axis and to touch the y-axis. The locus of the centre of the circle is

- a) $x^2 - y^2 = a^2$ b) $x^2 + y^2 = a^2$ c) $x^2 - y^2 = 2a^2$ d) $y^2 - x^2 = 2a^2$

Ans: (a)

Sol:



Let the equation of circle

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Circle touches y-axis

$$(i.e) f^2 = c$$

Length of the intercept made by the circle on x-axis is

$$2\sqrt{g^2 - c} = 2a$$

$$\Rightarrow g^2 - c = a^2$$

$$\Rightarrow g^2 - f^2 = a^2$$

Locus of the centre of the circle is

$$x^2 - y^2 = a^2$$

3. Tangent line equation :

(Position of a line w.r.t a circle in Circles, Condition for tangency in circles, Perpendicular distance from centre to the line is radius in circles, Equation of tangent in parametric form in circles, Point of contact of tangent in circles, Number of tangents from an external point in circles, Sum of slopes of tangents in circles, Equation of pair of tangents drawn from external point to circle in circles, Product of slopes of tangents in circles, Angle between pair of tangents drawn from external point to circle in circles, Director circle of a circle in circles)

- 1). If one of the diameters of the circle, given by the equation $x^2 + y^2 - 4x + 6y - 12 = 0$ is a chord of a circle S, whose centre is at $(-3, 2)$ then the radius of S is

(JEE MAIN 2020 SHIFT 1)

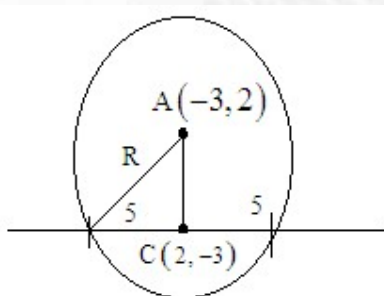
a) $5\sqrt{3}$

b) 5

c) 10

d) $5\sqrt{2}$

Ans : (a)



Sol :

Given circle $x^2 + y^2 - 4x + 6y - 12 = 0$

$$C = (2, -3)$$

$$r = \sqrt{4 + 9 + 12} = 5$$

$$AC = \sqrt{(2+3)^2 + (-3-2)^2} = \sqrt{50}$$

$$\begin{aligned} \text{from } \Delta ACB \quad AB(R) &= \sqrt{AC^2 + BC^2} \\ &= \sqrt{50 + 25} = 5\sqrt{3} \end{aligned}$$

- 1a. The line $3x - 4y = K$ will cut the circle $x^2 + y^2 - 4x - 8y - 5 = 0$ at distinct points if

(JEE MAIN 2020 SHIFT 2)

a) $-35 < K < 35$ b) $-35 < K < 15$ c) $-15 < K < 15$ d) $15 < K < 35$

Ans : B

Sol : Given $x^2 + y^2 - 4x - 8y - 5 = 0 \Rightarrow \text{Centre} = C(2, 4)$

$$\text{Radius (r)} = \sqrt{4 + 16 + 5} = 5$$

$$\text{Condition : } cp < 5 \Rightarrow \left| \frac{6-16-k}{5} \right| < 5 \Rightarrow -5 < \frac{-10-K}{5} < 5$$

$$\Rightarrow -35 < K < 15$$

- 1b. The straight line $x - y + 1 = 0$ meets the circle $x^2 + y^2 - 6x - 8y = 0$ at A and B. Then the equation of the circle of which AB is a diameter is _____ (JEE MAIN 2020 SHIFT 2)

a) $x^2 + y^2 - 2y - 6 = 0$

b) $x^2 + y^2 + 2y - 6 = 0$

c) $2(x^2 + y^2) + 2y - 6 = 0$

d) $3(x^2 + y^2) + 2y - 6 = 0$

Ans : (a)

Sol : Equation of circle $S + \lambda L = 0$

$$\Rightarrow x^2 + y^2 - 6x - 8y + \lambda(x + y - 1) = 0$$

$$\text{Centre} = \left(3 - \frac{\lambda}{2}, 4 - \frac{\lambda}{2} \right)$$

$$\text{Lies on } x + y - 1 = 0$$

$$\Rightarrow 3 - \frac{\lambda}{2} + 4 - \frac{\lambda}{2} - 1 = 0 \Rightarrow \lambda = 6$$

$$\text{Req. circle } x^2 + y^2 - 2y - 6 = 0$$

2. The radius of the circle which touches y-axis at (0,0) and passes through the point (b,c) is (2020)

a) $\frac{b^2 + c^2}{2|b|}$

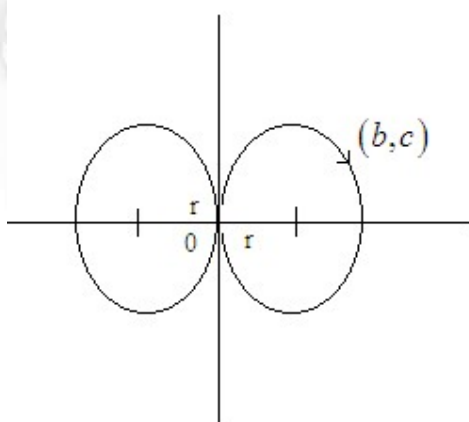
b) $\frac{b^2 + c^2}{2|c|}$

c) $\frac{b^2 + c^2}{2}$

d) $\frac{|b|}{2(b^2 + c^2)}$

Ans : (a)

Sol : Let r be the radius of circle then centre c (r,0)



$$\text{Equation of circle } (x - r)^2 + (y - 0)^2 = r^2$$

$$\Rightarrow (x - r)^2 + y^2 = r^2$$

Passes through (b,c)

$$\Rightarrow (b-r)^2 + c^2 = r^2$$

$$\Rightarrow b^2 + r^2 - 2br + c^2 = r^2$$

$$\Rightarrow b^2 + r^2 = 2br \Rightarrow r = \frac{b^2 + c^2}{2|b|}$$

2a) If the straight line $y = mx$ is outside the circle $x^2 + y^2 - 20y + 90 = 0$ then

(JEE MAIN 2020 SHIFT 1)

- a) $m > 3$ b) $m < 3$ c) $|m| > 3$ d) $|m| < 3$

Ans : (d)

Sol : $y = mx$ is outside the given circle then distance from centre of circle $>$ radius of circle

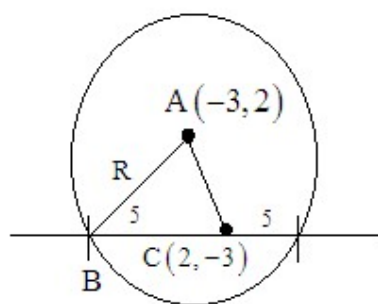
$$\Rightarrow \left| \frac{10}{\sqrt{1+m^2}} \right| > \sqrt{10} \Rightarrow 1+m^2 < 10 \Rightarrow |m| < 3$$

2b) If one of the diameters of the circle, given by the equation $x^2 + y^2 - 4x + 6y - 12 = 0$ is a chord of a circle S, whose centre is at $(-3, 2)$ then the radius of S is

(2020)

- a) $5\sqrt{3}$ b) 5 c) 10 d) $5\sqrt{2}$

Ans : (a)



Sol :

Given circle $x^2 + y^2 - 4x + 6y - 12 = 0$

$$C = (2, -3)$$

$$r = \sqrt{4+9+12} = 5$$

$$AC = \sqrt{(2+3)^2 + (-3-2)^2} = \sqrt{50}$$

$$\text{from } \Delta ACB \quad AB(R) = \sqrt{AC^2 + BC^2} = \sqrt{50 + 25} = 5\sqrt{3}$$

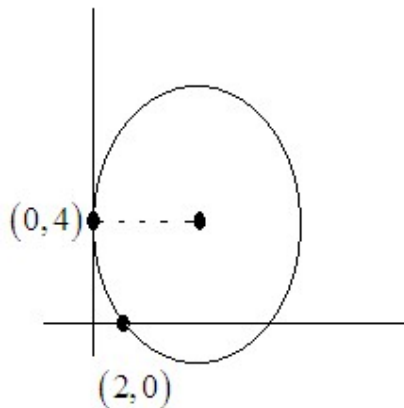
3. If a circle touches y-axis at $(0, 4)$ and passes through $(2, 0)$ then which of the following can-not be the tangent to the circle. (2020)

- a) $4x - 3y + 17 = 0$ b) $3x - 4y - 24 = 0$

c) $3x + 4y - 6 = 0$

d) $4x + 3y - 8 = 0$

Ans : (d)



Sol :

Equation of family of circle

$$(x-0)^2 + (y-4)^2 + \lambda x = 0$$

 $\Rightarrow$ passes (2,0)

$$4 + 16 + 2\lambda = 0 \Rightarrow \lambda = -10$$

$$x^2 + y^2 - 10x - 8y + 16 = 0$$

$$\text{Centre } (5, 4) \quad R = \sqrt{25 + 16 - 16} = 5$$

3a) If the circle touches y-axis at (0,3) and making intercept of length $2\sqrt{7}$ on x-axis then which of the following is a tangent to the circle is (JEE MAIN 2020 SHIFT 2)

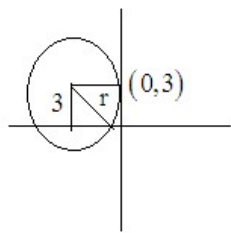
a) $x + 2y - 2 \pm 4\sqrt{5} = 0$

b) $x + 2y + 2 \pm 4\sqrt{5} = 0$

c) $x + 2y - 4 \pm 2\sqrt{5} = 0$

d) $x + 2y \mp 2\sqrt{5} + 4 = 0$

Ans : (a)

Sol : Equation of tangent at x_1, y_1 is $S_1 = 0$ 

$$r^2 = 9 + 7 = 16$$

$$r = 4$$

$$\therefore \text{Centres are } (\pm 4, 3)$$

$$\text{Equation of circle } (x \pm 4)^2 + (y - 3)^2 = 16$$

3b) If the circle touches x-axis at (5,0) and making an intercept of length 24 on y-axis then the length of the chord of the circle intercepted by the line $4x - 3y + 9 = 0$ is

(JEE MAIN 2020 SHIFT 2)

a) $2\sqrt{185}$

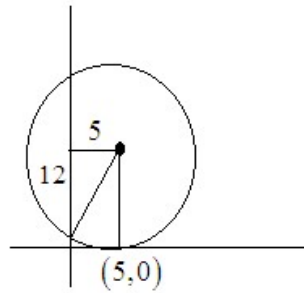
b) $\sqrt{85}$

c) $2\sqrt{165}$

d) $2\sqrt{65}$

Ans : (c)

Sol :



$$r^2 = 5^2 + 12^2 = 169$$

$$r = 13$$

$$\therefore C(5, 13) \quad (\text{or}) \quad (5, -13)$$

$$\text{Equation of circle } (x - 5)^2 + (y - 13)^2 = 13^2$$

(or)

$$(x - 5)^2 + (y + 13)^2 = 13^2$$

$$\text{Length of chord} = 2\sqrt{r^2 - d^2}$$

4) If a line, $y = mx + c$ is a tangent to the circle, $(x - 3)^2 + y^2 = 1$ and it is perpendicular to a line L_1 , where L_1 is the tangent to the circle, $x^2 + y^2 = 1$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

(JEE MAIN 2020 SHIFT 2)

a) $c^2 + 6c + 7 = 0$

b) $c^2 - 6c + 7 = 0$

c) $c^2 + 6c - 7 = 0$

d) $c^2 - 6c - 7 = 0$

Ans : (a)

Sol : Circle $x^2 + y^2 = 1$

Slope of the line joining point $P\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and centre $O(0, 0)$ is 1. So, slope of tangent at a point P to the circle is -1 .

Thus slope of L_1 is -1 . So the line $y = -x + c$ is tangent to $(x - 3)^2 + y^2 = 1$.

Distance of centre $(3, 0)$ from the line is equal to radius of the circle

$$\therefore \left| \frac{c + 3}{\sqrt{2}} \right| = 1$$

$$\Rightarrow c^2 + 6c + 7 = 0$$

4a) If a line $y = m\lambda + c$ is a tangent to the circle, $x^2 + y^2 - 6x + 8 = 0$ and it is perpendicular to a line L_1 where L_1 is the tangent to the circle $x^2 + y^2 = 2$ at the point (1,1) then find minimum value of C is (JEE MAIN 2020 SHIFT 1)

- a) 2 b) -8 c) -2 d) -4

Ans : (c)

Sol : Centre (3,0) radius $r = \sqrt{9-8} = 1$

Tangent at (1,1) is $S_1 = 0 \Rightarrow x + y - 2 = 0$

$$m = 1, r = d \Rightarrow 1 = \frac{3m + c}{\sqrt{1 + m^2}} \Rightarrow (c + 1)^2 = 2 \Rightarrow c^2 + 2c - 1 = 0$$

$$\text{Min. value} = \frac{4ac - b^2}{4a} = \frac{-4 - 4}{4} = -2$$

4b) If a line $y = mx + c$ is a tangent to the circle $x^2 + y^2 = 16$ and it is parallel to a line L_1 where L_1 is a tangent to the circle $x^2 + y^2 = 8$ at the point (2,2) then which of the following is not true (2020)

- a) $x + y - 4\sqrt{2} = 0$ b) $x + y + 4\sqrt{2} = 0$
c) $x + y - 4 = 0$ d) $x + y \pm 4\sqrt{2} = 0$

Ans : (d)

Sol : Centre (0,0) radius $r = 4$

Tangent at (2,2) is $S_1 = 0 \Rightarrow x + y - 4 = 0 \Rightarrow m = -1$

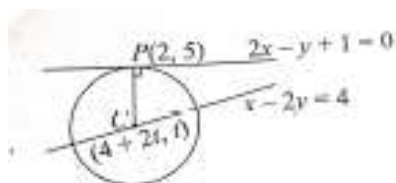
$$r = d \Rightarrow 4 = \frac{|C|}{\sqrt{1 + m^2}} \Rightarrow |C| = 4\sqrt{2}$$

5) The line $2x - y + 1 = 0$ is a tangent to the circle at the point (2,5) and the centre of the circle lies on $x - 2y = 4$. Then, the radius of the circle is: (JEE MAIN 2021 SHIFT 1)

- a) $3\sqrt{5}$ b) $5\sqrt{3}$ c) $5\sqrt{4}$ d) $4\sqrt{5}$

Ans: 1

Sol:



Centre C lies on the line $x - 2y = 4$.

$$\text{Let } C = (4 + 2t, t)$$

CP is perpendicular to the tangents line.

$$SO, m_1 m_2 = -1$$

$$\text{or } \left(\frac{t-5}{4+2t-2} \right) (2) = -1$$

$$\Rightarrow t - 5 = -1 - t \Rightarrow t = 2$$

$$\Rightarrow C = (8, 2)$$

$$\text{So, radius} = \sqrt{(8-2)^2 + (2-5)^2} = 3\sqrt{5}$$

5a) Equation of one of the tangents from (1,1) to a circle with its centre at (3,0) is $3x + y - 4 = 0$.

The equation of other tangent is

(JEE MAIN 2021 SHIFT 2)

a) $5x - y - 4 = 0$

b) $x - 3y + 2 = 0$

c) $3y + x - 4 = 0$

d) $3x - y - 2 = 0$

Ans: (d)

Sol: Let the tangent be

$$y - 1 = m(x - 1)$$

$$mx - y + (1 - m) = 0$$

by using $r = d$

$$\frac{|3m + 1 - m|}{\sqrt{m^2 + 1}} = \frac{|5|}{\sqrt{10}}$$

$$\frac{|2m + 1|}{\sqrt{m^2 + 1}} = \frac{5}{\sqrt{10}}$$

$$10(2m + 1)^2 = 25(m^2 + 1)$$

$$8m^2 + 2 + 8m = 5m^2 + 5$$

$$3m^2 + 8m - 3 = 0$$

$$m = -3, \frac{1}{3}$$

tangents are $y - 1 = -3(x - 1)$

$$3x + y - 4 = 0 \text{ (or)}$$

$$y - 1 = \frac{1}{3}(x - 1)$$

$$x - 3y + 2 = 0$$

5b) Equation of circle touching the line $|x - 1| + |y - 4| = 6$ is

(JEE MAIN 2021 SHIFT 2)

a) $x^2 + y^2 - 2x - 8y - 1 = 0$

b) $x^2 + y^2 - 2x - 8y - 18 = 0$

c) $x^2 + y^2 - 2x - 8y - 17 = 0$

d) $x^2 + y^2 = 4$

Ans: (a)

Sol: $C = (1, 4)$

consider one line $x - 1 + y - 4 = 6$

$$x + y - 11 = 0$$

$$r = d = \frac{|1+4-11|}{\sqrt{2}} = 3\sqrt{2}$$

$$\text{equation of circle } (x-1)^2 + (y-4)^2 = 18$$

$$= x^2 + y^2 - 2x - 8y - 1 = 0$$

- 6) Two tangents are drawn from a point P to the circle $x^2 + y^2 - 2x - 4y + 4 = 0$, such that the angle between these tangents is $\tan^{-1}\left(\frac{12}{5}\right)$, where $\tan^{-1}\left(\frac{12}{5}\right) \in (0, \pi)$. If the centre of the circle is denoted by C and these tangents touch the circle at points A and B, then the ratio of the areas of ΔPAB and ΔCAB is :

(JEE MAIN 2021 SHIFT 1)

a) 11 : 4

b) 9 : 4

c) 3 : 1

d) 2 : 1

Ans: (b)

Sol:



$$\frac{\text{Area of } \Delta PAB}{\text{Area of } \Delta CAB} = \frac{\frac{1}{2} \cdot PA \cdot PB \cdot \sin \theta}{\frac{1}{2} \cdot 1 \cdot 1 \cdot \sin(\pi - \theta)} = PA^2$$

$$\text{From } \Delta CAP, PA = \cot \frac{\theta}{2}$$

$$\text{So, } PA^2 = \cot^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{1 - \cos \theta} = \frac{1 + \frac{5}{13}}{1 - \frac{5}{13}} = \frac{9}{4}$$

- 6a) The area of the triangle formed by the tangents from the point (4,3) to the circle $x^2 + y^2 = 9$ and the line joining their points of contact is

a) $\frac{192}{25}$ sq units

b) 192 sq units

c) $\frac{192}{3}$ sq units

d) $\frac{192}{49}$ sq units

Ans : (a)

Sol: $x^2 + y^2 = 9, c = (0, 0), r = 3$

Area of triangle formed by the tangents at (h, k) to the circle $x^2 + y^2 = a^2$ and their point of

contact is a $\frac{(h^2 + k^2 - a^2)^{\frac{3}{2}}}{h^2 + k^2}$

$$a = 3, h = 4, k = 3$$

$$\text{area} = \frac{3(16 + 9 - 9)^{\frac{3}{2}}}{16 + 9} = \frac{192}{25} \text{ sq units}$$

6b) The area of the triangle formed by the tangent, normal at $(1, \sqrt{3})$ to the circle $x^2 + y^2 = 4$ and x-axis is

(JEE MAIN 2021 SHIFT 1)

a) $4\sqrt{3}$

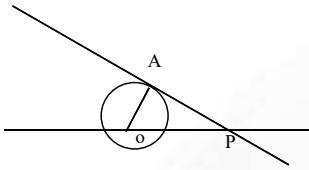
b) $\frac{7}{2}\sqrt{3}$

c) $2\sqrt{3}$

d) $\frac{\sqrt{3}}{2}$

Ans : (c)

Sol:



$$x^2 + y^2 = 4, C(0,0), r=2, A=(1, \sqrt{3})$$

Equation of the tangent at $(1, \sqrt{3})$ is $x + \sqrt{3}y = 4$

Tangent cuts x-axis at $P \Rightarrow P = (4, 0), AP = \sqrt{12}$

$$\text{Area of } \triangle OAP = \frac{1}{2}(OA)(AP) = \frac{1}{2} \times 2 \times \sqrt{12} = 2\sqrt{3}.$$

7) Let the tangents to the circle $x^2 + y^2 = 25$ at the point $R(3, 4)$ meet x-axis and y-axis at point P and Q, respectively. If r is the radius of the circle passing through the origin O and having centre at the incentre of the triangle OPQ, then r^2 is equal to:

(JEE MAIN 2021 SHIFT 1)

a) $\frac{529}{64}$

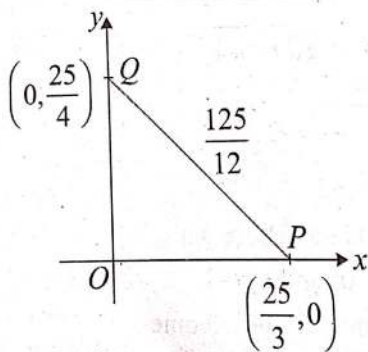
b) $\frac{125}{72}$

c) $\frac{625}{72}$

d) $\frac{585}{66}$

Ans: (c)

Sol: Tangent to circle $x^2 + y^2 = 25$ at $R(3, 4)$ is $3x + 4y = 25$



$$OP + OQ + PQ = 25$$

$$\text{So, Incentre} = \left(\frac{\frac{25}{4} \times \frac{25}{3}}{\frac{25}{4} + \frac{25}{3}}, \frac{\frac{25}{4} \times \frac{25}{3}}{\frac{25}{4} + \frac{25}{3}} \right) = \left(\frac{25}{12}, \frac{25}{12} \right)$$

$$\therefore r^2 = 2 \left(\frac{25}{12} \right)^2 = 2 \times \frac{625}{144} = \frac{625}{72}$$

7a) Tangents drawn from the point $P(1, 8)$ to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at A and B. the equation of circum circle of $\triangle PAB$ is

a) $x^2 + y^2 + 4x - 6y + 19 = 0$

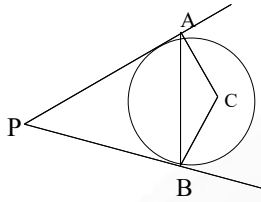
b) $x^2 + y^2 - 4x - 10y + 19 = 0$

c) $x^2 + y^2 - 2x + 6y - 20 = 0$

d) $x^2 + y^2 - 6x - 4y + 19 = 0$

Ans : (b)

Sol:



$$x^2 + y^2 - 6x - 4y - 11 = 0, C = (3, 2)$$

CA, CB are $\perp$ to PA, PB.

CP is the diameter of the required circle.

Then its equation is $(x-3)(x-1) + (y-2)(y-8) = 0$

$$x^2 + y^2 - 4x - 10y + 19 = 0$$

- 7b) The line $2x - y + 1 = 0$ is tangent to the circle at the point $(2, 5)$ and the center of the circle lies on $x - 2y = 4$. The radius of the circle is

a) $3\sqrt{5}$

b) $5\sqrt{3}$

c) $2\sqrt{5}$

d) $5\sqrt{2}$

Sol: $2x - y + 1 = 0$ is a tangent slope = 2 Let $A = (2, 5)$

$$\text{Slope of } CA = -\frac{1}{2}$$

$$\text{The equation of CA is, } y - 5 = -\frac{1}{2}(x - 2) \Rightarrow x + 2y - 12 = 0$$

$$\text{on solving } x + 2y - 12 = 0, x - 2y = 4, \text{ we get } C(8, 2)$$

$$\text{hence } r = CA = \sqrt{(8-2)^2 + (2-5)^2} = \sqrt{45} = 3\sqrt{5}$$

- 8) Two tangents are drawn from the point $P(-1, 1)$ to the circle $x^2 + y^2 - 2x - 6y + 6 = 0$. If these tangents touch the circle at point A and B, if D is a point on the circle such that length of the segments AB and AD are equal, then the area of the triangle ABD is equal to

(JEE MAIN 2021 SHIFT 1)

a) $(3\sqrt{2} + 2)$

b) $(3\sqrt{2} - 2)$

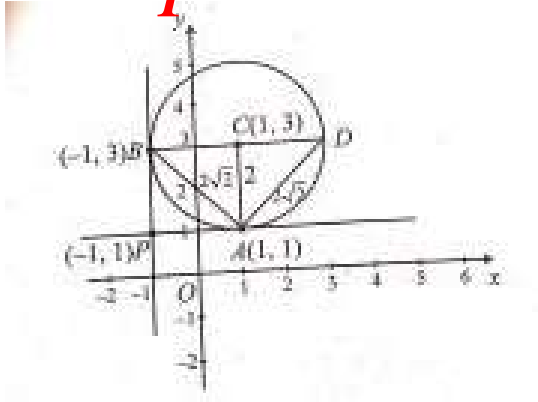
c) 4

d) 2

Ans: (c)

Sol: Equation of circle is

$$(x-1)^2 + (y-3)^2 = 4$$



from the figure $BD = 4$ and

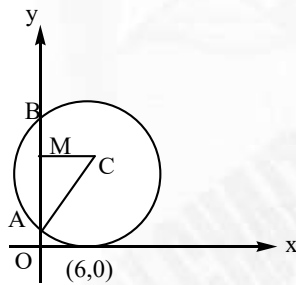
$$\text{Area of triangle } ABD = \frac{1}{2} \times 2 \times 4 = 4 \text{ sq. units}$$

8a) Consider a circle c which touches the x -axis at $(6,0)$ and cuts off an intercept $2\sqrt{3}$ on the y -axis then the coordinates at centre at 'c' is

- a) $(6, 4\sqrt{3})$ b) $(6, -4\sqrt{3})$ c) both 1 and 2 d) none

Ans : (c)

Sol:



$$\text{Let } c = (6, \pm\beta)$$

$$AB = 2l = 2\sqrt{3}$$

$$\beta = \sqrt{a^2 + l^2} = \sqrt{36 + 12}$$

$$= \sqrt{48} = 4\sqrt{3}$$

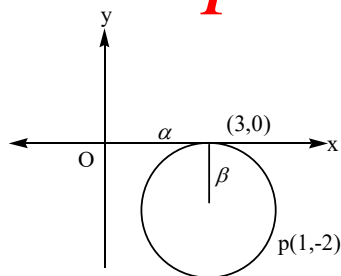
$$\therefore \text{centre} = (6, \pm 4\sqrt{3})$$

8b) The circle passing through $(1, -2)$ and touching the x -axis at $(3,0)$. also passes through the point is

- a) $(-5, 2)$ b) $(2, -5)$ c) $(5, -2)$ 4) $(-2, 5)$

Ans : (c)

Sol:



centre $c = (3, \beta)$

radius $r = |\beta|$

equation of circle $(x-3)^2 + (y-\beta)^2 = \beta^2$

$$\Rightarrow (1-3)^2 + (-2-\beta)^2 = \beta^2$$

$$4 + 4 + \beta^2 + 4\beta = \beta^2 \Rightarrow 4\beta = -8$$

$$\beta = -2$$

equation of circle $(x-3)^2 + (y+2)^2 = 4$

it is also passes through $(5, -2)$

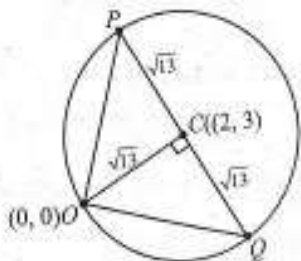
- 9) Let P and Q be two distinct points on a circle which has centre at $C(2,3)$ and which passes through origin O. If OC is perpendicular to both the line segment CP and CQ then the set $\{P, Q\}$ is equal to

(JEE MAIN 2021 SHIFT 1)

- a) $\{(-1,5), (5,1)\}$ b) $\{(2+2\sqrt{2}, 3-\sqrt{5}), (2-2\sqrt{2}, 3+\sqrt{5})\}$
 c) $\{(2+2\sqrt{2}, 3+\sqrt{5}), (2-2\sqrt{2}, 3-\sqrt{5})\}$ d) $\{(4,0), (0,6)\}$

Ans: (a)

Sol:



From the figure slope of PQ is $\tan \theta = -\frac{2}{3}$

Using symmetric form of straight line

$$P, Q = (2 \pm \sqrt{13} \cos \theta, 3 \pm \sqrt{13} \sin \theta)$$

$$= \left(2 \pm \sqrt{13} \left(-\frac{3}{\sqrt{13}} \right), 3 \pm \sqrt{13} \left(\frac{2}{\sqrt{13}} \right) \right)$$

$$= (-1, 5) \text{ and } (5, 1)$$

- 9a) If r_1 and r_2 are radii of the smallest and the largest circles, respectively, which pass through $(5,6)$ and touch the circle $(x-2)^2 + y^2 = 4$ then $r_1 r_2$ is

- a) $\frac{4}{41}$ b) $\frac{41}{4}$ c) $\frac{5}{41}$ d) $\frac{41}{6}$

Ans : (b)

Sol: the given circle is

$$(x-2)^2 + y^2 = 4$$

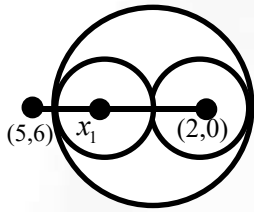
$$c_1 = (2,0) \text{ and } c_2 = (5,6), r = 2$$

$$c_1 c_2 = 3\sqrt{5}$$

$$r_1 = \frac{3\sqrt{5} - 2}{2}$$

$$r_2 = \frac{3\sqrt{5} + 2}{2}$$

$$r_1 r_2 = \frac{41}{4}$$



9b) The range of values of r for which the point $\left(-5 + \frac{r}{\sqrt{2}}, -3 + \frac{r}{\sqrt{2}}\right)$ is an interior point of the

major segment of the circle $x^2 + y^2 = 16$, cut-off by the line $x + y = 2$ is

- a) $(-\infty, 5\sqrt{2})$ b) $(4\sqrt{2} - \sqrt{14}, 5\sqrt{2})$ c) $(4\sqrt{2} - \sqrt{14}, 4\sqrt{2} + \sqrt{14})$ d) none

Ans: (b)

$$\text{Sol: } \left(-5 + \frac{r}{\sqrt{2}}, -3 + \frac{r}{\sqrt{2}}\right)^2 - 16 < 0$$

$$r^2 - 8\sqrt{2}r + 18 < 0$$

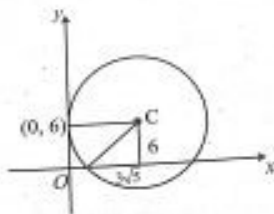
$$4\sqrt{2} - \sqrt{14} < r < 4\sqrt{2} + \sqrt{14}$$

10) Consider a circle C which touches the y -axis at $(0,6)$ and cuts off an intercept $6\sqrt{5}$ on the x -axis. Then the radius of the circle C is equal to (JEE MAIN 2021 SHIFT 2)

- a) 8 b) $\sqrt{82}$ c) $\sqrt{53}$ d) 9

Ans: (d)

Sol:



$$\text{from the figure, radius, } r = \sqrt{6^2 + (3\sqrt{5})^2} = \sqrt{36 + 45} = 9$$

10a) A Straight line with slope '2' and y -intercept '5' touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at point Q . then the coordinates at 'Q' are

a) (6, 11)

b) (-9, -13)

c) (-10, -15)

d) (-6, -7)

Ans : (d)

Sol:

$$y_1 = 2x_1 + 5 \text{ and } \frac{y_1 + 6}{x_1 + 8} \times 2 = -1$$

$$\Rightarrow x_1 = -6 \text{ and } y_1 = -7$$

10b) The line $y=x$ touches a circle at the points (1,1) .if the circle also passes through the point (1,-3), then its radius is

a) $\sqrt{2}$

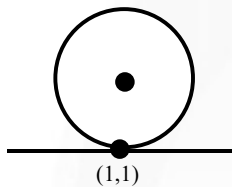
b) $2\sqrt{2}$

c) $3\sqrt{2}$

d) 2

Ans : (b)

Sol:



Equation of circle

$$(x-1)^2 + (y-1)^2 + \lambda(x-y) = 0$$

$$\Rightarrow \therefore \text{it passes through } (1, -3)$$

$$0 + 16 + \lambda(4) = 0$$

$$\Rightarrow \lambda = -4$$

equation of circle

$$(x-1)^2 + (y-1)^2 - 4(x-y) = 0$$

$$x^2 + y^2 - 6x + 2y + 2 = 0$$

$$r = \sqrt{9+1-2} = 2\sqrt{2}$$

11) If a tangent to the circle $x^2 + y^2 = 1$ intersects coordinate axes at distinct points P and Q, then the locus of the midpoint of PQ is

(JEE MAIN 2019 SHIFT 2)

a) $x^2 + y^2 - 4x^2y^2 = 0$

b) $x^2 + y^2 - 16x^2y^2 = 0$

c) $x^2 + y^2 - 2x^2y^2 = 0$

d) $x^2 + y^2 - 2xy = 0$

Ans : (a)

Sol : Let mid point (x_1, y_1)

Equation :

$$\frac{x}{x_1} + \frac{y}{y_1} = 2 \Rightarrow y_1x + x_1y = 2x_1y_1$$

It is tangent to $x^2 + y^2 = 1$

$$\Rightarrow r = d$$

$$1 = \frac{|-2x_1y_1|}{\sqrt{x_1^2 + y_1^2}} \Rightarrow x_1^2 + y_1^2 = 4x_1^2y_1^2$$

$$\text{Locus } x^2 + y^2 - 4x^2y^2 = 0.$$

11a) The equation of the circle with center at (4,3) and touches the line $5x - 12y - 10 = 0$ is

- a) $x^2 + y^2 - 4x - 6y + 4 = 0$ b) $x^2 + y^2 + 6x - 8y + 16 = 0$
 c) $x^2 + y^2 - 8x - 6y + 21 = 0$ d) $x^2 + y^2 - 24x - 10y + 144 = 0$

Ans: (c)

Sol: Equation of circle with center (4,3) and radius r is $(x-4)^2 + (y-3)^2 = r^2$

$5x - 12y - 10 = 0$ is tangent to the circle

$$r = d \Rightarrow r = \frac{|20 - 36 - 10|}{\sqrt{25 + 144}} = \frac{26}{13} = 2$$

$\Rightarrow$ equation of circle $x^2 + y^2 - 8x - 6y + 25 = 4$

$$x^2 + y^2 - 8x - 6y + 21 = 0$$

11 b) The locus of the point (l, m) , if the line $lx + my = 1$ touches the circle $x^2 + y^2 = a^2$

- a) $x^2 + y^2 = 2a^2$ b) $2x^2 + 2y^2 = a^2$ c) $a^2(x^2 + y^2) = 1$ d) $a^2(x^2 + y^2) = 2$

Ans: (c)

Solution: $lx + my = 1$ is tangent to the circle $x^2 + y^2 = a^2$

$$\therefore r = d$$

$$a = \frac{|-1|}{\sqrt{l^2 + m^2}} \Rightarrow l^2 + m^2 = \frac{1}{a^2}$$

Locus of (l, m) is $x^2 + y^2 = \frac{1}{a^2}$

$$\Rightarrow a^2(x^2 + y^2) = 1$$

12) A square is inscribed in the circle $x^2 + y^2 - 6x + 8y - 103 = 0$ with its sides parallel to the coordinate axes. Then the distance of the vertex of this square which is nearest to the origin is
(JEE MAIN 2019 SHIFT 1)

- a) 13 b) $\sqrt{137}$ c) 6 d) $\sqrt{41}$

Ans: (d)

Sol: Centre = (3, -4)

Radius $(r) = \sqrt{9 + 16 + 103} = 8\sqrt{2}$ since inscribed square has sides parallel to the coordinate axes, diagonals have slopes ± 1 . $\Rightarrow \theta = 45^\circ$ (or) 135° Coordinates of vertices are

$$(x_1 \pm r \cos \theta, y_1 \pm r \sin \theta) (3 \pm 8\sqrt{2} \cos 45^\circ, -4 \pm 8\sqrt{2} \sin 45^\circ)$$

$$\left(3 \pm 8\sqrt{2} \left(\frac{1}{\sqrt{2}} \right), -4 \pm 8\sqrt{2} \left(\frac{1}{\sqrt{2}} \right) \right)$$

$$A(11, 4), B(-5, -12)$$

$$\text{and } (3 \pm 8\sqrt{2} \cos 135^\circ, -4 \pm 8\sqrt{2} \sin 135^\circ)$$

$$\left(3 \pm 8\sqrt{2} \left(\frac{-1}{\sqrt{2}} \right), -4 \pm 8\sqrt{2} \left(\frac{1}{\sqrt{2}} \right) \right)$$

$$(3 \mp 8, -4 \pm 8)$$

$$C(-5, 4), D(11, -12)$$

$$OA = \sqrt{137}, OB = 13, OC = \sqrt{41}, OD = \sqrt{265}$$

12a) A square is inscribed in the circle $x^2 + y^2 - 6x - 2y - 8 = 0$. If its sides are perpendicular to the coordinate axes, then one vertex of the square is

a) $(3 + 3\sqrt{2}, 1)$ b) $(3 - 2\sqrt{2}, 1)$ c) $(6, -2)$ d) $(6, -4)$

Ans: (c)

Sol: Centre (c) = (3, 1) radius(r) = $\sqrt{9+1+8} = 3\sqrt{2}$

When $\theta = 45^\circ$

$$(x_1 \pm r \cos \theta, y_1 \pm r \sin \theta)$$

$$= \left(3 \pm 3\sqrt{2} \left(\frac{1}{\sqrt{2}} \right), 1 \pm 3\sqrt{2} \left(\frac{1}{\sqrt{2}} \right) \right)$$

$$= (6, 4), (0, -2)$$

When $\theta = 135^\circ$

$$(x_1 \pm r \cos \theta, y_1 \pm r \sin \theta)$$

$$= \left(3 \pm 3\sqrt{2} \left(\frac{-1}{\sqrt{2}} \right), 1 \pm 3\sqrt{2} \left(\frac{1}{\sqrt{2}} \right) \right)$$

$$= (3 \mp 3, 1 \pm 3)$$

$$= (0, 4), (6, -2)$$

12 b) A square is inscribed in the circle $x^2 + y^2 - 2x + 4y - 93 = 0$ with its sides parallel to the coordinate axes. The distance from origin to one of the vertex is

a) $\sqrt{69}$ b) $\sqrt{79}$ c) $\sqrt{89}$ d) $\sqrt{99}$

Ans: (c)

Sol: $C = (1, -2), r = \sqrt{1+4+93} = \sqrt{98} = 7\sqrt{2}$

When $\theta = 45^\circ$

$$(x_1 + r \cos \theta, y_1 \pm r \sin \theta)$$

$$= \left(1 \pm 7\sqrt{2} \left(\frac{1}{\sqrt{2}} \right), -2 \pm 7\sqrt{2} \left(\frac{1}{2} \right) \right)$$

$$= (1 \pm 7, -2 \pm 7)$$

$$= (8, 5), (-6, -9)$$

Distance from origin to (8, 5)

$$= \sqrt{8^2 + 5^2}$$

$$= \sqrt{89}$$

13) Let $y = x + 2$, $4y = 3x + 6$ and $3y = 4x + 1$ be three tangents lines to the circle

$(x-h)^2 + (y-k)^2 = r^2$. Then $h+k$ is

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{In centre, In circle, Tangents intersection Point}

- a) 5 b) $5(1+\sqrt{2})$ c) 6 d) $5\sqrt{2}$

Ans: (a)

Solution Intersection points of given tangents are $A(-2, 0)$, $B(2, 3)$, $C(5, 7)$

$$\text{In centre} = \left(\frac{15+4\sqrt{7}}{10+2\sqrt{7}}, \frac{35+6\sqrt{7}}{10+2\sqrt{7}} \right) = (h, k)$$

13A) Equation of the circle touching the three lines $x-\sqrt{3}y=0$, $x+\sqrt{3}y=0$, $x=5$ is

- a) $3x^2 + 3y^2 + 15x + 12y - 15 = 0$ b) $3x^2 + 3y^2 - 20x - 12y - 15 = 0$
c) $3x^2 + 3y^2 - 20x + 25 = 0$ d) $3x^2 + 3y^2 + 12y - 25 = 0$

Ans: C

Solution: Intersection points of given tangents are $A(0, 0)$, $B(5, 5/\sqrt{3})$, $C(5, -5/\sqrt{7})$, forms an equilateral triangle.

$$I = G = (10/3, 0)$$

14) Let a circle C_1 be obtained on rolling the circle $x^2 + y^2 - 4x - 6y + 11 = 0$ upwards 4 units on the tangent T to it at the point $(3, 2)$. Let C_2 be the image of C_1 in T . Let A and B be the centers of circles C_1 and C_2 respectively and M, N be respectively the feet of perpendiculars drawn from A, B on the x -axis. Then the area of the trapezium $AMNB$ is

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- a) $2(2+\sqrt{2})$ b) $4(1+\sqrt{2})$ c) $3+2\sqrt{2}$ d) $2(1+\sqrt{2})$

Ans: (b)

Solution: Centre $A(2, 3)$, $r = \sqrt{2}$, Tangent at $P(3, 2)$ is S_{11} is $x - y - 1 = 0$, has $m = 1$.

So centre of new circle is $A(2+2\sqrt{2}, 3+2\sqrt{2})$. Its image w.r.t $x - y - 1 = 0$ is $B(4+2\sqrt{2}, 1+2\sqrt{2})$

Hence $M(2+2\sqrt{2}, 0)$ and $N(4+2\sqrt{2}, 0)$. Area of trapezium $= \left(\frac{AM + BN}{2} \right) \cdot MN$

14A) Circle $x^2 + y^2 - 6x - 8y = 0$ is rolled on the tangent line $4x - 3y + 25 = 0$ until it touches the line $3x + 4y + 50 = 0$. Centre of the circle in new position is

- a) $(9, 12)$ b) $(3, 4)$ c) $(4, -3)$ d) $(-3, -4)$

Ans: (d)

Solution Given circle centre $A(3, 4)$, $r = 5$, New circle centre B , $r = 5$

and AB is parallel to $4x - 3y + 25 = 0$. So $B = (3 \pm d(3/5), 4 \pm d(4/5))$ Given new circle touches $3x + 4y + 50 = 0$ at say Q , then $d = AQ - r = 15 - 5 = 10$

15) If the line $3x - 4y - k = 0$, ($k > 0$) touches the circle $x^2 + y^2 - 4x - 8y - 5 = 0$ at (a, b) then

$k + a + b$ is equal to

(APRIL 2023)

- a) 20 b) 22 c) -30 d) -28

Ans: (a)

Solution: Since the given line touches the given circle, the length of the perpendicular from the centre $(2, 4)$ of the circle of the line $3x - 4y - k = 0$ is equal to the radius

$$\sqrt{4+16+5} = 5 \text{ of the circle.}$$

$$\Rightarrow \frac{3 \times 2 - 4 \times 4 - k}{\sqrt{9+16}} = \pm 5$$

$$\Rightarrow k = 15$$

Now equation of the tangent at (a, b) to the given circle is

$$xa + yb - 2(x+a) - 4(y+b) - 5 = 0$$

$$\Rightarrow (a-2)x + (b-4)y - (2a+4b+5) = 0$$

If it represents the given line $3x - 4y - k = 0$

$$\text{Then } \frac{a-2}{3} = \frac{b-4}{-4} = \frac{2a+4b+5}{k} = l \text{ (say)}$$

$$\text{Then } a = 3l + 2, b = 4 - 4l \text{ and } 2a + 4b + 5 = kl \text{ ----- (1)}$$

$$\Rightarrow 2(3l+2) + 4(4-4l) + 5 = 15l \quad (\because k=15)$$

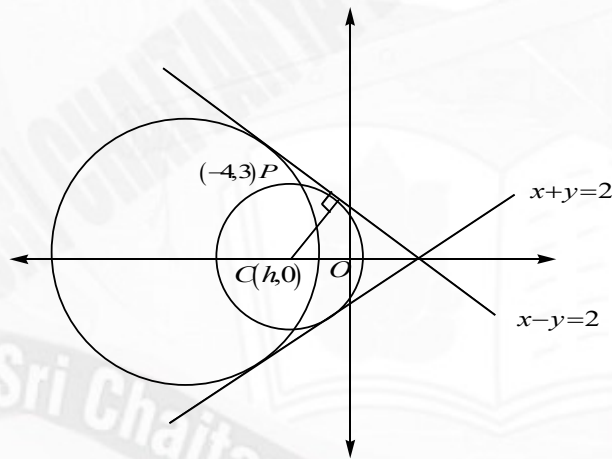
$$\Rightarrow l = 1 \Rightarrow a = 5, b = 0 \text{ and } k + a + b = 20$$

- 15a. Centre of the circles passing through the point $(-4, 3)$ and touching the lines $x + y = 2$ and $x - y = 2$ is

- a) $(-10 \pm 3\sqrt{6}, 0)$ b) $(0, -10 \pm 3\sqrt{6})$ c) $(-5 \pm 2\sqrt{6}, 0)$ d) none of these

Ans: A

Solution:



Centre of the circle is on x-axis which is $C(h, 0)$

Now, $CP = CM = \text{radius}$.

$$\Rightarrow \sqrt{(-4-h)^2 + (3-0)^2} = \frac{|h-2|}{\sqrt{2}}$$

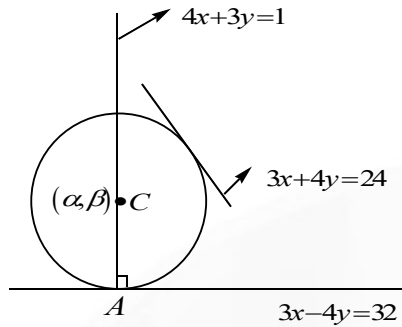
Solving, we get $h = -10 \pm 3\sqrt{6}$

- 15b. Let the centre of a circle C be (α, β) and its radius $r < 8$. Let $3x + 4y = 24$ and $3x - 4y = 32$ be two tangents and $4x + 3y = 1$ be a normal to C. Then $(\alpha - \beta + r)$ is equal to

- a) 7 b) 9 c) 5 d) 6

Ans: (a)

Solution:



first find point A by solving $4x + 3y = 1$ and $3x - 4y = 32$

After solving, point A is (4, -5)

Centre (α, β) lie on $4x + 3y = 1$

$$4\alpha + 3\beta = 1 \Rightarrow \beta = \frac{1 - 4\alpha}{3}$$

Now distance from centre to line $3x - 4y - 32 = 0$ and $3x + 4y - 24 = 0$ are equal.

$$\left| \frac{3\alpha - 4\left(\frac{1 - 4\alpha}{3}\right) - 32}{5} \right| = \left| \frac{3\alpha + 4\left(\frac{1 - 4\alpha}{3}\right) - 24}{5} \right|$$

After solving $\alpha = 1$ and $\alpha = \frac{28}{3}$

For $\alpha = 1$, centre (1, -1) $\Rightarrow$ radius = 5

For $\alpha = \frac{28}{3}$, centre $\left(\frac{28}{3}, \frac{-109}{2}\right)$

$\Rightarrow$ radius ≈ 49.78 (rejected)

Hence, $\alpha = 1, \beta = -1, r = 5$

$$\alpha - \beta + r = 7$$

CONCEPT : Equation of tangent

FORMULA : $S_1 = 0$

16. A circle C_1 passes through the origin O and has diameter 4 on the positive x-axis. The line $y = 2x$ gives a chord OA of circle C_1 . Let C_2 be the circle with OA as a diameter. If the tangent to C_2 at the point A meets the x-axis at P and y-axis at Q, then QA:AP is equal to

(2022_27 July - Shift 2)

a) 1:4

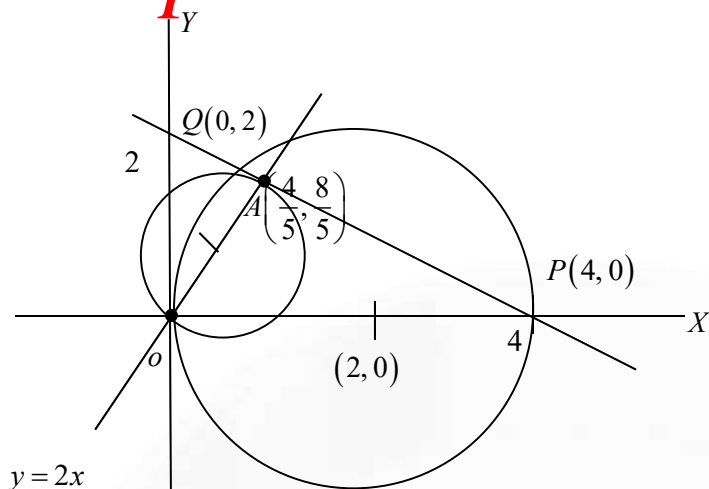
b) 1 :5

c) 2:5

d) 1:3

Key: (a)

Sol:



Solving $x^2 + y^2 - 4x = 0$ and $y = 2x$, we get $x^2 + 4x^2 - 4x = 0$

$$\Rightarrow x(5x - 4) = 0 \Rightarrow x = \frac{4}{5}, y = \frac{8}{5} \Rightarrow A\left(\frac{4}{5}, \frac{8}{5}\right)$$

Equation of the line PQ is equation of the line passing through A and perpendicular to

OA which will be $x\frac{4}{5} + y\frac{8}{5} = \frac{16 + 64}{25} = \frac{16}{5}$

$$\Rightarrow 4x + 8y = 16 \Rightarrow \frac{x}{4} + \frac{y}{2} = 1$$

So, we get $P(4, 0)$ and $Q(0, 2)$

A divides QP in the ratio $QA : AP = 0 - \frac{4}{5} : \frac{4}{5} - 4 = -1 : 1 - 5 = 1 : 4$

16a) From a point $R(5, 8)$ two tangents RP and RQ are drawn to a given circle $S=0$ whose radius is 5. If the circumcentre of ΔPQR is $(2, 3)$ then the equation of circle $S=0$ is _____

a) $x^2 + y^2 = 25$

b) $(x+1)^2 + (y+2)^2 = 25$

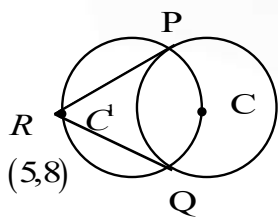
c) $(x-1)^2 + (y-1)^2 = 25$

d) $x^2 + y^2 = 1$

Key: B

$R(5, 8)C^1 = (2, 3) \Rightarrow C^1$ is midpoint of RC

$$\therefore C = (-1, -2)$$



Required equation of circle is

$$(x+1)^2 + (y+2)^2 = 25$$

CONCEPT: Pair of tangents to a circle

FORMULA: $S_1^2 = S \cdot S_{11}$ area of $\Delta^{le} = \frac{r \cdot S_{11}^{3/2}}{S_{11} + r^2}$

- 17) Let the tangents at two points A and B on the circle $x^2 + y^2 - 4x + 3 = 0$ meet at origin $O(0,0)$. Then the area of the triangle OAB is: [2022_28 July - Shift 2]

- a) $\frac{3\sqrt{3}}{2}$ b) $\frac{3\sqrt{3}}{4}$ c) $\frac{3}{2\sqrt{3}}$ d) $\frac{3}{4\sqrt{3}}$

Key: (b)

Sol: $S_{11} = 3$ $r = 1$

$$\begin{aligned} \text{Area of the triangle} &= \frac{r(S_{11})^{3/2}}{S_{11} + r^2} \\ &= \frac{3\sqrt{3}}{4} \end{aligned}$$

- 17) If the tangents at the points P and Q on the circle $x^2 + y^2 - 2x + y = 5$ meet at the point

$R\left(\frac{9}{4}, 2\right)$, then the area of the triangle PQR is (JEE MAIN Apr 2023 SHIFT 1)

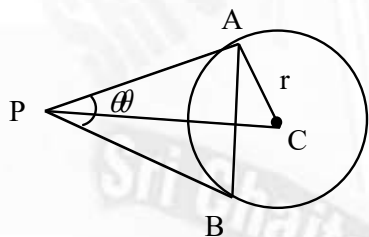
- a) $\frac{13}{4}$ b) $\frac{13}{8}$ c) $\frac{5}{4}$ d) $\frac{5}{8}$

- 17a) Let the tangents drawn from $P(-1, -1)$ to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ touch the circle at the points A and B then the area of triangle PAB (in square units) is

- a) $\frac{24}{13}$ b) $\frac{24}{7}$ c) $\frac{8}{13}$ d) $\frac{3}{13}$

Key: (a)

Sol: PA = length of tangent from P to the circle $= \sqrt{S_{11}} = 2$ radius $r = 3$



If $\angle APB = \theta \Rightarrow \tan \frac{\theta}{2} = \frac{r}{\sqrt{S_{11}}} = \frac{3}{2}$

$$\sin \theta = \frac{2 \tan \theta / 2}{1 + \tan^2 \theta / 2} = \frac{12}{13}$$

$$\text{Area of } \Delta PAB = \frac{1}{2}(PA)(PB)\sin \theta = \frac{1}{2} \times 2 \times 2 \times \frac{12}{13} = \frac{24}{13}$$

- 17b) Let the tangents drawn from the origin to the circle $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the points A and B then $(AB)^2 =$

- a) $\frac{32}{5}$ b) $\frac{52}{5}$ c) $\frac{56}{5}$ d) $\frac{64}{5}$

Key: (d)

Sol: radius $r = 2$, $S_{11} = 16$

$$\text{Length of chord } AB = \frac{2r\sqrt{S_{11}}}{\sqrt{S_{11} + r^2}} = \frac{2(2)(4)}{\sqrt{16+4}} = \frac{16}{\sqrt{20}} = \frac{8}{\sqrt{5}}$$

$$(AB)^2 = \frac{64}{5}$$

- 18) Circle 'c' touches the lines $L_1 = 4x - 3y + k_1 = 0$ and $L_2 = 4x - 3y + k_2 = 0$ where $k_1, k_2 \in R$. If a line passing through the centre of circle intersect at L_1 at $(-1, 2)$ and L_2 at $(3, -6)$ equation of circle is

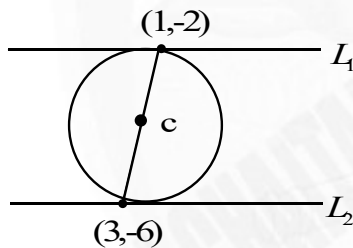
(JEE MAIN June 25 2022 shift – I)

- a) $(x-1)^2 + (y-2)^2 = 4$ b) $(x+1)^2 + (y-2)^2 = 4$
 c) $(x-1)^2 + (y+2)^2 = 16$ d) $(x-1)^2 + (y-2)^2 = 16$

Key : (c)

Sol: $P = (-1, 2), Q = (3, -6)$ $c = M.P of PQ$

$$= \left(\frac{-1+3}{2}, \frac{2-6}{2} \right) \\ = (1, -2)$$

'P' lies on $L_1 \Rightarrow -4 - 6 + k_1 = 0$

$$k_1 = 10$$

$$\therefore L_1 \rightarrow 4x - 3y + 10 = 0$$

'Q' lies on $L_2 \Rightarrow 12 + 18 + k_2 = 0$

$$\Rightarrow k_2 = -30$$

$$L_2 \text{ is } 4x - 3y - 30 = 0$$

$$2r = \text{distance between } L_1 \text{ \& } L_2 = \frac{|10+30|}{\sqrt{16+9}} = \frac{40}{5} = 8 \Rightarrow r = 4$$

$$\text{Circle is } (x-1)^2 + (y+2)^2 = (4)^2 = 16$$

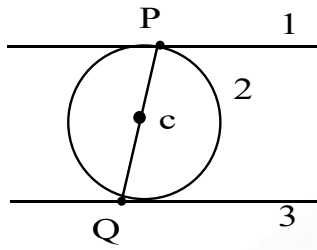
- 18a) $2x + y = 0$ is the equation of a diameter of the circle which touches the lines $4x - 3y + 10 = 0$ and $4x - 3y - 30 = 0$ then the equation of the circle is ____

- a) $x^2 + y^2 + 2x + 4y + 11 = 0$ b) $x^2 + y^2 - 2x + 4y - 11 = 0$
 c) $x^2 + y^2 - 2x - 4y - 11 = 0$ d) $x^2 + y^2 - 2x + 4y + 11 = 0$

Key: (b)

Sol: $4x - 3y + 10 = 0 \dots (1)$

$4x - 3y - 30 = 0 \dots (2)$



$2r = \text{distance between (1) and (2)}$

$$= \frac{|10 + 30|}{\sqrt{16 + 9}} = \frac{40}{5} = 8$$

$r = 4$

$2x + y = 0 \dots (3)$

Solve (1)&(3) $p = (-1, 2)$

Solve (2)&(3) $Q = (3, -6)$

$$c = M.P \text{ of } PQ = \left(\frac{-1+3}{2}, \frac{2-6}{2} \right)$$

$$= \left(\frac{2}{2}, \frac{-4}{2} \right) = (1, -2)$$

Circle is $(x-1)^2 + (y+2)^2 = 16$

$$x^2 + y^2 - 2x + 4y - 11 = 0$$

18b) The equation of the circle whose centre lies on $x + 2y = 0$ and touching $3x - 4y + 8 = 0$ and

$3x - 4y - 28 = 0$ is

a) $(x-2)^2 + (y+1)^2 = \left(\frac{18}{5} \right)^2$

b) $(x-2)^2 + (y-1)^2 = \left(\frac{18}{5} \right)^2$

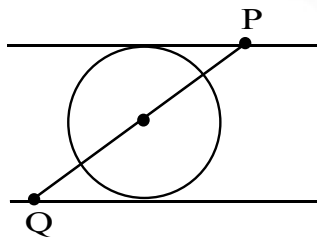
c) $(x-2)^2 - (y+1)^2 = \left(\frac{9}{5} \right)^2$

d) $(x-2)^2 + (y+1)^2 = \left(\frac{9}{5} \right)^2$

Key: (a)

Sol: $3x - 4y + 8 = 0 \dots (1)$

$3x - 4y - 28 = 0 \dots (2)$



Distance between (1)&(2) $2r = \frac{|8 + 28|}{\sqrt{9 + 16}} = \frac{36}{5}$

$$r = \frac{18}{5}$$

$$\text{Solve(1)\&(3) } p = \left(\frac{-8}{5}, \frac{4}{5} \right)$$

$$\text{Solve(2)\&(3) } Q = \left(\frac{28}{5}, \frac{-14}{5} \right)$$

$$c = M.P \text{ of } PQ = (2, -1)$$

$$\text{Circle is } (x-2)^2 + (y+1)^2 = \left(\frac{18}{5} \right)^2$$

19) If the tangents drawn at the point $O(0,0)$ and $P(1+\sqrt{5}, 2)$ on the circle $x^2 + y^2 - 2x - 4y = 0$ intersect at the point Q then the area of triangle OPQ is **(JEE MAIN 2022 SHIFT 1)**

$$\text{a) } \frac{3+\sqrt{5}}{2}$$

$$\text{b) } \frac{4+2\sqrt{5}}{2}$$

$$\text{c) } \frac{5+3\sqrt{5}}{2}$$

$$\text{d) } \frac{7+3\sqrt{5}}{2}$$

Ans : (c)

Sol : Slope of the tangent at the point $O(0,0)$

$$\frac{dy}{dx} = \frac{-1}{2}$$

Equation of tangent is

$$y - 0 = \frac{-1}{2}(x - 0)$$

$$\Rightarrow x + 2y = 0 \dots \dots \dots (1)$$

Similarly equation of tangent at P is $x(1+\sqrt{5}) + y(2) - (x+1+\sqrt{5}) - 2(y+2) = 0$

Put $x = -2y$ in the above equation

$$\Rightarrow y = -\left(\frac{\sqrt{5}+1}{2} \right)$$

From equation : 1

$$x = \sqrt{5} + 1$$

$$\therefore Q = \left(\sqrt{5} + 1, -\frac{\sqrt{5}+1}{2} \right)$$

$$\text{Area of triangle} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$= \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ \sqrt{5} + 1 & -\frac{\sqrt{5}+1}{2} & 1 \\ (\sqrt{5}+1) & 2 & 1 \end{vmatrix} = \frac{5+3\sqrt{5}}{2}$$

Concept: Area of triangle formed by the point and their chord of contact

Formula: $\frac{r(S_{11})^{3/2}}{S_{11} + r^2}$

19a) If the tangents drawn at the points P(1,7) Q(4,-2) on the circle $x^2 + y^2 - 2x - 4y - 20 = 0$ intersect at 'R'. Then area of $\Delta PQR = ?$

- a) 67.5 sq.units b) 57.5 sq.units c) 57 sq.units d) 67 sq.units

Ans : 67.5 sq units

Sol: Tangent at 'P' is $y = 7 \dots (1)$

Tangent at 'Q' is $3x - 4y - 20 = 0 \dots (2)$

From (1)&(2) $R = (16, 7)$

$$\text{Area of } \Delta PQR = \frac{135}{2} = 67.5 \text{ sq units}$$

19b) If the straight line $x - 2y + 1 = 0$ intersect the circle $x^2 + y^2 = 25$ at points 'P' and 'Q' the tangents at 'P' and Q to the circle intersect at 'R' then area of $\Delta PQR = ?$

a) $\frac{248\sqrt{27}}{5}$ sq units

b) $\frac{248\sqrt{21}}{5}$ sq.units

c) $\frac{248\sqrt{51}}{5}$ sq.units

d) $\frac{248\sqrt{31}}{5}$ sq. units

Ans: (d)

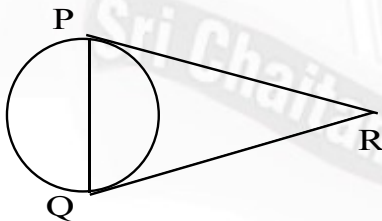
Sol: $x^2 + y^2 = 25 \dots (1)$

$x - 2y + 1 = 0 \dots (2)$

$$R = \left(\frac{-r^2 l}{n}, \frac{-r^2 m}{n} \right)$$

$$= \left(\frac{-25 \times 1}{1}, \frac{-25 \times -2}{1} \right)$$

$$= (-25, 50)$$



$$\text{Area of } \Delta PQR = \frac{r(S_{11})^{3/2}}{S_{11} + r^2}$$

$$= \frac{5((-25)^2 + (50)^2 - 25)^{3/2}}{(-25)^2 + (50)^2}$$

$$= \frac{5(3100)^{3/2}}{(25)^2 (1+4)}$$

$$= \frac{5 \times 31 \sqrt{31} (10^2)^{3/2}}{25 \times 25 \times 5}$$

$$= \frac{31 \sqrt{31} \times 1000}{25 \times 25} = \frac{248 \sqrt{31}}{5} \text{ squn}$$

- 20) The set of values of k for which the circle $C: 4x^2 + 4y^2 - 12x + 8y + k = 0$ lies inside the fourth quadrant and the point $\left(1, -\frac{1}{3}\right)$ lies on or inside the circle C is :

(JEE MAIN 2022 SHIFT 1)

- a) An empty set b) $\left(6, \frac{95}{9}\right]$ c) $\left[\frac{80}{9}, 10\right)$ d) $\left(9, \frac{92}{9}\right]$

Ans: (d)

Solution:

Given circle is $4x^2 + 4y^2 - 12x + 8y + k = 0$

$$\Rightarrow x^2 + y^2 - 3x + 2y + \left(\frac{k}{4}\right) = 0$$

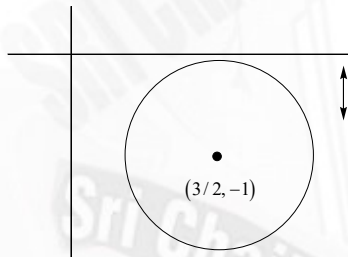
$$\text{Centre } \left(\frac{3}{2}, -1\right); r = \sqrt{\frac{13-k}{2}}$$

Here, $r \geq 0$, then $\sqrt{\frac{13-k}{2}} \geq 0$. So, $k \leq 13$(i)

(i) Point $\left(1, -\frac{1}{3}\right)$ lies on or inside circle C

$$\Rightarrow S_1 \leq 0 \Rightarrow k \leq \frac{92}{9} \quad \text{.....(ii)}$$

(ii) C lies in 4th quadrant



$$r < 1$$

$$\Rightarrow \frac{\sqrt{13-k}}{2} < 1 \Rightarrow k < 9 \quad \text{.....(iii)}$$

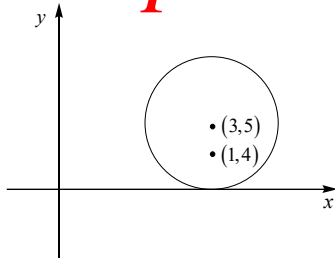
$$\text{Hence } (i) \cap (ii) \cap (iii) \Rightarrow k \in \left(9, \frac{92}{9}\right]$$

- 20a. The circle $x^2 + y^2 - 6x - 10y + k = 0$ does not touch (or), Intersect the co-ordinate axes and the pt(1,4) is inside the circle. Find the range of K

- a) $25 < K < 29$ b) $9 < K < 29$ c) $K < 9$ (or), $K > 25$ d) $K < 29$ (or) $K > 9$

Ans: (a)

Sol:



$$C = (3, 5)$$

$$r = \sqrt{9 + 25 - K}$$

$$= \sqrt{34 - K}$$

Here $r < 5$ and $r < 3$

$$34 - K < 25 \quad \text{and} \quad 34 - K < 9$$

$$K > 9 \quad K > 25$$

Here (1, 4) lies inside

$$\sqrt{(3-1)^2 + (5-4)^2} < \sqrt{34-K}$$

$$5 < 34 - K$$

$$K < 29$$

$$\therefore 25 < K < 29$$

20b. Let the circle S: $36x^2 + 36y^2 - 108x + 120y + C = 0$ be such that it neither intersects nor touches the co-ordinate axes. If the point of intersection of the lines, $x - 2y = 4$ and $2x - y = 5$ lies inside the circle S, then:

a) $\frac{25}{9} < C < \frac{13}{3}$ b) $100 < C < 165$ c) $81 < C < 156$ d) $100 < C < 156$

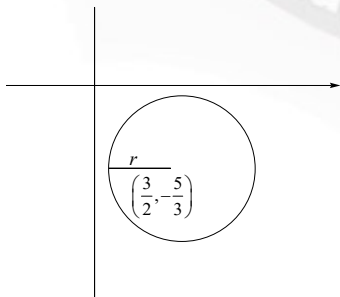
Ans: (d)

Solution: Since, S: $36x^2 + 36y^2 - 108x + 120y + C = 0$

$$\Rightarrow x^2 + y^2 - 3x + \frac{10}{3}y + \frac{C}{36} = 0$$

$$\text{Centre} = (-g, -f) = \left(\frac{3}{2}, -\frac{10}{6}\right)$$

$$\text{Radius} = r = \sqrt{\frac{9}{4} + \frac{100}{36} - \frac{C}{36}}$$



Since circle neither intersects nor touches the coordinates axes.

$$\therefore r < \frac{3}{2} \Rightarrow \frac{9}{4} + \frac{100}{36} - \frac{C}{36} < \frac{9}{4}$$

$$\Rightarrow C > 100 \quad \dots\dots\dots (i)$$

Now point of intersection of $x - 2y = 4$ and $2x - y = 5$ is $(2, -1)$, which lies inside the circle S.

$$\therefore S(2, -1) < 0$$

$$\Rightarrow (2)^2 + (-1)^2 - 3(2) + \frac{10}{3}(-1) + \frac{C}{36} < 0$$

$$\Rightarrow 4 + 1 - 6 - \frac{100}{3} + \frac{C}{36} < 0$$

$$C < 156 \quad \dots\dots\dots (ii)$$

From (i) & (ii), $100 < C < 156$

4) Normal to a circle (Equation of normal line at point of contact in circles, Condition for normal to circle, Equation of normal at P (theta))

- 1) The tangent and the normal lines at the point $(\sqrt{3}, 1)$ to the circle $x^2 + y^2 = 4$ and the x-axis form a triangle. The area of this (in square units) is (JEE MAIN 2019 SHIFT 1)

- a) $\frac{4}{\sqrt{3}}$ b) $\frac{2}{\sqrt{3}}$ c) $\frac{1}{3}$ d) $\frac{1}{\sqrt{3}}$

Ans : (b)

Given $x^2 + y^2 = 4$

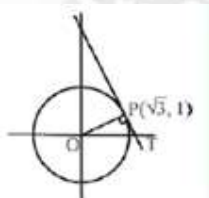
Equation of tangent

$$\Rightarrow \sqrt{3}x + y = 4$$

Equation of normal $x - \sqrt{3}y = 0$

Coordinate of T = $\frac{4}{\sqrt{3}}, 0$

Area of triangle = $\frac{2}{\sqrt{3}}$



- 1a) Let the tangents drawn to the circle $x^2 + y^2 = 16$ from the point $p(0, h)$ meet the x-axis at the point A and B. If the area of ΔAPB is minimum then h is equal to

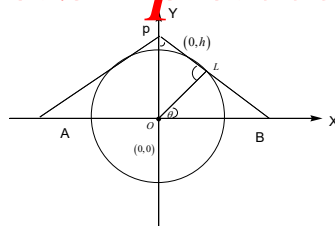
- a) $4\sqrt{2}$ b) $3\sqrt{3}$ c) $3\sqrt{2}$ d) $4\sqrt{3}$

Ans: (a)

Sol: Given circle $x^2 + y^2 = 16$

Centre = $(0, 0)$

$r = 4$



From $\triangle OPL$

$$\sin \theta = \frac{OL}{OP} = \frac{4}{OP} \Rightarrow OP = \frac{4}{\sin \theta}$$

$$\text{From } \triangle OLB \quad \cos \theta = \frac{OL}{OB} \Rightarrow OB = \frac{4}{\cos \theta}$$

$$\text{Now area} = OP \times OB = \frac{16}{\sin \theta \cos \theta} = \frac{32}{2 \sin \theta \cos \theta} = \frac{32}{\sin 2\theta}$$

Now the maximum value of $\sin 2\theta = 1$

$$2\theta = 90^\circ$$

$$\theta = 45^\circ$$

$$\text{Now } OP = h = \frac{4}{\sin \theta} = \frac{4}{\sin 45^\circ} = \frac{4}{\frac{1}{\sqrt{2}}}$$

$$= 4\sqrt{2}$$

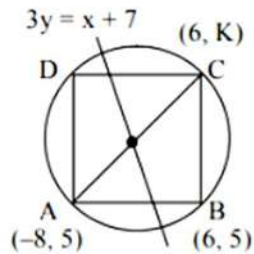
5) Chord of contact of a circle, Pole and polar of a circle

(Equation of chord of contact of a circle in circles, Equation of chord of a circle when mid-point is given in circles, Length of chord when mid-point is given in circles,, Definition of pole and polar in circles, Polar of a point with respect to circle Pole of a polar with respect to a circle, Conjugate points with respect to given circle, Conjugate lines with respect to circle, Properties of pole and polar of a circle, Problems based on locus)

- 1) A rectangle is inscribed in a circle with a diameter lying along the line $3y = x + 7$. If the two adjacent vertices of the rectangle are $(-8,5)$ and $(6,5)$, then the area of the rectangle (in sq. units) is (2019)
- a) 98 b) 84 c) 56 d) 72

Correct option is B)

Step 1 : finding vertex C



Let Vertex C is $(6, k)$

$\therefore$ center of circle is $(\frac{-8+6}{2}, \frac{5+k}{2}) = (-1, \frac{5+k}{2})$ which lies on diameter of circle $3y = x + 7$

$$\therefore 3(\frac{5+k}{2}) = -1 + 7$$

$$\therefore k = -1$$

$$\therefore \text{vertex of C} = (6, -1)$$

Step 2 : finding area of rectangle

Length of AB = 14

and Length of BC = 6

$$\therefore \text{area of rectangle} = AB \times BC$$

$$= 14 \times 6$$

$$\therefore \text{area of rectangle} = 84$$

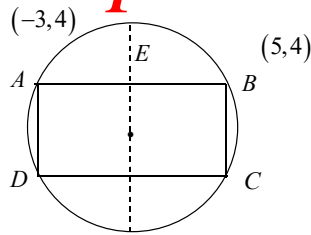
Therefore required area of rectangle is 84 sq. units

1a) If $A(-3,4)$, $B(5,4)$ are two vertices of a rectangle ABCD and $4y = x + 7$ is a diameter of the circle circumscribing ABCD. The width of the rectangle is

- a) 3 b) 6 c) 4 d) 2

Ans: (c)

Sol:



Let E be the midpoint of AB $E = (1, 4)$

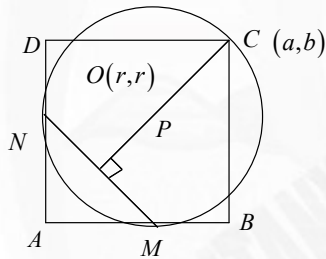
Equation of AB is $y - 4 = 0 \Rightarrow y = 4$ equation of line passing through E and perpendicular to AB is $x = 1$ and is passing through center O. Also equation of diameter is $4y = x + 7 \Rightarrow O = (1, 2)$ width of rectangle is $2 OE = 2\sqrt{0 + 2^2} = 2 \times 2 = 4$

- 2 b) ABCD is a rectangle. A circle passing through vertex C touches the sides AB and AD at M and N respectively. If the distance of the line MN from C is P units then the area of rectangle ABCD is (2019)

a) p b) p^2 c) $2p$ d) $2p^2$

Ans: (b)

Sol:



Let A as origin Let $AB = a$ and $AD = b$ and r be the radius of circle $O(r, r)$ equation of circle $(x - r)^2 + (y - r)^2 = r^2$

$$\Rightarrow (a - r)^2 + (b - r)^2 = r^2 \Rightarrow a^2 + b^2 - 2r(a + b) + r^2 = 0$$

Equation of line MN is $\frac{x}{r} + \frac{y}{r} = 1 \Rightarrow x + y - r = 0$ perpendicular distance from C to MN is

$$p = \frac{|a + b - r|}{\sqrt{2}} \Rightarrow (a + b - r)^2 = 2p^2$$

$$a^2 + b^2 + r^2 - 2r(a + b) + 2ab = 2p^2$$

$$\Rightarrow 2ab = 2p^2 \Rightarrow ab = p^2$$

Area of rectangle ABCD is $ab = p^2$

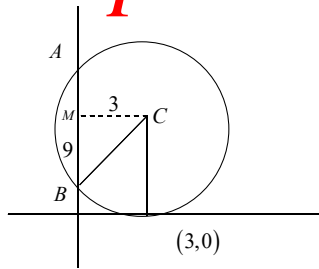
- 3) A circle touching the X-axis at (3, 0) and making an intercept of length 8 on the Y-axis passes through the point is

(JEE MAIN 2019 SHIFT 1)

a) (3, 10) b) (3, 5) c) (2, 3) d) (1, 5)

Ans : (a)

Sol:



Let centre of circle is 'C' and circle cuts the Y-axis at B and A. Let midpoint of chord BA is M.

$$CB = \sqrt{MC^2 + MB^2} = \sqrt{3^2 + 4^2} = 5 = r$$

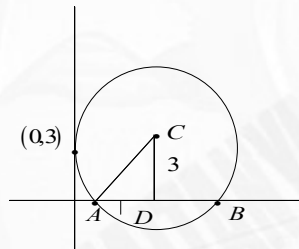
Equation of circle is $(x-3)^2 + (y-5)^2 = 5^2$

(3,10) satisfies this equation. Although there will be another circle satisfying the same conditions that will lie below the X-axis having equation $(x-3)^2 + (y-5)^2 = 5^2$

- 3a) The centre of the circle touching the Y-axis at (0,3) and making an intercept 2 Unit on positive X-axis is (2019)

- a) $(10, \sqrt{3})$ b) $(\sqrt{3}, 10)$ c) $(\sqrt{10}, 3)$ d) $(3, \sqrt{10})$

Ans : (c)



Sol :

In the figure $CD = 3$, $AB = 2 \Rightarrow AD = 1$

$$AC = \sqrt{1^2 + 3^2} = \sqrt{10}$$

$$\therefore C = (\sqrt{10}, 3)$$

- 4) A circle cuts a chord of length $4a$ on the x-axis and passes through a point on the y-axis, distant $2b$ from the origin. Then the locus of the centre of this circle is

(JEE MAIN 2019 SHIFT 1)

- a) a hyperbola b) a parabola c) a straight line d) an ellipse

Ans: (b)

Sol: Let the equation of circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ It passes through (0,2b)

$$\Rightarrow 4b^2 + 4bf + c = 0 \text{ -----(1)}$$

Intercept on the x-axis = $4a$

$$2\sqrt{g^2 - c} = 4a$$

$$\sqrt{g^2 - c} = 2a$$

$$g^2 - c = 4a^2$$

$$c = g^2 - 4a^2$$

$$\Rightarrow 4b^2 + 4bf + g^2 - 4a^2 = 0$$

$$[-g]^2 - 4b(-f) + 4(b^2 - a^2) = 0$$

Locus of centre $(-g, -f)$ is $x^2 - 4by + 4(b^2 - a^2) = 0$ parabola

- 4a) The locus of centre of circle of radius 2 units, if intercept cut on x-axis by circle is twice of intercept on y-axis is (JEE MAIN 2019 SHIFT 1)

a) $3x^2 - 3y^2 = 4$ b) $4x^2 - y^2 = 12$ c) $4y^2 - x^2 = 12$ d) $4y^2 - 3x^2 = 4$

Ans: (c)

Sol: Let (h, k) be the centre.

Intercept on x-axis $= 2\sqrt{h^2 - c}$ Intercept on y-axis $= 2\sqrt{k^2 - c}$

given condition

$$2\sqrt{h^2 - c} = 2[2\sqrt{k^2 - c}]$$

$$h^2 - c = 4(k^2 - c)$$

$$h^2 - 4k^2 = -3c \text{ -----(1)}$$

Given radius = 2

$$\sqrt{h^2 + k^2 - c} = 2$$

$$h^2 + k^2 - c = 4$$

$$c = h^2 + k^2 - 4 \text{ -----(2)}$$

From (1) and (2)

$$h^2 - 4k^2 = -3(h^2 + k^2 - 4)$$

$$4h^2 - k^2 = 12$$

Locus is $4x^2 - y^2 = 12$

- 4b) The locus of the centre of the circle which cuts a chord of length $2a$ from the positive x-axis and passes through a point on positive y-axis distant b from the origin is

(JEE MAIN 2019 SHIFT 2)

a) $x^2 + 2by = b^2 + a^2$ b) $x^2 - 2by = b^2 + a^2$ c) $x^2 - 2by = a^2 - b^2$ d) $x^2 - 2by = b^2 - a^2$

Ans: (c)

Sol: $2\sqrt{g^2 - c} = 2a$

$$g^2 - c = a^2 \text{ -----(1)}$$

And $x^2 + y^2 + 2gx + 2fy + c = 0$ passes through

$(0, b)$

$$\Rightarrow b^2 + 2fb + c = 0 \text{ -----(2)}$$

From (1) & (2)

$$b^2 + 2fb + g^2 - a^2 = 0$$

$$(-g)^2 - 2b(-f) = a^2 - b^2$$

Locus of the centre $(-g, -f)$ is $x^2 - 2by = a^2 - b^2$

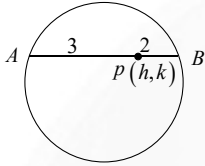
- 5) A Line segment AB of length ' λ ' moves such that the point A and B remain on the periphery of a circle of radius λ . Then the locus of the point, that divides the line segment AB in the ratio 2:3 is a circle of radius

(JEE MAIN 2023 SHIFT 1)

- a) $\frac{3}{5}\lambda$ b) $\frac{\sqrt{19}}{7}\lambda$ c) $\frac{2}{3}\lambda$ d) $\frac{\sqrt{19}}{5}\lambda$

Ans: (d)

Sol: $\left(\frac{\lambda}{\sqrt{2}}\sin\theta, \frac{-\lambda}{\sqrt{2}}\cos\theta\right) = A; B = \left(\frac{\lambda}{\sqrt{2}}\cos\theta, \frac{\lambda}{\sqrt{2}}\sin\theta\right)$



$$h = \frac{2\lambda\sin\theta}{\sqrt{2}} + 3 \times \frac{\lambda}{\sqrt{2}}\cos\theta; k = \frac{-2\lambda}{\sqrt{2}}\cos\theta + 1 \times \frac{3\lambda}{\sqrt{2}}\sin\theta$$

$$h^2 + k^2 = \frac{19\lambda^2}{5} \Rightarrow r = \frac{\sqrt{19}\lambda}{5}$$

- 5a) Let Q be a point on the circle B : $x^2 + y^2 = a^2$ and $p(h, k)$ be a fixed point. If the locus of the point which divides the join of P and Q in the ratio p : q in a circle C, then the centre of 'C' is (2023)

- a) $\left(\frac{p+q}{p}, \frac{p+q}{p}\right)$ b) $\left(\frac{hp+kq}{p}, \frac{hp+kq}{q}\right)$ c) $\left(\frac{hq}{p}, \frac{kp}{q}\right)$ d) $\left(\frac{hq}{p+q}, \frac{kq}{p+q}\right)$

Ans: (d)

Sol: Let $Q = (\alpha, \beta)$. The point which divides PQ in the ratio p:q is $\left(\frac{p\alpha + qh}{p+q}, \frac{p\beta + qk}{p+q}\right)$

Let $R(x_1, y_1)$ be a point in the locus

Then $x_1 = \frac{p\alpha + qh}{p+q}; y_1 = \frac{p\beta + qk}{p+q} \Rightarrow \alpha = \frac{(p+q)x_1 - qh}{p}; \beta = \frac{(p+q)y_1 - qk}{p}$

Q lies on the circle $x^2 + y^2 = a^2 \Rightarrow \left[\frac{(p+q)x_1 - qh}{p}\right]^2 + \left[\frac{(p+q)y_1 - qk}{p}\right]^2 = a^2$

$$(p+q)^2 x_1^2 + (p+q)^2 y_1^2 - 2qh(p+q)x_1 - 2qk(p+q)y_1 + q^2 h^2 + q^2 k^2 - p^2 a^2 = 0$$

Locus of R is $x^2 + y^2 - \frac{2qh}{p+q}x - \frac{2qk}{p+q}y + \frac{q^2(h^2 + k^2) - p^2 a^2}{(p+q)^2} = 0$

Centre of the circle = $\left(\frac{qh}{p+q}, \frac{qk}{p+q}\right)$

- 6) Let A be the point (1, 2) and B be any point on the curve $x^2 + y^2 = 16$. If the centre of the locus of the point 'P' which divides the line segment AB in the ratio 3:2 is the point $C(\alpha, \beta)$, then the length of the line segment AC is

a) $\frac{6\sqrt{5}}{5}$

b) $\frac{4\sqrt{5}}{5}$

c) $\frac{2\sqrt{5}}{5}$

d) $\frac{3\sqrt{5}}{5}$

Ans: (d)

$$A \bullet \xrightarrow{P} B (4\cos\theta, 4\sin\theta)$$

Sol: $(1, 2) \quad (h, k)$

$$h = \frac{12\cos\theta + 2}{5} \Rightarrow 12\cos\theta = 5h - 2 \text{ squaring and adding}$$

$$k = \frac{12\sin\theta + 4}{5} \Rightarrow 12\sin\theta = 5k - 4$$

$$144 = (5h - 2)^2 + (5k - 4)^2$$

$$\left(x - \frac{2}{5}\right)^2 + \left(y - \frac{4}{5}\right)^2 = \frac{144}{25}$$

$$\text{Centre} = \left(\frac{2}{5}, \frac{4}{5}\right) = (\alpha, \beta)$$

$$AC = \sqrt{\left(1 - \frac{2}{5}\right)^2 + \left(2 - \frac{4}{5}\right)^2} = \frac{3\sqrt{5}}{5}$$

6a) The locus of a point which divides the joint of $A(-1, 1)$ and a variable point p on the circle $x^2 + y^2 = 4$ in the ratio 3:2 is

a) $25(x^2 + y^2) + 20(x + y) + 28 = 0$

b) $25(x^2 + y^2) - 20(x + y) + 28 = 0$

c) $25(x^2 + y^2) + 20(x - y) + 28 = 0$

d) $25(x^2 + y^2) + 20(x - y) - 28 = 0$

Ans: (d)

Sol: Let $P(\alpha, \beta)$ & $Q(x_1, y_1)$ be a point in the locusNow Q divides AP in the ratio 3:2 where $A(-1, 1)$

$$\therefore Q\left(\frac{3\alpha - 2}{5}, \frac{3\beta + 2}{5}\right) \Rightarrow \frac{3\alpha - 2}{5} = x_1; \frac{3\beta + 2}{5} = y_1$$

$$\Rightarrow 3\alpha = 5x_1 + 2; 3\beta = 5y_1 - 2 \Rightarrow \alpha = \frac{5x_1 + 2}{3}; \beta = \frac{5y_1 - 2}{3}$$

$$P \text{ lies on } x^2 + y^2 = 4 \Rightarrow \left(\frac{5x_1 + 2}{3}\right)^2 + \left(\frac{5y_1 - 2}{3}\right)^2 = 4$$

$$\text{Locus of } Q \text{ is } 25(x^2 + y^2) + 20(x - y) - 28 = 0$$

7) The locus of the mid points of the chords of the circle $C : (x - 4)^2 + (y - 5)^2 = 4$ which subtend an angle θ_n at the centre of circle C , is a circle of radius r_n . If $\theta_1 = \pi/3$ and $\theta_3 = 2\pi/3$ and $r_1^2 = r_2^2 + r_3^2$ then θ_2 is equal to **(JAN 2023 SESSION - I)**

a) $\pi/4$

b) $3\pi/4$

c) $\pi/6$

d) $\pi/2$

Ans: (b)

$$\text{Solution: } \cos\left(\frac{\theta_n}{2}\right) = \frac{CM}{r} = \frac{\sqrt{(h-4)^2 + (k-5)^2}}{2}$$

$$\rightarrow (x-4)^2 + (y-5)^2 = \left(2\cos\frac{\theta_n}{2}\right)^2 \quad \text{So } r_n = 2\cos\frac{\theta_n}{2}$$

- 7A) Locus of mid point of chords of circle $x^2 + y^2 - 10x - 6y + 9 = 0$, which subtend 45° at any point on major part of the circle, is a circle of radius

- a) 5 b) $5\sqrt{2}$ c) $\frac{5}{\sqrt{2}}$ d) $2\sqrt{2}$

Ans : (c)

Solution: $C(5, 3)$ and $r = 5$ Hence $\cos(45^\circ) = \frac{1}{\sqrt{2}} = \frac{CM}{r} = \frac{\sqrt{(h-5)^2 + (k-3)^2}}{5}$

Hence $(x-5)^2 + (y-3)^2 = 25/2$

- 8) Let the tangents at the points $A(4, -11)$ and $B(8, -5)$ on the circle $x^2 + y^2 - 3x + 10y - 15 = 0$, intersect at the point C . Then the radius of the circle whose centre is C and the line joining A and B is tangent, is equal to

(JAN 2023 SESSION - I)

{ Pole or Intersection point of tangents at given points }

- a) $2\sqrt{13}$ b) $\frac{2\sqrt{13}}{3}$ c) $\sqrt{13}$ d) $\frac{3\sqrt{3}}{4}$

Ans: (b)

Solution: Tangents at A and B are $5x - 12y - 152 = 0$ and $13x + 0y - 104 = 0$

They intersect at $(8, -28/3)$, which is now centre of new circle.

That Circle touch AB line, $3x - 2y - 34 = 0 \implies radius = \frac{2\sqrt{13}}{3}$

- 8A) The line $ax + by = 1$ and circle $3x^2 + 3y^2 + 15x + 12y - 15 = 0$ intersect at A and B . If tangents at A, B intersect at $P(1, 2)$ then radius of the circle for which centre is P and AB

is a chord of length $\frac{2\sqrt{301}}{\sqrt{113}}$ is

- a) $\frac{31\sqrt{5}}{\sqrt{113}}$ b) $\frac{\sqrt{5}}{\sqrt{113}}$ c) $\frac{25}{\sqrt{113}}$ d) $\frac{31\sqrt{5}}{\sqrt{221}}$

Ans: C

Solution: $x^2 + y^2 + 5x + 4y - 5 = 0$, polar of $P(1, 2)$ is AB line, $7x + 8y - 5 = 0$

Distance of $P(1, 2)$ to this line $\frac{18}{\sqrt{113}}$ Now use length of chord is $= 2\sqrt{r^2 - d^2}$

- 9) Set of all values of a^2 for which the line $x + y = 0$ bisects two distinct chords drawn from a point $P\left(\frac{1+a}{2}, \frac{1-a}{2}\right)$ on the circle $2x^2 + 2y^2 - (1+a)x - (1-a)y = 0$ is equal to

{Eq. of chord, midpoint}

(JAN 2023 SESSION - I)

- a) $(8, \infty)$ b) $(4, \infty)$ c) $(0, 4]$ d) $(2, 12]$

Ans: (d)

Solution: circle centre $C\left(\frac{1+a}{4}, \frac{1-a}{4}\right) = (h, k)$. So $P\left(\frac{1+a}{2}, \frac{1-a}{2}\right) = (2h, 2k)$

- $\Rightarrow$ Hence circle equation $x^2 + y^2 - 2hx - 2ky = 0$. Mid point M is on $x + y = 0$, so $M(\lambda, -\lambda)$
 $\Rightarrow$ Chord equation is $S_1 = S_{11}$ and then substitute $P(2h, 2k)$
 $\Rightarrow 2\lambda^2 - (3h+3k)\lambda + 2h^2 + 2k^2 = 0$, is quadratic in λ .
 $\Rightarrow$ Given two such chords exists, hence $\Delta > 0$

- 9A) $P(a, a)$ is a point on $C_1: x^2 + y^2 - ax - ay = 0$. Locus of mid point of chords of circle C_1 through P is another circle C_2 . Equation of diameter of C_2 through P is
 a) $x + y = 0$ b) $x - y = 0$ c) $x + y = a$ d) $x - y = a$

Ans: (b)

Solution: chord mid point $(x_1, y_1) \Rightarrow$ chord equation is $S_1 = S_{11}$ and then substitute $P(a, a)$

$$\Rightarrow x_1^2 + y_1^2 - \frac{3a}{2}x_1 - \frac{3a}{2}y_1 + a^2 = 0 \quad \text{This circle has centre } A(3a/4, 3a/4)$$

$$\Rightarrow \text{So equation of AP is } x - y = 0$$

- 10) The locus of the mid point of the chord of contact of tangents drawn from points lying on the straight line $4x - 5y = 20$ to the circle $x^2 + y^2 = 9$ [IIT 2012]
 a) $20(x^2 + y^2) - 36x + 45y = 0$ b) $20(x^2 + y^2) + 36x - 45y = 0$
 c) $36(x^2 + y^2) - 20x + 45y = 0$ d) $36(x^2 + y^2) + 20x - 45y = 0$

Ans: (a)

Solution: Locus point = mid point $(x_1, y_1) \rightarrow$ hence chord equation is $S_1 = S_{11}$

$$xx_1 + yy_1 = x_1^2 + y_1^2 \rightarrow \text{Pole} = \left(\frac{-la^2}{n}, \frac{-ma^2}{n} \right) = \left(\frac{-x_1 \cdot 9}{-(x_1^2 + y_1^2)}, \frac{-y_1 \cdot 9}{-(x_1^2 + y_1^2)} \right)$$

It is on $4x - 5y = 20$

- 10A) Consider a fixed circle with radius 'a' and center on y axis. If a variable line through $(a, 0)$ cuts the given circle at P, Q and tangents to circle at P, Q intersect at R then locus of foot of perpendicular from R onto that line is
 a) a circle with centre $(a/2, a/2)$ b) a circle through origin
 c) a line through the origin d) a line through the point (a, a)

Ans: (b)

Solution: Circle $x^2 + (y - \alpha)^2 = a^2$ (here a, α are fixed constants)

Locus point $M(h, k)$ is foot of perpendicular from R onto PQ.

CM is perpendicular to MA, use $m_1 m_2 = -1$

- 11) A line $y = mx + 1$ intersects the circle $(x-3)^2 + (y+2)^2 = 25$ at the points P and Q. If the mid point of the line PQ has x-coordinate $-3/5$, then which one of the following options is correct? [IIT 2019]
 a) $4 \leq m < 6$ b) $-3 \leq m < -1$ c) $6 \leq m < 8$ d) $2 \leq m < 4$

Ans: (d)

Solution: Given chord PQ mid point is $M\left(\frac{-3}{5}, \frac{-3m}{5} + 1\right)$

CM is perpendicular to PQ, $\rightarrow m\left[\frac{-3m+15}{-18}\right] = -1 \Rightarrow m^2 - 5m + 6 = 0 \Rightarrow m = 2, 3$

11A) Consider a fixed circle with radius 'a' and center on y axis. If a variable line through (a, 0) cuts the given circle at P, Q and tangents to circle at P, Q intersect at R then locus of foot of perpendicular from R onto that line is

- a) a circle with centre (a/2, a/2) b) a circle through origin
c) a line through the origin d) a line through the point (a, a)

Ans: (b)

Solution: Circle $x^2 + (y - \alpha)^2 = a^2$ (here a, α are fixed constants)

Locus point $M(h, k)$ is foot of perpendicular from R onto PQ.

CM is perpendicular to MA, use $m_1 m_2 = -1$

2. Analysis of two circles

6) Relative positions of two circles

(Angle between intersecting circles in circles, Condition for orthogonal circles, Non-intersecting circles in circles, Intersecting circles in circles, Touching circles in circles, Circles touch internally in circles, Circles touch externally in circles, One circle lie inside the other in circles, Centres of similitude in circles, Different forms of common tangents to circles in circles, Direct common tangents in circles, Transverse common tangents in circles, Length of direct common tangent in circles, Length of transverse common tangent in circles, Equation of common tangent of two circles in circles)

1) If the curves, $x^2 - 6x + y^2 + 8 = 0$ and $x^2 - 8y + y^2 + 16 - k = 0$, ($k > 0$) touch each other at a point, then the largest value of k is

(JEE MAIN 2020 SHIFT 2)

- a) 15 b) 14 c) 36 d) 35

Ans: (c)

Sol : Two circles touches each other if $C_1 C_2 = |r_1 \pm r_2|$

Distance between $C_2(3, 0)$ and $C_1(0, 4)$ is either

$$\sqrt{k} + 1 \text{ (or) } |\sqrt{k} - 1| \quad (C_1 C_2 = 5)$$

$$\Rightarrow \sqrt{k} + 1 = 5 \text{ or } |\sqrt{k} - 1| = 5$$

$$\Rightarrow k = 16 \text{ or } k = 36$$

$\Rightarrow$ maximum value of k is 36

- 1a) If the curves, $x^2 + y^2 - 4x - 6y + 4 = 0$ and $x^2 + y^2 - 10x - 14y + 74 - k^2 = 0$ ($K > 0$) touch each other at a point, then sum of all possible values of K is _____

(JEE MAIN 2020 SHIFT 2)

- a) 8 b) 6 c) 12 d) 10

Ans : (d)

Sol : $C_1(2, 3)$ $r_1 = 3$ $C_2(5, 7)$ $r_2 = k$

Case I : $C_1 C_2 = r_1 + r_2$ Case II: $C_1 C_2 = r_2 - r_1$

$$5 = 3 + K$$

$$5 = K - 3$$

$$K = 2$$

$$K = 8$$

$$\therefore \text{Sum} = 2 + 8 = 10$$

- 1b) Two circles $x^2 + y^2 - 2Kx = 0$ and $x^2 + y^2 - 4x + 8y + 16 = 0$ touch each other externally then K is _____

- a) 4 b) 1 c) 2 d) -4

Ans : (c)

Sol : $C_1(K, 0) \Rightarrow r_1 = |K|$

$$C_2(2, -4) \Rightarrow r_2 = \sqrt{4 + 16 - 16} = 2$$

$$C_1 C_2 = r_1 + r_2 \Rightarrow \sqrt{(K-2)^2 + 16} = |K| + 2$$

Case i: $K > 0$

$$(K-2)^2 + 16 = K + 2$$

$$K^2 - 4K + 4 + 16 = K^2 + 4K + 4$$

$$K = 2$$

Case ii : $K < 0$

$$(K-2)^2 + 16 = -K + 2$$

$$K^2 - 4K + 4 + 16 = K^2 + 4 - 4K$$

$$8K = -16$$

$$K = -2$$

- 2) Choose the incorrect statement about the two circles whose equations are given below:

$$x^2 + y^2 - 10x + 10y + 41 = 0 \text{ and } x^2 + y^2 - 16x - 10y + 80 = 0$$

(JEE MAIN 2021 SHIFT 1)

- a) Distance between two centres is the average of radii of both the circle
 b) Both circles' centres lie inside region of one another
 c) Both circles pass through the centre of each other
 d) Circles have two intersection points.

Ans: (b)

Sol:

$$S_1 : x^2 + y^2 - 10x - 10y + 41 = 0 \text{ and}$$

$$S_2 : x^2 + y^2 - 16x - 10y + 80 = 0$$

$$r_1 = 3, C_1(5,5)$$

$$r_2 = 3, C_2(8,5)$$

$$C_1C_2 = 3, r_1 = 3, r_2 = 3$$

Clearly, option(1) is correct.

Also, C_1 lies on S_2 and C_2 lies on S_1 .

So, option(2) is not correct.

2a) Let

$$A = \{(x, y) \in R \times R \mid 2x^2 + 2y^2 - 2x - 2y = 1\},$$

$$B = \{(x, y) \in R \times R \mid 4x^2 - 4y^2 - 16y + 7 = 0\} \text{ and}$$

$$C = \{(x, y) \in R \times R \mid x^2 + y^2 - 4x - 2y + 5 \leq r^2\}$$

Then the minimum value of $|r|$ such that $A \cup B \subseteq C$ is equal to

a) $1 + \sqrt{5}$

b) $\frac{3+2\sqrt{5}}{2}$

c) $\frac{3+\sqrt{10}}{2}$

d) $\frac{2+\sqrt{10}}{2}$

Ans: (b)

Sol: $S_1 : x^2 + y^2 - x - y - \frac{1}{2} = 0$

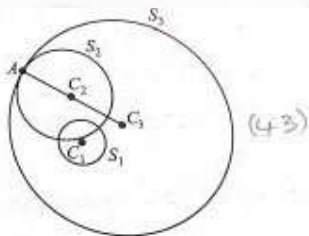
$$C_1\left(\frac{1}{2}, \frac{1}{2}\right) \text{ and } r_1 = \sqrt{\frac{1}{4} + \frac{1}{4} + \frac{1}{2}} = 1$$

$$S_2 : x^2 + y^2 - 4y + \frac{7}{4} = 0;$$

$$C_2 : (0, 2) \text{ and } r_2 = \sqrt{4 - \frac{7}{4}} = \frac{3}{2}$$

$$S_3 : x^2 + y^2 - 4x - 2y + 5 - r^2 = 0$$

$$C_3 : (2, 1) \text{ and } r_3 = \sqrt{4 + 1 - 5 + r^2} = |r|$$



$A \cup B \subseteq C$, means S_3 must contain S_1 and S_2 .

So, from the figure,

$$r_3 \geq C_3A = C_3C_2 + r_2$$

$$\therefore |r| \geq \sqrt{5} + \frac{3}{2}$$

2b) The locus of centers of the circles which touch the two circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = 4ax$ externally is

(JEE MAIN 2021 SHIFT 1)

a) $12x^2 - 4y^2 - 24ax + 9a^2 = 0$

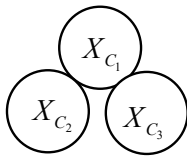
b) $12x^2 + 4y^2 + 24ax + 9a^2 = 0$

c) $12x^2 - 4y^2 + 24ax + 9a^2 = 0$

d) $12x^2 + 4y^2 - 24ax + 9a^2 = 0$

Ans: (a)

Sol:



$C_1 = (x, y)$

$C_2 = (0, 0)$

$C_3 = (2a, 0)$

$r_1 = r$

$r_2 = a$

$r_3 = 2a$

$C_1C_2 = a + r$

$C_1C_3 = 2a + r$

$C_1C_3 - C_1C_2 = a$

$\sqrt{(x-2a)^2 + y^2} - \sqrt{x^2 + y^2} = a$

by solving $12x^2 - 4y^2 - 24ax + 9a^2 = 0$

3) Choose the correct statement about two circles whose equations are given below:

$x^2 + y^2 - 10x - 10y + 41 = 0, x^2 + y^2 - 22x - 10y + 137 = 0$

(JEE MAIN 2021 SHIFT 1)

a) Circles have same centre

b) Circles have no meeting point

c) Circles have only one meeting point

d) Circles have two meeting points

Ans: 3

Sol: $x^2 + y^2 - 10x - 10y + 41 = 0$

$A(5, 5), R_1 = 3$

$x^2 + y^2 - 22x - 10y + 137 = 0$

$B(11, 5), R_2 = 3$

$AB = 6 = R_1 + R_2$

Touch each other externally

 $\Rightarrow$ Circles have only one meeting point.3a) The circles $x^2 + y^2 - 2x - 4y - 20 = 0, x^2 + y^2 + 4x - 2y + 4 = 0$ are (JEE MAIN 2021 SHIFT 1)

a) One lies outside the other circle

b) One lies completely inside the other circle

c) Touch externally

d) Touch internally

Sol: (b)

$C_1 = (1, 2), r_1 = \sqrt{1+4+20} = 5$

$C_2 = (-2, 1), r_2 = \sqrt{4+1-4} = 1$

$C_1C_2 = \sqrt{9+1} = \sqrt{10}, |r_1 - r_2| = 4$

$C_1C_2 < |r_1 - r_2|$

3b) Two circles $x^2 + y^2 - 2g_1x + 2f_1y = 0, x^2 + y^2 + 2g_2x + 2f_2y = 0$ touch each other externally then

(JEE MAIN 2021 SHIFT 1)

a) $f_1 g_1 = f_2 g_2$

b) $\frac{f_1}{g_1} = \frac{f_2}{g_2}$

c) $f_1 f_2 = g_1 g_2$

d) $\frac{f_1}{g_2} = \frac{f_2}{g_1}$

Sol: (b)

$$C_1 C_2 = r_1 + r_2$$

$$C_1 = (-g_1 - f_1), r_1 = \sqrt{g_1^2 + f_1^2}$$

$$C_2 = (-g_2 - f_2), r_2 = \sqrt{g_2^2 + f_2^2}$$

$$\sqrt{(-g_2 + g_1)^2 + (-f_2 + f_1)^2} = \sqrt{g_1^2 + f_1^2} + \sqrt{g_2^2 + f_2^2}$$

SBS

$$g_2^2 + g_1^2 - 2g_1 g_2 + f_1^2 + f_2^2 - 2f_1 f_2 = g_1^2 + f_1^2 + g_2^2 + f_2^2 + 2\sqrt{(g_1^2 + f_1^2)(g_2^2 + f_2^2)}$$

$$\Rightarrow -2(g_1 g_2 + f_1 f_2) = 2\sqrt{g_1^2 g_2^2 + g_1^2 f_2^2 + f_1^2 g_2^2 + f_1^2 f_2^2}$$

$$\Rightarrow \frac{f_1}{g_1} = \frac{f_2}{g_2}$$

4) Let Z be the set of all integers,

(JEE MAIN 2021 SHIFT 1)

$$A = \{(x, y) \in Z \times Z : (x-2)^2 + y^2 \leq 4\},$$

$$B = \{(x, y) \in Z \times Z : x^2 + y^2 \leq 4\} \text{ and}$$

$$C = \{(x, y) \in Z \times Z : (x-2)^2 + (y-2)^2 \leq 4\}$$

If the total number of relations from $A \cap B$ to $A \cap C$ is 2^p , then the value of P is

a) 25

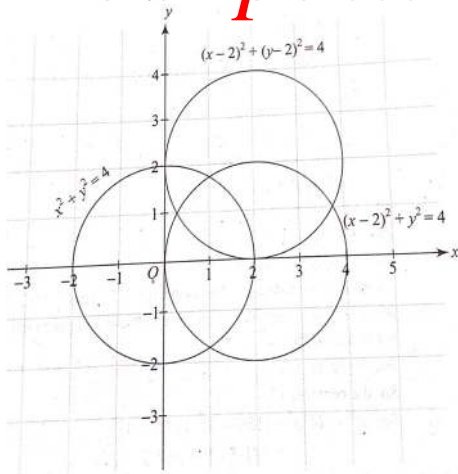
b) 16

c) 49

d) 9

Ans: (a)

Sol:



$$C_1 : (x-2)^2 + y^2 \leq 4$$

$$C_2 : x^2 + y^2 \leq 4$$

$$C_3 : (x-2)^2 + (y-2)^2 \leq 4$$

No. of points common in C_1 and C_2 is 5.

$$(0,0), (1,0), (2,0), (1,1), (1,-1)$$

Similarly, no. of points common in C_2 and C_3 is 5.

$$\text{So, number of relations} = 2^{5 \times 5} = 2^{25}.$$

- 4a) If the radius of the circumcircle of the triangle TPQ, where PQ is chord of contact corresponding to point T with respect to circle $x^2 + y^2 - 2x + 4y - 11 = 0$, is 6 units, then minimum distance of T from the director circle of the given circle is

(JEE MAIN 2021 SHIFT 1)

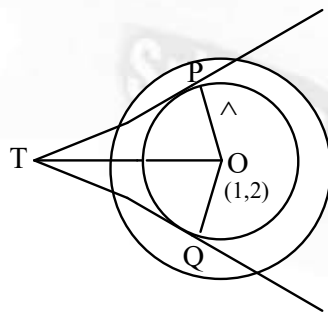
- a) 6 b) 12 c) $6\sqrt{2}$ d) $12 - 4\sqrt{2}$

Sol: (d)

$$\text{given circle } (x-1)^2 + (y+2)^2 = 16$$

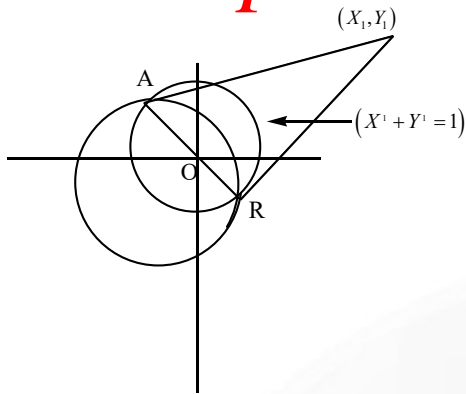
$$\text{its director circle is } (x-1)^2 + (y+2)^2 = 36$$

$$\Rightarrow OS = 4\sqrt{2}$$



Therefore, required distance,

$$TS = OT - SO = 12 - 4\sqrt{2}$$



Locus of point of intersection of tangents chord of contact of $((X_1, Y_1))$ w. r. t

$$x^2 + y^2 = 1 \text{ is } xx_1 - yy_1 = 1(AB)$$

AB is also common chord between two circles $\therefore -1 + (\lambda + 6)x - (8 - 2\lambda)y + 3 = 0$.

4b) If the circle $(x-a)^2 + y^2 = 25$ intersects the circle $x^2 + (y-b)^2 = 16$ in such a way that common chord is of maximum length, then value of $a^2 + b^2$ is **(JEE MAIN 2021 SHIFT 1)**

a)6

b)7

c)5

d)9

Ans : (d)

Sol: radical axis:

$$-2ax + a^2 + 2by - b^2 = 9$$

which passes through $(0, b)$

$$\text{i.e. } 0 + a^2 + 2b^2 - b^2 = 9 \Rightarrow a^2 + b^2 = 9$$

6) Three circles of radii a, b and c ($a < b < c$) touch each other externally. If they have x-axis as a common tangent, then **(JEE MAIN 2019 SHIFT 2)**

a) $\frac{1}{\sqrt{a}} = \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}}$ b) a, b, c are in A.P c) $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are in A.P d) $\frac{1}{\sqrt{b}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{c}}$

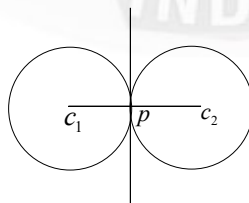
6a) The two circles $x^2 + b^2 + 2ax + c = 0$ and $x^2 + b^2 + 2by + c = 0$ touching each other then

A) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$ B) $a^2 + b^2 = c^2$ C) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c}$ D) None of these

Ans: C

Sol:

$$\text{Let } c_1(-a, 0) \quad c_2(0, -b) \quad c_1 c_2 = r_1 \pm r_2$$



$$\sqrt{a^2 + b^2} = \sqrt{a^2 - c} \pm \sqrt{b^2 - c}$$

$$c = \pm \sqrt{(a^2 - c)(b^2 - c)}$$

$$c^2 = a^2 b^2 - a^2 c - b^2 c + c^2$$

$$a^2b^2 = a^2c + b^2c$$

$$\Rightarrow \frac{1}{c} = \frac{1}{a^2} + \frac{1}{b^2}$$

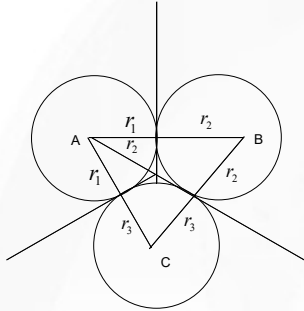
- 6b) Three circles are such that each touch the other two externally. The common tangents are concurrent at p. The length of tangent to each circle is p. The ratio of product of their radii to sum of their radii is

(JEE MAIN 2021 SHIFT 1)

- a) $\frac{p}{2}$ b) $\frac{p^2}{2}$ c) p d) p^2

Ans : (d)

Sol:



Let A, B, C be centres

r_1, r_2, r_3 be radii

$$\tan \frac{A}{2} = \frac{p}{r_1} \quad \tan \frac{B}{2} = \frac{p}{r_2}$$

$$\tan \frac{C}{2} = \frac{p}{r_3}$$

$$A + B + C = 180^\circ \text{ then } \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$$

$$\frac{r_1}{p} + \frac{r_2}{p} + \frac{r_3}{p} = \frac{r_1}{p} \cdot \frac{r_2}{p} \cdot \frac{r_3}{p}$$

$$\frac{1}{p}(r_1 + r_2 + r_3) = \frac{r_1 r_2 r_3}{p \cdot p^2}$$

$$\frac{r_1 r_2 r_3}{r_1 + r_2 + r_3} = p^2$$

- 7) If the circles $x^2 + y^2 - 16x - 20y + 164 = r^2$ and $(x-4)^2 + (y-7)^2 = 36$ intersect at two distinct points, then

(JEE MAIN 2019 SHIFT 2)

- a) $0 < r < 1$ b) $1 < r < 11$ c) $r > 11$ d) $r = 11$

Ans: b

Solution: Given

$$(x-8)^2 + (y-10)^2 = r^2$$

$$(x-4)^2 + (y-7)^2 = 36$$

$$c_1 = (8, 10)$$

$$c_2 = (4, 7)$$

$$r_1 = r$$

$$r_2 = 6$$

Here $|r_1 - r_2| < c_1 c_2$

$$|r - 6| < 5$$

$$\Rightarrow -5 < r - 6 < 5$$

$$\Rightarrow 1 < r < 11$$

7a) If two circles $(x-2)^2 + (y+3)^2 = \lambda^2$ and $x^2 + y^2 - 4x + 4y - 1 = 1$ intersect in two distinct points then (JEE MAIN 2019 SHIFT 1)

- a) $\lambda < -2$ b) $-2 < \lambda < 4$ c) $2 < \lambda < 4$ d) $\lambda = -2$

Ans: B

Sol: $c_1(2, -3)$ $r_1 = \lambda$

$c_2(2, -2)$ $r_2 = \sqrt{4+4+1} = 3$

$c_1 c_2 = \sqrt{0+1} = 1$

$r_1 - r_2 < c_1 c_2 < r_1 + r_2$

$\lambda - 3 < 1 < \lambda + 3$

$\lambda < 4, \quad -2 < \lambda$

$-2 < \lambda < 4$

7b) The circle $x^2 + y^2 - 6x - 10y + k = 0$ does not touch or intersects the co-ordinate axes and point (1,4) lies inside the circle then limits of k are (JEE MAIN 2019 SHIFT 1)

- A) $25 < k < 29$ B) $25 < k < 27$ C) $28 < k < 29$ D) $0 < k < 29$

Ans: (a)

Sol: $x^2 + y^2 - 6x - 10y + k = 0$

$c(3, 5)$ $r = \sqrt{9+25-k} = \sqrt{34-k}$

(1,4) lies inside $\Rightarrow s_{11} < 10$

$1+16-6-40+k < 0$

$k < 29$ -----(1)

The circle neither touch not intersect area

$\therefore r < 3$

$r^2 < 9$

$34-k < 9 \Rightarrow 25 < k$ -----(2)

From (1) and (2) $25 < K < 29$

8) The common tangent to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 + 6x + 8y - 24 = 0$ also passes through the point (JEE MAIN 2019 SHIFT 1)

- a) $(-4, 6)$ b) $(6, -2)$ c) $(4, -2)$ d) $(-6, 4)$

Ans: A

Solution: $s - s^1 = 0$

$\Rightarrow 6x + 8y - 24 = 0$

Above equation passes through $(-4, 6)$

- 8a) The external center of similitude of the circle $x^2 + y^2 - 12x + 8y + 48 = 0$ and $x^2 + y^2 - 4x + 2y - 4 = 0$ divides the segment joining centres in the ratio

(JEE MAIN 2019 SHIFT 1)

- a) 2 : 3 b) 3 : 2 c) -2 : 3 d) -3 : 2

Ans: (c)

Sol:

$$x^2 + y^2 - 12x + 8y + 48 = 0 \quad x^2 + y^2 - 4x + 2y - 4 = 0$$

$$c_1 = (6, -4) \quad c_2 = (2, -1)$$

$$r_1 = \sqrt{36 + 16 - 48} = 2 \quad r_2 = \sqrt{4 + 1 + 4} = 3$$

$$c_1 c_2 = \sqrt{16 + 9} = 5 \quad r_1 + r_2 = 2 + 3 = 5$$

$$c_1 c_2 = r_1 + r_2$$

The two circles touches each other externally. The external centre of similitude divides the centers in the ratio $r_1 : r_2$ externally

$$= -2 : 3$$

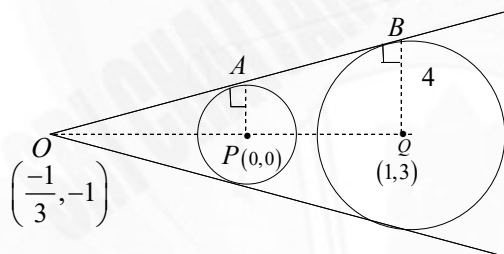
- 8b) If $\left(-\frac{1}{3}, -1\right)$ is a center of similitude for the circles $x^2 + y^2 = 1$ and $x^2 + y^2 - 2x - 6y - 6 = 0$

then the length of common tangent to the circle is

- a) $\frac{1}{3}$ b) $\frac{4}{3}$ c) 1 d) $\frac{2}{3}$

Ans: (c)

Sol:



$$c_1 = (0, 0), \quad c_2 = (1, 3), \quad r_1 = 1, \quad r_2 = 4$$

$$OP = \sqrt{\frac{1}{9} + 1} = \frac{\sqrt{10}}{3}, \quad OQ = \sqrt{\left(1 + \frac{1}{3}\right)^2 + (3+1)^2} = \sqrt{\frac{16}{9} + 16} = \frac{4\sqrt{10}}{3}$$

$$OB^2 = OQ^2 - BQ^2 = \left(\frac{4\sqrt{10}}{3}\right)^2 - 4^2 = \frac{160 - 144}{9} = \frac{16}{9}$$

$$OB = \frac{4}{3}$$

$$OA^2 = OP^2 - AP^2 = \frac{10}{9} - 1 = \frac{1}{9} \Rightarrow OA = \frac{1}{3}$$

$$\text{Length of common tangent} = AB = OB - OA = \frac{4}{3} - \frac{1}{3} = 1$$

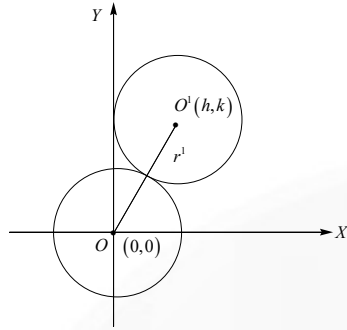
- 9) The locus of the centre of the circles which touch the circle $x^2 + y^2 = 1$ externally, also touch the Y-axis and lies in the first quadrant is

(JEE MAIN 2019 SHIFT 1)

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a) $x = \sqrt{1+4y}, y \geq 0$ b) $y = \sqrt{1+2x}, x \geq 0$ c) $y = \sqrt{1+4x}, x \geq 0$ d) $x = \sqrt{1+2x}, y \geq 0$

Ans : (b)



Sol :

Let centre of required circle is (h,k)

$$OO' = r + r'$$

Diagram

By the diagram $\sqrt{h^2 + k^2} = 1 + h$

$$h^2 + k^2 = 1 + h^2 + 2h$$

$$k^2 = 2h + 1$$

Locus is $y = \sqrt{1+2x}$; $x \geq 0$

- 9a) The lines represented by $5x^2 - xy - 5x + y = 0$ are normals to a circle $S = 0$. If this circle touches the circle $S' = x^2 + y^2 - 2x + 2y - 7 = 0$ externally, then the equation of the chord of contact of centre of $S' = 0$ with respect to $S = 0$ is **(JEE MAIN 2019 SHIFT 1)**

- a) $2y - 7 = 0$ b) $x - 1 = 0$ c) $3x + 4y - 7 = 0$ d) $x + y = 5$

Ans : (a)

$$\text{Sol : } 5x^2 - xy - 5x + y = 0 \Rightarrow (x-1)(5x-y) = 0$$

Normals of the circle $S = 0$ are $x = 1$ & $5x = y$

Centre of circle $S = 0$ is the P.I of normals . $C_1(1,5)$; r_1 be it radius .

For the circle $S' = 0$; Centre $C_2(1,-1)$; $r_2 = 3$

$$C_1C_2 = r_1 + r_2 \Rightarrow 6 = r_1 + 3 \Rightarrow r_1 = 3.$$

Equation of circle $S = 0$ is $(x-1)^2 + (y-5)^2 = 9$

$$\Rightarrow x^2 + y^2 - 2x - 10y + 17 = 0$$

Equation of chord of contact of $(1,-1)$ w.r.t $S = 0$ is

$$S_1 = 0 \Rightarrow x(1) + y(-1) - 1(x+1) - 5(y-1) + 17 = 0$$

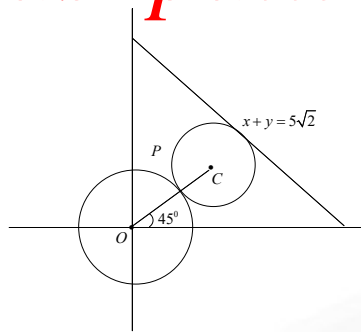
$$\Rightarrow 2y - 7 = 0$$

- 9b) The co-ordinates of the centre of the smallest circle touching the circle $x^2 + y^2 = 4$ and the line $x + y = 5\sqrt{2}$ are. **(JEE MAIN 2019 SHIFT 1)**

- a) $\left(\frac{7}{2\sqrt{2}}, \frac{7}{2\sqrt{2}}\right)$ b) $\left(\frac{3}{2}, \frac{3}{2}\right)$ c) $\left(\frac{-7}{2\sqrt{2}}, \frac{-7}{2\sqrt{2}}\right)$ d) $\left(\frac{-3}{2}, \frac{-3}{2}\right)$

Ans : A

Sol:



$$OC = \frac{7}{2} \text{ \& Slope of } OC = 1$$

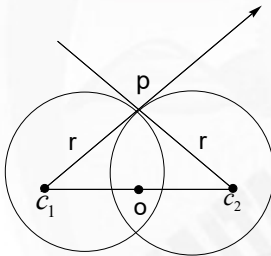
$$C = \left(0 + \frac{7}{2} \cdot \frac{1}{\sqrt{2}}, 0 + \frac{7}{2} \cdot \frac{1}{\sqrt{2}} \right)$$

- 10) Two circles with equal radii are intersecting at the points (0, 1) and (0, -1). The tangent at the point (0, 1) to one of the circles passes through the centre of the other circle. Then the distance between the centres of these circles is **(JEE MAIN 2019 SHIFT 2)**

- a) 1 b) $\sqrt{2}$ c) $2\sqrt{2}$ d) 2

Ans: (d)

Sol:



From fig.

$$OP = OC_1 = OC_2 = 1$$

$$\therefore C_1C_2 = 1 + 1 = 2$$

- 10 a) Centre of the circle cutting $x^2 + y^2 - 2x + 4y = 0$ orthogonally and passing through (0,0), (2,0) is **(JEE MAIN 2019 SHIFT 2)**

- A) $\left(\frac{3}{2}, 1\right)$ B) $\left(1, \frac{1}{2}\right)$ C) $\left(1, -\frac{3}{4}\right)$ D) $(-1, -3)$

Ans: B

Sol: Circle $x^2 + y^2 + 2gx + 2fy + c = 0$ ----- (1)

Passing through (0,0)

$$\Rightarrow c = 0$$

Passing through (2,0)

$$\Rightarrow 4 + 4g = 0$$

$$\Rightarrow g = -1$$

$$x^2 + y^2 - 2x + 4y = 0 \text{ ----- (2)}$$

$$g^1 = 1, f^1 = +2, c^1 = 0$$

Orthogonal condition

$$2gg^1 + 2ff^1 = c + c^1$$

$$2(-1)(-1) + 2f(+2) = c + 0$$

$$+2 + 4f = 0$$

$$4f = -2$$

$$f = -\frac{1}{2}$$

$$\text{Centre} = (-g, -f)$$

$$= \left(1, \frac{1}{2}\right)$$

- 10 b) If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = 4$ orthogonally then the locus of its centre is (JEE MAIN 2019 SHIFT 1)

a) $2ax + 2by + a^2 + b^2 + 4 = 0$

b) $2ax - 2by - a^2 - b^2 - 4 = 0$

c) $2ax - 2by + a^2 + b^2 + 4 = 0$

d) $2ax + 2by - a^2 - b^2 - 4 = 0$

Ans: (d)

Sol: Let $c_1 = (h, k)$ circle passing through A(a, b)

$$\therefore r_1 = c_1A = \sqrt{(h-a)^2 + (k-b)^2}$$

$$x^2 + y^2 = 4 \Rightarrow c_2 = (0, 0), r_2 = 2$$

Orthogonal condition

$$(c_1c_2)^2 = r_1^2 + r_2^2$$

$$h^2 + k^2 = (h-a)^2 + (k-b)^2 + 2^2$$

$$h^2 + k^2 = h^2 + a^2 - 2ah + k^2 + b^2 - 2bk + 4$$

$$2ah + 2bk - a^2 - b^2 - 4 = 0$$

Locus of (h, k) is

$$2ax + 2by - a^2 - b^2 - 4 = 0$$

- 11) Consider three circles:

$$C_1 : x^2 + y^2 = r^2$$

$$C_2 : (x-1)^2 + (y-1)^2 = r^2$$

$$C_3 : (x-2)^2 + (y-1)^2 = r^2$$

If a line $L : y = mx + c$ be a common tangent to C_1, C_2 and C_3 such that C_1 and C_3 lie on one side of line L while C_2 lies on other side, then the value of $20(r^2 + c)$ is equal to :

a) 23

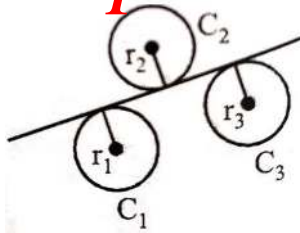
b) 15

c) 12

d) 6

Ans: d

Solution:



Given $L : y = mx + c$ be a common tangent

Let r_1 for c_1 is, $r_1 = \left| \frac{c}{\sqrt{1+m^2}} \right|$

$$r_3 = \left| \frac{2m-1+c}{\sqrt{1+m^2}} \right|$$

$$\therefore m = \frac{1}{2} [as \ r_1 = r_3 = r_2]$$

$$\text{For } r_2 = \left| \frac{m-1+c}{\sqrt{1+m^2}} \right| \Rightarrow 2c = 1-m \Rightarrow c = \frac{1}{4}$$

$$\text{Now } 20(r^2 + c) \Rightarrow 20\left(\frac{1}{20} + \frac{5}{20}\right) = 6$$

11a. If the lengths of tangents drawn to the circles

$x^2 + y^2 - 8x + 40 = 0$, $5x^2 + 5y^2 - 25x + 80 = 0$, $x^2 + y^2 - 8x + 16y + 160 = 0$ from the point p are equal then p =

- a) $\left(8, \frac{15}{2}\right)$ b) $\left(-8, \frac{15}{2}\right)$ c) $\left(8, \frac{-15}{2}\right)$ d) $\left(-8, \frac{-15}{2}\right)$

Ans: (c)

Sol: Radical axis of (1)&(2) is $3x - 24 = 0$

$$\Rightarrow x = 8$$

Radical axis of (1)&(3) is $16y + 120 = 0$

$$\Rightarrow y = \frac{-15}{2}$$

11b. $C_1 : x^2 + y^2 = r^2$ and $C_2 : \frac{x^2}{16} + \frac{y^2}{9} = 1$ intersect at four distinct points A, B, C and D. Their common tangents form a parallelogram A'B'C'D'. If A'B'C'D' is a square then r is equal to

- a) $\frac{13\sqrt{2}}{5}$ b) $\frac{12\sqrt{2}}{5}$ c) $\frac{12\sqrt{3}}{5}$ d) $\frac{13\sqrt{3}}{5}$

Ans: (b)

Sol: $C_1 : x^2 + y^2 = r^2$ -----(1)

$$C_2 : \frac{x^2}{16} + \frac{y^2}{9} = 1$$

$$\Rightarrow \frac{r^2 - y^2}{16} + \frac{y^2}{9} = 1$$

$$\Rightarrow 9(r^2 - y^2) + 16y^2 = 144$$

$$\Rightarrow y^2 = \frac{144 - 9r^2}{7}$$

If A'B'C'D' is a square the $x^2 = y^2$

$$\frac{16r^2 - 144}{7} = \frac{144 - 9r^2}{7}$$

$$\Rightarrow 25r^2 = 2(144)$$

- 12) The number of common tangents, to the circles $x^2 + y^2 - 18x - 15y + 131 = 0$ and $x^2 + y^2 - 6x - 6y - 7 = 0$, is :

(15 APRIL 2023 SHIFT - I)

- a) 3 b) 2 c) 1 d) 4

Ans : (a)

$$\text{Sol: } C_1\left(9, \frac{15}{2}\right) \quad r_1 = \sqrt{81 + \frac{225}{4} - 131} = \frac{5}{2}$$

$$C_2(3, 3) \quad r_2 = 5$$

$$C_1C_2 = \sqrt{6^2 + \frac{81}{4}} = \frac{15}{2}$$

$$r_1 + r_2 = \frac{15}{2}$$

$$C_1C_2 = r_1 + r_2$$

Number of common tangents = 3

- 12A. The number of common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is

- a) 0 b) 1 c) 3 d) 4

Ans: (b)

$$\text{Sol: } x^2 + y^2 = 4 \quad x^2 + y^2 - 6x - 8y - 24 = 0$$

$$C_1 = (0, 0) \quad C_2 = (3, 4)$$

$$r_1 = 2 \quad r_2 = \sqrt{9 + 16 + 24} = 7$$

$$C_1C_2 = \sqrt{9 + 16} = 5$$

$$r_1 + r_2 = 9 \quad (r_1 - r_2) = 5$$

$$C_1C_2 = |r_1 - r_2|$$

Number of common tangents = 1

- 12B. The number of common tangents to the circles $x^2 + y^2 - 12x + 8y + 48 = 0$ and

$$x^2 + y^2 - 4x + 2y - 4 = 0 \text{ is}$$

- a) 0 b) 1 c) 2 d) 3

Ans: (d)

$$\text{Sol: } C_1 = (6, -4) \quad C_2 = (2, -1)$$

$$r_1 = \sqrt{36 + 16 - 48} = 2 \quad r_2 = \sqrt{4 + 1 + 4} = 3$$

$$C_1 C_2 = \sqrt{16+9} = 5$$

$$r_1 + r_2 = 5$$

$$C_1 C_2 = r_1 + r_2$$

Number of common tangents = 3

7) System of circles, Radical axis

(Family of circles passing through P.O.I of circle and line in circles, Family of circles touching the circle at point $(x_1; y_1)$ in circles, Family of circles passing through P.O.I of two circles in circles, Equation of radical axis, Properties of radical axis, Equation of common chord of two circles in circles, Co-axial system of circles, Radical centre in circles, Properties of radical centre, Limiting points of a co-axial system of circles in circles)

- 1) The circle passing through the intersection of the circles $x^2 + y^2 - 6x = 0$ and $x^2 + y^2 - 4y = 0$ having centre on the line $2x - 3y + 12 = 0$ also passes through the point

(JEE MAIN 2020 SHIFT 2)

- a) $(-3, 6)$ b) $(-1, 3)$ c) $(-3, 1)$ d) $(1, -3)$

Ans : (a)

Sol : The req eq of circle $S_1 + \lambda S_2 = 0$

$$x^2 + y^2 - 6x + \lambda(x^2 + y^2 - 4y) = 0 \dots\dots(1)$$

$$\Rightarrow (1 + \lambda)x^2 + (1 + \lambda)y^2 - 6x - 4\lambda y = 0$$

$$\Rightarrow x^2 + y^2 - \frac{6}{1 + \lambda}x - \frac{4\lambda}{1 + \lambda}y = 0$$

$$\text{Centre } \left(\frac{3}{1 + \lambda}, \frac{2\lambda}{1 + \lambda} \right) \text{ lies on } 2x - 3y + 12 = 0$$

$$\Rightarrow 2\left(\frac{3}{1 + \lambda}\right) - 3\left(\frac{2\lambda}{1 + \lambda}\right) + 12 = 0$$

$$\Rightarrow 6 - 6\lambda + 12 + 12\lambda = 0$$

$$\Rightarrow 6\lambda + 18 = 0 \Rightarrow \lambda = -3$$

Sub in (1)

$$x^2 + y^2 - 6x - 3(x^2 + y^2 - 4y) = 0$$

$$\Rightarrow x^2 + y^2 - 6x - 3x^2 - 3y^2 + 12y = 0$$

$$\Rightarrow -2x^2 - 2y^2 - 6x + 12y = 0$$

$$\Rightarrow x^2 + y^2 + 3x - 6y = 0$$

$(-3, 6)$ satisfies the circle.

1a) The equation of circle passing through point of intersection of the circles

$x^2 + y^2 - 6x + 2y + 4 = 0$ and $x^2 + y^2 + 2x - 4y - 6 = 0$ and having centre on $y = x$ is _____ (JEE MAIN 2020 SHIFT 1)

- a) $3(x^2 + y^2) - 5x - 5y + 2 = 0$ b) $x^2 + y^2 - 4x + 3y + 7 = 0$
 c) $7(x^2 + y^2) - 10x - 10y - 12 = 0$ d) $5(x^2 + y^2) - 2x + 3y + 6 = 0$

Ans : (c)

Sol : The req. equation circle $S_1 + \lambda S_2 = 0$

$$x^2 + y^2 - 6x + 2y + 4 + \lambda(x^2 + y^2 + 2x - 4y - 6) = 0 \dots\dots(1)$$

$$\Rightarrow (1 + \lambda)x^2 + (1 + \lambda)y^2 + (-6 + 2\lambda)x + (2 - 4\lambda)y + 4 - 6\lambda = 0$$

$$\Rightarrow x^2 + y^2 + \frac{2(\lambda - 3)}{1 + \lambda}x + 2\frac{(1 - 2\lambda)}{1 + \lambda}y + \frac{2(2 - 3\lambda)}{1 + \lambda} = 0$$

Centre $\left(\frac{3 - \lambda}{1 + \lambda}, \frac{2\lambda - 1}{1 + \lambda}\right)$ lines on $y = x$

$$\frac{2\lambda - 1}{1 + \lambda} = \frac{3 - \lambda}{1 + \lambda} \Rightarrow 2\lambda - 1 = 3 - \lambda \Rightarrow 3\lambda = 4 \Rightarrow \lambda = \frac{4}{3}$$

Sub in (1)

$$7(x^2 + y^2) - 10x - 10y - 12 = 0$$

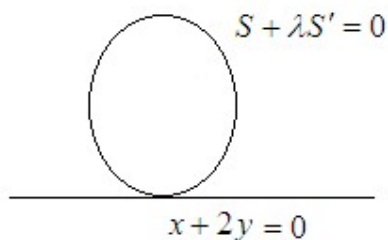
1b) The equation of circle passing through point of intersection of $x^2 + y^2 - 4 = 0$ and $x^2 + y^2 - 2x - 4y + 4 = 0$ and touching the line $x + 2y = 0$ is _____

(JEE MAIN 2020 SHIFT 2)

- a) $x^2 + y^2 - x - 2y = 0$ b) $x^2 + y^2 + x + 3y = 0$
 c) $x^2 + y^2 - 3x - 6y = 0$ d) None

Ans : A

Sol :



The req circle $S + \lambda S' = 0$

$$x^2 + y^2 - 2x - 4y + 4 + \lambda(x^2 + y^2 - 4) = 0$$

$$(1 + \lambda)x^2 + (1 + \lambda)y^2 - 2x - 4y + 4 - 4\lambda = 0$$

$$x^2 + y^2 - \frac{2x}{1+\lambda} - \frac{4y}{1+\lambda} + \frac{4-4\lambda}{1+\lambda} = 0$$

$$\text{Centre} \left(\frac{1}{1+\lambda}, \frac{2}{1+\lambda} \right), r = \sqrt{\frac{1}{(1+\lambda)^2} + \frac{4}{(1+\lambda)^2} - \frac{4-4\lambda}{1+\lambda}}$$

$$r = \sqrt{\frac{4\lambda^2 + 1}{(1+\lambda)^2}}$$

$$r = d$$

$$d = \left| \frac{\frac{1}{1+\lambda} + \frac{4}{1+\lambda}}{\sqrt{1^2 + 2^2}} \right| = \sqrt{\frac{4\lambda^2 + 1}{(1+\lambda)^2}}$$

S.O.b.

$$\Rightarrow \lambda = \pm 1 \Rightarrow \lambda = 1$$

$$x^2 + y^2 - 2x - 4y + 4 + 1(x^2 + y^2 - 4) = 0$$

$$\Rightarrow 2x^2 + 2y^2 - 2x - 4y = 0$$

$$\Rightarrow x^2 + y^2 - x - 2y = 0$$

2) For the four circles M, N, O and P, following four equations are given:

Circle M: $x^2 + y^2 = 1$

Circle N: $x^2 + y^2 - 2x = 0$

Circle O: $x^2 + y^2 - 2x - 2y + 1 = 0$

Circle P: $x^2 + y^2 - 2y = 0$

If the centre of circle M is joined with centre of the circle N, further centre of circle N is joined with centre of circle O, centre of circle O is joined with the centre of circle P and lastly, centre of circle P is joined with centre of circle M, Then these lines from the sides of a:

(JEE MAIN 2021 SHIFT 1)

- a) Rhombus b) Square c) Rectangle d) Parallelogram

Ans: (b)

Sol: $M : x^2 + y^2 = 1, \text{Centre}, C_1 \equiv (0,0)$

$N : x^2 + y^2 - 2x = 0, \text{Centre}, C_2 \equiv (1,0)$

$O : x^2 + y^2 - 2x - 2y + 1 = 0, \text{Centre}, C_3 \equiv (1,1)$

$P : x^2 + y^2 - 2y = 0, \text{Centre}, C_4 \equiv (0,1)$

Clearly, $C_1 C_2 C_3 C_4$ is a square.

2a) $c_1 c_2 c_3$ are circles of radii 5, 3, 2 respectively and $c_1 c_2$ touches externally and C internally.

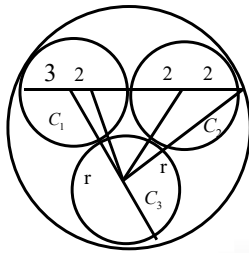
The radius of the circle C which touches C internally and $c_1 c_2$ externally is

(JEE MAIN 2021 SHIFT 1)

- a) $\frac{3}{2}$ b) $\frac{20}{9}$ c) $\frac{35}{19}$ d) $\frac{30}{19}$

Ans : (d)

Sol:

from $\Delta c_1 c_2 c_3$

$$\cos \theta = \frac{25 + (3+r)^2 - (2+r)^2}{2(5)(3+r)}$$

$$\text{from } \Delta c c_1 c_3, \cos \theta = \frac{4 + (3+r)^2 - (5-r)^2}{2(2)(3+r)}$$

$$\text{solving above two, } r = \frac{30}{9}$$

- 2b) If $c_1 : x^2 + y^2 - 20x + 64 = 0$, $c_2 : x^2 + y^2 + 30x + 144 = 0$. Then the length of the shortest line segment of a line PQ which touches c_1 at P and c_2 at Q is **(JEE MAIN 2021 SHIFT 2)**
- a) 20 b) 15 c) 22 d) 27

Sol:

$$x^2 + y^2 - 20x + 64 = 0 - c_1 = (100) \quad r_1 = 6$$

$$x^2 + y^2 + 30x + 144 = 0 - c_2 = (-150) \quad r_2 = 9$$

$$d = c_1 c_2 = 25, PQ = \sqrt{d^2 - (r_1 + r_2)^2} = 20$$

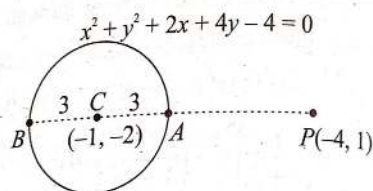
- 3) Let r_1 and r_2 be the radii of the largest and smallest circles, respectively, which pass through the point $(-4, 1)$ and having their centres on the circumference of the circle

$$x^2 + y^2 + 2x + 4y - 4 = 0. \text{ If } \frac{r_1}{r_2} = a + b\sqrt{2}, \text{ then } a + b \text{ is equal to } \quad \textbf{(JEE MAIN 2021 SHIFT 1)}$$

- a) 5 b) 7 c) 3 d) 11

Ans: (a)

Sol:



Centre of smallest circle is A.

Centre of largest circle is B.

$$r_2 = |CP - CA| = 3\sqrt{2} - 3$$

$$r_1 = CP + CB = 3\sqrt{2} + 3$$

$$\therefore \frac{r_1}{r_2} = \frac{3\sqrt{2} + 3}{3\sqrt{2} - 3} = \frac{(3\sqrt{2} + 3)^2}{9} = (\sqrt{2} + 1)^2 = 3 + 2\sqrt{2}$$

$$\therefore a = 3, b = 2$$

- 3a) Let r_1, r_2 be the radii of the tangent and smallest circles, which passes through $(-4, 1)$ and their centers on the circumference of the circle $x^2 + y^2 + 2x + 4y - 4 = 0$.

(JEE MAIN 2021 SHIFT 1)

- a) 3 b) 7 c) 5 d) 11

Ans : (a)

Sol:

$$x^2 + y^2 + 2x + 4y - 4 = 0$$

$$x^2 + 2x + 1 + y^2 + 4y + 4 = 4 + 1 + 4$$

$$(x+1)^2 + (y+2)^2 = 3^2 \rightarrow c = (-1, -2), r = 3, P(-4, 1)$$

$$r_1 = 6 + r_2, CP = 3\sqrt{2}$$

$$r_2 = 3\sqrt{2} - 3, r_1 = 3 + 3\sqrt{2}$$

$$\frac{r_1}{r_2} = \frac{3\sqrt{2} + 3}{3\sqrt{2} - 3} = (\sqrt{2} + 1)3 + 2\sqrt{2}$$

$$a = 3, b = 2$$

$$a + b = 5$$

- 3b) r and r^1 are the radii of the circles $s = 0, s^1 = 0$ respectively then $\frac{s}{r} \pm \frac{s^1}{r^1} = 0$ intersect at an angle of

(JEE MAIN 2021 SHIFT 1)

- a) $\frac{s}{r} \pm \frac{s^1}{r^1} = 0$ b) $\frac{s}{r} \pm \frac{s^1}{r^1} = 0$ c) $\frac{\pi}{2}$ d) $\frac{\pi}{6}$

Sol: Conceptual

- 4) A circle C touches the line $x=2y$ at the point $(2, 1)$ and intersects the circle

$C_1 : x^2 + y^2 + 2y - 5 = 0$ at two points P and Q such that PQ is a diameter of C_1 . Then the diameter of C is

(JEE MAIN 2021 SHIFT 1)

- a) $4\sqrt{15}$ b) $\sqrt{285}$ c) 15 d) $\frac{7}{2}\sqrt{5}$

Ans: (d)

Sol: Equation of family of circles touching line $x - 2y = 0$ at point $(2, 1)$ is given by

$$(x-2)^2 + (y-1)^2 + \lambda(x-2y) = 0$$

$$\text{or } x^2 + y^2 + x(\lambda-4) + y(-2-2\lambda) + 5 = 0$$

Equation of common chord with the circle C_1 is

$$(\lambda-4)x - (2\lambda+4)y + 10 = 0$$

Common chord is diameter of the circle C_1

So, it passes through the point $(0, -1)$

$$\Rightarrow \lambda = -7$$

$$(x-2)^2 + (y-1)^2 - 7(x-2y) = 0$$

$$x^2 + y^2 - 11x + 12y + 5 = 0$$

$$r = \sqrt{\frac{121}{4} + 36 - 5}$$

$$= \frac{7}{2}\sqrt{5}$$

- 4a) A circle 'c' touches the line $x=3y$ at the point $(3,1)$ and intersect the circle $c_1: x^2 + y^2 + 2y - 5 = 0$ at the two points 'P' and 'Q'. Such that PQ is a diameter of c_1 .

Then the diameter of 'c'.

(JEE MAIN 2019 SHIFT 1)

a) $\frac{5\sqrt{10}}{2}$

b) $5\sqrt{10}$

c) $5\sqrt{5}$

d) $\frac{5\sqrt{5}}{2}$

Sol: equation of 'c'

$$(x-3)^2 + (y-1)^2 + \lambda(x-3y) = 0$$

$$\Rightarrow x^2 + y^2 + (\lambda-6)x - (2+3\lambda)y + 10 = 0$$

$$c_1: x^2 + y^2 + 2y - 5 = 0$$

$$\text{equation of } PQ \Rightarrow c - c_1 = 0$$

$$(\lambda-6)x - 3\lambda y + 15 = 0$$

$\therefore PQ$ as diameter at circle c_1

$$0 - 3\lambda(-1) + 15 = 0$$

$$\Rightarrow 3\lambda + 15 = 0 \Rightarrow \lambda = -5$$

$$\text{equation of } c: x^2 + y^2 - 11x + 13y + 10 = 0$$

$$r = \sqrt{\frac{121}{4} + \frac{169}{4} - 10} = \sqrt{\frac{290-40}{4}}$$

$$= \sqrt{\frac{250}{4}} = \frac{5\sqrt{10}}{2}$$

- 4b) If the equation of the circle passing through $(1,1)$ and point of intersection of the circles $x^2 + y^2 + 13x - 3y = 0$ and $2x^2 + 2y^2 + 4x - 7y - 25 = 0$ is $4x^2 + 4y^2 + ax + 6y + c = 0$ then $a+b+c=$

(JEE MAIN 2021 SHIFT 1)

a)-8

b)8

c)10

d) -10

Sol: required circle

$$(x^2 + y^2 + 13x - 3y) + \lambda(2x^2 + 2y^2 + 4x - 7y - 25) = 0$$

If this circle passing through (1,1) then $\lambda = \frac{1}{2}$

Equation of required circle is $4x^2 + 4y^2 + 30x - 13y - 25 = 0$

$A=30, b=-13, c=-25$

$a+b+c=30-13-25=-8$

- 5) If a circle 'c' passing through (4, 0) touches the circle $x^2 + y^2 + 4x - 6y = 12$ externally at (1,-1) then radius of 'c' is **(JEE MAIN 2021 SHIFT 1)**

- a) $\sqrt{57}$ b) $2\sqrt{5}$ c) 4 d) 5

Ans : (d)

Equation of tangent at (1,-1) is

$$x - y + 2(x+1) - 3(y-1) - 12 = 0$$

$$3x - 4y - 7 = 0$$

$$x^2 + y^2 + 4x - 6y - 12 + \lambda(3x - 4y - 7) = 0$$

It passes through (4,0)

$$\lambda = -4$$

$$x^2 + y^2 + 4x - 6y - 12 + (-4)(3x - 4y - 7) = 0$$

$$\therefore x^2 + y^2 - 8x + 10y + 16 = 0$$

- 5a) If a circle passes through (0,0) touches circle $x^2 + y^2 - 10x - 10y - 2 = 0$ externally at (1,-1) then radius of circle is **(JEE MAIN 2021 SHIFT 2)**

- a) $\sqrt{31}$ b) $\sqrt{13}$ c) $\sqrt{30}$ d) 13

Ans: (d)

Sol: $x^2 + y^2 - 10x - 10y - 2 = 0$

Equation of tangent at (1,-1) is

$$x + y(-1) - 5(x+1) - 5(y-1) - 2 = 0$$

$$x - y - 5x - 5 - 5y + 5 - 2 = 0$$

$$-4x - 6y - 2 = 0$$

$$2x + 3y + 1 = 0$$

$\therefore$ Required circle is $S + \lambda L = 0$

$$x^2 + y^2 - 10x - 10y - 2 + \lambda(2x + 3y + 1) = 0$$

It passes (0,0)

$$-2 + \lambda(0 + 0 + 1) = 0$$

$$\lambda = 2$$

$$x^2 + y^2 - 10x - 10y - 2 + 4x + 6y + 2 = 0$$

$$x^2 + y^2 - 6x - 4y = 0$$

$$\text{Radius} = \sqrt{9+4} = \sqrt{13}$$

5b) The radius of circle which touches $x^2 + y^2 - 6x + 6y + 17 = 0$ externally and having the co-ordinate axes as their normal's is

(JEE MAIN 2019 SHIFT 2)

- a) $\sqrt{2} - 1$ b) $4\sqrt{2} - 1$ c) $3\sqrt{2} - 1$ d) $3\sqrt{2} + 1$

Ans: (c)

Sol: Given $x^2 + y^2 - 6x + 6y + 17 = 0$

$$c_1(3, -3) \quad r_1 = \sqrt{9 + 9 - 17} = 1$$

$$c_2(0, 0) \quad r_2 = ?$$

Condition $c_1c_2 = r_1 + r_2$

$$\sqrt{9 + 9} = 1 + r_2$$

$$\sqrt{18} - 1 = r_2$$

$$r_2 = 3\sqrt{2} - 1$$

6) Let C_1 and C_2 be the centres of the circles $x^2 + y^2 - 2x - 2y - 2 = 0$ and $x^2 + y^2 - 6x - 6y + 14 = 0$ respectively. If P and Q are the points of intersection of these circles then the area (in sq. units) of the quadrilateral PC_1QC_2

(JEE MAIN 2019 SHIFT 1)

- a) 8 b) 6 c) 9 d) 4

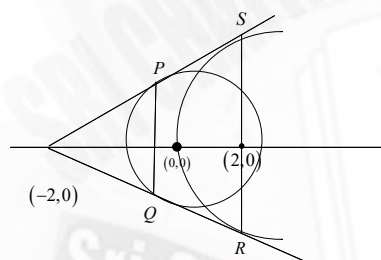
6a) The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the points P, Q and the parabola at the points R, S. Then the area of the quadrilateral PQRS is

(JEE MAIN 2019 SHIFT 1)

- a) 3 b) 6 c) 9 d) 15

Ans: (d)

Sol:



Equation of tangent to

$$y^2 = 8x \text{ is}$$

$$y = mx + \frac{2}{m} \quad \text{----- (1)}$$

Above equation is also a tangent to given circle $x^2 + y^2 = 2$ (I,e) $r = d$

$$\Rightarrow \sqrt{2} = \frac{\left| \frac{2}{m} \right|}{\sqrt{m^2 + 1}}$$

$$\Rightarrow \frac{4}{m^2(1 + m^2)} = 2$$

$$\Rightarrow m^4 + m^2 - 2 = 0$$

$$\Rightarrow (m^2 + 2)(m^2 - 1) = 0$$

$$\Rightarrow m^2 = 1$$

$$(i.e) m = \pm 1$$

From (1)

$$y = \pm x \pm 2$$

(I,e) $y = x + 2$ and $y = -x - 2$

By solving above equation we get $x = -2$ and $y = 0$

Tangents are drawn from $(-2, 0)$; chord of contact PQ to circle is $x(-2) + y(0) = 2$

$$\Rightarrow x = -1$$

Chord of contact RS to the parabola is

$$y(0) = 4(x - 2)$$

$$\Rightarrow x = 2$$

$$\therefore P(-1, 1)$$

$$Q(-1, -1)$$

$$R(2, -4)$$

$$S(2, 4)$$

$$\therefore \text{area} = \frac{1}{2} \begin{vmatrix} x_1 - x_3 & x_2 - x_4 \\ y_1 - y_3 & y_2 - y_4 \end{vmatrix}$$

$$= \frac{1}{2}(30)$$

$$= 15$$

- 6b) Let c_1 and c_2 denote the centres of the circles $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 1$ respectively and let P and Q be their points of intersection. Then the ratio of the areas of triangles C_1PQ and C_2PQ are **(JEE MAIN 2019 SHIFT 1)**

- a) 3 : 1 b) 5 : 1 c) 7 : 1 d) 9 : 1

Ans: (c)

Sol: $c_1 = (0, 0)$, $r_1 = 2$

$$c_2 = (2, 0)$$
, $r_2 = 1$

Equation of common chord $s - s' = 0$

$$x = \frac{7}{4}$$

Substitute x value in $x^2 + y^2 = 4$

$$\Rightarrow \frac{49}{16} + y^2 = 4$$

$$\Rightarrow y^2 = \frac{15}{16}$$

$$\Rightarrow y = \pm \frac{\sqrt{15}}{4}$$

$$\therefore p\left(\frac{7}{4} - \frac{\sqrt{15}}{4}\right)$$

$$Q\left(\frac{7}{4} - \frac{\sqrt{15}}{4}\right)$$

$$\text{Area of } \Delta C_1PQ = \frac{1}{2}|x_1y_2 - x_2y_1|$$

$$= \frac{1}{2}\left| -7\sqrt{15} - \frac{7\sqrt{15}}{16} \right|$$

$$= \frac{7\sqrt{15}}{16}$$

$$\text{Area of } \Delta C_2PQ = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$= \frac{\sqrt{15}}{16}$$

$$\text{Required ratio} = \frac{7\sqrt{15}}{16} : \frac{\sqrt{15}}{16} = 7 : 1$$

7) If the circles $x^2 + y^2 + 5kx + 2y + k = 0$ and $2(x^2 + y^2) + 2kx + 3y - 1 = 0, (k \in R)$, intersect at the points P and Q, then the line $4x + 5y - k = 0$ passes through P and Q for

(JEE MAIN 2019 SHIFT 1)

- a) exactly one value of k b) exactly two values of k
c) no value of k d) infinitely many values of k

7a) The equation of the circle which passes through the point $(2a, 0)$ and whose radical axis with the circle $x^2 + y^2 = a^2$ is the line $x = \frac{a}{2}$ is

(JEE MAIN 2019 SHIFT 1)

- a) $x^2 + y^2 - 2ax = 0$ b) $x^2 + y^2 + 2ax = 0$ c) $x^2 + y^2 - ax = 0$ d) $x^2 + y^2 + ax = 0$

Ans: (a)

Sol:

Center of the circle whose radical axis with $x^2 + y^2 = a^2$ is $x = \frac{a}{2}$ lies on x-axis ($y = 0$)

$$\Rightarrow x^2 + y^2 + 2gx + c = 0$$

$$\text{Radical axis is } 2gx + c + a^2 = 0$$

$$\text{Given radical axis is } x = \frac{a}{2}$$

$$\Rightarrow \frac{2g}{1} = \frac{c + a^2}{-\frac{a}{2}} \Rightarrow -ag = c + a^2$$

$$c = -ag - a^2 \text{ ----- (1)}$$

$$(2a, 0) \text{ lies on the circle } x^2 + y^2 + 2gx + c = 0$$

$$\Rightarrow 4a^2 + 4ag + c = 0$$

$$\text{From (1)} \quad 4a^2 + 4ag - ag - a^2 = 0$$

$$3a^2 + 3ag = 0 \Rightarrow g = -a$$

$$(1) \Rightarrow c = -a(-a) - a^2 = 0$$

Equation of circle is $x^2 + y^2 - 2ax = 0$

7 b) If the circle $x^2 + y^2 + 2ax + cy + a = 0$ and $x^2 + y^2 - 3ax + dy - 1 = 0$ intersect tow distinct points P and Q then the line $5x + by - a = 0$ passes through P and Q for **(JEE MAIN 2019 SHIFT 1)**

- a) exactly one value of a b) no real value of a
c) infinitely many real values of a d) exactly two real values of a

Ans: (b)

Sol: Equation radical axis is $5ax + (c-d)y + a+1 = 0$

Given radical axis is $5x + by - a = 0$

$$\Rightarrow \frac{5a}{5} = \frac{c-d}{b} = \frac{a+1}{-a}$$

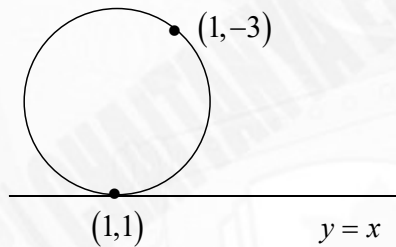
$$a = \frac{a+1}{-a} \Rightarrow -a^2 = a+1 \Rightarrow a^2 + a + 1 = 0$$

$\Rightarrow$ No real values of a

8) The line $y = x$ touches a circle at the point (1,1). If the circle also passes through the point (1,-3) then its radius is **(JEE MAIN 2019 SHIFT 1)**

- a) 3 b) $2\sqrt{2}$ c) 2 d) $3\sqrt{2}$

Ans : (b)



Sol :

Equation of circle which touches the line $y=x$ at (1,1) is $(x-1)^2 + (y-1)^2 + \lambda(y-x) = 0$. This circle passes through (1,-3)

$$\therefore 0 + 16 + \lambda(-3-1) = 0$$

$$16 + \lambda(-4) = 0 \Rightarrow \lambda = 4$$

$$(x-1)^2 + (y-1)^2 + 4y - 4x = 0$$

$$x^2 + y^2 - 6x + 2y + 2 = 0$$

$$\therefore \text{Radius} = \sqrt{9+1-2} = 2\sqrt{2}$$

8a) A variable circle $S=0$ touches the line $y = x$ and passes through the point (0,0), then the fixed point that lies on the common chord of the circles $x^2 + y^2 + 6x + 8y - 7 = 0$ and $S=0$ is

(JEE MAIN 2019 SHIFT 1)

- a) $\left(\frac{1}{2}, \frac{1}{2}\right)$ b) $\left(-\frac{1}{2}, -\frac{1}{2}\right)$ c) $\left(\frac{1}{2}, -\frac{1}{2}\right)$ d) $\left(-\frac{1}{2}, \frac{1}{2}\right)$

Ans : (a)

Sol :

$S=0$ touches the line $y=x$ and passes through $(0,0)$. Equation of $S=0$, is

$x^2 + y^2 + \lambda(x - y) = 0$. Common chord of $x^2 + y^2 + 6x + 8y - 7 = 0$; $x^2 + y^2 + \lambda(x - y) = 0$ is

$$(6\lambda)x + (8 + \lambda)y - 7 = 0$$

$$\Rightarrow (6x + 8y - 7) + \lambda(-x + y) = 0$$

This line passes through the P.I of $6x + 8y - 7 = 0$ & $-x + y = 0$

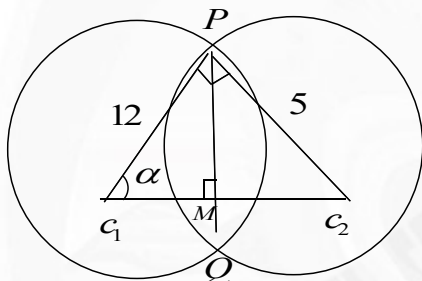
Required point $\left(\frac{1}{2}, \frac{1}{2}\right)$

- 9) If the angle of intersection at a point where the two circles with radii 5cm and 12 cm intersect is 90° , then the length (in cm) of their common chord is (2019)

- a) $\frac{13}{5}$ b) $\frac{120}{13}$ c) $\frac{60}{13}$ d) $\frac{13}{2}$

Ans : (b)

Sol :



According to the diagram in ΔPC_1C_2 , $\tan \alpha = \frac{5}{12} \Rightarrow \sin \alpha = \frac{5}{13}$. In

$$\Delta PC_1M, \sin \alpha = \frac{PM}{12} \Rightarrow \frac{5}{13} = \frac{PM}{12}$$

$$\Rightarrow PM = \frac{60}{13}$$

Hence the length of common chord $(PQ) = \frac{120}{13}$

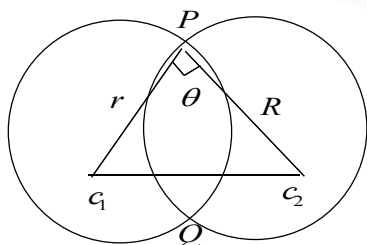
- 9a) Two circles of radii r and R intersect at an acute angle ' θ '. The length of their common chord is

(JEE MAIN 2019 SHIFT 1)

- a) $\frac{2rR \sin \theta}{\sqrt{r^2 + R^2 - 2rR \cos \theta}}$ b) $\frac{2rR \sin \theta}{\sqrt{r^2 + R^2}}$ c) $\frac{2rR \cos \theta}{\sqrt{R^2 + r^2}}$ d) None of these

Ans : (a)

Sol :



Let C_1, C_2 be the centres of two circles with radii r, R intersecting at P and Q

$$C_1P = r, C_2P = R$$

From ΔPC_1C_2 , by cosine rule

$$\begin{aligned}(C_1C_2)^2 &= (C_1P)^2 + (C_2P)^2 - 2C_1P.C_2P \cos\theta \\ &= r^2 + R^2 - 2rR \cos\theta\end{aligned}$$

$$\text{Area of } \Delta PC_1C_2 \text{ is } \frac{1}{2}C_1P.C_2P \sin\theta = \frac{1}{2}rR \sin\theta \dots\dots\dots(1)$$

$$\begin{aligned}\text{If 'h' is the length of the chord, then Area of } \Delta PC_1C_2 &= \frac{1}{2}C_1C_2 \times \frac{h}{2} = \frac{h}{4}(C_1C_2) \\ &= \frac{h}{4}\sqrt{r^2 + R^2 - 2rR \cos\theta} \dots\dots\dots(2)\end{aligned}$$

$$\begin{aligned}\text{From (1),(2)} \quad \frac{1}{2}rR \sin\theta &= \frac{h}{4}\sqrt{r^2 + R^2 - 2rR \cos\theta} \\ \Rightarrow h &= \frac{2rR \sin\theta}{\sqrt{r^2 + R^2 - 2rR \sin\theta}}\end{aligned}$$

9b) The length of the common chord of the two circles $(x-a)^2 + (y-b)^2 = c^2$ and

$$(x-b)^2 + (y-a)^2 = c^2 \text{ is}$$

(JEE MAIN 2019 SHIFT 1)

$$\text{a) } \sqrt{4c^2 - 2(a-b)^2} \quad \text{b) } \sqrt{4c^2 - (a-b)^2} \quad \text{c) } \sqrt{2c^2 - (a-b)^2} \quad \text{d) } \sqrt{c^2 - (a-b)^2}$$

Ans : (a)

Sol : Equation of common chord is $S - S' = 0$

$$2(a-b)x - 2(a-b)y = 0 \Rightarrow 2x - 2y = 0$$

Equation of a circle coaxial with the given circle is

$$x^2 + y^2 - 2ax - 2by + a^2 + b^2 - c^2 + \lambda(2x - 2y) = 0$$

$$x^2 + y^2 + (2\lambda - 2a)x + (-2\lambda - 2b)y + a^2 + b^2 - c^2 = 0 \dots\dots\dots(1)$$

Centre of (1) is $(a - \lambda, b + \lambda)$

If the common chord $x - y = 0$ is a diameter of circle then centre of (1) lies on $x - y = 0$

$$(a - \lambda) - (b + \lambda) = 0 \Rightarrow \lambda = \frac{a - b}{2}$$

$$r^2 = (a - \lambda)^2 + (b + \lambda)^2 - (a^2 + b^2 - c^2) = \left(a - \frac{a - b}{2}\right)^2 + \left(b + \frac{a - b}{2}\right)^2 - a^2 - b^2 + c^2$$

$$= \frac{1}{4}(4c^2 - 2(a - b)^2)$$

$$\text{Length of the common chord} = 2(\text{radius}) = 2 \times \frac{1}{2}\sqrt{4c^2 - 2(a - b)^2}$$

10) Let the mirror image of a circle $c_1 : x^2 + y^2 - 2x - 6y + \alpha = 0$ in line $y = x + 1$ be

$c_2 : 5x^2 + 5y^2 + 10gx + 10fy + 38 = 0$. If r is the radius of circle c_2 , then $\alpha + 6r^2$ is equal to _____.

[29 July 2022 - Shift 1]

a) 12

b) 11

c) 10

d) 13

Ans: (a)

Sol: Image of centre $c_1 \equiv (1,3)$ in $x - y + 1 = 0$ is given by

$$\frac{x_1 - 1}{1} = \frac{y_1 - 3}{-1} = \frac{-2(1 - 3 + 1)}{1^2 + 1^2}$$

$$\Rightarrow x_1 = 2, y_1 = 2 ; c_2 \equiv (2,2)$$

Now radius of c_2 is $r = \sqrt{\frac{2}{5}}$

$$\Rightarrow 10 - \alpha = \frac{2}{5} \Rightarrow \alpha = \frac{48}{5}$$

$$\therefore \alpha + 6r^2 = \frac{48}{5} + \frac{12}{5} = 12$$

CONCEPT: Image of a circle w.r.t a line

FORMULA: Image formula $\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{-2(ax_1 + by_1 + c)}{a^2 + b^2}$

10a) Equation of the circle, which is the mirror image of the circle $x^2 + y^2 - 2x = 0$ with respect to the line $x + y = 2$ is

a) $x^2 + y^2 - 2x - 4y + 3 = 0$

b) $2(x^2 + y^2) - x + y + 1 = 0$

c) $x^2 + y^2 - 2x + 4y + 3 = 0$

d) $x^2 + y^2 - 4x - 2y + 4 = 0$

Key: (d)

Sol: For the circle $x^2 + y^2 - 2x = 0$ centre $(1,0)$; radius = 1

Let $P(h,k)$ be the image of 'C' w.r.t the line $x + y - 2 = 0$

$$\frac{h - 1}{1} = \frac{k - 0}{1} = \frac{-2(1 - 0 - 2)}{1 + 1} \Rightarrow h = 2, k = 1$$

The equation of the mirror image of given circle is $(x - 2)^2 + (y - 2)^2 = 1$.

$$x^2 + y^2 - 4x - 2y + 4 = 0$$

10b) The common tangent of the two circles $x^2 + y^2 + 2x + 4y + 7 = 0$ and $x^2 + y^2 + 4x + 2y + 5 = 0$ subtends of a right angle at $(-2, -1)$ on their line of centres. The other similar point on the line of centres is

a) $(3,0)$

b) $(0,3)$

c) $(-3,0)$

d) $(0,-3)$

Key: (d)

Sol: radical axis is $2x - 2y - z = 0 \Rightarrow x - y - 1 = 0$

Since common tangent subtends a right angle at the limiting points, the other similar point is the other limiting point. ie image of $(-2, -1)$ w.r.t $x - y - 1 = 0$

If (α, β) is the image then $\frac{\alpha + 2}{1} = \frac{\beta + 1}{-1} = \frac{-2(-2 + 1 - 1)}{1 + 1}$

$$\Rightarrow \alpha = 0, \beta = -3$$

Concept: Circle touching two parallel lines

Formula: $2r = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$

11) Concept: Circles touching conditions

$$\text{Formula: } c_1 c_2 = |r_1 \pm r_2|$$

Let the locus of the centre $(\alpha, \beta), \beta > 0$, of the circle which touches the circle $x^2 + (y-1)^2 = 1$ externally and also touches the x-axis be L. then the area bounded by L and the line $y = 4$ is :

(25 July 2022 Shift – 1)

a) $\frac{32\sqrt{2}}{3}$ b) $\frac{40\sqrt{2}}{3}$ c) $\frac{64}{3}$ d) $\frac{32}{3}$

Ans : (c)

Sol : $C_1 = (0, 1) \quad r_1 = 1$

$$C_2 = (\alpha, \beta) \quad r_2 = |\beta|$$

Here $C_1 C_2 = r_1 + r_2$

$$\alpha^2 = 4\beta$$

Locus of centre (α, β) is $x^2 = 4y$

$$A = 2 \int_0^4 \left(4 - \frac{x^2}{4} \right) dx = \frac{64}{3}$$

11a) Let the locus of the centre $(\alpha, \beta), \beta > 0$ of the circle which touches the circle $x^2 + (y-2)^2 = 4$ externally and also touches the x-axis be 'L' then the area bounded by 'L' and the line $y=9$ is _____

a) $72\sqrt{2}$ b) $36\sqrt{2}$ c) $70\sqrt{2}$ d) $71\sqrt{2}$

Ans : (a)

Sol: $c_1 = (0, 2) \quad r_1 = 2$

$$c_2 = (\alpha, \beta) \quad r_2 = |\beta|$$

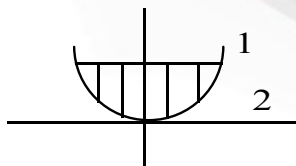
$$c_1 c_2 = r_1 + r_2 \Rightarrow (c_1 c_2)^2 = (r_1 + r_2)^2$$

$$\Rightarrow (\alpha - 0)^2 + (\beta - 2)^2 = (2 + \beta)^2$$

$$\alpha^2 + \beta^2 - 4\beta + 4 = 4 + 4\beta + \beta^2$$

$$\alpha^2 = 8\beta$$

Locus of (α, β) is $x^2 = 8y$(1)



Line $y = 9$(2)

$$\text{Area} = 2 \int_0^9 \sqrt{8y} dy$$

$$= 2\sqrt{8} \left(\frac{y^{3/2}}{3/2} \right)_0^9 = \frac{4\sqrt{8}}{3} (9)^{3/2}$$

$$= \frac{8\sqrt{2}}{3} (3^2)^{3/2}$$

$$= \frac{8\sqrt{2}}{3} \times 2 = 72\sqrt{2} \text{ sq units}$$

11b) Let the locus of the centre (α, β) $\alpha > 0$ of the circle which touches the circle $(x-4)^2 + y^2 = 16$ externally and also touches the y-axis be 'L' then the area bounded by 'L' and the line $x = 4$ is _____

a) $\frac{1}{2}$

b) $\frac{28}{3}$

c) $\frac{158}{3}$

d) $\frac{128}{3}$

Ans: (d)

Sol: $c_1 = (4, 0)$ $r_1 = 4$

$c_2 = (\alpha, \beta)$ $r_2 = |\alpha|$

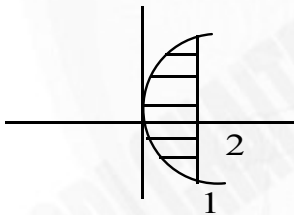
$$c_1 c_2 = r_1 + r_2 \Rightarrow (c_1 c_2)^2 = (r_1 + r_2)^2$$

$$\Rightarrow (\alpha - 4)^2 + (\beta - 0)^2 = (4 + |\alpha|)^2$$

$$\Rightarrow \alpha^2 - 8\alpha + 16 + \beta^2 = 16 + 2^2 + 8\alpha$$

$$\beta^2 = 16\alpha$$

Locus of (α, β) is $y^2 = 16x$(1)



Line is $x=4$(2)

$$\text{Area} = 2 \int_0^4 \sqrt{16x} dx$$

$$= 2 \times 4 \int_0^4 \sqrt{x} dx$$

$$= 8 \left(\frac{x^{3/2}}{3/2} \right)_0^4 = \frac{16}{3} (4^{3/2}) = \frac{16}{3} (2^2)^{3/2}$$

$$= \frac{16}{3} \times 8 = \frac{128}{3} \text{ sq units}$$

Yashpatil TG~ @bohring_bot

Exercise : II

(Numerical / Integer Value based Questions Including PYQ's)

I. Standard form of the Circle

1. Let a point P be such that its distance from the point (5, 0) is thrice the distance of P from the point (-5, 0). If the locus of the point P is a circle of radius r, then $4r^2 = \underline{\hspace{2cm}}$

(2021 Feb Main)

Ans: 56.25

Sol:

Let P(x, y) be any point on locus

A = (5, 0) B = (-5, 0)

PA = 3 PB

$$PA^2 = 9PB^2$$

$$= x^2 + 25 - 10x + y^2 = 9[x^2 + 25 + 10x + y^2]$$

$$= (x-5)^2 + (y-0)^2 = 9[(x+5)^2 + (y+0)^2]$$

$$= x^2 + y^2 - 10x + 25 = 9x^2 + 225 + 90x + 9y^2$$

$$= 8x^2 + 8y^2 + 100x + 200 = 0$$

$$= x^2 + y^2 + \frac{25}{2}x + 25 = 0$$

$$r = \sqrt{\frac{625}{16} - 25} = \sqrt{\frac{625 - 400}{16}} = \frac{15}{4}$$

$$4r^2 = 4\left(\frac{225}{16}\right) = 56.25$$

2. Let a point P be such that its distance from the point (3, 0) is twice of the distance from the point (-3, 0). If the locus of the point of circle of radius r. Then $r^2 = \underline{\hspace{2cm}}$

Ans: 4

Sol: PA = 2PB

$$PA^2 = 4PB^2$$

$$= (x-3)^2 + y^2 = 4[(x+3)^2 + (y-0)^2]$$

$$= x^2 + 9 - 6x + y^2 = 4[x^2 + 9 + 6x + y^2]$$

$$= x^2 + y^2 - 6x + 9 = 4x^2 + 4y^2 + 24x + 36$$

$$= 3x^2 + 3y^2 + 30x + 27 = 0$$

$$= x^2 + y^2 + 10x + 9 = 0$$

$$\text{Radius } r = \sqrt{25 - 9} = 4$$

$$r^2 = 16$$

Concyclic points of the circle

3. If a circle passes through the points of intersection of the coordinate Axis with the line $\lambda x - y + 1 = 0$ and $x - 2y + 3 = 0$, then find the value of $\lambda = ?$

Ans: $2, \frac{1}{3}$

Sol: Points of intersection with coordinate axes are

$$\left(-\frac{1}{\lambda}, 0\right)(0, 1)(-3, 0)\left(0, \frac{3}{2}\right)$$

The eq of circle passes through (0, 1) (-3, 0)

(0, 3/2) is

$$x^2 + y^2 + \frac{7}{2}x - \frac{5}{2}y + \frac{3}{2} = 0$$

It passes through

$$\left(-\frac{1}{\lambda}, 0\right)$$

$$\frac{1}{\lambda^2} - \frac{7}{2\lambda} + \frac{3}{2} = 0 \Rightarrow 3\lambda^2 - 7\lambda + 2 = 0$$

$$\lambda = 2, \frac{1}{3}$$

Diameter of the circle

4. The lines $2x-3y=5$ and $3x-4y=7$ are the diameters of the circle of area 154 square units.

Then the equation of circle is

Ans: 7^2

Sol: Centre = P.I of diameters

$$= (1, -1)$$

$$\text{Area} = \pi r^2 = 154$$

$$r = 7$$

Hence eq of required circle

$$(x-1)^2 + (y+1)^2 = 7^2$$

Standard form of the circle

5. The locus of a point, which moves such that the sum of squares of its distance from the point (0, 0), (1, 0), (0, 1), (1, 1) is 18 units is a circle of diameter d. Then $d^2 =$ _____

(2021 Aug Main)

Ans: 16

Sol: Let $P = (x, y)$

Given points $A = (0, 0)$ $B = (1, 0)$ $C = (0, 1)$ $D = (1, 1)$

$$PA^2 + BP^2 + CP^2 + DP^2 = 18$$

$$= x^2 + y^2 + (x-1)^2 + y^2 + x^2 + (y-1)^2 + (x-1)^2 + (y-1)^2 = 18$$

$$= x^2 + y^2 + x^2 + 1 - 2x + y^2 + x^2 + y^2 + 1 - 2y + x^2 + 1 - 2x + y^2 + 1 - 2y = 18$$

$$= 4x^2 + 4y^2 - 4x - 4y - 14 = 0$$

$$= x^2 + y^2 - x - y - \frac{7}{2} = 0$$

$$\text{radius } (r) = \sqrt{\frac{1}{4} + \frac{1}{4} + \frac{7}{2}} = \sqrt{\frac{1+1+14}{4}} = 2$$

$$d = 2r = 4$$

$$d^2 = 4^2 = 16$$

6. The locus of a point which moves such that the sum of square of its distances from the points (0, 0), (2, 0), (0, 2), (2, 2) is 20 units is a circle of radius r. Then $r^2 =$ ____

Ans: 3

Sol:

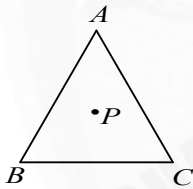
$$\begin{aligned}
 AP^2 + BP^2 + CP^2 + DP^2 &= 20 \\
 = x^2 + y^2 + (x-2)^2 + y^2 + x^2 + (y-2)^2 + (x-2)^2 + (y-2)^2 &= 20 \\
 = x^2 + y^2 + x^2 + 4 - 4x + x^2 + y^2 + y^2 + 4 - 4y + x^2 + 4 - 4x + y^2 + 4 - 4y &= 20 \\
 = 4x^2 + 4y^2 - 8x - 8y - 16 &= 0 \\
 = 4x^2 + 4y^2 - 8x - 8y - 4 &= 0 \\
 = x^2 + y^2 - 2x - 2y - 1 &= 0 \\
 r = \sqrt{1+1+1} &= \sqrt{3} \\
 r^2 &= 3
 \end{aligned}$$

Equilateral triangle inscribed in a circle

7. For $t \in (0, 2\pi)$ if ABC is an equilateral triangle with vertices $A(\sin t, -\cos t)$, $B(\cos t, \sin t)$ and C (a, b) such that its ortho centre lies with center $(1, 1/3)$ then $(a^2 - b^2) =$ ____?

Ans: 8

Sol:



Let P (h, k) be the orthocentre of $\triangle ABC$

$$h = \frac{\sin t + \cos t + a}{3} \dots\dots(1)$$

$$k = \frac{-\cos t + \sin t + b}{3} \dots\dots(2)$$

(orthocentre & circumcentre coincide)

from 1 & 2

$$(3h - a)^2 + (3k - b)^2 = 2$$

$$\left(h - \frac{a}{3}\right)^2 + \left(k - \frac{b}{3}\right)^2 = \frac{2}{9}$$

Orthocentre lies on the centre $(1, \frac{1}{3})$

$$\frac{a}{3} = 1, \frac{b}{3} = \frac{1}{3}$$

$$a = 3, b = 1$$

$$a^2 - b^2 = 9 - 1 = 8$$

8. The equation of the circle with origin as centre and passing through the vertices of an equilateral triangle whose median is of length $3a$ is $x^2 + y^2 = ka^2$ then $K =$?

Ans: 4

Sol: radius of circle = $\frac{2}{3}$ (median) = $\frac{2}{3}(3a)$

$$r = 2a$$

eq of circle is $x^2 + y^2 = 4a^2$

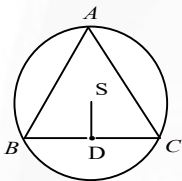
$$k = 4$$

Circumcircle of the Triangle

9. Consider a triangle having vertices $A=(-2,3)$, $B=(1,9)$, and $C=(3,8)$, If the line L passes through the circumcenter of triangle meets y -axis at point $\left(0, \frac{\alpha}{2}\right)$ then the value of real number α is _____

Ans: 9

Sol:



$D = \text{Midpoint of } \overline{BC}$

$$= \left(\frac{1+3}{2}, \frac{9+8}{2} \right) = \left(2, \frac{17}{2} \right)$$

$$\text{Slope of } \overline{BC} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{8-9}{3-1} = \frac{-1}{2}$$

$$\text{equation of } \overline{SD} \text{ is } y - y_1 = \frac{-1}{m}(x - x_1)$$

$$= y - \frac{17}{2} = \frac{1}{\frac{1}{2}}(x - 2)$$

$$= \frac{2y - 17}{2} = 2x - 4$$

$$= 2y - 17 = 4x - 8$$

$$= 4x - 2y + 9 = 0$$

This line intersects y -axis at $\left(0, \frac{\alpha}{2}\right)$

$$0 - 2y + 9 = 0$$

$$2y = 9$$

$$y = \frac{9}{2}$$

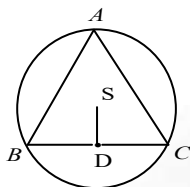
$$\therefore \frac{\alpha}{2} = \frac{9}{2}$$

$$\alpha = 9$$

10. Consider a triangle having vertices $C = (-4, 2)$. If a line passing through the circumcentre of triangle ABC, bisects the line BC and intersects y-axis is at a point $(0, +\frac{\alpha}{2})$ then the real value of α is _____

Ans: 7

Sol: Midpoint of BC = $\left(-1, \frac{5}{2}\right)$



$$\text{BC slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - 3}{-2 - 4} = \frac{1}{6}$$

$$\text{eq of SD is } y - \frac{5}{2} = \frac{1}{1/6}(x + 1)$$

$$\frac{2y - 5}{2} = -6x - 6$$

$$2y - 5 = -12x - 12$$

$$12x - 2y + 7 = 0$$

This intersects y-axis at $\left(0, \frac{\alpha}{2}\right)$

$$-2y + 7 = 0$$

$$y = \frac{7}{2}$$

$$\frac{\alpha}{2} = \frac{7}{2}$$

$$\alpha = 7$$

Length of Tangent

11. The length of tangent drawn from the point $(1, 2)$ to the circle $x^2 + y^2 + 2x + 5y + k = 0$ is $\sqrt{37}$ then $K = ?$

Ans: 20

Sol: Length of tangent = $\sqrt{S_{11}}$

$$\Rightarrow \sqrt{1 + 4 + 2 + 10 + k} = \sqrt{37}$$

$$\Rightarrow 17 + k = 37$$

$$\Rightarrow k = 20$$

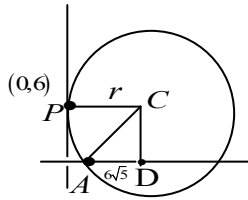
Chord of the circle and touching y-axis

12. Consider a circle C which touches the y-axis at $(0, 6)$ and cut off an intercept $6\sqrt{5}$ on the x-axis. Then the radius of the circle C is equal to

(2021 July Main)

Ans: 9

Sol:



$$AD = 3\sqrt{5}$$

$$CA^2 = CD^2 + AD^2$$

$$= 36 + 45$$

$$CA^2 = 81$$

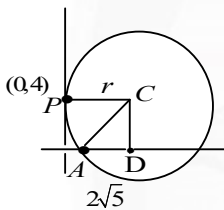
$$CA = 9$$

$$\text{Radius } (r) = 9$$

13. Consider a circle C which touches the y-axis at (0, 4) and cut off an intercept $2\sqrt{5}$ on the x-axis then the radius of the circle C is equal to

Ans: 6

Sol:



$$AD = 2\sqrt{5}$$

$$CA^2 = CD^2 + AD^2$$

$$= 4^2 + (2\sqrt{5})^2$$

$$CA^2 = 16 + 20$$

$$CA = 6$$

$$\text{Radius } (r) = 6$$

II. Intercepts made by a circle on axes

14. Let the length of intercepts on x-axis and y-axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$ be $2\sqrt{2}$ and $2\sqrt{5}$ respectively, then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is equal to ____

(2021 March Main)

Ans: $\sqrt{6}$

$$\text{Sol: Length of x-intercept} = 2\sqrt{g^2 - c}$$

$$\text{Length of y-intercept} = 2\sqrt{f^2 - c}$$

$$= 2\sqrt{\frac{a^2}{4} - c} = 2\sqrt{2} = \frac{a^2}{4} - c = 2 = a^2 - 4c = 8$$

$$= 2\sqrt{a^2 - c} = 2\sqrt{5} \Rightarrow a^2 - c = 5 \Rightarrow a^2 = 5 + c$$

$$5 + c - 4c = 8$$

$$5 + 3c = 8 \Rightarrow 3c = +3 \Rightarrow c = -1$$

$$a^2 = 5 - 1 = 4 \Rightarrow a = 2$$

$$\text{Equation of tangent } y = 2x + k$$

$$\text{distance} = \frac{|c|}{\sqrt{a^2 + b^2}} = \frac{|k|}{\sqrt{5}}$$

$$\frac{|k|}{\sqrt{5}} = \sqrt{\frac{a^2}{4} + a^2} - c = \sqrt{\frac{4}{4} + 4 + 1} = \sqrt{6}$$

$$k = \sqrt{30}$$

$$\text{distance} = \frac{\sqrt{30}}{\sqrt{5}} = \sqrt{6}$$

15. Let the length of intercepts on x-axis and y-axis made by the circle $x^2 + y^2 + 2x + by + c = 0$ be 2 and 4 respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to $x + 7y = 0$ is _____

Ans: $\sqrt{5}$

Sol: Length of x-axis = $2\sqrt{g^2 - c} = 2$

$$= g^2 - c = 1$$

$$= 1 - c = 1 \Rightarrow c = 0$$

Length of y-axis = $2\sqrt{f^2 - c} = 4$

$$= f^2 - c = 4$$

$$= \frac{b^2}{4} = 4 \Rightarrow b = 0$$

Eq of tangent parallel to $x + 7y = 0$ is $y = 7x + k$ distance from origin to the tangent = $\frac{|c|}{\sqrt{a^2 + b^2}}$

$$\frac{|k|}{\sqrt{49 + 1}} \Rightarrow \frac{|k|}{\sqrt{50}} = \sqrt{5}$$

$$k = \sqrt{50} \cdot \sqrt{5}$$

$$\text{distance} = \frac{\sqrt{50} \cdot \sqrt{5}}{\sqrt{50}} = \sqrt{5}$$

Position of point w.r.t the circle

16. Find the value of K, If the point $(-2, k)$ lies on the circle $x^2 + y^2 + x - 4y + 2 = 0$

Ans: 2

Sol: $S_{11} = 0$

$$= 4 + k^2 - 2 - 4k + 2 = 0$$

$$= k^2 - 4k + 4 = 0$$

$$= (k - 2)^2 = 0$$

$$= K = 2$$

III & IV. TANGENT AND NORMAL

(Position of a line w.r.t a circle in circles, Condition for tangency in circles,

Perpendicular distance from centre to the line is radius in circles, Equation of tangent in

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parametric form in circles, Point of contact of tangent in circles, Number of tangents from an external point in circles, Sum of slopes of tangents in circles, Equation of pair of tangents drawn from external point to circle in circles, Product of slopes of tangents in circles, Angle between pair of tangents drawn from external point to circle in circles, Director circle of a circle in circles)
(Equation of normal line at point of contact in circles, Condition for normal to circle, Equation of normal at P (theta))

17. The circle passing through the point $(-1, 0)$ and touching the y-axis at $(0, 2)$ also passes through the point (α, β) then, the value of $\beta - \alpha$

Ans: 4

Sol:

Equation of circle passing through a point (x_1, y_1) and touching the straight line L, is given by

$$(x - x_1)^2 + (y - y_1)^2 + \lambda L = 0$$

The circle is passing through the point $(0, 2)$

$$\Rightarrow (x - 0)^2 + (y - 2)^2 + \lambda L = 0$$

$$\Rightarrow x^2 + (y - 2)^2 + \lambda x = 0 \dots (i)$$

Circle passes through $(-1, 0)$.

$$\Rightarrow 1 + 4 - \lambda = 0$$

$$\Rightarrow \lambda = 5$$

Put $\lambda = 5$ in (i)

$$x^2 + (y - 2)^2 + 5x = 0$$

$$\Rightarrow x^2 + y^2 - 4y + 4 + 5x = 0$$

$$\Rightarrow x^2 + y^2 + 5x - 4y + 4 = 0$$

Put $y = 0$

$$\Rightarrow x = -4, x = -1$$

Check the options.

18. If the tangent at $(1, 7)$ to the curve $x^2 = y - 6$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$, then the value of c is

Ans: 95

Sol: Let the family of circles be $S_1 + \lambda S_2 = 0$

$$x^2 + y^2 - 6x + \lambda(x^2 + y^2 - 4y) = 0$$

$$\Rightarrow (1+\lambda)x^2 + (1+\lambda)y^2 - 6x - 4\lambda y = 0 \dots (i)$$

$$\text{Centre } (-g, -f) = \left(\frac{3}{(1+\lambda)}, \frac{2}{(1+\lambda)} \right)$$

Centre lies on $2x - 3y + 12 = 0$

$$\text{Then } \left(\frac{6}{(\lambda+1)} \right) - \left(\frac{6}{\lambda+1} \right) + 12 = 0$$

$$\Rightarrow \lambda = -3$$

Equation of circle (i)

$$-2x^2 - 2y^2 - 6x + 12y = 0$$

$$\Rightarrow x^2 + y^2 + 3x - 6y = 0$$

Check options.

$(-3, 6)$ satisfy equation (ii).

19. Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at points A and B. Then $5(AB)^2$ is equal to

Ans: 64

Sol: Length of tangent, $L = \sqrt{S_1} = \sqrt{16} = 4$

$$R = \sqrt{(16 + 4 - 16)} = 2$$

$$\text{Length of chord of contact} = \frac{2LR}{\sqrt{L^2 + R^2}} = \frac{16}{\sqrt{20}}$$

$$\text{Square of length of chord of contact} = \frac{64}{5}$$

20. The nearest point on circle $x^2 + y^2 - 6x + 4y - 12 = 0$ from a point $(-5, 4)$ is (α, β) , then $\alpha + \beta =$

Ans: 0

Sol: given point $P(-5, 4)$

For given circle, centre $C(3, 2)$, & radius $r=5$

Now

$$CP = \sqrt{64 + 36} = 10 = 2(5) = 2r$$

$\therefore$ nearest point Q is midpoint of $\overline{CP}$

$$\therefore Q = \left(\frac{3-5}{2}, \frac{4-2}{2} \right) = (-1, 1) = (\alpha, \beta) \text{ given}$$

$$\text{required} = \alpha + \beta = -1 + 1 = 0$$

21. If the tangent at the point P on the circle is $x^2 + y^2 + 6x + 6y = 2$ meets a straight line $5x - 2y + 6 = 0$ at a point Q on the y-axis, then the length of PQ is

Ans: 5

Sol: Given that the line $5x - 2y + 6 = 0$ is intersected by tangent at P to the circle

$$x^2 + y^2 + 6x + 6y - 2 = 0 \text{ on y axis at Q.}$$

On y axis, $x = 0$

$$\Rightarrow 2y = 6$$

$$\Rightarrow y = 3$$

So Q is $(0, 3)$

Tangent passes through $(0, 3)$.

PQ = length of tangent to the circle from $(0, 3)$

$$= \sqrt{(0+9+0+18-2)} = 5$$

22a. Let ABCD be square of side of unit length. Let a circle C_1 centered at A with unit radius is drawn. Another circle C_2 which touches C_1 and the lines $\overline{AD}$ and $\overline{AB}$ are tangent to it, is also drawn. Let a tangent line from the point C to circle C_2 meet the side $\overline{AD}$ at E. if area of ΔCDE is $a + \frac{\sqrt{3}}{2}b$. Then $a + b =$

Ans: 0

Sol: Area

$$\Delta = \frac{1}{2}(CD)(DE) = \frac{1}{2}(1)(2 - \sqrt{3}) - 1 - \frac{1}{2}\sqrt{3} = a + \frac{\sqrt{3}}{2}b \text{ (given)}$$

$$= a = 1, b = -1$$

$$\therefore \text{required} = a + b = 1 + (-1) = 0$$

23. Let the normals at all points on given curve pass through a fixed point (a, b). If the curve passes through two points (2, 0) and (0, 3) and given that $3a - 2b = 1$, then the value $a^2 + b^2$ is

Ans: 2

Sol: Let A(2,0), B(0,3) be given points & center C(a, b). Given curve is circle

$$\Rightarrow CA^2 = CB^2 \Rightarrow (a - 2)^2 + b^2 = a^2 + (b - 3)^2$$

$$\Rightarrow 4a - ab + 5 = 0 \quad \text{_____ (1)}$$

$$\text{Given } 3a - 2b = 1 \quad \text{_____ (2)}$$

$$\text{Solving 1 \& 2} \Rightarrow a = 1, b = 1$$

$$\therefore \text{required value} = a^2 + b^2 = (1)^2 + (1)^2 = 1 + 1 = 2$$

23a. Let B be the center of the circle $x^2 + y^2 + 2x - 2y + 3 = 0$. Let the tangent at points P and Q on the circle intersect at the point A(1,3). Then $\frac{\text{area } \Delta APQ}{\text{area } \Delta BPQ} = ?$

Ans: 9

Sol: equation of chord contact $s_1 = 0$

$$\text{Center} = (-1, 1)$$

$$\Rightarrow x + 3y + 1(x + 1) - 1(y + 3) + 3 = 0$$

$$\Rightarrow x + 3y + x + 1 - y - 3 + 3 = 0$$

$$\Rightarrow 2x + 2y + 1 = 0$$

$$\text{AR distance} = \frac{|ax_1 - by_1 - c|}{\sqrt{a^2 + b^2}} = \frac{|2 + 6 + 1|}{\sqrt{8}} = \frac{9}{\sqrt{8}}$$

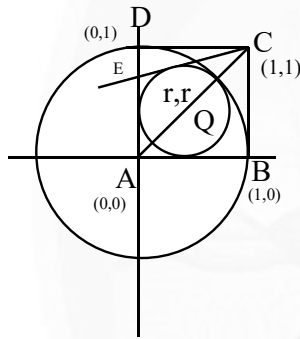
$$\text{BR distance} = \frac{|-2+2+1|}{\sqrt{8}} = \frac{1}{\sqrt{8}}$$

$$\Rightarrow \frac{\frac{9}{\sqrt{8}}}{\frac{1}{\sqrt{8}}} = 9$$

24. Let ABCD be square of side of unit length. Let a circle C_1 centered at A with unit radius is drawn. Another circle C_2 which touches C_1 and the lines $\overrightarrow{AD}$ and $\overrightarrow{AB}$ are tangent to it, is also drawn. Let a tangent line from the point C to circle C_2 meet the side $\overrightarrow{AD}$ at E. If the length of ED is $A + \sqrt{3}b$, where a, b are integers, then a+b is equal to

Ans: 1

Sol: From diagram $AQ + QP = 1$



$$\Rightarrow \sqrt{2}r + r = 1$$

$$\Rightarrow (\sqrt{2}r + 1)r = 1 \Rightarrow r = \sqrt{2} - 1$$

{ $\because$ r is radius of $c_2 \Rightarrow$ center of c_2 is $Q(r,r)$ }

Now equation of line through point C(1,1) is $y-1=m(x-1)$

$$\Rightarrow mx - y - m + 1 = 0 \quad \text{_____ (1)}$$

$$\text{Of 1 is tangent to } c_2 = \left| \frac{mr - r - m + 1}{\sqrt{m^2 + 1}} \right| = r$$

$$\Rightarrow \left| \frac{(m-1)(r-1)}{\sqrt{m^2 + 1}} \right| = r \Rightarrow \frac{(m-1)^2 (r-1)^2}{m^2 + 1} = r^2 \quad \text{_____ (2)}$$

$$\text{sub } r = \sqrt{2} - 1 \text{ in (2)} \Rightarrow m = 2 \pm \sqrt{3}$$

For line $\overline{CE}$ slope $m = 2 - \sqrt{3}$

$$\therefore \text{equation of line } \overline{CE} \text{ is } y-1 = (2-\sqrt{3})(x-1) \quad \text{_____ (3)}$$

$$\text{Putting } x=0 \text{ in (3)} \Rightarrow y-1 = (2-\sqrt{3})(-1)$$

$$\Rightarrow 1-y = 2-\sqrt{3}$$

$$\therefore ED = AD - AE = 1 - y = 2 - \sqrt{3} = a + \sqrt{3}b \text{ (given)}$$

$$\Rightarrow a = 2, b = -1$$

$$\therefore \text{required} = a + b = 2 + (-1) = 1.$$

25. If the curves, $x^2 - 6x + y^2 + 8 = 0$ and $x^2 - 8y + y^2 + 16 - k = 0$, ($k > 0$) touch each other at a point, then the largest value of k is _____

(JEE MAIN 2020 SHIFT-1)

Key: 36**Sol:** since two circles are touching, common tangent at point of contact is $S_1 - S_2 = 0$

$$\text{or } -6x + 8y - 8 + k = 0$$

using condition of tangency, we get

$$\frac{|-18 - 8 + k|}{10} = 1$$

$$\Rightarrow k = 36 \text{ or } 16$$

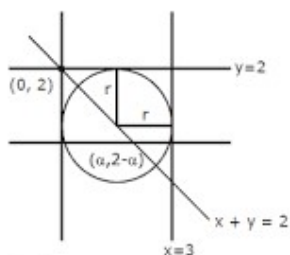
$$\Rightarrow k_{\max} = 36$$

26. The diameter of the circle, whose centre lies on the line $x + y = 2$ in the first quadrant and which touches both the lines $x = 3$ and $y = 2$, is _____

(JEE MAIN 2020 SHIFT-1)

Key: 3**Sol:** center of the circle lies on the line $x + y = 2$. so, centre is of the form $(\alpha, 2 - \alpha)$.Circle touches the lines $x = 3$ and $y = 2$.

$$\text{So radius} = |3 - \alpha| = |2 - (2 - \alpha)|$$



$$\Rightarrow 3 - \alpha = \alpha$$

$$\Rightarrow \alpha = \frac{3}{2}$$

$$\therefore \text{Diameter} = 2\alpha = 3.$$

27. If the area of the triangle formed by the positive x -axis, the normal and the tangent to the circle $(x-2)^2 + (y-3)^2 = 25$ at the point $(5, 7)$ is A , then $24A$ is equal to _____

(JEE MAIN 2021 SHIFT -2)

Key: 1225**Sol:** Equation of normal to circle at point $(5, 7)$ is $4x - 3y + 1 = 0$ Equation of tangent to circle at point $(5, 7)$ is $3x + 4y - 43 = 0$ Normal meets x -axis at $A(-1/4, 0)$ and tangent meets x -axis at $(0, \frac{43}{3})$

$$\therefore \text{area of triangle, } A = \frac{1}{2} \left(\frac{43}{3} + \frac{1}{4} \right) \times (7)$$

$$= \frac{1}{2} \left(\frac{172 + 3}{12} \right) \times 7$$

$$\therefore 24A = 1225.$$

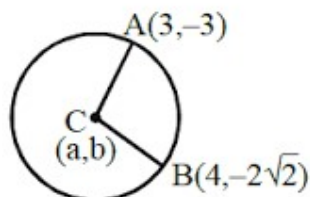
28. Let the normal's at all the points on a given curves pass through a fixed point (a, b). If the curve passes through (3,-3) and (4, $-2\sqrt{2}$), and given that $a - 2\sqrt{2}b = 3$, then $(a^2 + b^2 + ab)$ is equal to _____

(JEE MAIN 2021 SHIFT-2)

Key: 9

Sol: If all the normal's of the curve pass through fixed point (a, b), then the curve must be circle. In circle, all the normal pass through the center.

From the figure, $CA^2 = CB^2$



$$\Rightarrow (a-3)^2 + (b+3)^2 = (a-4)^2 + (b+2\sqrt{2})^2$$

$$\Rightarrow a + (3 - 2\sqrt{2})b = 3$$

$$a - 2\sqrt{2}b + 3b = 3$$

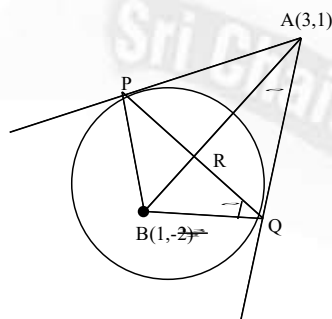
Given that $a - 2\sqrt{2}b = 3$

Solving, we get $a=3$, $b=0$

$$\therefore a^2 + b^2 + ab = 9$$

29. Let B be the centre of the circle $x^2 + y^2 - 2x + 4y + 1 = 0$. Let the tangents at two points P and Q on the circle intersect at the point A (3,1). Then $8 \left(\frac{\text{area } \triangle APQ}{\text{area } \triangle BPQ} \right)$ is equal to _____

(JEE MAIN 2021 SHIFT-I)

Key: 18**Sol:**

Equation of chord of contact PQ is $3x + y - (x+3) + 2(y+1) + 1 = 0$ or $2x + 3y = 0$

Distance, $AR = \frac{|2(3) + 3(1)|}{\sqrt{13}} = \frac{9}{\sqrt{13}}$

Distance, $BR = \frac{|2(1) + 3(-2)|}{\sqrt{13}} = \frac{4}{\sqrt{13}}$

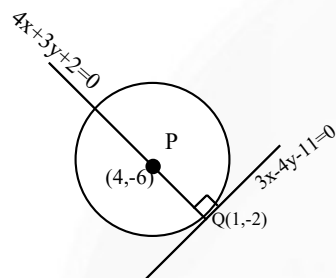
$$\text{So, } 8 \left(\frac{\text{area } \triangle APQ}{\text{area } \triangle BPQ} \right) = 8 \frac{AR}{BR} = 18$$

30. Let a circle C of radius 5 lie below the x-axis. The line $L_1 = 4x + 3y + 2 = 0$ passes through the center P of the circle C and intersects the line $L_2 : 3x - 4y - 11 = 0$ at Q. The line L_2 touches C at the point Q. then the distance of P from the line $5x - 12y + 51 = 0$ is _____

(JEE MAIN 2022 SHIFT-2)

Key: 11

Sol:



From the figure, $P \equiv (1 - 5 \cos \theta, -2 - 5 \sin \theta)$, where $\tan \theta = -\frac{4}{3}$.

$$\therefore P = \left(1 - 5 \left(-\frac{3}{5} \right), -2 - 5 \left(\frac{4}{5} \right) \right)$$

$\equiv (4, -6)$ (as circle lies below x-axis)

$$\therefore \text{Distance of P from the line } 5x - 12y + 51 = 0 \text{ is, } d = \left| \frac{5(4) - 12(-6) + 51}{13} \right| = \left| \frac{20 + 72 + 51}{13} \right| = \frac{143}{13} = 11$$

31. Let the lines $y + 2x = \sqrt{11} + 7\sqrt{7}$ and $2y + x = 2\sqrt{11} + 6\sqrt{7}$ be normal to a circle

$C : (x - h)^2 + (y - k)^2 = r^2$. if the line $\sqrt{11}y - 3x = \frac{5\sqrt{77}}{3} + 11$ is tangent to the circle C, then the value of $(5h - 8k)^2 + 5r^2$ is equal to _____

(JEE MAIN 2022 SHIFT-1)

Key: 816

Sol: Normal's are

$$y + 2x = \sqrt{11} + 7\sqrt{7} \quad \dots(1)$$

$$2x + x = 22\sqrt{11} + 6\sqrt{7} \quad \dots(2)$$

On solving the above equations, we get center of circle as $\left(\frac{8\sqrt{7}}{3}, \sqrt{11} + \frac{5\sqrt{7}}{3} \right)$

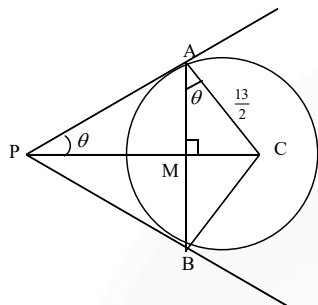
$$\text{Tangent is } \sqrt{11}y - 3x = \frac{5\sqrt{77}}{3} + 11$$

$$\text{Radius} = \text{perpendicular distance of tangent from center} = 4\sqrt{\frac{7}{5}}$$

$$\therefore (5h - 8k)^2 + 5r^2 = 816.$$

32. Let AB be a chord of length 12 of the circle $(x-2)^2 + (y+1)^2 = \frac{169}{4}$. If tangents drawn to the circle at points A and B intersect at the point P, then five times the distance of point P from chord AB equal to _____

(JEE MAIN 2022 SHIFT-2)

Key: 72**sol:**

from the figure ,

$$\cos \theta = \frac{6}{\frac{13}{2}} = \frac{12}{13}$$

$$\therefore \cot \theta = \frac{12}{5}$$

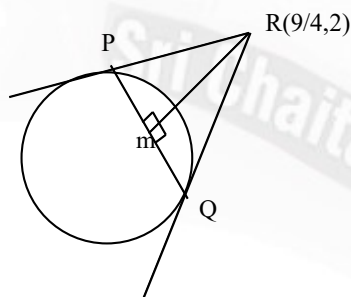
$$\text{so, } 5PM = 5AM \cot \theta = 5 \times 6 \left(\frac{12}{5} \right) = 72$$

33. If the tangents at the points P and Q on the circle $x^2 + y^2 - 2x + y = 5$ meet at the point $R\left(\frac{9}{4}, 2\right)$, then the area of the triangle PQR is

- a) $\frac{13}{4}$ b) $\frac{13}{8}$ c) $\frac{5}{4}$ d) $\frac{5}{8}$

Ans:4

Sol: Equation of circle is $x^2 + y^2 - 2x + y - 5 = 0$ $R = \frac{5}{2}$



$$\text{Length of } PR = QR = \sqrt{S_1}$$

$$= \sqrt{\frac{81}{16} + 4 - \frac{2 \times 9}{4} + 2 - 5} = \frac{5}{4}$$

$$\text{Area of triangle PQR } PQR = \frac{RL^3}{R^2 + L^2} = \frac{\frac{5}{2} \cdot \frac{125}{64}}{\frac{25}{4} + \frac{25}{16}} = \frac{5}{8}$$

34. Let the tangent to the curve $x^2 + 2x - 4y + 9 = 0$ at the point P (1, 3) on it meet the y-axis at A. Let the line passing through P and parallel to the line $x - 3y = 6$ meet the parabola $y^2 = 4x$ at B lies on the line $2x - 3y = 8$. Then $(AB)^2$ is equal to

Ans: 292

Sol: Equation of tangent at P (1, 3) to the curve $x^2 + 2x - 4y + 9 = 0$ is $y - x = 2$.

Then the point A is (0, 2) equation of the line passing through P and parallel to the line $x - 3y = 6$.

The possible coordinate of B are (4, 4) or (16, 8) but (4, 4) does not satisfy $2x - 3y = 8$

Thus the point B is (16, 8)

Then $(AB)^2 = 292$

V. Chord of contact of a circle, Pole and polar of a circle

(Equation of chord of contact of a circle in circles, Equation of chord of a circle when mid point is given in circles, Length of chord when mid point is given in circles, Definition of pole and polar in circles, Polar of a point with respect to circle Pole of a polar with respect to a circle, Conjugate points with respect to given circle, Conjugate lines with respect to circle, Properties of pole and polar of a circle, Problems based on locus)

35. If the pole of a line w.r.t to circle $x^2 + y^2 = a^2$ lies on circle $x^2 + y^2 = a^4$. Then the line touches circle $x^2 + y^2 = r^2$, then $r^2 =$

Ans: 1

Sol: let pole (α, β) lines on $x^2 + y^2 = a^4 \Rightarrow \alpha^2 + \beta^2 = a^4$ _____(1)

equation of polar is $s_1 = 0 \Rightarrow \alpha x + \beta y - a^2 = 0$ _____(2)

If (2) touches a circle

$$x^2 + y^2 = r^2 \Rightarrow r = \frac{|-a^2|}{\sqrt{\alpha^2 + \beta^2}}$$

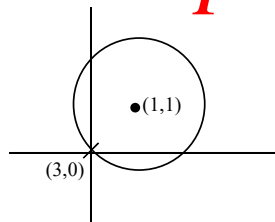
$$\Rightarrow r = \frac{a^2}{\sqrt{a^4}} \{ \text{from (1)} \} = 1$$

36. If the equation of the locus of the middle point of a chord of the circle $x^2 + y^2 = 2(x + y)$ such that the pair of lines joining the origin to the point of intersection of the chord and the circle are equally inclined to the x-axis is $ax + by = 1$ then $a + b$ equals ____

key: 1

sol: $S_1 = S_{11} \Rightarrow xx_1 + yy_1 - (x + x_1) - (y + y_1) = 0$

$$\Rightarrow (x_1 - 1)x + (y_1 - 1)y = x_1 + y_1$$



$$x^2 + y^2 - 2(x+y) \left\{ \frac{(x_1-1)x + (y_1-1)y}{x_1 + y_1} \right\} = 0$$

Equally inclined to x-axis $\rightarrow$ coefficient of

$$xy = 0 \Rightarrow \frac{-2}{x_1 + y_1} \{y_1 - 1 + x_1 - 1\} = 0$$

Locus is $x + y = 2 \Rightarrow a + b = 1$.

37. The radius of the circle which pass through the point (2, 3) and cut off equal chords of length 6 units along the line $y-x-1=0$ and $y+x-5=0$ is 'r' then $[r]$ is (where $[.]$ denotes greatest integer function)

key: 1

sol: The given two lines pass through the point (2,3) and are inclined at 45° and 135° to the x-axis the order ends of chords can easily be calculate as $(2 + 3\sqrt{2}, 3 + 3\sqrt{2})$ and $(2 - 3\sqrt{2}, 3 - 3\sqrt{2})$

There is symmetry about the line $x=2$ and therefore the centers of circles lie on $x=2$ as the chord subtend right angles at the centre. $\therefore 2r^2 = 6^2 \Rightarrow r = 3\sqrt{2}$

38. Two distinct chords each bisected by x-axis are drawn to the circle

$$2(x^2 + y^2) - 2ax - by = 0 \text{ from the point } (a, \frac{b}{2}). \text{ then the least possible integer of } \frac{a^2}{b^2} = \underline{\hspace{1cm}}$$

Key: 2

Sol: chord with midpoint $(x_1, 0)$ is $S_1 = S_{11}$

But it passes through $(a, \frac{b}{2})$

From a quadratic in x_1

For real and distinct root $\Delta > 0$

$$\Rightarrow a^2 > 2b^2$$

$$\frac{a}{b^2} > 2$$

$\therefore$ Possible least integer of $\frac{a^2}{b^2} = 2$

Pole and polar

39. The locus of poles of chord of contacts of with respect to the circle $x^2 + y^2 - 4x - 2y + 5 = 0$ are tangents to the circle $x^2 + y^2 = 25$ is $[ax_1 + by_1 - c]^2 = d(x-a)^2 + (y-b)^2$ [where H.C.F. of a, b, c, d is '1'] then $a+b+c+d=$

Key: 33**Sol:**

Let $p(x_1, y_1)$ is pole equation of chord of contact of $p(x_1, y_1)$ is $S_1 = 0$

$$xx_1 + yy_1 - 2(x + x_1) - 1(y + y_1) + 5 = 0$$

$$\Rightarrow (x_1 - 2)x + (y_1 - 1)y + (-2x_1 - y_1 + 5) = 0$$

$$\Rightarrow (x_1 - 2)x + (y_1 - 1)y + (-2x_1 + y_1 + 5) = 0 \quad \dots(1)$$

Equation (1) is tangent to $x^2 + y^2 = 25$

$$r = d$$

$$5 = \frac{|2x_1 + y_1 - 5|}{\sqrt{(x_1 - 2)^2 + (y_1 - 1)^2}}$$

$$\Rightarrow (2x_1 + y_1 - 5)^2 = 25[(x_1 - 2)^2 + (y_1 - 1)^2]$$

$$a = 2, b = 1, c = 5, d = 25$$

$$a + b + c + d = 33$$

Chord of contact

40. If the radius of the circumcircle of the triangle TPQ, where PQ is chord of contact corresponding to point T with respect to circle $x^2 + y^2 - 2x + 4y - 11 = 0$, is 6 units, then minimum distance of T from the director circle of the given circles is $3a - a\sqrt{2}$ then $a =$

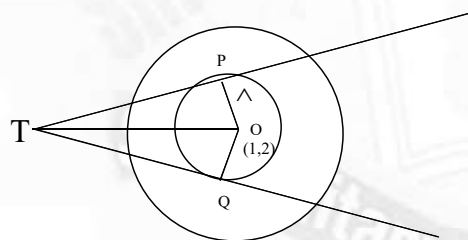
key: 4

sol: Given circle $(x-1)^2 + (y+2)^2 = 16$

its director circles is $(x-1)^2 + (y+2)^2 = 32$

$$(x-1)^2 + (y+2)^2 = 32$$

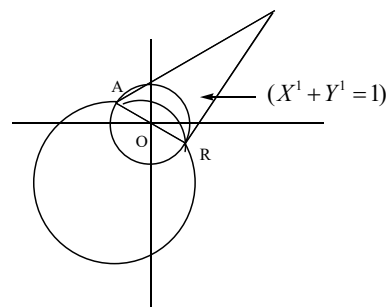
$$\Rightarrow OS = 4\sqrt{2}$$



Therefore, required distance,

$$TS = OT - SO = 12 - 4\sqrt{2}$$

$$(X_1 + Y_1)$$



Locus of point of intersection of tangents chord of contact of (x_1, y_1) w.r.t

$$x^2 + y^2 = 1 \text{ is } xx_1 + yy_1 = 1(AB)$$

AB is also common chord between two circles

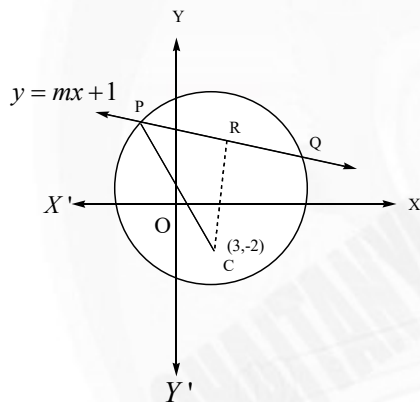
$$\therefore -1 + (\lambda + 6)x - (8 - 2\lambda)y + 3 = 0$$

41. A line $y = mx + 1$ intersects the circle $(x - 3)^2 + (y + 2)^2 = 25$ at the points p and Q. if the midpoint of the line segment PQ has x-coordinate $-\frac{3}{5}$, and m_1, m_2 are possible values of m, then $m_1 + m_2$

Key: 5

Sol: Circle $(x - 3)^2 + (y + 2)^2 = 25$, with centre C(3, -2) a radius 5 is intersected by a line $y = mx + 1$ at P & Q such that midpoint R of PQ has its x-coordinate as $-\frac{3}{5}$.

$$\text{Let } R\left(-\frac{3}{5}, \frac{-3m}{5} + 1\right)$$



$$\text{Then } CR \perp PQ \Rightarrow \frac{-\frac{3}{5} + 1 + 2}{-\frac{3}{5} - 3} \times m = -1$$

$$\Rightarrow \frac{(-3m + 15)m}{-18} = -1 \Rightarrow m^2 - 5m + 6 = 0$$

$$\Rightarrow m = 2, 3$$

Alternately:-

The coordinates of intersection points P and Q of circle and line can be given by solving their equations for x. putting $y = mx + 1$ in

$$(x - 3)^2 + (y - 2)^2 = 25, \text{ we get}$$

$$(x - 3)^2 + (mx + 3)^2 = 25$$

$$\Rightarrow (m^2 + 1)x^2 + 6(m - 1)x - 16 = 0$$

$$\Rightarrow x_1 + x_2 = \frac{-6(m - 1)}{m^2 + 1}, \text{ where } x_1 \text{ and } x_2 \text{ are x-coordination of P and Q respectively.}$$

$$\text{As per question, x-coordinate of midpoint of PQ is } \frac{-3}{5} \therefore \frac{x_1 + x_2}{2} = -\frac{3}{5} \Rightarrow \frac{-3(m - 1)}{m^2 + 1} = -\frac{3}{5}$$

$$\Rightarrow m^2 - 5m + 6 = 0 \Rightarrow m = 2, 3.$$

Conjugate points

42. The points $(-2, -1)$ and $(2, k)$ are conjugate points with respect to the circles:

$x^2 + y^2 - 4x - 2y + 5 = 0$ and the circle ' c_2 ' cuts the circle ' c_1 ' orthogonally then square of radius ' c_2 ' is _____

Key: 5

Sol:

$$C_1 : x^2 + y^2 - 4x - 2y + 5 = 0 \dots (1)$$

If $P(-2, -1)$, $Q(2, k)$ are conjugate points w.r. to C_1 then $S_{12} = 0$

$$x_1x_2 + y_1y_2 + g(x_1 + x_2) + f(y_1 + y_2) + c = 0$$

$$-4 - k - 2(-2 + 2) - 1(-1 + k) + 5 = 0$$

$$-4 - k + 1 - k + 5 = 0$$

$$\Rightarrow -2k + 2 = 0 \Rightarrow k = +1 \Rightarrow Q = (2, 1)$$

We know that the circle 'PQ' as diameter cuts the circle ' C_1 ' orthogonally

$$\text{Equation of } C_2: (x + 2)(x - 2) + (y + 1)(y - 1) = 0$$

$$\Rightarrow x^2 + y^2 - 4 - 1 = 0$$

$$x^2 + y^2 = 5.$$

Pair of Tangents

43. If the lines $2x - ky - 2\sqrt{2}a = 0$ and $2x - 2y - 2\sqrt{2}a = 0$ are tangents of the circle $x^2 + y^2 = a^2$, and if the area formed by given tangents and chord of contact is $\frac{\lambda}{2}a^2$ then the value of

$6\lambda + k$ equals

a) 1

b) 4

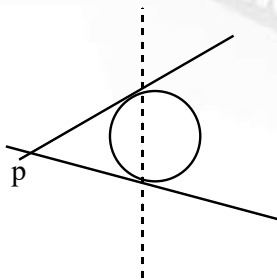
c) 0

d) 2

Key: B

$$\text{sol: } r = d \Rightarrow a \frac{2\sqrt{2}a}{\sqrt{4 + K^2}} \Rightarrow K^2 + 4 = 8$$

$$\Rightarrow K = \pm 2 \Rightarrow K = -2$$



$$\left. \begin{array}{l} 2x - 2y - 2\sqrt{2}a = 0 \\ 2x + 2y - 2\sqrt{2}a = 0 \end{array} \right\} P(\sqrt{2}a, 0)$$

$$\text{Area} = \frac{r \cdot S_{11}^{3/2}}{S_{11} + r^2} = \frac{a \cdot (a^2)^{3/2}}{a^2 + a^2} = \frac{a^2}{2}$$

$$\therefore \left. \begin{array}{l} \lambda = 1 \\ K = -2 \end{array} \right\} \Rightarrow 6\lambda + K = 6(1) - 2 = 4.$$

Pair of Tangents

44 : The chords of contact of the pair of tangents drawn from each point on the line $2x+y=4$ to the circle $x^2+y^2=1$ pass through the point (a, b) then $4(a+b)=$ ____

(1997 IIT Question)

Ans: 3

Sol: Let $P(\alpha, 4 - 2\alpha)$ is any point on the line $2x+y=4$

Equation of chord of contact of 'P' is $S_1=0$

$$\alpha x + (4 - 2\alpha)y - 1 = 0$$

$$\alpha(x - 2y) + (4y - 1) = 0$$

$$x - 2y = 0, 4y - 1 = 0 \Rightarrow y = \frac{1}{4}$$

$$x = \frac{1}{2}$$

$$P = \left(\frac{1}{2}, \frac{1}{4} \right)$$

$$4(a+b) = 3$$

44a: The chord of contacts of the pair of tangents drawn from each point on the line $x-y=1$ to the circle $x^2+y^2-2x-4y+5=0$ passes through the point (a, b) then $a+b=$ ____

Ans: 9

Sol: Let $P(\alpha, \alpha - 1)$ be a point on the line $x-y=1$

Equation of chord of contact of 'P' is $S_1=0$

$$\alpha x + (\alpha - 1)y - 1(x + \alpha) - 2(y + \alpha - 1) - 5 = 0$$

$$(-x - 3y - 3) + \alpha(x + y - 3) = 0 \dots (1)$$

Equation 1 passing through $(6, -3) = (a, b)$

$$a - b = 9$$

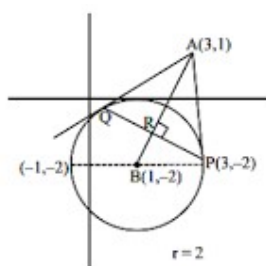
45. Let B be the centre of the circle $x^2 + y^2 - 2x + 4y + 1 = 0$. Let the tangents at two points P

and Q on the circle intersect at the point $A(3,1)$. Then $8 \left(\frac{\text{area } \triangle APQ}{\text{area } \triangle BPQ} \right)$ is equal to ____

(JEE MAIN 2021 SHIFT-2)

Key: 18

Sol:



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Equation of chord of contact PQ is $3x+y-(x+3)+2(y+1)+1=0$ or $2x+3y=0$

$$\text{Distance, } AR = \frac{|2(3)+3(1)|}{\sqrt{13}} = \frac{9}{\sqrt{13}}$$

$$\text{Distance, } BR = \frac{|2(1)+3(-2)|}{\sqrt{13}} = \frac{4}{\sqrt{13}}$$

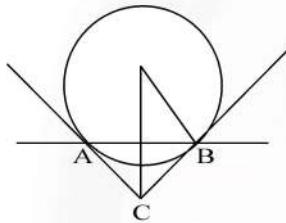
$$\text{So, } 8 \left(\frac{\text{area } \Delta APQ}{\text{area } \Delta BPQ} \right) = 8 \frac{AR}{BR} = 18$$

45a. AB is a chord of the circle $x^2 + y^2 = 25$ the tangent at A and B intersect at C. if (2,3) is

the midpoint of AB, the area of quadrilateral OACB. Is $\sqrt[k]{\frac{3}{13}}$ then $k = \underline{\hspace{2cm}}$

Key: 50

Sol:



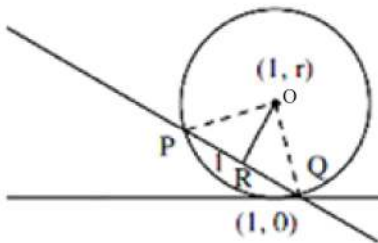
$$\Delta = OB \cdot BC$$

46. Let a circle $C : (x-h)^2 + (y-k)^2 = r^2, k > 0$, touch the x-axis at (1, 0). If the line $x+y=0$ intersect the circle C at P and Q such that the length of the chord PQ is 2, then the value of $h+k+r$ is equal to

(JEE MAIN 2019 SHIFT-2)

Key: 7

Sol:



Circle $(x-h)^2 + (y-k)^2 = r^2, k > 0$ touches the x-axis at (1,0).

$$\therefore k=r \text{ and } h=1$$

In the figure $OP=r$, $PR=1$ and $OR = \frac{|r+1|}{\sqrt{2}}$

$$\therefore r^2 = 1 + \frac{(r+1)^2}{2}$$

$$\Rightarrow r^2 - 2r - 3 = 0$$

$$\Rightarrow (r-3)(r+1) = 0$$

$$\Rightarrow r = 3 \text{ (as } r = k > 0)$$

$$\Rightarrow h+k+r=1+3+3=7.$$

46a. Tangents drawn from the point P(1,8) to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A&B if R is the radius of circum circle of triangle PAB then[R]= ____

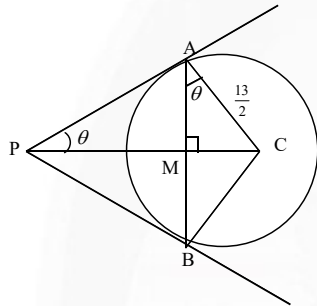
Key: 3

Sol: circum circle of triangle PAB is $x^2 + y^2 - 4x - 10y + 19 = 0$

47. Let AB be a chord of length 12 of the circle $(x-2)^2 + (y+1)^2 = \frac{169}{4}$. If tangents drawn to the circle at points A and B intersect at the point P, then five times the distance of point P from chord AB equal to ____
(JEE MAIN 2022 SHIFT-2)

Key: 72

sol:



from the figure,

$$\cos \theta = \frac{6}{\frac{13}{2}} = \frac{12}{13}$$

$$\therefore \cot \theta = \frac{12}{5}$$

$$\text{So, } 5P.M = 5A.M \cot \theta = 5 \times 6 \left(\frac{12}{5} \right) = 72$$

47a. Let A be the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$. Suppose that the tangents at the points B(1,7) and D(4,-2) on the circle meet at the point C. the area of quadrilateral ABCD is (in sq.units) ____

Key: 75

Sol: $r\sqrt{s_{11}}$.

VI. Relative positions of two circles

(Angle between intersecting circles in circles, Condition for orthogonal circles, Non-intersecting circles in circles, Intersecting circles in circles, Touching circles in circles, Circles touch internally in circles, Circles touch externally in circles, One circle lie inside the other in circles, Centres of similitude in circles, Different forms of common tangents to circles in circles, Direct common tangents in circles, Transverse common tangents in circles, Length of direct common tangent in circles, Length of transverse common tangent in circles, Equation of common tangent of two circles in circles)

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48. The number of common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is

Ans: 1

Sol:

Given circle $x^2 + y^2 = 4$ with centre $C_1(0, 0)$ and $R_1 = 2$

Also $x^2 + y^2 - 6x - 8y - 24 = 0$ with centre $C_2(3, 4)$ and $R_2 = 7$

Distance between centres $= C_1C_2 = 5 = R_2 - R_1$

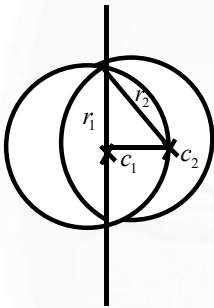
So the circles touch internally and they can have just one common tangent at the point of contact.

49. If one of the diameters of the circle $x^2 + y^2 - 2\sqrt{2}x - 6\sqrt{2}y + 14 = 0$ is a chord of the circle $(x - 2\sqrt{2})^2 + (y - 2\sqrt{2})^2 = r^2$ then $r^2 =$ _____

(JEE MAIN 2022 SHIFT-II)

Ans: 10

Sol: $C_1 = (\sqrt{2}, 3\sqrt{2})$



$$r_1 = \sqrt{2 + 18 - 14} = \sqrt{6}$$

$$C_2 = (2\sqrt{2}, 2\sqrt{2})$$

$$r = ?$$

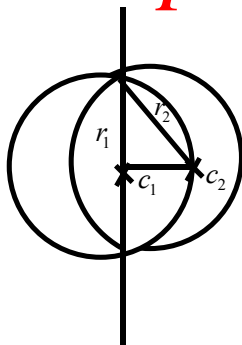
$$C_1C_2 + r_1^2 = r_2^2$$

$$4 + 6 = r_2^2$$

$$r^2 = 10$$

49a. If one of the diameter of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord to the circle $S=0$ with centre $(2, 1)$ then radius of $S=0$ is _____

Ans: 3



Sol: $C_1 = (1, 3)$

$$C_2 = (2, 1)$$

$$r_1 = \sqrt{1+9-6} = 2$$

$$r_2 = ?$$

$$C_1C_2^2 + r_1^2 = r_2^2$$

$$5 + 4 = r_2^2$$

50 . If the circle $x^2 + y^2 + 6x + 8y + 16 = 0$ and $x^2 + y^2 + 2(3 - \sqrt{3})x + 2(4 - \sqrt{6})y = k + 6\sqrt{3} + 8\sqrt{6}$,

$K > 0$ touch internally at the $P(\alpha, \beta)$ then $(\alpha + \sqrt{3})^2 + (\beta + \sqrt{6})^2 = \underline{\hspace{2cm}}$

(2022 JULY SHIFT I)

Ans: 25

Sol: $C_1 = (-3, -4)$

$$r_1 = \sqrt{9+16-16} = 3$$

$$C_2 = (\sqrt{3}-3, \sqrt{6}-4)$$

$$C_1C_2 = \sqrt{3+6} = 3$$

$$r^2 = \sqrt{3+9-6\sqrt{3}+6+16-8\sqrt{6}+k+6\sqrt{3}+8\sqrt{6}} = \sqrt{34+k}$$

$$C_1C_2 = |r_1 - r_2|$$

$$3 = |3 - \sqrt{34+k}|$$

$$\pm 3 = 3 - \sqrt{34+k}$$

$$34+k = 6^2 = 36$$

$$k = 2$$

$$r_2 = 6$$

$$r_1 : r_2 = 3 : 6 = 1 : 2$$

P divides C_1C_2 in the ratio 1: 2 externally

$$P = \left(\frac{\sqrt{3}-3+6}{-1}, \frac{\sqrt{6}-4+8}{-1} \right) = (-3-\sqrt{3}, -4-\sqrt{6}) = (\alpha, \beta)$$

$$\begin{aligned} & (\alpha + \sqrt{3})^2 + (\beta + \sqrt{6})^2 \\ & = (-3)^2 + (-4)^2 = 25 \end{aligned}$$

51. If the two circles $x^2+y^2-6x-9y+13=0$ and $x^2+y^2-2x-16y=0$ touch each internally then point of contact is

Ans: (-1, -1)

Sol: $C_2 = (1, 8) \quad r_2 = \sqrt{65}$

$$C_1 = \left(3, \frac{9}{2} \right) \quad r_1 = \sqrt{9 + \frac{81}{4} - 13} = \frac{\sqrt{65}}{2}$$

$$C_1C_2 = \sqrt{4 + \frac{49}{9}} = \frac{\sqrt{65}}{2}$$

$$C_1C_2 = |r_1 - r_2|$$

P divides C_1C_2 in the ratio

$$\frac{\sqrt{65}}{2} : \sqrt{65} \Rightarrow \frac{1}{2} : 1 \Rightarrow 1 : 2$$

$$P = (-1, -1)$$

52. Circles with radii 3, 4, 5 touch each other externally. If P is point of intersection of tangent to these circles at their points of contact, find the distance of P from the point of contact **(JEE Adv 2005)**

Ans: $\sqrt{5}$

Sol: $a = 3 + 4 = 7, \quad b = 4 + 5 = 9$

$$c = 3 + 5 = 8$$

$$s = \frac{7 + 8 + 9}{2} = 12$$

$$\Delta = \sqrt{12 \cdot 5 \cdot 3 \cdot 4} = 12\sqrt{5}$$

$$r = \frac{\Delta}{s} = \sqrt{5}$$

52a. Three circles are such that touch each other two externally. The common tangent are concurrent at P the length of the tangent to each circle is P. The ratio of the product of their radii to the sum of their radii is _____

Ans: 1

Sol: $a = r_1 + r_2 \quad b = r_2 + r_3 \quad c = r_3 + r_1$

$$r = r_1 + r_2 + r_3$$

$$D = \sqrt{(r_1 + r_2 + r_3)r_1r_2r_3}$$

$$D = \frac{\sqrt{(r_1 + r_2 + r_3)r_1r_2r_3}}{r_1 + r_2 + r_3} = \sqrt{\frac{r_1r_2r_3}{r_1 + r_2 + r_3}}$$

$$P^2 = \frac{r_1r_2r_3}{r_1 + r_2 + r_3}$$

$$\text{Put } r_1 = 1, r_2 = 2, r_3 = 3$$

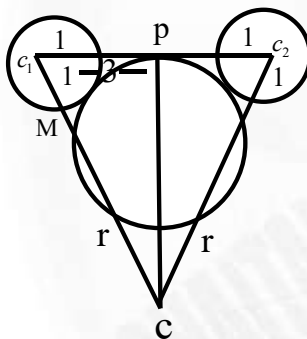
$$P^2 = 1$$

53. The centre of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be midpoint of line segment joining the centre of C_1 and C_2 and C be a circle touching circles C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C then the radius of the circle is ____

(Jee Adv 2009)

Ans: 8

Sol: Let r be the radius of required of clearly in ΔC_1CC_2 , $C_1C = C_2C = r + 1$ and P be midpoint of C_1C_2



$CP \perp C_1C_2$ and $PM \perp CC_1$

from similar

Δ les ΔPMC_1 , & ΔPCP_1

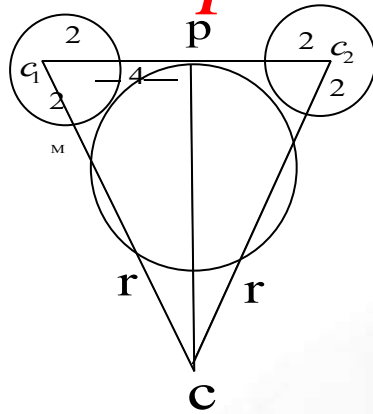
$$\frac{MC_1}{PC_1} = \frac{PC_1}{CC_1} \Rightarrow \frac{1}{3} = \frac{3}{r+1}$$

$$r = 8$$

53a. The centre of two circles C_1 and C_2 each of two radius are at a distance of 8 units from each other. Let P be midpoint of line segment joining the centre of C_1 and C_2 and C be a circle touching circles C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C then the radius of the circle is ____

Ans: 7

Sol:



Let r be the radius of required of clearly $\Delta C_1 C C_2$, $C_1 C = C_2 C = r + 1$ and P in midpoint of $C_1 C_2$

$$CP \perp C_1 C_2 \text{ and } PM \perp C C_1$$

from similar

$$\frac{2}{4} = \frac{4}{r+1}$$

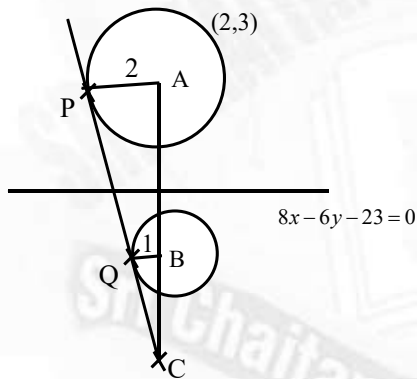
$$\Rightarrow r = 7$$

54 . Let the point B be the reflection of the $A(2, 3)$ wrt line $8x - 6y - 23 = 0$ Let T_A and T_B be circles of radii 2 and 1 with centre A and B respectively. Let T be a common tangent to the circles T_A and T_B such that both the circles are on the same side of T . If C is point of intersection of T and line passing through A and B , then the length of line segment AC is

(Jee adv 2019)

Ans: 10

Sol:



$$AN = \frac{5}{2} = BN$$

ΔCPA similar ΔCQB

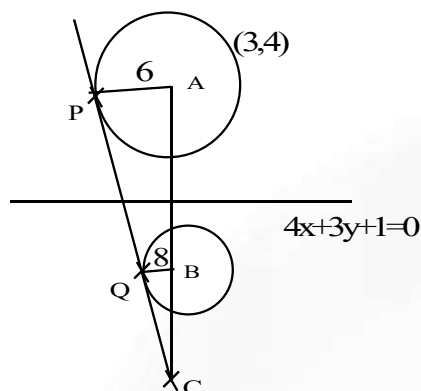
$$\frac{CA}{CB} = \frac{PA}{QB} = \frac{CA}{CA - 5} = \frac{2}{1}$$

$$CA = 10.$$

54. Let the point B be the reflection of the A(3, 4) and radii 6 & 8 and line $4x+3y+1=0$ then

$$\frac{CA}{10} = ?$$

Ans: 4



Sol: $AN = \frac{25}{5} = BN$

$\triangle CPA$ similar $\triangle CQB$

$$\frac{CA}{CB} = \frac{PA}{QB} \Rightarrow \frac{CA}{CA-10} = \frac{8}{6}$$

$$6CA = 8CA - 80$$

$$80 = 2CA$$

$$CA = 40$$

$$\frac{CA}{10} = 4$$

55. If the circles $x^2 + y^2 - 6x + 8 = 0$ and $x^2 + y^2 - 8y + 16 - k^2 = 0$ ($k > 0$) touch each other at a point then largest value of K is _____

(JEE Main 2020 Shift-1)

Ans: 36

Sol: $C_1 = (3, 0) \quad r_1 = \sqrt{9-8} = 1$

$$C_2 = (0, 4) \quad r_2 = \sqrt{k}$$

$$C_1 C_2 = 5$$

$$C_1 C_2 = 1 + \sqrt{k} \Rightarrow 1 + \sqrt{k} = 5 \Rightarrow \sqrt{k} = 4 = k = 16$$

$$C_1 C_2 = |1 - \sqrt{k}| \Rightarrow 5 = |1 - \sqrt{k}|$$

$$\sqrt{k} = 6$$

Largest value = 36

55a. If the circle $x^2 + y^2 = 2$ and $x^2 + y^2 - 4x - 4y + \lambda = 0$ have exactly three real common tangents then $\lambda =$ _____

Ans: 6

Sol: $C_1 = (0, 0) \quad r_1 = \sqrt{2}$

$$C_2 = (2, 2) \quad r_2 = \sqrt{8 - \lambda}$$

$$C_1 C_2 = 2\sqrt{2}$$

$$2\sqrt{2} = \sqrt{2} + \sqrt{8 - \lambda}$$

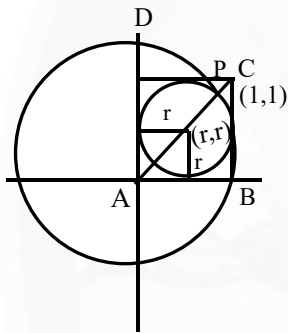
$$\sqrt{2} = \sqrt{8 - \lambda}$$

$$\lambda = 6$$

56. Let ABCD be square of side of unit length. Let a circle C_1 centered at A with unit radius is drawn. Another circle C_2 which touches C_1 and the line AD and AB are tangent to it is also drawn let a tangent line from the point C to the circle C_2 meet the side AB at E if the length of EB is $\alpha + \sqrt{3}\beta$ where α, β are integers then $\alpha + \beta =$ _____ (Jee Main 2021 Shift-1)

Ans: 1

Sol:



from the fig $AQ + QP = 1$

$$(\sqrt{2} + 1)r = 1$$

$$r = \sqrt{2} - 1$$

Equation of line through $C(1,1)$ is

$$y - 1 = m(x - 1)$$

$$mx - y + 1 - m = 0$$

By using $r = d$ to C_2

$$\left| \frac{mr - r + 1 + m}{\sqrt{1 + m^2}} \right| = r$$

$$(m - 1)^2 (r - 1)^2 = r^2 (m^2 + 1)$$

$$\text{put } r = \sqrt{2} - 1$$

$$m = 2 + \sqrt{3}$$

$$\text{Equation of } CE = y - 1 = (2 + \sqrt{3})(x - 1)$$

$$\text{Put } y = 0 \Rightarrow x = -1 + \sqrt{3}$$

$$E = (-1 + \sqrt{3}, 0)$$

$$EB = 2 - \sqrt{3} = \alpha + \sqrt{3}\beta$$

$$\alpha + \beta = 1$$

56a. Let ABCD be square of side of two unit length. Let a circle C_1 contained at A with unit radius is drawn. Another circle C_2 which touches C_1 and the line AD and AB are tangent to it is also drawn let a tangent line from the point C to the circle C_2 meet the side AB at E if the length of EB is $\alpha + \sqrt{3}\beta$ where α, β are integers then $\alpha + \beta =$ ____
same as above but $r=2$

Ans: 1

Sol: By using above figure

$$AQ + QP = 2$$

$$\sqrt{2}r + r = 2$$

$$r = \frac{2}{\sqrt{2}+1} 2(\sqrt{2}-1)$$

equation of line passing through C = (2,2) in

$$y-2=m(x-2)$$

$$mx-y+2-2m=0$$

by using $r = d \text{ to } C_2$

$$\frac{|mr - r + 2 - 2m|}{\sqrt{m^2 + 1}} = r$$

$$\frac{(m-1)^2(r-2)^2}{m^2+1} = r^2$$

$$\text{put } r = 2(\sqrt{2}-1)$$

$$\frac{(m-1)^2[2(\sqrt{2}-1)-2]^2}{m^2+1} = r^2$$

$$\frac{(m-1)^2 4[\sqrt{2}-2]^2}{m^2+1} = 4(\sqrt{2}-1)^2$$

$$\frac{(m-1)^2 4.2[\sqrt{2}-1]^2}{m^2+1} = 4(\sqrt{2}-1)^2$$

$$2m^2 - 4m = m^2 + 1$$

$$m^2 - 4m + 1 = 0$$

$$m = \frac{4 \pm \sqrt{12}}{2} = 2 \pm \sqrt{3}$$

$$\text{IF } m = 2 + \sqrt{3}$$

$$y-2 = (2+\sqrt{3})(x-2), \text{ Put } y=0$$

$$-2 = (2+\sqrt{3})(x-2)$$

$$2-x = 2-\sqrt{3} \quad E = (\sqrt{3}, 0) \Rightarrow EB = \sqrt{(2-\sqrt{3})^2} = 2-\sqrt{3} = \alpha + \sqrt{3}\beta$$

$$\alpha + \beta = 1$$

57 . Two circles each of radius 5 units touch each other at the point (1, 2). If the equation of their common tangent in $4x + 3y = 10$ and $C_1 = (\alpha, \beta)$ and $C_2 = (\gamma, \delta)$, $C_1 \neq C_2$ are their centre

then $|(\alpha + \beta)(\gamma + \delta)| = \underline{\hspace{2cm}}$

(JEE Main – 2021 Shift-I)

Ans: 40

Sol: Slope of tangent = $-\frac{4}{3}$

Slope of line of centre = $\frac{3}{4} = \tan \theta$

we know that

$$(1 \pm 5 \cos \theta, 2 \pm 5 \sin \theta)$$

$$= \left(1 \pm 5 \cdot \frac{4}{5}, 2 \pm 5 \cdot \frac{3}{5} \right)$$

$$= (5, 5), (-3, -1)$$

$$\alpha + \beta = 10, \gamma + \delta = -4$$

$$|(\alpha + \beta)(\gamma + \delta)| = 40$$

57a. Centre of circle whose radius is 5 and which touches the circle $x^2 + y^2 - 2x - 4y - 20 = 0$ at

$(5, 5)$ is (h, k) then $|h + k| = \underline{\hspace{2cm}}$

Ans: 17

Sol: $C_1 = (h, k), r_1 = 5$

$$C_2 = (1, 2) \quad r_2 = 5$$

$$r_1 : r_2 = 1 : 1$$

$(5, 5)$ in mid point of $C_1 C_2$

$$\left(\frac{h+1}{2}, \frac{k+2}{2} \right) = (5, 5)$$

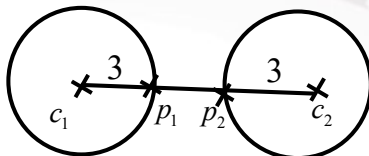
$$h = 9, k = 8$$

$$|h + k| = 17$$

58. The minimum distance between any two points P_1 and P_2 while considering point P_1 on are circle and point P_2 on the other circle for given circle equation

$x^2 + y^2 - 10x - 10y + 41 = 0$ and $x^2 + y^2 - 24x - 10y + 160 = 0$ is $\underline{\hspace{2cm}}$ (Jee Main 2021 Shift-2)

Ans: 1



Sol: $C_1 = (5, 5) \quad r_1 = \sqrt{50 - 41} = 3$

$$C_2 = (12, 5) \quad r_2 = \sqrt{144 + 25 - 160} = 3$$

$$C_1 : C_2 = \sqrt{49} = 7$$

$$P_1P_2 = 7 - 6 = 1$$

58a. The maximum distance between any two points P_1 and P_2 while considering point P_1 on are circle and point P_2 on the other circle for given circle equation

$$x^2 + y^2 - 10x - 10y + 41 = 0 \text{ and } x^2 + y^2 - 24x - 10y + 160 = 0 \text{ is } \underline{\hspace{2cm}}$$

$$\text{max of } P_1P_2 = 3 + 7 + 3 = 13$$

(Not covered in mains and Advance)

59. If the angle between two equal circles with centre $(-2,0)$ and $(2,3)$ is 120° then radius of circle is $\underline{\hspace{2cm}}$

Ans: 5

$$\text{Sol: } C_1 = (-2,0), C_2 = (2,3)$$

$$r_1 = r_2 = r, C_1C_2 = 5$$

$$\cos 120^\circ = \frac{25 - 2r^2}{2r^2}$$

$$-\frac{1}{2} = \frac{25 - 2r^2}{2r^2}$$

$$r = 5$$

60. The point of intersection of two equal circles which cut orthogonally are $(2, 3)$ and $(5, 4)$, if r is radius of circle then $[r] = \underline{\hspace{2cm}}$

Ans: 2

$$\text{Sol: } L.C = \frac{2r_1r_2}{\sqrt{r_1^2 + r_2^2}}$$

$$\sqrt{10} = \frac{2r^2}{\sqrt{2}r} = \sqrt{2}r$$

$$r = \sqrt{5} \Rightarrow [r] = 2$$

61. A circle S passes through the point $(0, 1)$ and is orthogonal to circles $(x-1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$ then radius of circle is $\underline{\hspace{2cm}}$

Ans: 7

$$\text{Sol: Let the circle be } x^2 + y^2 + 2gx + 2fy + c = 0 \dots\dots\dots(1)$$

$$\text{given } x^2 + y^2 - 2x - 15 = 0 \dots\dots\dots(2)$$

$$x^2 + y^2 - 1 = 0 \dots\dots\dots(3)$$

$\therefore$ 1 & 2 cut orthogonally

$$2(g)(-1) = C - 15$$

$$2g + c = 15$$

$\therefore$ 1, 2, 3 cut orthogonally

$$2(g)(0) = 2f(0) = c - 1$$

$$C = 1$$

$$G = -1$$

$\therefore$ 1 passes through (0, 1)

$$1 + 2f + c = 0$$

$$f = -1$$

$$C = (-7, -1), r = \sqrt{49 + 1} = 7$$

62. For the circles $x^2 + y^2 + 2\lambda x + c = 0$, $x^2 + y^2 + 2my - c = 0$, the number of common tangent is

Ans: 2

Sol: Clearly given circles intersect orthogonally and they intersect

$\therefore$ no. of common tangents = 2

63. If the distance between the centers of the circles of radii 3, 4 is 25 then length of transverse common tangent is _____

Ans: 24

Sol:

$$L.T = \sqrt{d^2 - (r_1 + r_2)^2}$$

$$= \sqrt{25^2 - 7^2} = 24$$

64. A circle passing through the point (3, 4) and with the circle $x^2 + y^2 = a^2$ orthogonally. The locus of its centre is a straight line. If the distance of the straight line from the origin is 817 then $a^2 =$ _____

Ans: 8145

Sol: Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ _____ (1)

given circle $x^2 + y^2 = a^2$ _____ (2)

$\therefore$ 1 & 2 cut orthogonally

$$C = a^2$$

$\therefore$ 1 passes through (3, 4)

$$6g + 8f + a^2 + 25 = 0$$

$$\text{Locus of centre } 6x + 8y - a^2 - 25 = 0$$

$$\text{Parallel distance} = \frac{|a^2 + 25|}{10} = 817$$

$$a^2 = 8145$$

65. If $\left(-\frac{1}{3}, -1\right)$ is centre of similitude for the circle $x^2 + y^2 = 1$ and $x^2 + y^2 - 2x - 6y - 6 = 0$ then length of common tangent of the circle is _____

Ans: 1

Sol: $C_1 = (0, 0), \quad r_1 = 1$

$C_2 = (1, 3), \quad r_2 = \sqrt{1+9+6} = 4$

$r_1 : r_2 = 1 : 4$

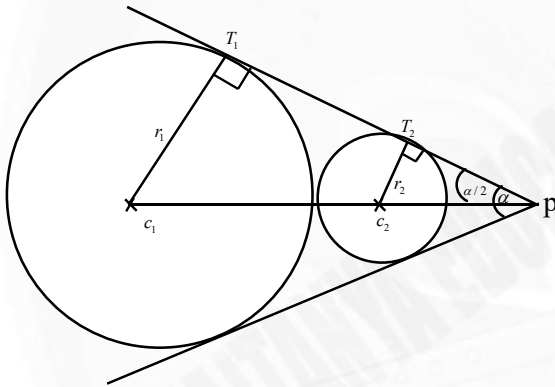
$$ECS = \left(\frac{1}{-3}, \frac{3}{-3} \right) = \left(-\frac{1}{3}, -1 \right)$$

Length of direct common tangent

$$= \sqrt{d^2 - (r_1 - r_2)^2} = \sqrt{10 - 9} = 1$$

66. Two circles with radii r_1 and r_2 , $r_1 > r_2 \geq 2$ touch each other externally. If α is angle between direct common tangent then

Ans: $\frac{1}{7}$

Sol:

$$\sin \frac{\alpha}{2} = \frac{r_2}{PC_2} = \frac{r_1}{PC_1}$$

$$\frac{PC_1}{PC_2} = \frac{r_1}{r_2}$$

$$\frac{C_1C_2 + C_2P}{PC_2} + 1 = \frac{r_1}{r_2}$$

$$\frac{r_1 + r_2}{PC_2} + 1 = \frac{r_1}{r_2}$$

$$\frac{r_1 + r_2}{PC_2} = \frac{r_1 - r_2}{r_2}$$

$$PC_2 = \frac{r_2(r_1 + r_2)}{r_1 - r_2}$$

$$\therefore \sin \frac{\alpha}{2} = \frac{(r_1 - r_2)}{r_1 + r_2}$$

$$\alpha = 2 \sin^{-1} \left(\frac{r_1 - r_2}{r_1 + r_2} \right)$$

if $r_1 = 4, r_2 = 3$

then $\sin \frac{\alpha}{2} = \frac{1}{7}$

67. The set of integral values of λ for which exactly two common tangents can be drawn to the circles $x^2 + y^2 - 4x - 4y + 6 = 0$ and $x^2 + y^2 - 10x - 10y + \lambda = 0$ is _____

Ans: 22

Sol: $C_1 = (2, 2)$ $r_1 = \sqrt{2}$

$C_2 = (5, 5)$, $r_2 = \sqrt{50 - \lambda}$

$C_1 C_2 = 3\sqrt{2}$

By using $|r_1 - r_2| < C_1 C_2 < r_1 + r_2$

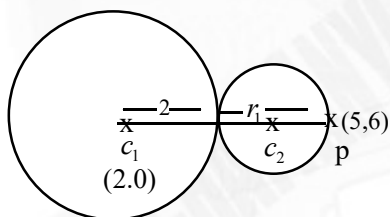
$\lambda \in (18, 42)$

No. of integral values 23

68. If r_1 and r_2 are radii smallest and largest circles which passes through $(5, 6)$ and touches the circle $(x-2)^2 + y^2 = 4$ then $[r_1 r_2] = \underline{\hspace{2cm}}$

Ans: 12

Sol: Case (i)

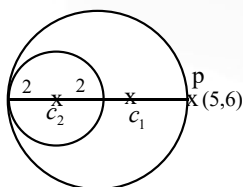


$$C_1 P = \sqrt{45}$$

$$2 + 2r_2 = \sqrt{45}$$

$$r_1 = \frac{\sqrt{45} - 2}{2}$$

Case –(ii)



$$C_2 P = \sqrt{45}$$

$$2r_2 - 2 = \sqrt{45}$$

$$r_2 = \frac{\sqrt{45} + 2}{2}$$

$$r_1 r_2 = \frac{41}{4}$$

$$\left[\frac{41}{4} \right] = 10$$

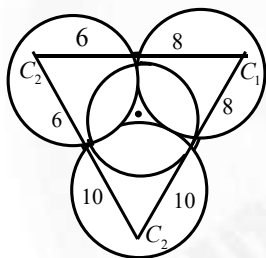
VII. System of Circles

(Family of circles passing through P.O.I of circle and line in circles, Family of circles touching the circle at point (x1; y1) in circles, Family of circles passing through P.O.I of two circles in circles, Equation of radical axis, Properties of radical axis, Equation of common chord of two circles in circles, Co-axial system of circles, Radical centre in circles, Properties of radical centre, Limiting points of a co-axial system of circles in circles)

69. Circles with radii 6, 8, 10 are touching each other externally. If "P" is the point of intersection of tangents to these circles at their point of contact. The distance of 'P' from the points of contact is ____

Ans: 4.50

Sol:



P is in centre of $\Delta C_1 C_2 C_3$. The distance from centre (P) to the points of contact = in radius

$$2S = 6 + 8 + 8 + 10 + 10 + 6 = 48$$

$$2S = 48$$

$$S = 24$$

$$r = \frac{\Delta}{S} = \sqrt{\frac{24(24-14)(24-18)(24-16)}{24 \times 24}}$$

$$= \sqrt{\frac{10 \times 6 \times 8}{24}} = \sqrt{20} = \sqrt{4 \times 5}$$

$$= 2\sqrt{5} = 2(2.25)$$

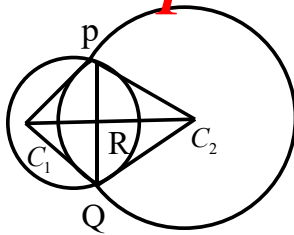
$$= 4.50$$

Angle between Two Circles:

70. If the angle of intersection at a point where the two circles with radii 5cm and 12cm intersect is 90° , then the length of their common chord is ____

Ans: 9.2307

Sol:



$$\angle C_1PC_2 = 90^\circ$$

$$= (C_1C_2)^2 = (C_1P)^2 + (C_2P)^2$$

$$= 5^2 + 12^2$$

$$= 169$$

$$C_1C_2 = 13$$

$$\text{Let } PR=QR=x, \text{ Then } C_1R = \sqrt{25-x^2}, C_2R = \sqrt{144-x^2}$$

$$\sqrt{25-x^2} + \sqrt{144-x^2} = C_1R + C_2R = C_1C_2 = 13$$

$$\sqrt{144-x^2} = 13 - \sqrt{25-x^2}$$

$$144 - x^2 = 169 + 25 - x^2 - 26\sqrt{25-x^2}$$

$$26\sqrt{25-x^2} = 50$$

$$13(\sqrt{25-x^2}) = 25$$

$$169(25-x^2) = 625$$

$$169x^2 = 169 \times 25 - 625$$

$$= 25(169 - 25)$$

$$= 25 \times 144$$

$$169x^2 = 25 \times 144$$

$$x^2 = \frac{25 \times 144}{169}$$

$$x = \frac{5 \times 12}{13} = \frac{60}{13}$$

$$\text{Length of common chord} = 2x$$

$$2 \times \frac{60}{13} = \frac{120}{13}$$

$$= 9.2307$$

Common Chord

71. If the circle $x^2+y^2+4x+22y+c=0$ bisects the circumference of the circle $x^2+y^2-2x+8y-d=0$ then $c+d$ is equal to _____

Ans: 50

Sol: The common chord must be a diameter of the second circle. The equation to the common chord

$$S - S^1 = 0$$

$$6x + 14y + c + d = 0$$

$$C_2(11-4) \Rightarrow 6-56+c+d=0$$

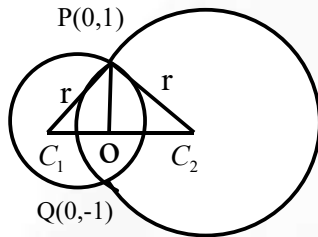
$$c+d=50$$

System of Circles

72. Two circles with equal radii are intersecting at the point (0,1) and (0,-1) the tangents at the point (0, 1) to one of the circle passes through the centre of the other circle. Then the distance between the centre of these circle is _____

Ans: 2

Sol:



$$OP=1, OC_1=r, \sin 45^\circ = \frac{r}{\sqrt{2}}$$

$$\text{from } \triangle OCP_1 \quad OC_1^2 + OP^2 = r^2$$

$$\frac{r^2}{2} + 1 = r^2 \Rightarrow 1 = \frac{r^2}{2}$$

$$r^2 = 2$$

$$r = \sqrt{2}$$

$$C_1C_2 = 2OC_1 = 2 \times \frac{r}{\sqrt{2}} = 2 \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$C_1C_2 = 2$$

Length of the Common Chord

73. If $\frac{2}{\sqrt{5}}$ is the length of the common chord of the circle $x^2+y^2+2x+2y+1=0$ and $x^2+y^2+\alpha x+3y+2=0$ then $\alpha =$ _____ ($\alpha \neq 0$)

Ans: 4

Sol: Equation of the common chord is

$$(\alpha - 2)x + y + 1 = 0$$

For the circle $x^2+y^2+2x+2y+1=0$

$$C(-1, -1) \quad r = \sqrt{1+1-1} = 1$$

d = perpendicular distance from (-1, -1) to common chord

$$= \frac{|(\alpha - 2)(-1) - 1 + 1|}{\sqrt{(\alpha - 2)^2 + 1}} = \frac{|\alpha - 2|}{\sqrt{(\alpha - 2)^2 + 1}}$$

$$\text{Given length of common chord} = \frac{2}{\sqrt{5}}$$

$$2\sqrt{r^2 - d^2} = \frac{2}{\sqrt{5}} \Rightarrow \sqrt{r^2 - d^2} = \frac{1}{\sqrt{5}}$$

$$1 - d^2 = \frac{1}{5} \Rightarrow 1 - \frac{1}{5} = \frac{5-1}{5}$$

$$\frac{(\alpha - 2)^2}{(\alpha - 2)^2 + 1} = \frac{4}{5} = \frac{2^2}{2^2 + 1}$$

$$(\alpha - 2)^2 = 2$$

$$\alpha - 2 = 2$$

$$\alpha = 4$$

Orthogonally

74. If the equation of the circle which passes through the point (1, 1) and cuts the circle $x^2 + y^2 - 4x - 6y + 4 = 0$ and $x^2 + y^2 + 6x - 4y + 15 = 0$ orthogonally is $x^2 + y^2 + 2gx + 2fy + c = 0$ then $5g + 2f + c = \underline{\hspace{2cm}}$

Ans: 2

Sol:

$$S = 0 \Rightarrow x^2 + y^2 + 2gx + 2fy + c = 0 \quad (i)$$

$$(1, 1) \Rightarrow 2g + 2f + c + 2 = 0 \Rightarrow 2g + 2f = -c - 2$$

$$x^2 + y^2 + 6x - 4y + 15 = 0 \dots\dots\dots(2)$$

$$x^2 + y^2 - 4x - 6y + 4 = 0 \dots\dots\dots(3)$$

$$(1) \& (2) \text{ cut orthogonally } \Rightarrow 6g - 4f = c + 15 \dots\dots\dots(4)$$

$$(1) \& (3) \text{ cut orthogonally } \Rightarrow -4g - 6f = c + 4 \dots\dots\dots(8)$$

$$g = \frac{4}{3}, f = \frac{-7}{6}, c = \frac{-7}{3}$$

$$5g + 2f + c = 5\left(\frac{4}{3}\right) + 2\left(\frac{-7}{6}\right) - \frac{7}{3}$$

$$= \frac{20 - 7 - 7}{3} = \frac{6}{3} = 2$$

75. If the circle $x^2 + y^2 + 2x + 2ky + 6 = 0$ and $x^2 + y^2 + 2ky + k = 0$ intersect orthogonally, then K is $(2k + 3 \neq 0)$

Ans: 2

Sol:

$$2(gg^1 + ff^1) = C + C^1$$

$$2(1.0 + k.k) = 6 + k$$

$$2k^2 - k - 6 = 0$$

$$2k^2 - 4k + 3k - 6 = 0$$

$$2k(k - 2) + 3(k - 2) = 0$$

$$k - 2 = 0$$

$$k = 2$$

Angle between the Circles

76. If the angle between two circles with centre $(-2, 0)$, $(2, 3)$ is 120° then the radius of the circle is _____

Ans: 5

Sol:

$$\cos \theta = \frac{d^2 - r_1^2 - r_2^2}{2r_1r_2} \quad \theta = 120^\circ, r_1 = r_2 = r$$

$$\cos 120 = \frac{25 - 2r^2}{2r^2} \quad d = \sqrt{(2+2)^2 + (3-0)^2} = \sqrt{16+9} = \sqrt{25} = 5$$

$$\cos(90 + 30) = \frac{25 - 2r^2}{2r^2}$$

$$\frac{-1}{2} = \frac{25 - 2r^2}{2r^2}$$

$$-r^2 = 25 - 2r^2$$

$$r^2 = 25$$

$$r = 5$$

Radical Axis

77. The radical centre of the three circle $x^2 + y^2 - 2x - 1 = 0$, $x^2 + y^2 - 3y - 1 = 0$ and $2x^2 + 2y^2 - x - 7y - 2 = 0$, is Q then $Qx + Qy =$ _____

Ans: 0

Sol:

$$x^2 + y^2 - 2x - 1 = 0 \dots\dots(1)$$

$$x^2 + y^2 - 3y - 1 = 0 \dots\dots(2)$$

$$x^2 + y^2 - \frac{1}{2}x - \frac{7}{2}y - 1 = 0 \dots\dots(3)$$

radical axes of 1 & 2 and $(1) - (2) = 0$

$$-2x + 3y = 0 \Rightarrow 2x - 3y = 0 \dots\dots(4)$$

$$\text{radical axis of 2 \& 3 is } \frac{1}{2}x + \frac{1}{2}y = 0$$

$$x + y = 0 \dots\dots(5)$$

From (4) and (5)

$$x = 0, y = 0$$

$$Q(x, y) = (0, 0)$$

$$Qx + Qy = 0 + 0 = 0$$

78. The distance of the point $(1, 2)$ from the common chord of the circle

$$x^2 + y^2 + 6x - 16 = 0 \text{ and } x^2 + y^2 - 2x - 6y - 6 = 0 \text{ is } \underline{\hspace{2cm}}$$

Ans: 1

Sol: common chord = $S - S^1 = 0$

$$8x + 6y - 10 = 0$$

$$4x + 3y - 5 = 0$$

$$\text{distance} = \left| \frac{4(1) + 3(2) - 5}{\sqrt{4^2 + 3^2}} \right| = \frac{5}{5} = 1$$

79. A circle passes through the point (3, 4) and cuts the circle $x^2 + y^2 = a^2$ orthogonally. The locus of its centre is a straight line. If the distance of this straight line from the origin is 25. Then $a^2 =$ ____

Ans: 225

Sol: (x_1, y_1) is a point on the locus

circle equation

$$(x - x_1)^2 + (y - y_1)^2 = (x - 3)^2 + (y - 4)^2$$

$$x^2 + y^2 - 2xx_1 - 2yy_1 + x_1^2 + y_1^2 - (x_1 - 3)^2 - (y_1 - 4)^2 = 0$$

Cuts orthogonally $x^2 + y^2 = a^2$

$$-a^2 + x_1^2 + y_1^2 - (x_1 - 3)^2 - (y_1 - 4)^2 = 0$$

$$6x_1 + 8y_1 - 25 - a^2 = 0$$

Locus is $6x + 8y - 25 - a^2 = 0$

$$\text{Distance from angle} = \frac{|25 + a^2|}{\sqrt{36 + 64}} = 25$$

$$25 + a^2 = 250$$

$$a^2 = 225$$

EXERCISE – III

MORE THAN ONE ANSWER TYPE QUESTIONS & PASSAGE TYPE QUESTIONS INCLUDING PYQ'S

1. The equation of the tangents drawn from the origin to the circle $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ are (IIT JEE-1988)

a) $x = 0$

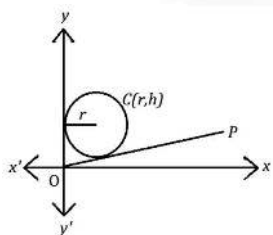
b) $y = 0$

c) $(h^2 - r^2)x - 2rhy = 0$

d) $(h^2 - r^2)x + 2rhy = 0$

Key: ac 9866316227

Sol:



Centre of circle $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ is (r, h) and radius is r . Equation of any line through the origin $(0, 0)$ is $y = mx$... (i)

If (i) is tangent to the circle then the length of perpendicular from the centre (r, h) on (i) equal to the radius of the circle.

$$\therefore \frac{|mr - h|}{\sqrt{1 + m^2}} = \sqrt{r^2 + h^2 - h^2}$$

$$\Rightarrow (mr - h)^2 = (m^2 + 1)r^2 \Rightarrow 0m^2 + (2hr)m + (r^2 - h^2) = 0$$

($\because$ From external point two tangents are drawn, so two slopes should be exist)

$$\Rightarrow m = \infty, \frac{h^2 - r^2}{2hr}$$

Substituting these values in eq. (i), we get tangents as $x = 0$ and $(h^2 - r^2)x - 2rhy = 0$

Duplicate:

2. Find the equation of tangents to circle $x^2 + y^2 - 2x + 4y - 4 = 0$ drawn from point $P(2, 3)$.

a) $y - 3 = \left(\frac{-5 - \sqrt{153}}{8} \right)(x - 2)$

b) $y - 3 = \left(\frac{-5 + \sqrt{153}}{8} \right)(x - 2)$

c) $y - 3 = \left(\frac{5 + \sqrt{153}}{8} \right)(x - 2)$

d) $y - 3 = \left(\frac{5 - \sqrt{153}}{8} \right)(x - 2)$

Key: ab

Sol. Equation of the circle is $x^2 + y^2 - 2x + 4y - 4 = 0$. Centre of the circle is $C(1, -2)$ and radius is $\sqrt{(-1)^2 + 2^2 - (-4)} = 3$. Tangents are drawn to the circle from point $P(2, 3)$. Equation of the line through point P having slope m is

$$y - 3 = m(x - 2)$$

$$\text{or } mx - y - 2m + 3 = 0$$

If the line touches the circle, then the distance from the centre of the circle to this line will be radius of the circle.

$$\therefore \left| \frac{m(1) + 2 - 2m + 3}{\sqrt{m^2 + 1}} \right| = 3$$

$$\Rightarrow (5 - m)^2 = 9(m^2 + 1)$$

$$\Rightarrow 8m^2 + 10m - 16 = 0$$

$$\Rightarrow 4m^2 + 5m - 8 = 0$$

$$\Rightarrow m = \frac{-5 \pm \sqrt{153}}{8}$$

Therefore, required equations of tangents are $y - 3 = \left(\frac{-5 \pm \sqrt{153}}{8} \right) (x - 2)$

3. Let L_1 be a straight line passing through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 and L_2 are equal, then which of the following equations can represent L_1 ? (IIT JEE 1999)

a) $x + y = 0$

b) $y - x = 0$

c) $y + 7x = 0$

d) $x - 7y = 0$

Key: bc

Sol: Let equation of line L_1 be $y = mx$. Intercept made by the circle on L_1 and L_2 are equal if L_1 and L_2 are at the same distance from the centre of the circle.

centre of the given circle is $\left(\frac{1}{2}, -\frac{3}{2} \right)$

$$\left| \frac{\frac{1}{2} - \frac{3}{2} - 1}{\sqrt{1+1}} \right| = \frac{\frac{3m}{2} + \frac{1}{2}}{\sqrt{m^2 + 1}} \Rightarrow \frac{2}{\sqrt{2}} = \frac{3m+1}{2\sqrt{m^2 + 1}}$$

$$\Rightarrow 8(m^2 + 1) = (3m + 1)^2$$

$$\Rightarrow m^2 + 6m - 7 = 0$$

$$\Rightarrow (m + 7)(m - 1) = 0 \Rightarrow m = -7, m = 1$$

Thus, two chords are $y + 7x = 0$, and $y - x = 0$.

Duplicate:

4. Which of the following lines have the intercepts of equal lengths on the circle $x^2 + y^2 - 2x + 4y = 0$?

a) $3x - y = 0$

b) $x + 3y = 0$

c) $x + 3y + 10 = 0$

d) $3x - y - 10 = 0$

Key: abcd

Sol: Chords equidistant from the center are equal.

5. A straight line through the vertex P of a triangle PQR intersects the side QR at the point S and the circumcircle of the triangle PQR at the point T. If S is not the centre of the circumcircle, then (IIT JEE 2008)

a) $\frac{1}{PS} + \frac{1}{ST} < \frac{2}{\sqrt{QS \times SR}}$

b) $\frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \times SR}}$

$$c) \frac{1}{PS} + \frac{1}{ST} < \frac{4}{QR}$$

$$d) \frac{1}{PS} + \frac{1}{ST} > \frac{4}{QR}$$

Key: bd

Sol: we know that $ps.st = qs.sr$

$$\frac{ps + st}{2} > \sqrt{ps.st}$$

(By $AM > GM$. Here)

Inequality does not occur since $ps \neq st$ since S is not centre)

$$ps + st > 2 \cdot \frac{ps.st}{\sqrt{ps.st}}$$

$$\frac{ps}{ps} + \frac{st}{st} > \frac{2}{\sqrt{qs.sr}}$$

$$= \frac{ps + st}{ps.st} > \frac{2}{\sqrt{qs.sr}} (\because ps.st = qs.sr)$$

$$\Rightarrow \frac{1}{ps} + \frac{1}{st} > \frac{2}{\sqrt{qs.sr}}$$

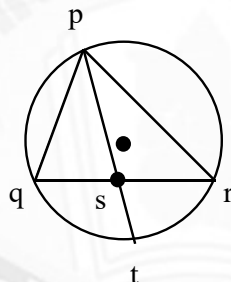
$$\text{Again } pt = ps + st > 2\sqrt{ps.st}$$

$$qr = qs + sr > 2\sqrt{qs.sr}$$

$$\text{Multiply } pt.qr > 4\sqrt{ps.st}\sqrt{qs.sr}$$

$$\Rightarrow pt.qr > 4(ps.st)$$

$$\Rightarrow \frac{(ps + st)}{ps.st} > \frac{4}{qr} \Rightarrow \frac{1}{ps} + \frac{1}{st} > \frac{4}{qr}$$



Duplicate:

6. Triangle ABC is right angled at A. The circle with centre A and radius AB cuts BC and AC internally at D and E respectively. If $BD = 20$ and $DC = 16$ then the length AC equals.

a) $6\sqrt{21}$

b) $6\sqrt{26}$

c) 30

d) 32

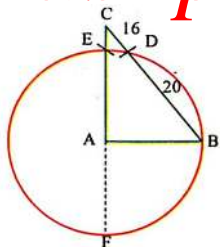
Key: b

$$\text{Sol: } (AC)^2 + r^2 = (36)^2 \dots\dots\dots(1)$$

$$(CE)(CF) = (CD)(BC)$$

$$\Rightarrow (AC - r)(AC + r) = 16 \times 36 \dots\dots\dots(2)$$

On adding (1) and (2), we get



$$\Rightarrow 2(AC)^2 = 36 \times 52$$

$$\Rightarrow AC = 6\sqrt{26}$$

7. Circle(s) touching x-axis at a distance 3 from the origin and having an intercept of length (IIT ADV-2013)

a) $x^2 + y^2 - 6x + 8y + 9 = 0$

b) $x^2 + y^2 - 6x + 7y + 9 = 0$

c) $x^2 + y^2 - 6x - 8y + 9 = 0$

d) $x^2 + y^2 - 6x - 7y + 9 = 0$

Key: ac

Sol. Let the equation be $x^2 + y^2 + 2gx + 2fy + c = 0$ Then $g^2 = c$ (touching x-axis) And

$$2\sqrt{f^2 - c} = 2\sqrt{7}.$$

$$\therefore f^2 - c = 7$$

$$\text{Now, } 9 + 6g + g^2 = 0 \Rightarrow (g+3)^2 = 0.$$

$$\therefore g = -3$$

$$\text{Thus } c = 9 \text{ and } f^2 = 16.$$

$$\therefore f = \pm 4$$

$$\text{Circles are } x^2 + y^2 - 6x + 8y + 9 = 0 \text{ and } x^2 + y^2 - 6x - 8y + 9 = 0$$

Duplicate:

8. The equation of the circles which pass through the origin and makes intercepts of lengths 4 and 8 on the x and y-axis respectively are

a) $x^2 + y^2 + 4x + 8y = 0$

b) $x^2 + y^2 + 4x - 8y = 0$

c) $x^2 + y^2 - 4x + 8y = 0$

d) $x^2 + y^2 - 4x - 8y = 0$

Key: abcd

Sol: Circles passes through origin then $c = 0$

$$2\sqrt{g^2 - c} = 4, \quad 2\sqrt{f^2 - c} = 8$$

$$\Rightarrow \sqrt{g^2 - c} = 2 \Rightarrow \sqrt{f^2 - c} = 4$$

$$\Rightarrow g^2 - 0 = 4 \Rightarrow f^2 - 0 = 16$$

$$\Rightarrow g = \pm 2 \Rightarrow f = \pm 4$$

$$\text{The equation of required circle is } x^2 + y^2 \pm 4x \pm 8y = 0$$

UNCOVERED SUB TOPIC QUESTIONS

9. If $H \equiv (3, 4)$ and $C \equiv (1, 2)$ are orthocenter and circumcentre of ΔPQR and equation of side PQ is $x - y + 7 = 0$, then

a) Equation of circum circle is $(x-1)^2 + (y-2)^2 = 80$

b) Equation of circum circle is $(x-1)^2 + (y-2)^2 = 70$

c) Centroid is $\left(\frac{5}{3}, \frac{8}{3}\right)$

d) Circum radius $= \sqrt{70}$

Key: ac

Sol: Image of $H(3,4)$ in $x - y + 7 = 0$ is $P(-3,10)$. It lies on circum circle. Hence circum radius $R = SP = \sqrt{80}$ where $S = (1,2)$ Also G divides 'HS' in ratio 2:1.

10. Tangents are drawn to the circle $x^2 + y^2 = 50$ from a point 'P' lying on the x-axis. These tangents meet the y-axis at points 'P₁' and 'P₂'. Possible coordinates of 'P' so that area of triangle PP₁P₂ is minimum, are-

- a) (10,0) b) $(10\sqrt{2}, 0)$ c) $(-10, 0)$ d) $(-10\sqrt{2}, 0)$

Key: ac

Sol: Let $P = (h, 0)$. Equation of tangent is $y = mx \pm 5\sqrt{2}\sqrt{1+m^2}$

put $(h, 0) \Rightarrow -mh = \pm 5\sqrt{2}\sqrt{1+m^2}$

SOBS $\Rightarrow m^2 h^2 = 50(1+m^2)$.

$$\Rightarrow m^2 = \frac{50}{h^2 - 50} \Rightarrow m = \frac{5\sqrt{2}}{\sqrt{h^2 - 50}}$$

Area of $\Delta PP_1 O \Delta OPP_2$ is $\frac{c^2}{2|ab|} = \frac{50(1+m^2)}{2|m|} = 25 \left[m + \frac{1}{m} \right]$

it is least for $m = 1 \Rightarrow h = \pm 10$

11. A circle passes through the points $(-1,1), (0,6)$ and $(5,5)$. The point(s) on this circle, the tangents (S) at which is/are parallel to the straight line joining the origin to its centre is/are
- a) $(1, -5)$ b) $(5, 1)$ c) $(-5, -1)$ d) $(-1, 5)$

Key: bd

Sol: Equation of circle will be $x^2 + y^2 - 4x - 6y = 0$.

Centre $= (2, 3)$ $O = (0, 0)$

slope $= 3/2$

$\therefore$ equation of tangent to circle with tangent slope $3/2$ is

$$3x - 2y \pm 13 = 0$$

Their points of contact are $(5, 1)(-1, 5)$

12. Suppose two perpendicular tangents can be drawn from the origin to the circle $x^2 + y^2 - 6x - 2py + 17 = 0$ for some real p, then possible value(s) of p may be.
- a) 0 b) 5 c) -5 d) $\frac{1}{5}$

Key: bc

Sol: $(0, 0)$ lies on director circle $s = r^2 \Rightarrow p^2 = 25$

13. A circle touches the lines $x \pm y = 0$ and passes through $(0, 1)$. Equation of the circle is

a) $x^2 + y^2 - 2\sqrt{2}(\sqrt{2} + 1)y + (3 - 2\sqrt{2}) = 0$

b) $x^2 + y^2 - 2\sqrt{2}(\sqrt{2} - 1)y + (3 + 2\sqrt{2}) = 0$

$$c) x^2 + y^2 - 2\sqrt{2}(\sqrt{2}+1)y + (3+2\sqrt{2}) = 0$$

$$d) x^2 + y^2 - 2\sqrt{2}(\sqrt{2}-1)y + (3-2\sqrt{2}) = 0$$

Key: cd

Sol: $c = (0, k) (k > 0)$, $P = (0, 1)$

$$r = CP = |1 - k|$$

Also $r = d$ (since $x - y = 0$ is tangent)

$$\Rightarrow |1 - k| = \frac{|0 - k|}{\sqrt{2}}$$

$$\Rightarrow 1 - K = \pm \frac{k}{\sqrt{2}} \Rightarrow k$$

$$\Rightarrow \sqrt{2} - \sqrt{2}K = \pm k \Rightarrow K = \sqrt{2}(\sqrt{2} \pm 1)$$

Now find circle equations.

14. Normal to the circle $x^2 + y^2 = 4$ divide the circle centred at $(2, 4)$ and having radius 2 in the area ratio of $(\pi - 2) : (3\pi + 2)$ then normal may be –

a) $y = x$

b) $y = 3x$

c) $y = 5x$

d) $y = 7x$

Key: ad

Sol: Let chord be $y = mx$ make angle θ at centre.

Area of one part of circle is

$$\frac{r^2\theta}{2} - \frac{1}{2}r^2 \sin \theta = 2\theta - 2 \sin \theta \dots\dots\dots(1)$$

Remaining point of circle area is

$$\pi r^2 - (2\theta - 2 \sin \theta)$$

$$= 4\pi - 2\theta + 2 \sin \theta \dots\dots\dots(2)$$

$$\text{Given } \frac{(1)}{(2)} = \frac{\pi - 2}{3\pi + 2}$$

$$\Rightarrow \frac{2\theta - 2 \sin \theta}{4\pi - 2\theta + 2 \sin \theta} = \frac{\pi - 2}{3\pi + 2}$$

$$\text{On simplifying we get } \theta - \sin \theta = \frac{\pi}{2} - 1$$

$$\Rightarrow \theta = \pi / 2$$

Now chord $y = mx$ subtends $\frac{\pi}{2}$ at centre.

By solving we get $m = 1, 7$

15. An isosceles triangle is inscribed in the circle $x^2 + y^2 - 6x - 8y = 0$ with vertex at $(0, 0)$ and one of equal sides along x-axis. Then slope of other side cannot be

a) $\frac{-24}{7}$

b) $\frac{-1}{2}$

c) $\frac{4}{3}$

d) $\frac{12}{5}$

Key: bcd

Sol: Solve circle and x-axis

$$\Rightarrow (0, 0)(6, 0)$$

Length of chord=6

Let slope of other side be m

Equation is $y=mx$

Apply length of chord=6

$$\Rightarrow 2\sqrt{r^2 - d^2} = 6$$

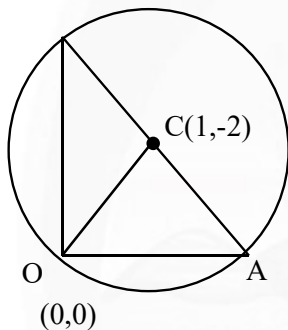
$$\text{It gives } m = 0, \frac{-24}{7}$$

16. Two perpendicular chords are drawn through $(0,0)$ on the circle $x^2 + y^2 - 2x + 4y = 0$. Then slopes of chords are

- a) 3 b) 4 c) $\frac{-1}{3}$ d) $\frac{-1}{4}$

Key: ac

Sol: Let slope of $OA = m$



Slope of $OC = -2$

$\triangle OAC$ becomes right angled isosceles

$$\Rightarrow \text{Angle } AOC = \frac{\pi}{4}$$

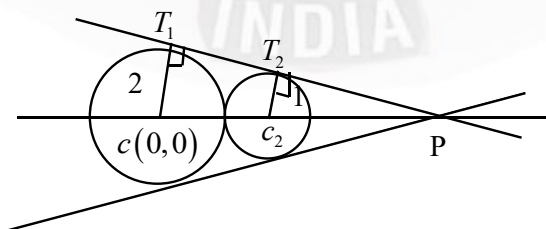
$$\tan \frac{\pi}{4} = \left| \frac{m - (-2)}{1 - 2m} \right|$$

$$\Rightarrow m = 3, \frac{-1}{3}$$

17. A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $p(\sqrt{3}, 1)$ straight line L , perpendicular to PT is a tangent to the circle $(x-3)^2 + y^2 = 1$. The possible equations of L are

- a) $x - \sqrt{3}y = 1$ b) $x + \sqrt{3}y = 1$ c) $x - \sqrt{3}y = -1$ d) $x - \sqrt{3}y = 5$

(IIT ADV 2012)



Key: ad

Sol: Equation of tangent PT to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3}, 1)$ is $x\sqrt{3} + y = 4$

Let the line L, perpendicular to tangent PT be $x - \sqrt{3}y + \lambda = 0$

As it is tangent to the circle $(x-3)^2 + y^2 = 1$

Length of perpendicular from centre of circle to the tangent = Radius of circle.

$$\Rightarrow \left| \frac{3+\lambda}{2} \right| = 1 \Rightarrow \lambda = -1 \text{ or } -5$$

Equation of L can be $x - \sqrt{3}y = 1$ (or) $x - \sqrt{3}y = 5$

Duplicate:

18. Let there be two circles with one touching x-axis at $A(2,0)$ and has radius 1 unit and other touching x-axis at $B(6,0)$. Let C and D be the centres of first and second circles respectively. Again let $P(6,2)$ be a point on a line perpendicular to CD and bisecting AB. If PA and PB intersect circles at E and F respectively, then

a) $\frac{PE}{PF} = \frac{1}{\sqrt{5}}$

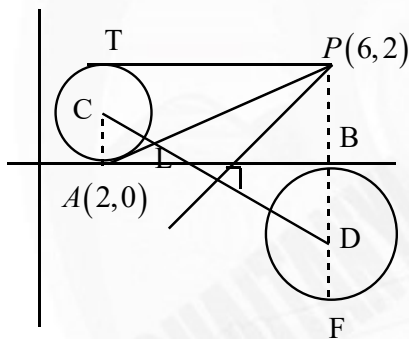
b) $\frac{PE}{PF} = \sqrt{5}$

c) $PE = \frac{8}{\sqrt{5}}$

d) $PE = 8$

Key: ac

Sol:



Clearly P lies on radical axis

$$\Rightarrow PE \cdot PA = PB \cdot PF \Rightarrow \frac{PE}{PF} = \frac{PB}{PA} = \frac{2}{2\sqrt{5}} = \frac{1}{\sqrt{5}}$$

Again $PE \cdot PA = PT^2$

$$\Rightarrow PE = \frac{4^2}{2\sqrt{5}} = \frac{8}{\sqrt{5}}$$

19. A circle 'S' passes through the point $(0,1)$ and is orthogonal to the circle $(x-1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$ then (IIT ADV 2014)

a) Radius of 'S' is 8

b) Radius of 'S' is 7

c) Radius of 'S' is $(-7,1)$

d) Centre of 'S' is $(-8,1)$

Key: bc

Sol: $x^2 + y^2 + 2gx + 2fy + c = 0$

$$x^2 + y^2 = 1 \dots (2)$$

$$x^2 + y^2 - 2x - 15 = 0 \dots (3)$$

Passing through $(0,1) \Rightarrow 1 + 2f + c = 0$

(1)&(2) cut of orthogonally $\Rightarrow c = 1$

$$2f = 15 - c$$

$$\text{Centre} = (-g, -f) = (-7, 1)$$

$$\text{Radius} = \sqrt{g^2 - f^2 - c} = 7$$

Duplicate:

20. A circle 'S' passing through the point (1,1) and cuts orthogonally to the circles

$$x^2 + y^2 - 8x - 2y + 16 = 0 \text{ and } x^2 + y^2 - 4x - 4y - 1 = 0$$

a) radius of 's' is $\sqrt{\frac{905}{6}}$

b) radius of 's' is $\sqrt{\frac{905}{6}}$

c) Radius of 's' is $\left(\frac{7}{3}, \frac{-23}{6}\right)$

d) centre 's' is $\left(\frac{-7}{3}, \frac{23}{6}\right)$

Key: ac

Sol: Let S is $x^2 + y^2 + 2gx + 2fy + c = 0$(1)

$$x^2 + y^2 - 8x - 2y + 16 = 0$$
.....(2)

$$x^2 + y^2 - 4x - 4y - 1 = 0$$
.....(3)

Passing through (1,1) $\Rightarrow 1 + 1 + 2g + 2f + c = 0$

$$2g + 2f + c = -2$$
.....(4)

$$(1) \& (2) \text{ cut orthogonally } \Rightarrow 8g + 2f + c = -16$$
.....(5)

$$(1) \& (3) \text{ cut orthogonally } \Rightarrow 4g + 4f + c = 1$$
.....(6)

$$\text{From (4) \& (5) } g = \frac{-7}{3}$$

$$\text{From (4) \& (6) } 2g + 2f = 3$$

$$2f = 3 - 2g = 3 + \frac{14}{3} = \frac{23}{3}$$

$$f = \frac{23}{6}$$

$$\text{From (1) } c = -2 - 2g - 2f$$

$$= -2 + \frac{14}{3} - \frac{23}{3}$$

$$= \frac{-6 + 14 - 23}{3}$$

$$= \frac{15}{3} = -5$$

$$\text{Centre} = \left(\frac{7}{3}, \frac{-23}{6}\right)$$

$$r = \sqrt{\frac{905}{36}}$$

21. Two circles each of radius 5 units touch each other at (1,2). If the equation of their common tangent is $4x + 3y = 10$. Find the equations of the circles. (IIT ADV 1991)

a) $x^2 + y^2 - 10x - 10y + 25 = 0$

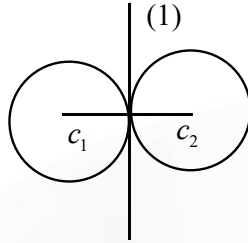
b) $x^2 + y^2 + 6x + 2y - 15 = 0$

c) $x^2 + y^2 + 10x + 10y - 25 = 0$

d) $x^2 + y^2 - 6x - 2y + 15 = 0$

Key: ab

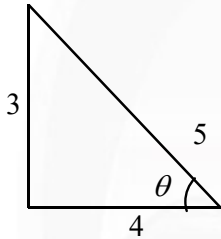
Sol:



$P(1,2)$ line is $4x + 3y - 10 = 0$(1)

$$R=5 \text{ slope of (1)} = \frac{-4}{3}$$

$$\text{Slope } c_1 c_2 = \frac{3}{4}$$



$$\sin \theta = \frac{3}{5} \cos \theta = \frac{4}{5}$$

$$c_1 c_2 \text{ are } = (x_1 \pm r \cos \theta, y_1 \pm r \sin \theta)$$

$$= \left(1 \pm 5 \left(\frac{4}{5} \right), 2 \pm 5 \left(\frac{3}{5} \right) \right)$$

$$(1 \pm 4, 2 \pm 3)$$

$$\text{Circles are } (x-5)^2 + (y-5)^2 = 25$$

$$(x+3)^2 + (y+1)^2 = 25$$

Duplicate:

22. Two circles each of radius $\sqrt{2}$ units touch each other at $(2, -1)$. If the equation of their common tangent is $x + y = 1$. Find the equations of the circles.

a) $x^2 + y^2 - 4x + 2y + 3 = 0$

b) $x^2 + y^2 - 6x + 7 = 0$

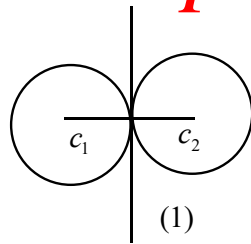
c) $x^2 + y^2 - 2x + 4y + 3 = 0$

d) $x^2 + y^2 - 6y + 7 = 0$

Key: bc

Sol: $P = (2, -1)$ given line is $x + y = 1$(1)

$$\text{Slope (1)} = -1 \quad r = \sqrt{2}$$



Slope $c_1c_2 = 1 \Rightarrow \tan \theta = 1$

$$\theta = \frac{\pi}{4}$$

$$c_1c_2 = (x_1 \pm r \cos \theta, y_1 \pm r \sin \theta)$$

$$= \left(2 \pm \sqrt{2} \frac{1}{2}, -1 \pm \sqrt{2} \frac{1}{\sqrt{2}} \right)$$

$$= (2 \pm 1, -1 \pm 1)$$

$$= (3, 0)(1, -2)$$

$$\text{Circles are } (x-3)^2 + (y-0)^2 = 2 \Rightarrow x^2 + y^2 - 6x + 7 = 0$$

$$(x-1)^2 + (y+2)^2 = 2 \Rightarrow x^2 + y^2 - 2x + 4y + 3 = 0$$

23. Three circles touch one another externally the tangents at their points of contact meet at a point whose distance from a point of contact is '4' the ratio of the product of the radii to the sum of the radii of the circles cannot be (IIT ADV 1992)

a) 17

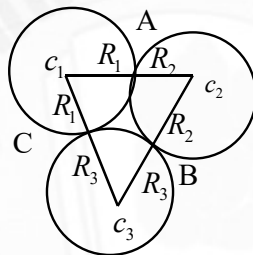
b) 18

c) 16

d) 15

Key: abd

Sol:



Suppose the circles have centres at c_1, c_2 & c_3 with radius R_1, R_2, R_3 respectively. Let the circles touch at A, B, C. Let the common tangent at A, B, C meet at 'O' we have $OA = OB = OC = 4$ now the circle with centre at 'O' and passing through A, B, C is the incircle of the triangle $c_1c_2c_3$

In radius of $\Delta c_1c_2c_3$ is 4

$$r = \frac{\Delta}{S}$$

Perimeter of a triangle

$$2s = 2(R_1 + R_2 + R_3)$$

$$S = R_1 + R_2 + R_3$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{(R_1 + R_2 + R_3)R_1R_2R_3}$$

$$r = \frac{\Delta}{s} \Rightarrow 4 = \frac{\sqrt{(R_1 + R_2 + R_3) R_1 R_2 R_3}}{R_1 + R_2 + R_3}$$

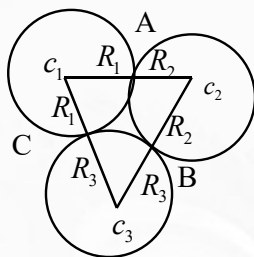
$$\Rightarrow 16 = \frac{(R_1 + R_2 + R_3) R_1 R_2 R_3}{(R_1 + R_2 + R_3) 2}$$

Duplicate:

24. Three circles touch one another externally the tangents at their points of contact meet at a point whose distance from a point of contact is '6' the ratio of the product of the radii to the sum of the radii of the circles cannot be _____
- a)39 b)36 c)6 d)63

Key:acd

Sol:



Suppose the circles have centres at c_1, c_2 and c_3 with radius R_1, R_2, R_3 respectively.

Let the circles touch at A, B, C. Let the common tangents at A, B, C meet at 'O' we have $OA = OB = OC = 6$. Now the circle with centre at 'O' and passing through A, B, C is the incircle of triangle c_1, c_2, c_3

In circle of c_1, c_2, c_3 is 6

$$\text{And } r = \frac{\Delta}{s}$$

$$\text{Perimeter of } \Delta = 2s = 2(R_1 + R_2 + R_3)$$

$$S = R_1 + R_2 + R_3$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{(R_1 + R_2 + R_3) R_1 R_2 R_3}$$

$$= \sqrt{\frac{(R_1 + R_2 + R_3) R_1 R_2 R_3}{R_1 + R_2 + R_3}}$$

$$36 = \frac{(R_1 + R_2 + R_3) R_1 R_2 R_3}{(R_1 + R_2 + R_3)}$$

$$36 = \frac{R_1 + R_2 + R_3}{R_1 + R_2 + R_3}$$

25. Lines $5x + 12y - 10 = 0$ and $5x - 12y - 40 = 0$ touch a circle ' c_1 ' lies in the fourth quadrant. then the centre and radius of the circle ' c_2 ' which is concentric with ' c_1 ' and cuts intercepts of length 8 on these lines
- (IIT ADV 1986)

- a) centre = $\left(5, \frac{-9}{2}\right)$ b) centre = $(5, -9)$ c) radius = 5 d) radius = 25

Key: ac

Sol: Let 'c' be the centre of the circle $c_1 = (a, b)$

$$2r = 6 \Rightarrow r = 3$$

$$\left| \frac{5a+12b-10}{13} \right| = \left| \frac{5a-12b-40}{13} \right| = 3$$

$$5a+12b-10 = \pm 39, 5a-12b-40 = \pm 39$$

$$5a+12b = -29 \dots (1), 5a-12b = 79 \dots (2)$$

$$\text{Solve (1)\&(2) } a=5, b=\frac{-9}{2} \Rightarrow \text{centre} = \left(5, \frac{-9}{2}\right)$$

$$r = \sqrt{3^2 + 4^2} = 5$$

Duplicate:

26. Lines $5x+12y-10=0$ and $5x-12y-40=0$ touch a circle ' c_1 ' of diameter '6'. If the centre of ' c_1 ' lies in the first quadrant. Then the centre and radius of the circle ' c_2 ' which is concentric with ' c_1 ' and cuts intercepts of length '8' on these lines.

- a) centre = $(5, 2)$ b) centre = $(2, 5)$ c) radius = 5 d) radius = 25

Key: c

Sol: Let 'c' be the centre of the circle $c_1 = (a, b)$

$$2r = 6 \Rightarrow r = 3$$

' c_1 ' touches the given lines

$$\left| \frac{5a+12b-10}{13} \right| = \left| \frac{5a-12b-40}{13} \right| = 3$$

$$5a+12b-10 = \pm 39, 5a-12b-40 = \pm 39$$

$$5a+12b = 49 \dots (1), 5a-12b = 1 \dots (2)$$

$$\text{Solve (1)\&(2) } a=5, b=2 \Rightarrow \text{centre} = (5, 2)$$

$$r = \sqrt{3^2 + 4^2} = 5$$

27. The equations of the circles passing through $(-4, 3)$ and touching the lines

$$x+y=2 \text{ and } x-y=2$$

(IIT ADV 1982)

$$\text{a) } x^2 + y^2 + 2(10+3\sqrt{6})x + (55+24\sqrt{6}) = 0 \quad \text{b) } x^2 + y^2 + 2(10-3\sqrt{6})x + (55-24\sqrt{6}) = 0$$

$$\text{c) } x^2 + y^2 + (10+3\sqrt{6})x + (55+\sqrt{6}) = 0 \quad \text{d) } x^2 + y^2 + (10+3\sqrt{6})y + (55+\sqrt{6}) = 0$$

Key: ab

Sol: Let the required equation of circle be $x^2 + y^2 + 2gx + 2fy + c = 0 \dots (1)$

$$\text{It passes through } (-4, 3) \quad 25 - 8g + 6f + c = 0 \dots (2)$$

Since the circle touches two circles we get $f = 0$ or $g = -2$

$$\text{When } f = 0 \text{ we get } g = 10 \pm 3\sqrt{6} \text{ and } c = 55 \pm 24\sqrt{6}$$

Req eq of circles are

$$x^2 + y^2 + 2(10 \pm 3\sqrt{6})x + (55 \pm 24\sqrt{6}) = 0$$

Duplicate:

28. $y-1=M_1(x-3)$ and $y-3=M_2(x-1)$ are two family of straight lines, at right angles to each other. The locus of their point of intersection is $x^2 + y^2 - \alpha x - \alpha y + \beta = 0$.

a) $\alpha = 4$

b) $\beta = 6$

c) $\alpha = -4$

d) $\beta = -6$

Key: ab

Sol: $m_1 m_2 = -1$

Equation of circle is $(y-1)(y-3) + (x-1)(x-3) = 0$

29. ABCD is a square of side length 2 units, C_1 is the circle touching all the sides of the square ABCD and C_2 is the circumcircle of square ABCD, L is a fixed line in the same plane and it is a fixed point. If P is any point of C_1 and Q is another point on C_2 then

$$\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2} \text{ is not equal to } \underline{\hspace{2cm}}$$

(IIT ADV 2006)

a) 0.75

b) 1.25

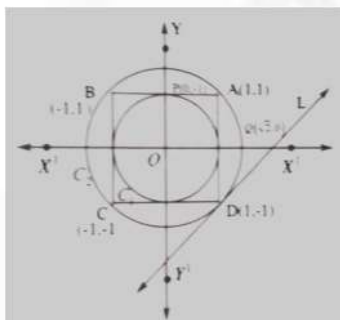
c) 1

d) 0.5

Key: bcd

Sol: Without loss of generality we can assume the square ABCD with its vertices $A(1,1), B(-1,1), C(-1,-1), D(1,-1)$

P to be the point $(0,1)$ and Q as $(\sqrt{2}, 0)$



Then $\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2}$

$$= \frac{1+1+5+5}{2[(\sqrt{2}-1)^2 + 1] + 2[(\sqrt{2}+1)^2 + 1]} = \frac{12}{16} = 0.75$$

Duplicate:

30. ABCD is a square of side length 2 units, C_1 is the circle touching all the sides of the square ABCD and C_2 is the circumcircle of square ABCD, L is a fixed line in the same plane and it is a fixed point. If P is any point of C_1 and Q is another point on C_2 then

$$\frac{QA^2 + QB^2 + QC^2 + QD^2}{PA^2 + PB^2 + PC^2 + PD^2} \text{ is not equal to } \underline{\hspace{2cm}}$$

a) $1.\bar{3}$

b) $1.\bar{4}$

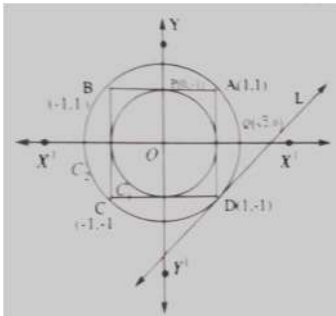
c) $1.\bar{5}$

d) $1.\bar{6}$

Key: bcd

Sol: Without loss of generality we can assume the square ABCD with its vertices $A(1,1), B(-1,1), C(-1,-1), D(1,-1)$

P to be the point $(0,1)$ and Q as $(\sqrt{2},0)$



$$\text{Then } \frac{QA^2 + QB^2 + QC^2 + QD^2}{PA^2 + PB^2 + PC^2 + PD^2} = \frac{16}{12} = 1.\bar{3}$$

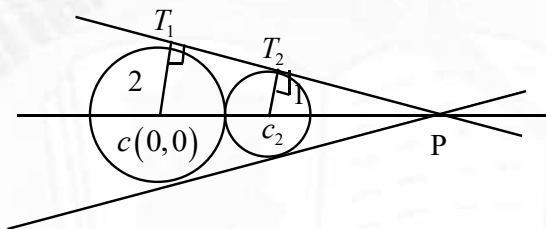
31. A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3},1)$. A straight line L, perpendicular to PT is a tangent to the circle $(x-3)^2 + y^2 = 1$. A common tangent to the two circles is

(IIT ADV 2012)

- a) $x + 2\sqrt{2}y = 6$ b) $x - 2\sqrt{2}y = 6$ c) $x = 4$ d) $x - \sqrt{3}y = 4$

Key: ab

Sol:



From diagram $PC_1 : PC_2 = 2 : 1$

Or P divides c_1c_2 in the ratio 2:1 externally

Co ordinates of P are $(6,0)$

Let the equations of tangent through P be $y = m(x-6)$

As it touches $x^2 + y^2 = 4$

$$\left| \frac{6m}{\sqrt{m^2 - 1}} \right| = 2 \Rightarrow m = \pm \frac{1}{2\sqrt{2}}$$

Equation of common tangents as $y = \pm \frac{1}{2\sqrt{2}}(x-6)$

Duplicate:

32. The line $x + 2y + a = 0$ intersects the circle $x^2 + y^2 = 4$ at two distinct points A and B.

Another line $12x - 6y - 41 = 0$ intersects the circle $x^2 + y^2 - 4x - 2y + 1 = 0$ at two distinct points C and D. Then which of the following are true.

a) The value of a for which A, B, C, D are concyclic is 2

b) If ABCD are concyclic then its centre is $\left(\frac{4}{8}, \frac{8}{3}\right)$

c) If $C_1 = 0$ is the circle on which A, B, C, D lie then the locus of points of intersection of perpendicular tangents to $C_1 = 0$ is a circle whose radius is $\frac{6\sqrt{7}}{5}$

d) There is a chord of maximum length, of the circle passing through A, B, C, D lies on the line $3x + y - 4 = 0$

Key: abd

Sol: the circle through A, B is of the form $x^2 + y^2 - 4 + \lambda_1(x + 2y + a) = 0$ and a circle through C, D is of the form $x^2 + y^2 - 4x - 2y + 1 + \lambda_2(12x - 6y - 41) = 0$

If A, B, C, D lie on same circle then the two equations must be same which gives $a = 2$ and equation of circle as $5x^2 + 5y^2 - 8x - 16y - 36 = 0$

33. Let G be a circle of radius $R > 0$. Let $G_1, G_2, \dots, G_n$ be n circles of equal radius $r > 0$. Suppose each of the n circles $G_1, G_2, \dots, G_n$ touches the circle G externally. Also, for $i = 1, 2, \dots, n-1$, the circle G_i touches G_{i+1} externally, and G_n touches G_1 externally. Then, which of the following statements is/are true? (IIT ADV 2022)

a) If $n = 4$, then $(\sqrt{2} - 1)r < R$

b) If $n = 5$, then $r < R$

c) If $n = 8$, then $(\sqrt{2} - 1)r < R$

d) If $n = 12$, then $\sqrt{2}(\sqrt{3} + 1)r > R$

Key: cd

Sol: We have $\sin \frac{\pi}{n} = \frac{r}{R+r}$

$$\rightarrow 1 + \frac{R}{r} = \operatorname{cosec} \frac{\pi}{n}$$

$$\text{Thus, } \frac{r}{R} = \frac{1}{\operatorname{cosec} \frac{\pi}{n} - 1}$$

$$\text{For } n=4; \frac{r}{R} = \frac{1}{\sqrt{2}-1}$$

$$\text{For } n=5, \frac{r}{R} = \frac{1}{\operatorname{cosec} \frac{\pi}{5} - 1} > \frac{1}{\operatorname{cosec} \frac{\pi}{6} - 1}$$

$$\Rightarrow r > R$$

For $n=8$

$$\frac{r}{R} = \frac{1}{\operatorname{cosec} \frac{\pi}{8} - 1} < \frac{1}{\operatorname{cosec} \frac{\pi}{4} - 1} \Rightarrow r < \frac{R}{\sqrt{2} - 1}$$

For $n=12$

$$\begin{aligned} \frac{r}{R} &= \frac{1}{\operatorname{cosec} \frac{\pi}{12} - 1} = \frac{1}{\frac{2\sqrt{2}}{\sqrt{3}-1} - 1} \\ &= \frac{1}{\sqrt{2}(\sqrt{3}+1) - 1} > \frac{1}{\sqrt{2}(\sqrt{3}+1)} \\ &\Rightarrow \sqrt{2}(\sqrt{3}+1)r > R \end{aligned}$$

33a. Let G be a circle of radius $R > 0$. Let $G_1, G_2, \dots, G_n$ be n circles of equal radius $r > 0$. Suppose each of the n circles $G_1, G_2, \dots, G_n$ touches the circle G externally. Also, for $i = 1, 2, \dots, n-1$, the circle G_i touches G_{i+1} externally, and G_n touches G_1 externally. Then, which of the following statements is/are true?

a) If $n=4$, then $(\sqrt{2}-1)r = R$

b) If $n=6$, then $r = R$

c) If $n=8$, then $(\sqrt{2}-1)r > R$

d) If $n=12$, then $\sqrt{2}(\sqrt{3}+1)r < R$

Key: ab

Sol: We have $\sin \frac{\pi}{n} = \frac{r}{R+r}$

$$\rightarrow 1 + \frac{R}{r} = \operatorname{cosec} \frac{\pi}{n}$$

$$\text{Thus, } \frac{r}{R} = \frac{1}{\operatorname{cosec} \frac{\pi}{n} - 1}$$

For $n=4$

$$\frac{r}{R} = \frac{1}{\sqrt{2} - 1}$$

$$\text{For } n=5, \frac{r}{R} = \frac{1}{\operatorname{cosec} \frac{\pi}{5} - 1} > \frac{1}{\operatorname{cosec} \frac{\pi}{6} - 1}$$

$$\Rightarrow r > R$$

For $n=8$

$$\frac{r}{R} = \frac{1}{\operatorname{cosec} \frac{\pi}{8} - 1} < \frac{1}{\operatorname{cosec} \frac{\pi}{4} - 1} \Rightarrow r < \frac{R}{\sqrt{2} - 1}$$

For $n=12$

$$\frac{r}{R} = \frac{1}{\operatorname{cosec} \frac{\pi}{12} - 1} = \frac{1}{\frac{2\sqrt{2}}{\sqrt{3}-1} - 1}$$

$$= \frac{1}{\sqrt{2}(\sqrt{3}+1)-1} > \frac{1}{\sqrt{2}(\sqrt{3}+1)}$$

$$\Rightarrow \sqrt{2}(\sqrt{3}+1)r > R$$

PASSAGE TYPE QUESTIONS

PASSAGE - 01

Let $M = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 \leq r^2\}$ where $r > 0$.

Consider the geometric progression $a_n = \frac{1}{2^{n-1}}, n = 1, 2, 3, \dots$. Let

$S_0 = 0$ and, for $n \geq 1$, let S_n denote the sum of the first n terms of this progression. For $n \geq 1$, let

C_n denote the circle with center $(S_{n-1}, 0)$ and radius a_n , and D_n denote the circle with center (S_{n-1}, S_{n-1}) and radius a_n . [Adv. 2021]

1. Consider M with $r = \frac{1025}{513}$. Let k be the number of all those circles C_n that are inside M .

Let l be the maximum possible number of circles among these k circles such that no two circles intersect. Then

- a) $k + 2l = 22$ b) $2k + l = 26$ c) $2k + 3l = 34$ d) $3k + 2l = 40$

Key: d

Sol: $\because a_n = \frac{1}{2^{n-1}}$

$$S_n = 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}}$$

$$= 2 \left(1 - \frac{1}{2^n} \right) = 2 - \frac{1}{2^{n-1}}$$

For circles C_n to inside M

$$S_{n-1} + a_n < \frac{1025}{513}$$

$$\Rightarrow 2 - \frac{1}{2^{n-2}} + \frac{1}{2^{n-1}} < \frac{1025}{513}$$

$$\Rightarrow 1 - \frac{1}{2^n} < \frac{1025}{1026} = 1 - \frac{1}{1026}$$

$$\Rightarrow 2^n < 1026 \Rightarrow n \leq 10 \Rightarrow k = 10$$

Also $l = 5$

$$3k + 2l = 30 + 10 = 40$$

2. Consider M with $r = \frac{(2^{199} - 1)\sqrt{2}}{2^{198}}$. The number of all those circles D_n that are inside M is

- a) 198 b) 199 c) 200 d) 201

Key: b

Sol: $\because r = \frac{(2^{199} - 1)\sqrt{2}}{2^{198}}$

$$\text{Now, } \sqrt{2}S_{n-1} + a_n < \frac{2^{199}-1}{2^{198}}\sqrt{2}$$

$$\Rightarrow 2\sqrt{2}\left(1 - \frac{1}{2^{n-1}}\right) + \frac{1}{2^{n-1}} < \frac{2^{199}-1}{2^{198}}$$

$$\Rightarrow \frac{2\sqrt{2}-1}{2 \cdot 2^{n-2}} > \frac{\sqrt{2}}{2^{198}}$$

$$\Rightarrow 2^{n-2} < \left(2 - \frac{1}{\sqrt{2}}\right)2^{197}$$

$$\therefore n \leq 199 \Rightarrow n = 199$$

PASSAGE - 02

Let S be the circle in the xy-plane defined by the equation $x^2 + y^2 = 4$. (Adv. 2018)

3. Let E_1E_2 and F_1F_2 be the chords of S passing through the point $P_0(1,1)$ and parallel to the x-axis and the y-axis, respectively. Let G_1G_2 be the chord of S passing through P_0 and having slope -1. Let the tangents to S at E_1 and E_2 meet at E_3 , the tangents to S at F_1 and F_2 meet at F_3 , and the tangents to S at G_1 and G_2 meet at G_3 . Then, the points E_3, F_3 , and G_3 lie on the curve

a) $x + y = 4$

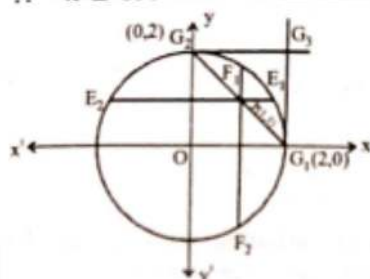
b) $(x-4)^2 + (y-4)^2 = 16$

c) $(x-4)(y-4) = 4$

d) $xy = 4$

Key: a

Sol:



Equation of E_1E_2 is $y = 1$

Equation of F_1F_2 is $x = 1$

Equation of G_1G_2 is $x + y = 2$

By symmetry, tangents at E_1 and E_2 will meet on y-axis and tangents at F_1 and F_2 will meet on x-axis

$$E_1 = (\sqrt{3}, 1) \text{ and } F_1 = (1, \sqrt{3})$$

Equation of tangent at E_1 is $\sqrt{3}x + y = 4$

Equation of tangent at F_1 is $x + \sqrt{3}y = 4$

$$\therefore \text{Points } E_3(0, 4) \text{ and } F_3(4, 0)$$

Tangents at G_1 and G_2 are $x = 2$ and $y = 2$ respectively intersecting each other at $G_3(2, 2)$.

Clearly E_3, F_3 and G_3 lie on the curve $x + y = 4$.

4. Let P be a point on the circle S with both coordinates being positive. Let the tangent to S at P intersect the coordinate axes at the points M and N. Then, the mid-point of the line segment MN must lie on the curve

a) $(x + y)^2 = 3xy$

b) $x^{2/3} + y^{2/3} = 2^{4/3}$

c) $x^2 + y^2 = 2xy$

d) $x^2 + y^2 = x^2y^2$

Key: d

Sol: Let point P be $(2\cos\theta, 2\sin\theta)$

Tangent at P is $x\cos\theta + y\sin\theta = 2$

$\therefore M\left(\frac{2}{\cos\theta}, 0\right)$ and $N\left(0, \frac{2}{\sin\theta}\right)$

$\therefore$ Mid point of MN $= \left(\frac{1}{\cos\theta}, \frac{1}{\sin\theta}\right)$

For locus of mid point (x, y) of MN,

$x = \frac{1}{\cos\theta}, y = \frac{1}{\sin\theta}$

$\Rightarrow \frac{1}{x^2} + \frac{1}{y^2} = 1 \Rightarrow x^2 + y^2 = x^2y^2$

PASSAGE-03

5. A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3}, 1)$. A straight line L, perpendicular to PT is a tangent to the circle $(x-3)^2 + y^2 = 1$. (Adv.2012)

A possible equation of L is

a) $x - \sqrt{3}y = 1$

b) $x + \sqrt{3}y = 1$

c) $x - \sqrt{3}y = -1$

d) $x + \sqrt{3}y = 5$

Key: a

Sol: Equation of tangent PT to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3}, 1)$ is $x\sqrt{3} + y = 4$

Let the line L, perpendicular to tangent PT be $x - y\sqrt{3} + \lambda = 0$

As it is tangent to the circle $(x-3)^2 + y^2 = 1$

$\therefore$ Length of perpendicular from centre of circle to the Tangent = radius of circle .

$\Rightarrow \left| \frac{3 + \lambda}{2} \right| = 1 \Rightarrow \lambda = -1 \text{ or } -5$

$\therefore$ Equation of L can be $x - \sqrt{3}y = 1$ or $x - \sqrt{3}y = 5$

6. A common tangent of the two circles is

a) $x = 4$

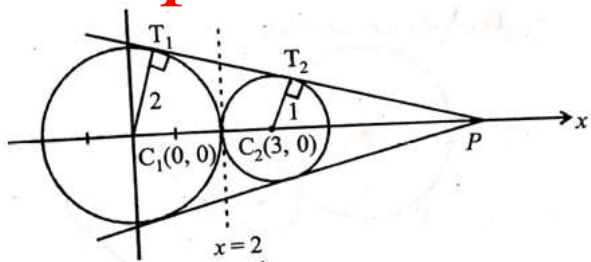
b) $y = 2$

c) $x + \sqrt{3}y = 4$

d) $x + 2\sqrt{2}y = 6$

Key: d

Sol:



From the figure it is clear that the intersection point of two direct common tangents lies on x-axis.

Also $\Delta PT_1C_1 \sim \Delta PT_2C_2$

$$\therefore PC_1 : PC_2 = 2 : 1$$

or P divides C_1C_2 in the ratio 2:1 externally

$\therefore$ Coordinates of P are (6,0)

Let the equation of tangent through P be

$$y = m(x - 6)$$

As it touches $x^2 + y^2 = 4$

$$\therefore \left| \frac{6m}{\sqrt{m^2 + 1}} \right| = 2 \Rightarrow 36m^2 = 4(m^2 + 1)$$

$$\therefore m = \pm \frac{1}{2\sqrt{2}}$$

PASSAGE-04

ABCD is a square of side length 2 units. C_1 is the circle touching all the sides of the square ABCD and C_2 is the circumcircle of square ABCD. L is a fixed line in the same plane.

(Adv.2006)

7. If P is any point of C_1 and Q is another point on C_2 , then

$\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2}$ is equal to

a) 0.75

b) 1.25

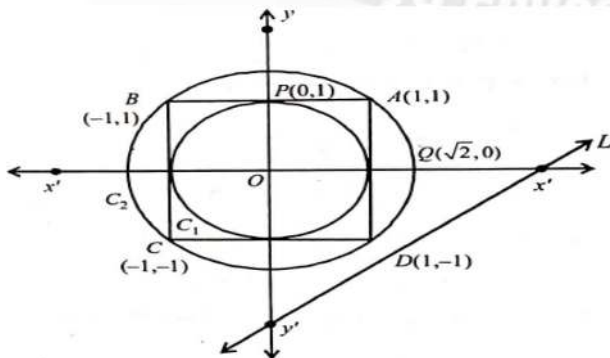
c) 1

d) 0.5

Key: a

Sol: According to the given question, we can assume the square ABCD with its vertices

$A(1,1), B(-1,1), C(-1,-1), D(1,-1)$. P be the point (0,1) and Q be the point $(\sqrt{2}, 0)$.



$$\begin{aligned} \text{Then, } & \frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2} \\ &= \frac{1+1+5+5}{2[(\sqrt{2}-1)^2+1] + 2[(\sqrt{2}+1)^2+1]} = \frac{12}{16} = 0.75 \end{aligned}$$

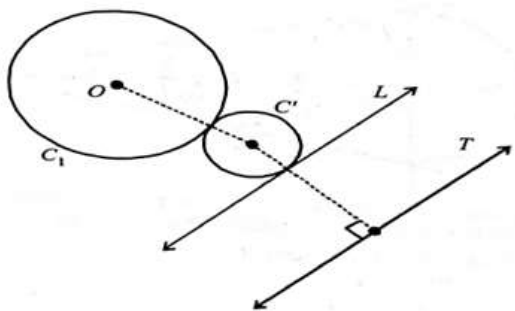
8. If a circle is such that it touches the line L and the circle C_1 externally, such that both the circles are on the same side of the line, then the locus of centre of the circle is
 a) ellipse b) hyperbola c) parabola
 d) pair of straight line

Key: c

Sol: Let C' be the said circle touching C_1 and L, so that C_1 and C' are on the same side of L.

Let us draw a line T on opposite side of L. Then the centre of C' and from line T.

$\Rightarrow$ Locus of centre of C' is a parabola



9. A line L' through A is drawn parallel to BD. Point S moves such that its distances from the line BD and the vertex A are equal. If locus of S cuts L' at T_2 and T_3 and AC at T_1 , then area of $\Delta T_1 T_2 T_3$ is

- a) $\frac{1}{2}$ sq. units b) $\frac{2}{3}$ sq. units c) 1sq. Units d) 2 sq. units

Key: c

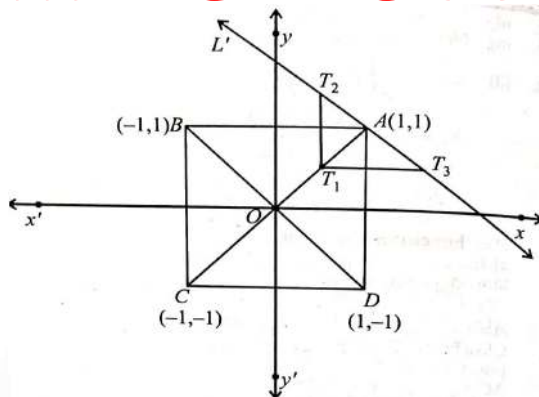
Sol: Since S is equidistant from A and line BD, it traces a parabola. Clearly, AC is the axis, $A(1,1)$ is the focus and

$T_1\left(\frac{1}{2}, \frac{1}{2}\right)$ is the vertex of the parabola.

$$AT_1 = \frac{1}{\sqrt{2}}.$$

$T_2 T_3$ = latus rectum of parabola

$$= 4 \times \frac{1}{\sqrt{2}} = 2\sqrt{2}$$



$$\therefore \text{Area } (\Delta T_1 T_2 T_3) = \frac{1}{2} \times \frac{1}{\sqrt{2}} \times 2\sqrt{2} = 1 \text{ sq. units}$$

PASSAGE-05

For each positive real number 'k', let C_k denotes the circle with centre at origin and radius 'k' unit. On a circle C_k , a particle moves 'k' unit in the counter-clockwise direction. After completing its motion on C_k , the particle moves onto the circle C_{k+r} in some well defined manner, $r > 0$. The motion of the particle continues in this manner.

10. Let k be a positive integer and $r=1$ and the particle moves in the radial direction from C_k to C_{k+1} . If particle starts from the point $(-1,0)$, then :
- it will cross X-axis again at $(4,0)$
 - it will cross X-axis again at $(6,0)$
 - it will cross the positive Y-axis at $(0,5)$
 - it will cross the positive Y-axis at $(0,7)$

Key:a

Sol: On a circle the particle travels 1 radian angular displacement. To cross the X-axis again it should cover π radian. As $\pi < 4$, particle will be travelling on C_4 when it crosses X-axis again. Thus the point is $(4,0)$.

11. If k be a positive integer and $r \in \mathbf{R}$ and particles moves tangentially from C_k to C_{k+r} such that the length of tangent is equal to 'k' unit itself. If particle starts from the point $(2,0)$, then :
- it will cross the positive X-axis again for $2\sqrt{2} < x < 4$
 - it will cross the positive X-axis again at $x = 3$
 - it will cross the positive x-axis again for $y < x < 4\sqrt{2}$
 - it will cross the positive x-axis again at $x = 4\sqrt{2}$

Key:d

Sol: The particle will travel an angular displacement of $\left(\frac{\pi}{4} + 1\right)$ radian while moving from one circle to another. It will cross the positive X-axis again when it travels 2π radian angular

displacement. We have $\left[\frac{2\pi}{\frac{\pi}{4} + 1} \right] = 3$, where $[x]$ represents the integral part of x. Thus, the

particle will be on the fourth circle while crossing the positive X-axis again. The radius of the fourth circle will be given by $r = 2(\sqrt{2})^3 = 4\sqrt{2}$.

Hence the particle crosses the positive X-axis at $(4\sqrt{2}, 0)$.

12. Let k be a positive integer and $r \in R$ and the particle moves tangentially from C_k to C_{k+r} such that the length of tangent is equal to k unit itself. If particle starts from the point $(3, 0)$, then the number of circles of integral radii it travels upon before passing the positive x-axis again is
- a) 1 b) 2 c) 3 d) 4

Key: b

Sol: The particle moves an angular displacement of $\left(\frac{\pi}{4} + 1\right)$ radian and starts at C_3 . As obtained previously, it will be travelling on the fourth circle when it crosses the positive X-axis again. The radius of these four circles are given by $3(\sqrt{2})^r$, where $r = 0, 1, 2, 3$. Thus there are two circles of integral radii.

PASSAGE-06

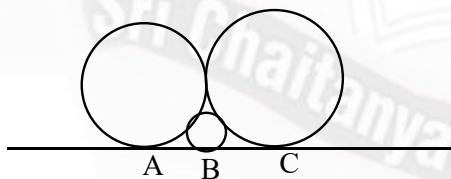
When two circles neither intersect nor touch and are far apart, then four common tangents can be drawn to two circles, two of which are direct and other two are transverse. Let the distance between the centers of the circles be 'd' and their radii are r_1 and r_2 . The lengths of the direct common tangents and the transverse common tangents are given by

$$L_D = \sqrt{d^2 - (r_1 - r_2)^2} \text{ and } L_T = \sqrt{d^2 - (r_1 + r_2)^2} \text{ respectively.}$$

13. Two circles of radii a and b ($a > b$) touch each other externally. The radius of a circle, which touches both the circles externally and also their common tangents, is equal to :
- a) $\frac{a+b}{2}$ b) $\frac{a^2+b^2}{a-b}$ c) $\frac{ab}{(\sqrt{a}+\sqrt{b})^2}$ d) $\frac{ab}{a+b}$

Key: c

Sol: Let the desired radius be 'r', then



$$AB = \sqrt{(r+a)^2 - (r-a)^2} = 2\sqrt{ra}$$

$$BC = \sqrt{(r+b)^2 - (r-b)^2} = 2\sqrt{rb}$$

$$AC = \sqrt{(a+b)^2 - (a-b)^2} = 2\sqrt{ab}$$

Now, $AB + BC = AC$

$$\Rightarrow r = \frac{ab}{(\sqrt{a} + \sqrt{b})^2}$$

14. Two circles have radii equal to 5 and 3. If the distance between their centres is twice the sum of their radii, then the length of the transverse common tangent of the circles is equal to:

a) $\sqrt{192}$ b) $\sqrt{128}$ c) $\sqrt{217}$ d) $\sqrt{60}$

Key: a

Sol: The distance between the centers, $d = 2(r_1 + r_2)$

$\therefore$ Length of the transverse common tangent

$$= \sqrt{d^2 - (r_1 + r_2)^2} = \sqrt{3(r_1 + r_2)^2}$$

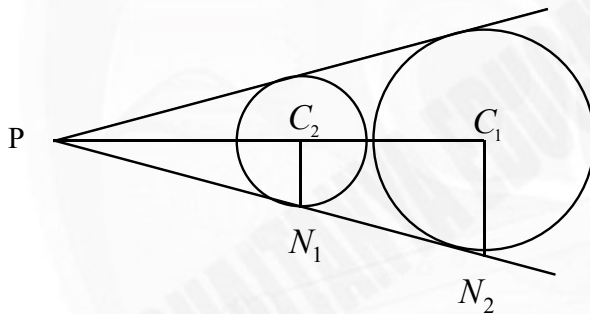
$$= \sqrt{3(5+3)^2} = \sqrt{192}$$

15. Two circles of radii 'a' and 'b' ($a < b$) touch externally. The distance of the point of intersection of direct common tangent from the smaller circle is :

a) $\frac{2a^{1/2}b^{3/2}}{b-a}$ b) $\frac{2a^{3/2}b^{1/2}}{b-a}$ c) $\frac{2ab}{\sqrt{b}-\sqrt{a}}$ d) $\frac{2a^{3/2}b^{3/2}}{b^2-a^2}$

Key: b

Sol: Let the direct common tangent intersect at P and $PN_1 = l$



ΔPC_1N_1 and PC_2N_2 are similar, so,

$$\frac{PN_1}{C_1N_1} = \frac{PN_2}{C_2N_2} \Rightarrow \frac{l}{a} = \frac{l + L_D}{b}$$

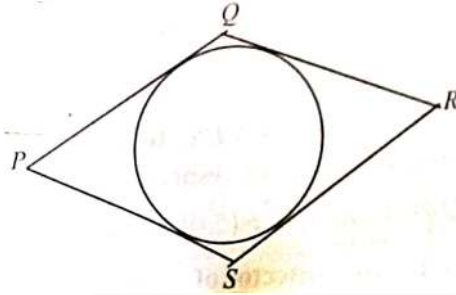
[where $L_D = N_1N_2 =$ length of direct common tangent]

$$\text{But } L_D = \sqrt{d^2 - (a-b)^2} = \sqrt{(a+b)^2 - (a-b)^2} = 2\sqrt{ab}$$

$$\therefore l = \frac{2a\sqrt{ab}}{b-a}$$

PASSAGE-07

In a variable parallelogram PQRS radius of circle 'C' touching all its sides internally is $r = 256$, Now circles of integer radius are drawn on right side of C, Such that every circle touches externally the circle present to its left and also both the sides RQ, RS of the parallelogram and now same is repeated on left side of C.i.e. Every circle on left side of C touches the circle to its right and also the sides PQ, PS



16. If in (n, r) , n = number of circles that can be drawn to the right of 'C' and r is the least radius ($r > 1$) among them, Find sum of all possible values of n
- a) 14 b) 7 c) 16 d) 8
17. As defined in the above question if (n_1, r_1) and (n_2, r_2) denote the right side and left side of C respectively, given $\frac{1}{r_1} + \frac{1}{r_2} = \frac{k}{r}$ and $n_1 + n_2 = l$ Which of the following is not true?
- a) For some value of k , two distinct values of l are possible
 b) For some value of l , two distinct values of k are possible
 c) l is always even
 d) $\frac{k}{l}$ is always an integer
18. If C_1, C_2 represent circles touching C on either side and the sides PQ, RS respectively and C_3, C_4 (with radii r_3, r_4) represent circles touching the pairs (C, C_1) and (C, C_2) and also their common tangents respectively then a possible rational value of
- $$\sqrt{\frac{7\sqrt{r_3} - 4\sqrt{r_4}}{\sqrt{r_3} \cdot \sqrt{r_4}}} =$$
- a) $\frac{3}{4}$ b) $\frac{9}{16}$ c) $\frac{4}{3}$ d) $\frac{16}{9}$

Key:acd

Sol: From the given description Radii of circles form two geometric progression on each side of C let common ratio be λ

$$\lambda = 2 \Rightarrow n = 7, r = 2$$

$$\lambda = 4 \Rightarrow n = 3, r = 4$$

$$\lambda = 8 \Rightarrow n = 2, r = 4$$

$$\lambda = 16 \Rightarrow n = 1, r = 16$$

$$\lambda = 32 \Rightarrow n = 1, r = 32$$

$$\lambda = 64 \Rightarrow n = 1, r = 64$$

$$\lambda = 128 \Rightarrow n = 1, r = 128$$

$$r_1 = r_2 \text{ and } n_1 = n_2 \Rightarrow k = \frac{256}{2r_1} \quad 2n_1 = l$$

$$(n_1, r_1) = (7, 2) \Rightarrow k = \frac{64}{7}, l = 4$$

$$= (3, 4) \Rightarrow k = 32, l = 6 = (2, 4) \Rightarrow k = 32, l = 4$$

$$= (1, 16) \Rightarrow k = 8, l = 2(1, 32) \Rightarrow k = 4, l = 2$$

$$= (1, 64) \Rightarrow k = 2, l = 2(1, 128) \Rightarrow k = 1, l = 2$$

Hence d is incorrect option

From the description

$$\frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{256}} = \frac{1}{\sqrt{r_3}}; \frac{1}{\sqrt{r_2}} + \frac{1}{\sqrt{256}} = \frac{1}{\sqrt{r_4}}$$

$$\text{Also } r_1 = r_2 \text{ and } r_3 = r_4 \Rightarrow \sqrt{\frac{3\sqrt{r_3}}{r_3}} = \sqrt{\frac{3}{\sqrt{r_3}}}$$

$\therefore$ for possible rational values of

$$\frac{1}{\sqrt{r_3}} = \frac{1}{16} + \frac{1}{2} = \frac{9}{16}$$

$$= \frac{1}{16} + \frac{1}{4} = \frac{5}{16} = \frac{1}{16} + \frac{1}{8} = \frac{3}{16}$$

$$\Rightarrow \sqrt{\frac{3}{\sqrt{r_3}}} = \sqrt{\frac{9}{16}} = \frac{3}{4} \text{ When } \frac{1}{\sqrt{r_3}} = \frac{3}{16}$$

PASSAGE-08

Consider a circle $x^2 + y^2 = a^2$. Let $A(a, 0)$ and $D(\alpha, \beta)$ be the given interior points of the circle. BC be an arbitrary chord of the circle through point D inclined at an angle θ with the positive x-axis. Also, $DB = r_1$ and $DC = r_2$.

19. The harmonic mean of BD and DC is

a) $\frac{\alpha^2 + \beta^2 - a^2}{2(\alpha \cos \theta + \beta \sin \theta)}$

b) $-\frac{\alpha^2 + \beta^2 - a^2}{(\alpha \cos \theta + \beta \sin \theta)}$

c) $\frac{\alpha^2 + \beta^2 - a^2}{\alpha \sin \theta + \beta \cos \theta}$

d) $\frac{2(\alpha^2 + \beta^2 - a^2)}{\alpha \cos \theta + \beta \sin \theta}$

20. The locus of the centroid of $\triangle ABC$ is

a) $\left(x - \frac{a}{3}\right)^2 + (y)^2 = \left(\frac{r_1 - r_2}{3}\right)^2$

b) $\left(x + \frac{a}{3}\right)^2 + (y)^2 = \left(\frac{r_1 + r_2}{3}\right)^2$

c) $\left(y - \frac{a}{3}\right)^2 + (x)^2 = \left(\frac{r_1 + r_2}{3}\right)^2$

d) $\left(x - \frac{a}{3}\right)^2 + (y)^2 = \left(\frac{r_1 + r_2}{3}\right)^2$

21. The locus of the centroid of $\triangle ABC$ has radius

a) less than $2a/3$

b) less than $a/3$

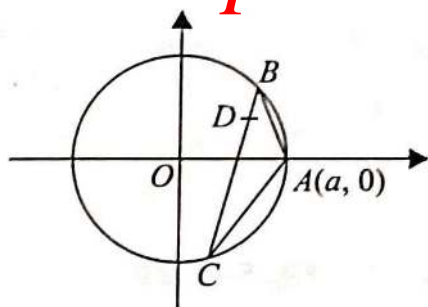
c) more than $2a/3$

d) more than $a/3$

Key: abd

Sol: Let $D(\alpha, \beta)$ be an interior point of a circle $x^2 + y^2 = a^2$. Then,

$$\alpha^2 + \beta^2 < a^2$$



Now, the equation of BC is

$$\frac{x - \alpha}{\cos \theta} = \frac{y - \beta}{\sin \theta}$$

Let $DB = r$. Then the coordinates of B are $(\alpha + r \cos \theta, \beta + r \sin \theta)$. It lies on $x^2 + y^2 = a^2$.

Therefore,

$$(\alpha + r \cos \theta)^2 + (\beta + r \sin \theta)^2 = a^2 \text{ or } r^2 + 2r(\alpha \cos \theta + \beta \sin \theta) + \alpha^2 + \beta^2 - a^2 = 0 \quad (\text{ii})$$

This is quadratic in r, which gives the values of r, one each for points B and C.

Let $DB = r_1$ and $DC = r_2$.

Then r_1 and r_2 are the roots of (ii). Therefore,

$$r_1 + r_2 = -2(\alpha \cos \theta + \beta \sin \theta) \quad (\text{iii})$$

$$\text{Also, } r_1 r_2 = \alpha^2 + \beta^2 - a^2$$

The harmonic mean of r_1 and r_2 is

$$\begin{aligned} \frac{2r_1 r_2}{r_1 + r_2} &= -\frac{2(\alpha^2 + \beta^2 - a^2)}{2(\alpha \cos \theta + \beta \sin \theta)} \\ &= -\frac{(\alpha^2 + \beta^2 - a^2)}{(\alpha \cos \theta + \beta \sin \theta)} \end{aligned}$$

Now, let $G(x_1, y_1)$ be the centroid of $\triangle ABC$. Then

$$\therefore 3x_1 = a + r_1 \cos \theta + r_2 \cos \theta \text{ and } 3y_1 = r_1 \sin \theta + r_2 \sin \theta$$

Eliminating θ , we get the required equation of locus as

$$(3x - a)^2 + (3y)^2 = (r_1 + r_2)^2$$

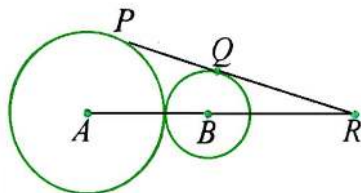
$$\text{or } \left(x - \frac{a}{3}\right)^2 + (y)^2 = \left(\frac{r_1 + r_2}{3}\right)^2$$

Now radius of circle is

$$\begin{aligned} \left|\frac{r_1 + r_2}{3}\right| &= \left|\frac{2(\alpha \cos \theta + \beta \sin \theta)}{3}\right| \\ &\leq \frac{2}{3} \sqrt{\alpha^2 + \beta^2} \\ &< \frac{2a}{3} \end{aligned}$$

PASSAGE - 09

In the given figure, there are two circles with centers A and B. The common tangent to the circles touches them, respectively, at P and Q. AR is 40cm and AB is divided by the point of contact of the circles in the ratio 5:3.



22. What is ratio of the length of AB to that of BR?

- a) 1:4 b) 2:3 c) 2:5 d) 7:4

Key: b

23. The radius of the circle with center B is

- a) 10cm b) 3cm c) 6cm d) 8cm

Key: c

24. The length of QR is

- a) $10\sqrt{15}$ cm b) $5\sqrt{15}$ cm c) $4\sqrt{15}$ cm d) $6\sqrt{15}$ cm

Key: d

Sol:

$$22. \frac{r_1}{r_2} = \frac{5}{3} = \frac{AR}{BR}$$

$$\text{or} \quad BR = \frac{3 \times 40}{5} = 24 \text{ cm}$$

$$\frac{AB}{BR} = \frac{40 - 24}{24} = \frac{16}{24} = \frac{2}{3}$$

$$23. \frac{r_1}{r_2} = \frac{5}{3}$$

$$\therefore \frac{r_1 + r_2}{r_2} = \frac{8}{3}$$

$$\therefore r_2 = \frac{(40 - 24)3}{8} = 6 \text{ cm}$$

$$24. QR = \sqrt{(BR)^2 - r_2^2} = \sqrt{(24)^2 - 36} = 6\sqrt{15} \text{ cm}$$

Passage -10

An equation of the family of circles passing through a given pair of points (x_1, y_1) real parameter. If a member of this family satisfies some other condition then that enables us to determine k and hence the member.

1. There exist exactly two circles that pass through $(3, -5)$ and $(5, -3)$ and touch the line $2x + 2y + 13 = 0$. Let the ratio of radii of the two circles be m/n with $m(>0)$ and $n(>0)$ having no common factors except 1. Then $(m+n)$ equals

- a) 2 b) 3 c) 5 d) 7

Key: a

Sol: Circles passing through (3, -5) and (5, -3) is

$$(x-3)(x-5) + (y+5)(y+3) + \lambda(x-y-8) = 0$$

$$\Rightarrow x^2 + y^2 + x(\lambda-8) - y(\lambda-8) + 30 - 8\lambda = 0$$

$2x + 2y + 13 = 0$ is a tangent is Perpendicular to

$$x - y - 8 = 0,$$

$\therefore$ The two circles have same radii

$$\therefore \frac{m}{n} = 1$$

$$\Rightarrow m + n = 2$$

2. Consider the set S of all possible circles that pass through (4, -3) and (-3, 4) and touch the line $x + y - 5\sqrt{2} = 0$. Then which of the following statements is correct?
- A) S consists of exactly one circle, having radius 5
 B) S consists of exactly one circle, having radius 10
 C) S consists of exactly two circles, the ratio of radii being 1: 2
 D) S consists of exactly two circles, the ratio of radii being 1: 3

Key: a

Sol: The circle passing through (4, -3) and (-3, 4) is

$$(x-4)(x+3) + (y+3)(y-4) + \lambda(x+y-1) = 0$$

$$x^2 + y^2 + x(\lambda-1) + y(\lambda-1) - (\lambda+24) = 0$$

$$x + y - 5\sqrt{2} = 0 \text{ is Parallel to } x + y - 1 = 0$$

$\therefore$ only one circle is possible centre Posses through $x = y$

$\therefore$ centre (0,0)

$$r = 5$$

Passage -11

Consider two circles S_1 and S_2 (externally touching) having centres at points A and B whose radii are 1 and 2 respectively. A tangent to circle S_2 from point B intersects the circle S_1 at point C. D is chosen on circle S_2 so that AC is parallel to BD and the two segments BC and AD do not intersect. S the circle S_1 at E. The line through B and E intersects the circle S_1 at another point F.

1. The length of segment EF is
- a) $2\sqrt{2}$ b) $\frac{2\sqrt{2}}{3}$ c) $\frac{2\sqrt{3}}{3}$ d) $\sqrt{3}$

Key: c

2. The area of triangle ABD is
- a) 2 b) $\sqrt{3}$ c) $2\sqrt{2}$ d) $\sqrt{5}$

Key: C

3. The length of the segment DE is
- a) $\sqrt{3}$ b) $2\sqrt{2}$ c) 2 d) 3

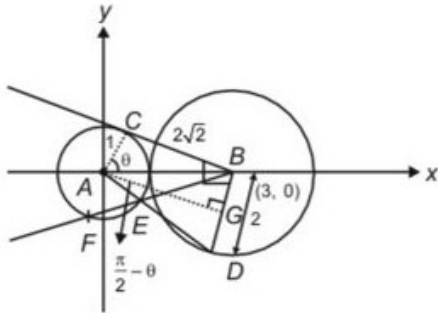
Key: c

Sol: for Q. Nos. 14, to 16

$$\tan \theta = 2\sqrt{2}$$

$$AG = 2\sqrt{2}$$

$$m_{BD} = 2\sqrt{2}$$



$$D = \left(3 - 2\left(\frac{1}{3}\right), 0 - 2\left(\frac{2\sqrt{2}}{3}\right) \right)$$

$$\equiv \left(\frac{7}{3}, -\frac{4\sqrt{2}}{3} \right)$$

$$m_{AE} = -\tan(\pi - 2\theta) = \tan 2\theta$$

$$= \frac{2(2\sqrt{2})}{1-8} = -\frac{4\sqrt{2}}{7}$$

$$E = \left(0 - 1\left(-\frac{7}{9}\right), 0 - 1\left(\frac{4\sqrt{2}}{9}\right) \right)$$

$$= \left(\frac{7}{9}, -\frac{4\sqrt{2}}{9} \right)$$

Equation of BE is $y = \frac{\sqrt{2}}{5}(x-3)$, solving intersection of line

BE with circle S_1 ,

$$x^2 + \frac{2}{25}(x-3)^2 = 1$$

$$\Rightarrow 27x^2 - 12x - 7 = 0$$

$$\Rightarrow (9x-7)(3x+1) = 0$$

$$\Rightarrow F \equiv \left(-\frac{1}{3}, \frac{-2\sqrt{2}}{3} \right)$$

Passage -12

Given a line segment AB , $A \equiv (0,0)$ and $B(a,0)$. Three circles S_1, S_2, S_3 of radius 'R' are centered at the end points and the midpoint of the line segment AB . A fourth circles S_4 is drawn touching the 3 given circles.

1. If $0 < R < \frac{a}{4}$, then sum of all possible distinct values of radius of S_4 is

- a) $\frac{a^2}{16R}$ b) $\frac{a^2}{7R}$ c) $\frac{3a^2}{16R}$ d) $\frac{a^2}{4R}$

Key: c

2. If $0 < R < \frac{a}{4}$, then number of possible circles S_4 is

- a) 2 b) 4 c) 6 d) 8

Key: c

3. If $\frac{a}{4} < R < \frac{a}{2}$, then radius of circle S_4 is

- a) $\frac{a^2}{16R}$ b) $\frac{a^2}{8R}$ c) $\frac{3a^2}{16R}$ d) $\frac{a^2}{4R}$

Key: a

Sol: Let $(k, 0)$ be centre of S_4

If S_4 touches two circles internally one externally

$$|k - 0| = |R - r| \Rightarrow k^2 = R^2 + r^2 - 2Rr \quad \text{---(A)}$$

$$|k - a/2| = |R - r| \Rightarrow k^2 + a^2/4 - ak = R^2 + r^2 - 2Rr \quad \text{---(B)}$$

$$|k - a| = |R + r| \Rightarrow k^2 + a^2 - 2ak = R^2 + r^2 + 2Rr \quad \text{---(C)}$$

Solving 1, 2, 3 $r = \frac{a^2}{8R}$ [Two such circles are possible]

If S_4 touches two circle Externally one internally

$$|k - 0| = |R + r| \Rightarrow k^2 = R^2 + r^2 + 2Rr \quad \text{---(A)}$$

$$|k - a/2| = |R - r| \Rightarrow k^2 + a^2/4 - ak = R^2 + r^2 - 2Rr \quad \text{---(B)}$$

$$|k - a| = |R + r| \Rightarrow k^2 + a^2 - 2ak = R^2 + r^2 + 2Rr \quad \text{---(C)}$$

solving (A) (B) ,(C)

$$r = \frac{a^2}{16R} \quad \text{[four such circles are possible]}$$

$$\text{sum of distinct values of } r = \frac{a^2}{8R} + \frac{a^2}{16R} = \frac{3a^2}{16R}$$

3. In this case, S_4 touching touching two circles Externally (S_1 & S_3) and S_2 internally is

$$\text{possible } \therefore r = \frac{a^2}{16R}$$

Passage -13

Let $C_1 : x^2 + y^2 = r^2$ and $C_2 : (x - p)^2 + (y - q)^2 = r^2$ be 2 circles with radius $r (r > 0)$ and have n points of intersection, (x_i, y_i) for $i \in \{1, 2, \dots, n\}$. If C_1 and C_2 are orthogonal at all points of intersection, then

1. As we move the centre of C_2 along $p + bq = 0$ for some constant $b \neq 0$, then $\frac{dr}{dq}$ is equal to

- a) $\frac{b^2q + q}{r}$ b) $\frac{b^2q + q}{2r}$ c) $\frac{b^2q + q}{3r}$ d) $\frac{b^2q + q}{4r}$

Key: b

Sol: As circles intersect orthogonally

$$d^2 = r_1^2 + r_2^2$$

$$p^2 + q^2 = 2r^2, \quad p + bq = 0$$

$$b^2 q^2 + q^2 = 2r^2 \quad \Rightarrow p = -bq$$

Differentiate w.r.t q

$$b^2(2q) + 2q = 4r \frac{dr}{dq}$$

$$\frac{(b^2 + 1)q}{2r} = \frac{dr}{dq}$$

2. As we move the centre of C_2 along the curve $q = a$ for some constant $a \neq 0$, then $\frac{dr}{dp}$ is equal to

a) $\frac{p}{4r}$

b) $\frac{p}{3r}$

c) $\frac{p}{2r}$

d) $\frac{p}{r}$

Key: c

Sol: $q = a$

$$p^2 + a^2 = 2r^2$$

Differentiate w.r.t p

$$2p + 0 = 4r \frac{dr}{dp}$$

$$\frac{dr}{dp} = \frac{p}{2r}$$

EXERCISE –IV

(Matrix Matching)

01. Answer Q1, Q2 and Q3 by appropriately matching the information given in the three columns of the following table.

	COLUMN I		COLUMN II		COLUMN III
I)	$x^2 + y^2 + 3x + y + 2 = 0$	A)	Does not meet X-axis or Y-axis	P)	(-1, -1) lies inside circle
II)	$x^2 + y^2 + x + 3y + 2 = 0$	B)	Cuts X-axis at two points	Q)	(-1, -1) lies outside circle
III)	$x^2 + y^2 + 4x + 4y + 4 = 0$	C)	Touches both X, Y- axes	R)	(-1, -1) lies on circle
IV)	$x^2 + y^2 - 4x - 6y + 12 = 0$	D)	Cuts Y-axis at two points	S)	(1, 1) lies inside circle

- Which of the following combination is correct ?
a) I-B-R b) II-B-R c) I-B-S d) II-B-S
- Which of the following combination is correct ?
a) III-A-P b) III-B-Q c) III-C-P d) III-C-Q
- Which of the following combination is correct ?
a) II-C-R b) IV-A-Q c) IV-A-S d) II-C-R

Key: 1-a,2-c,3-b

Sol: $S_{11} > 0 \Rightarrow$ point lies outside circle

$S_{11} = 0 \Rightarrow$ point lies on the circle

$S_{11} < 0 \Rightarrow$ point lies inside the circle

$$x^2 + y^2 + 2gx + 2fy + C = 0$$

$g^2 - C > 0 \Rightarrow$ circle cuts x-axis at 2 points

$g^2 - C = 0 \Rightarrow$ circle touches x-axis

$g^2 - C < 0 \Rightarrow$ circle doesn't meet x-axis

$f^2 - C > 0 \Rightarrow$ circle cuts y-axis at 2 points

$f^2 - C = 0 \Rightarrow$ circle touches y-axis

$f^2 - C < 0 \Rightarrow$ circle doesn't meet y-axis

- 02 Answer Q, 1, Q, 2 and Q, 3 by appropriately matching the information given in the three columns of the following table.

	COLUMN I		COLUMN II		COLUMN III
I)	$x^2 + y^2 - 6x - 8y + 21 = 0$	A)	Number of tangents passing through point (3,5) is 0	P)	Number of normals passing through point (3,5) is
II)	$x^2 + y^2 + 6x + 8y + 21 = 0$	B)	Number of tangents passing through point (-5,4) is 1	Q)	Number of tangents passing through point (-3,-2) is 1
III)	$x^2 + y^2 - 6x + 8y + 21 = 0$	C)	Number of tangents passing through point (-3,-2) is 2	R)	Number of normals passing through point (3,-4) is greater than 3
IV)	$x^2 + y^2 + 6x - 8y + 21 = 0$	D)	Number of tangents passing through point (-1,-4) is 1	S)	Number of normals passing through point (-1,-4) is 0

- Which of the following combination is correct ?
a) 1-B-Q b) 1-A-Q c) I-C-R d) I-D-R
- Which of the following combination is correct ?

a) II-D-Q

b) II-D-R

c) IV-B-P

d) IV-C-S

3. Which of the following combination is correct?

a) III-C-P

b) III-A-Q

c) III-A-R

d) III-C-R

Key: 1-b, 2-a, 3-d

Sol: No. of normals from centre is infinite

No. of normals from any point other than centre is 1

No. of tangents from a point outside circle = 2

No. of tangents from a point inside circle = 0

No. of tangents from a point on the circle = 1

03. Answer Q, 1, Q, 2 and Q, 3 by appropriately matching the information given in the three columns of the following table.

Consider the circle $(x-2)^2 + (y-3)^2 = 25$ and a point $P(x, y)$ on itLet $N = \sqrt{(x-8)^2 + (y-11)^2}$ Consider the line $L: x - y + 1 = 0$ and $Q_1, Q_2, Q_3, \dots, Q_n$ are the points on the line L from which perpendicular tangents can be drawn to the given circle circle is more than two

	COLUMN I		COLUMN II		COLUMN III
P	Slope of tangent drawn to the given circle from Q_1 can be zero	I	Quadrilateral formed by the tangents drawn to the given circle from Q_1 and Q_2 is a rectangle of area 100 square units	K	When N is maximum P is (6,6)
Q	Slope of tangent drawn to the given circle from Q_1 can be $-\frac{3}{4}$	II	Quadrilateral formed by the tangents drawn to the given circle from Q_1 and Q_2 is a square of area 100 square units	L	When N is maximum P is (6,6)
R	Number of points $Q_1, Q_2, Q_3, \dots, Q_n$ on the line L from which perpendicular tangents can be drawn to the given circle is exactly two	III	Quadrilateral formed by the tangents drawn to the given circle from Q_1 and Q_2 can have (7,-2) as one of its vertex	M	When N is maximum P is (-1,-1)
S	Number of points $Q_1, Q_2, Q_3, \dots, Q_n$ on	IV	One of the side of Quadrilateral formed	N	When N is maximum P is

	the line L from which perpendicular tangents can be drawn to the given circle is more than two		by the tangents drawn to the given circle from Q_1 and Q_2 can have 10 units as its length.		(5,7)
--	--	--	---	--	-------

- Which of the following option is correct?
a) Q I L b) S I K c) R IV L d) P II M
- Which of the following option is correct?
a) Q II N b) P I L c) S III L d) R IV N
- Which of the following option is correct?
a) P II L b) P IV N c) R III L d) S IV M

Key: 1-d, 2-d, 3-b

Sol: Distance between the points (8,11) and (2,3) is 10

When N is maximum, P is $(-1, -1)$ and when N is minimum, P is (5,7), using section

formula Slope of the tangent drawn to the circle from $\left(\frac{39}{7}, \frac{46}{7}\right)$ at $\frac{-3}{4}$ & $\frac{-4}{3}$ and the points of contact of tangents

04. Answer Q1, Q2 and Q3 by appropriately matching the information given in the three column its of the following table. Consider the four circles

$$S_1 : x^2 + y^2 + 4x - 4y + 4 = 0$$

$$S_1 : x^2 + y^2 - 2x - 2y + 1 = 0$$

$$S_3 : x^2 + y^2 - 12x - 12y + 36 = 0$$

$$S_4 : x^2 + y^2 - 6x + 6y + 9 = 0$$

director circles of $S_1 = 0, S_2 = 0, S_3 = 0, S_4 = 0$ are $S'_1 = 0, S'_2 = 0, S'_3 = 0, S'_4 = 0$ respectively.

	COLUMN I		COLUMN II		COLUMN III
P	$S'_1 = 0$ is $x^2 + y^2 + 4x - 4y = 0$	I	Circles $S'_2 = 0$ and $S'_3 = 0$ intersect each other at exactly two points	K	Length of the tangent drawn from origin to the circle $S_1 = 0$ is 4
Q	$S'_2 = 0$ is $x^2 + y^2 - 2x - 2y + 2 = 0$	II	Circles $S'_1 = 0$ and $S'_2 = 0$ touches each other internally	L	Length of the tangent drawn from (4,0) to the circle $S_2 = 0$ is 3
R	$S'_3 = 0$ is $x^2 + y^2 - 12x - 12y = 0$	III	Circles $S'_2 = 0$ and $S'_4 = 0$	M	Length of the

			touch each other externally		tangent drawn from (0,3) to the circle $S_3 = 0$ is 9
S	$S'_4 = 0$ is $x^2 + y^2 - 6x + 6y - 6 = 0$	IV	Circles $S'_1 = 0$ and $S'_3 = 0$ are neither intersecting nor touching each other	N	Length of the tangent drawn from $(2\sqrt{2}, 2\sqrt{2})$ to the circle $S_4 = 0$ is 5

- Which of the following option is correct
a) Q1L b) SIK c) RIVN d) P II M
- Which of the following option is incorrect
a) P IV L b) P I N c) S III L d) R IV N
- Which of the following option is correct
a) P II L b) P IV N c) R III L d) R IV M

Key: 1-c, 2-c, 3-b

Sol: Conceptual

05. C_1, C_2 are two circles of radii a, b ($a < b$) respectively touching both the coordinate axes and have their centres in the first quadrant.

	Column-I		Column-II
A)	C_1, C_2 touches each other then $\frac{b}{a}$ is	p	$2 + \sqrt{2}$
B)	If C_1, C_2 are orthogonal then $\frac{b}{a}$ is	q	$2 + \sqrt{3}$
C)	If C_1, C_2 intersect in such a way that their common chord has maximum length then $\frac{a}{b}$ is	r)	3
D)	If C_1 passes through centre of C_2 then $\frac{b}{a}$ is	s)	$3 + 2\sqrt{2}$

Key: A $\rightarrow$ s, B $\rightarrow$ r, C $\rightarrow$ q, D $\rightarrow$ p

Sol: A) $C_1 : x^2 + y^2 - 2a(x + y) + a^2 = 0$,

Centre : (a, a) , radius : a , $C_2 : x^2 + y^2 - 2b(x + y) + b^2 = 0$

Centre : (b, b) , radius b

$$C_1 \text{ and } C_2 \text{ touch} \Rightarrow \sqrt{2}(b-a) = a+b \Rightarrow \frac{b}{a} = (\sqrt{2}+1)^2 = 3+2\sqrt{2}.$$

$$B) C_1 \text{ and } C_2 \text{ intersect orthogonally} \Rightarrow 2(b-a)^2 = a^2 + b^2 \Rightarrow \frac{b}{a} = 2 + \sqrt{3}$$

$$C) \text{ The common chord is } 2(b-a)(x+y) = b^2 - a^2. \text{ If passes through } (a, a) \Rightarrow b/a = 3.$$

$$D) C_2 \text{ passes through } (a, a) \Rightarrow 2a^2 - 4ab + b^2 = 0 \Rightarrow \frac{b}{a} = 2 + \sqrt{2}.$$

06. A variable straight line through $A(-1,1)$ is drawn to cut the circle $x^2 + y^2 = 1$ at the points B and C. A point P is chosen on the line ABC satisfying the condition given in column I. Let d be the minimum distance of origin from the locus of P.

a) AB, AP, AC are in A.P.

p)

b) AB, AP, AC are in G.P.

q) $\frac{1}{\sqrt{2}}$

c) AB, AP, AC are in H.P.

r) $\sqrt{2}$

d) $AB, AP/2, AC$ are in A.P.

s) $\sqrt{2}-1$

Key: (a) $\rightarrow$ (p), (b) $\rightarrow$ (s), (c) $\rightarrow$ (q), (d) $\rightarrow$ (r)

Sol: (a) The equation of the line is $x = -1 + r \cos \theta, y = 1 + r \sin \theta$. The equation of the circle gives

$$(-1 + r \cos \theta)^2 + (1 + r \sin \theta)^2 = 1.$$

$$\Rightarrow r^2 - 2r(\cos \theta - \sin \theta) + 1 = 0. \text{ If } AB = r_1, AC = r_2$$

$$r_1 + r_2 = 2(\cos \theta - \sin \theta), r_1 r_2 = 1$$

$$AP = r = 2r = r_1 + r_2 \Rightarrow r = \cos \theta - \sin \theta, r^2 = r \cos \theta - r \sin \theta$$

$$\text{The Locus of P is } (x+1)^2 + (y-1)^2 = x+1 - y+1 = x - y + 2$$

$$\Rightarrow x^2 + y^2 + x - y = 0 \text{ It is a circle through the origin } \therefore d = 0$$

$$(b) r^2 = r_1 r_2 = 1 \Rightarrow (x+1)^2 + (y-1)^2 = 1 \Rightarrow d = \sqrt{2} - 1$$

$$(c) r = \frac{2r_1 r_2}{r_1 + r_2} \Rightarrow r \cos \theta - r \sin \theta = 1 \Rightarrow x+1 - y+1 = 1 \text{ or } x - y + 1 = 0 \Rightarrow d = \frac{1}{\sqrt{2}}$$

$$(d) r = r_1 + r_2 \Rightarrow r^2 = 2(r \cos \theta - r \sin \theta)$$

$$\Rightarrow (x+1)^2 + (y-1)^2 = 2(x+1 - y+1) \Rightarrow x^2 + y^2 = 2 \Rightarrow d = \sqrt{2}$$

08. Match the statements in Column I with properties in Column 2.

(a) Two intersecting circles (p) have a common tangent

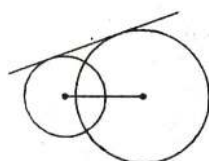
(b) Two mutually external circles (q) have a common normal

(c) Two circles, one strictly inside the other (r) do not have a common tangent

(d) Two branches of a hyperbola (s) do not have a common normal

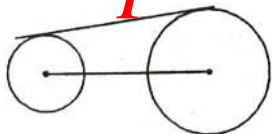
Key: (a) $\rightarrow$ (p), (q); (b) $\rightarrow$ (p), (q); (c) $\rightarrow$ (q), (r); (d) $\rightarrow$ (p), (q)

Sol:



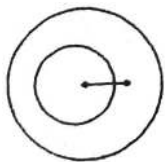
(a)

have a tangent and a normal



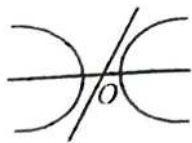
(b)

have a tangent and a normal



(c)

no tangent, a normal



(d)

have a normal and a tangent (asymptote)

09. Consider the circles $x^2 + y^2 - 4x - 6y + 12 = 0$ and $(x-5)^2 + (y-6)^2 = r^2 > 1$. Column 2 gives the values of r

(a) They touch internally

(p) $3\sqrt{2} + 1$

(b) They touch externally

(q) $3\sqrt{2} - 1$

(c) They intersect orthogonally

(r) $\sqrt{19}$

(d) The common chord is tangent

(s) $\sqrt{17}$

Key: (a) $\rightarrow$ (p), (b) $\rightarrow$ (q), (c) $\rightarrow$ (s), (d) $\rightarrow$ (r)

Sol: (2,3), (5,6) are centres. The distance between them is $\sqrt{9+9} = 3\sqrt{2}$

a) $3\sqrt{2} = r - 1 \Rightarrow r = 3\sqrt{2} + 1$ b) $3\sqrt{2} = r + 1 \Rightarrow r = 3\sqrt{2} - 1$ c) $(3\sqrt{2})^2 = r^2 + 1 \Rightarrow r = \sqrt{17}$ d) Common chord is $6(x+y) - 49 + r^2 = 0$

It passes through (2,3) $r = \sqrt{19}$.

10. Let C_1 and C_2 be two circles of centres O_1, O_2 and radii r_1 and $r_2 > r_1$. They have common tangents $x + y = 2$ and $x - y = 2$. They touch the circle $x^2 + y^2 = 1$ externally.

(a) $r_1 =$ (p) $4(1 + \sqrt{2})$ (b) $r_2 =$ (q) $3(1 + \sqrt{2})$ (c) $O_1O_2 =$ (r) $4 + 3\sqrt{2}$ (d) OO_2 (s) $\sqrt{2} - 1$

Key: (a) $\rightarrow$ (s), (b) $\rightarrow$ (q), (c) $\rightarrow$ (p), (d) $\rightarrow$ (r)

Sol: The centres O_1, O_2 lie on the x-axis. Let $(\alpha, 0)$ be the centre of a circle

$$\left(\frac{\pm\alpha-2}{\sqrt{2}}\right)^2 = (\alpha-1)^2 \Rightarrow \alpha^2 = 2, (\alpha-4)^2 = 18, O_1 = (\sqrt{2}, 0), O_2 = (-4-3\sqrt{2}, 0)$$

a) $r_1 = \sqrt{2} - 1$ b) $r_2 = -1 + 4 + 3\sqrt{2} = 3(1 + \sqrt{2})$ c) $O_1O_2 = 4(1 + \sqrt{2})$ d) $OO_2 = 4 + 3\sqrt{2}$

11.

	Column-I		Column-II
p.	The circle $x^2 + y^2 - 4x - 4y + 4 = 0$ is inscribed in a triangle which has two sides along coordinate axes. The locus of circumcentre of triangle is $x + y - xy + k\sqrt{x^2 + y^2} = 0$. Then $2K =$	1.	4
q.	A variable circle passes through $A(a, b)$ and touches x-axis. Locus of other end of diameter through A is $(x - a)^2 = \lambda y$ where $\left \frac{\lambda}{b} \right =$	2.	2
r.	If two circle $(x - 1)^2 + (y - 3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct points the number of integer values taken by radius r is.....	3.	3
s.	Area of the region $ 3x - 4y + 4x + 3y = 5\sqrt{2}$ in (sq.units)	4.	5

a)p-2;Q-1;R-3;S-4;

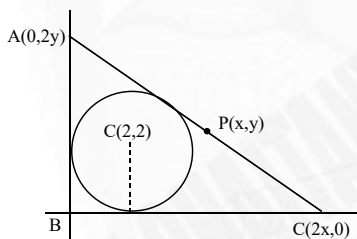
b)P-2;Q-1;R-4;S-1;

c)P-3;Q-1;R-4;S-2

d)P-4;Q-2;R-4;S-2;

Key: b

Sol: p)



Given, radius of circle = 2 C = centre of circle = (2, 2)

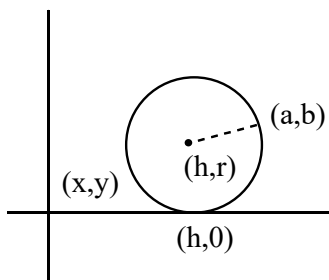
Equation of AC is $\frac{X}{2x} + \frac{Y}{2y} = 1$

Circumcentre is midpoint of AC i.e (x, y) = P

$$\perp^2 \text{ distance from C to AC is radius} \Rightarrow 2 = \frac{\frac{1}{x} + \frac{1}{y}}{\sqrt{\frac{1}{4x^2} + \frac{1}{4y^2}}}$$

Locus of P(x, y) is $\Rightarrow x + y - xy + \sqrt{x^2 + y^2} = 0$

q)



Let centre of circle be (h,r) radius of circle is r

(x,y) be the other end of diameter

$$\frac{a+x}{2} = h; \frac{b+y}{2} = r \Rightarrow 2r - y = b$$

Equation of circle $(x-h)^2 + (y-r)^2 = r^2$

(x,y) satisfy the given equation

$$\left(x - \left(\frac{a+x}{2}\right)\right)^2 + (y-r)^2 = r^2$$

$$\left(\frac{x-a}{2}\right)^2 + y^2 - 2yr = 0$$

$$\left(\frac{x-a}{2}\right)^2 = 2yr - y^2$$

$$= y(2r - y)$$

$$\frac{(x-a)^2}{4} = by$$

$$(x-a)^2 = 4by$$

$$\lambda = 4b \Rightarrow \frac{\lambda}{b} = 4$$

r) Given, equations of circles are $(x-1)^2 + (y-3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$

Let, $S_1 \equiv (x-1)^2 + (y-3)^2 = r^2$

$$C_1 = (1,3) \quad r_1 = r$$

$$S_2 \equiv x^2 + y^2 - 8x + 2y + 8 = 0$$

$$C_2 = (4,-1) \quad r_2 = 3$$

Condition for intersecting in two points is

$$|r_1 - r_2| < c_1c_2 < r_1 + r_2$$

$$|r - 3| < 5 < r + 3$$

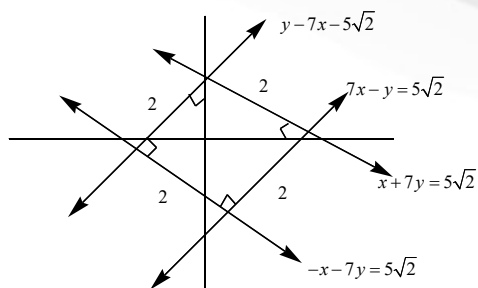
$$|r - 3| < 5 \quad 5 < r + 3$$

$$r - 3 < 5 \quad r > 2$$

$$r < 8$$

$\therefore r \in (2,8)$ 6 possible values of r

s)



$$|3x - 4y| + |4x + 3y| = 5\sqrt{2}$$

Possible lines are

$$(3x-4y)+(4x+3y)=5\sqrt{2} \Rightarrow 7x-y=5\sqrt{2} \Rightarrow \frac{x}{\left(\frac{5\sqrt{2}}{2}\right)} + \frac{y}{(-5\sqrt{2})} = 1$$

$$(3x-4y)-(4x+3y)=5\sqrt{2} \Rightarrow -x-7y=5\sqrt{2} \Rightarrow \frac{x}{(-5\sqrt{2})} + \frac{y}{\left(-\frac{5\sqrt{2}}{7}\right)} = 1$$

$$-(3x-4y)+(4x+3y)=5\sqrt{2} \Rightarrow x+7y=5\sqrt{2} \Rightarrow \frac{x}{5\sqrt{2}} + \frac{y}{\frac{5\sqrt{2}}{7}} = 1$$

$$-(3x-4y)-(4x+3y)-5\sqrt{2} \Rightarrow y=7x=5\sqrt{2}$$

$$\Rightarrow \frac{x}{\left(\frac{5\sqrt{2}}{7}\right)} + \frac{y}{5\sqrt{2}} = 1$$

They form a square with side length of 2 units

$$\text{Area} = (2)^2 \text{ sq. units}$$

$$\text{Area} = 4 \text{ sq. units}$$

12.

	Column-I		Column-II
A)	The length of the common chord of two circles of radii 3 units and 4 units which intersect orthogonally is $k/5$. Then k is equal to	p)	1
B)	The circumference of the circle $x^2 + y^2 + 4x + 12y + p = 0$ is bisected by the circle $x^2 + y^2 - 2x + 8y - q = 0$ then $p + q$ is equal to	q)	24
C)	the number of distinct chords of the circle $2x(x - \sqrt{2}) + y(2y - 1) = 0$ where the chords are passing through the point $(\sqrt{2}, 1/2)$ and are bisected on the x-axis is	r)	32
D)	One of the diameters of the circle circumscribing the rectangle ABCD is $4y = x + 7$. If A and B are the points (3,4) and	s)	36

(5,4) respectively, then the area of the rectangle is

- a) A-Q; B-S; C-P; D-R;
c) A-Q; B-S; C-P; D-S

- b) A-P; B-S; C-Q; D-R
d) A-R; B-P; C-Q; D-S

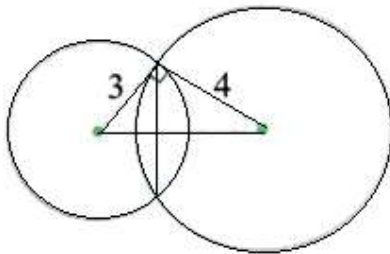
Key: a

Sol: Let the length of the common chord be $2a$. Then,

$$\sqrt{9-a^2} + \sqrt{16-a^2} = 5$$

$$\text{or } \sqrt{16-a^2} = 5 - \sqrt{9-a^2}$$

$$\text{or } 16 - a^2 = 25 + 9 - a^2 - 10\sqrt{9-a^2}$$



$$\text{or } 10\sqrt{9-a^2} = 18$$

$$\text{or } 100(9-a^2) = 324$$

$$\text{i.e., } 100a^2 = 576$$

$$\text{or } a = \sqrt{\frac{576}{100}} = \frac{24}{10}$$

$$\therefore 2a = \frac{24}{5} = \frac{k}{5} \text{ or } k = 24$$

b. The equation of common chord is $6x + 4y + p + q = 0$. The common chord passes through the center $(-2, -6)$ of the circle $x^2 + y^2 + 4x + 12y + p = 0$. Therefore,
 $p + q = 36$

c. The equation of the circle is $2x^2 + 2y^2 - 2\sqrt{2}x - y = 0$. Let $(\alpha, 0)$ be the midpoint of a chord. Then, the equation of the chord is $2\alpha x - \sqrt{2}(x + \alpha) - \frac{1}{2}(y + 0) = 2\alpha^2 - 2\sqrt{2}\alpha$

Since it passes through the point $(\sqrt{2}, 1/2)$, we have

$$2\sqrt{2}\alpha - \sqrt{2}(\sqrt{2} + \alpha) - \frac{1}{4} = 2\alpha^2 - 2\sqrt{2}\alpha$$

$$\text{i.e., } 8\alpha^2 - 12\sqrt{2}\alpha + 9 = 0$$

$$\text{i.e., } (2\sqrt{2}\alpha - 3)^2 = 0$$

$$\text{i.e., } \alpha = \frac{3}{2\sqrt{2}}$$

Therefore, the number of chords is 1.

d. Midpoint of $AB \equiv (1, 4)$ The equation perpendicular bisector of AB is $x=1$. A diameter of the circle is $4y = x + 7$. Therefore, the center of the circle is $(1, 2)$. Hence, the sides of the rectangle are 8 and 4. Therefore,

$$\text{Area} = 32$$

13. If the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ cuts each of the circles $x^2 + y^2 - 4 = 0$, $x^2 + y^2 - 6x - 8y + 10 = 0$ and $x^2 + y^2 + 2x - 4y - 2 = 0$ at the extremities of a diameter, then

- | | |
|-------------------------|-------|
| A) $c = -4$ | p) 17 |
| B) $g + f + c =$ | q) -4 |
| C) $g^2 + f^2 - c = 17$ | r) 6 |
| D) $gf = 6$ | s) -9 |

Key: A-q, B-s, C-p, D-r

Sol: Let $S_1 = x^2 + y^2 + 2gx + 2fy + c = 0$

$$S_2 = x^2 + y^2 - 4 = 0$$

$$S_3 = x^2 + y^2 - 6x - 8y + 10 = 0$$

$$S_4 = x^2 + y^2 + 2x - 4y - 2 = 0$$

Equation of the common chords $S_1 - S_2 = 0$,

$$S_1 - S_3 = 0, S_1 - S_4 = 0$$

Satisfy these equations by the coordinates of the centre of corresponding circles.

14. Let $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$ be 3 distinct points lying on circle $S: x^2 + y^2 = 1$, such that

$$x_1x_2 + y_1y_2 + x_2x_3 + y_2y_3 + x_3x_1 + y_3y_1 = -\frac{3}{2}.$$

Column -I

A) Let P be any arbitrary point lying on S , then

$$(PA)^2 + (PB)^2 + (PC)^2 =$$

B) Let the perpendicular dropped from point 'A'

to BC meets S at Q and $\angle OBQ = \frac{\pi}{k}$, where 'O' is origin, then $k =$

C) Let R be the point lying on line $x + y = 2$ at

The minimum distance from S and the square of

the maximum distance of R from S is $a + b\sqrt{b}$,

then $a + b =$

D) Let I and G represent incentre and centroid of

$\triangle ABC$ respectively, then $IA + IB + IC + GA + GB + GC =$

Column-II

P) 3

Q) 4

R) 5

S) 6

Key: A-s, B-p, C-r, D-s

Sol: $(2x_1x_2 + 2x_2x_3 + 2x_3x_1 + x_1^2 + x_2^2 + x_3^2) + (y_1^2 + y_2^2 + y_3^2 + 2y_1y_2 + 2y_2y_3 + 2y_3y_1) = 0$

$$\Rightarrow (x_1 + x_2 + x_3)^2 + (y_1 + y_2 + y_3)^2 = 0$$

$$\Rightarrow x_1 + x_2 + x_3 = 0 \text{ and } y_1 + y_2 + y_3 = 0$$

$\Rightarrow$ circumcenter and centroid of $\triangle ABC$ coincide

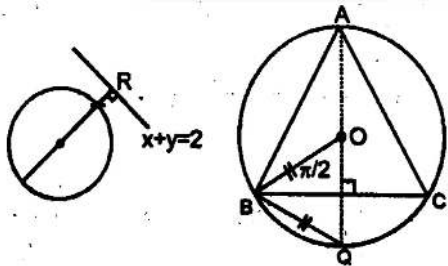
$\Rightarrow \triangle ABC$ is equilateral

A) $(PA)^2 + (PB)^2 + (PC)^2$

$$= (x - x_1)^2 + (y - y_1)^2 + (x - x_2)^2 + (y - y_2)^2 + (x - x_3)^2 + (y - y_3)^2$$

$$= 3(x^2 + y^2 + 1) = 6$$

B) $\angle OBQ = \frac{\pi}{3}, k = 3$



C) Max. distance of R from S, $d = \frac{2}{\sqrt{2}} + 1 = \sqrt{2} + 1$

$$\Rightarrow d^2 = 3 + 2\sqrt{2}$$

$$a + b = 3 + 2 = 5$$

D) I and G coincide with origin $IA = IB = IC = GA = GB = GC = 1$

$$IA + IB + IC + GA + GB + GC = 6$$

15. In the parallelogram ABCD with angle $A = 60^\circ$, the bisector of angle B is drawn which cuts the side CD at a point E. A circle S_1 of radius 'r' is inscribed in the $\triangle ECB$. Another circle ' S_2 ' is inscribed in the trapezium ABED.

Column - I

- A) The value of radius of S_1 is
- B) The value of distance between the centres of S_1 and S_2 is
- C) The value of the length of common tangent of S_1 and S_2 is
- D) The value of the length CE is

Column - II

P) $2\sqrt{3}r$

Q) $\frac{\sqrt{3}}{2}r$

R) $\sqrt{7}r$

S) $\frac{3}{2}r$

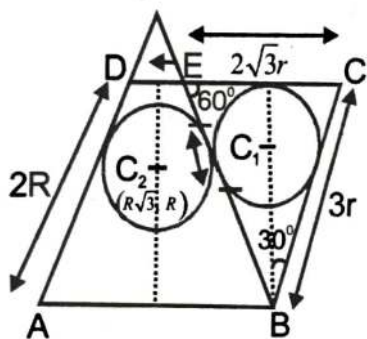
Key: A-s, B-r, C-q, D-p

Sol: $\perp$ ar distance between AB and CD

$$2R = 3r \Rightarrow R = \frac{3r}{2}$$

$$C_1 = (2R\sqrt{3}, 2r) = (3\sqrt{3}r, 2r)$$

$$C_2 = (R\sqrt{3}, R) = \left(\frac{3\sqrt{3}r}{2}, \frac{3r}{2} \right)$$



$$C_1 C_2 = \sqrt{\frac{r^2}{4} + \frac{27r^2}{4}} = r\sqrt{7}$$

$$\text{Length of common tangent} = \sqrt{7r^2 - (R+r)^2}$$

$$= \sqrt{7r^2 - \frac{25r^2}{4}} = \frac{\sqrt{3}}{2}r$$

16. Match the statements of column I with values of column II

Column – I

A) Let S_1 and S_2 be two variable circles

whose centres lie in different members of family of circles

$x^2 + y^2 - 2x \cos \theta - 2y \sin \theta - 3 = 0$ and pass through a fixed

point $(9, \sqrt{243})$. The maximum difference of the radii

of S_1 and S_2 is

B) From a point $(h, 0)$ common tangents are drawn to the circles

$x^2 + y^2 = 1$ and $(x-2)^2 + y^2 = 4$, the value of $|h|$ is

C) If the circle $x^2 + y^2 + 4x + 22y + 1 = 0$ bisects the

Circumference of the circle $x^2 + y^2 - 2x + 8y - m = 0$

, then $1+m$ is equal to

D) The value of $|a|$ for which the common chord of the circles

$x^2 + y^2 - 8 = 0$ & $(x-a)^2 + y^2 - 8 = 0$ subtends a right angle at origin

Column – II

p) -2

q) 6

r) ± 4

s) 50

Key: A-q, B-p, C-s, D-r

Sol: A) Distance between variable radii of two circles passing through a fixed point will be maximum when the fixed point and respective centres are collinear. So, the maximum difference of radius will be 6.

B) Equation of tangent to the circle $x^2 + y^2 = 1$, is

$y = mx \pm \sqrt{1+m^2}$. This also touches the circle

$$(x-2)^2 + y^2 = 4 \text{ So, } \left| \frac{2m \pm \sqrt{1+m^2}}{\sqrt{1+m^2}} \right| = 2 \Rightarrow m = \pm \frac{1}{\sqrt{3}}$$

$\Rightarrow$ Common tangents are $y = \frac{1}{\sqrt{3}}x + \frac{2}{\sqrt{3}}$ and $y = -\frac{1}{\sqrt{3}}x - \frac{2}{\sqrt{3}}$

On putting $y = 0$, from both the equations, we get $x = -2 \Rightarrow h = -2$

C) The common chord of the given circles is $6x + 14y + 1 + m = 0$

Since $x^2 + y^2 + 4x + 22y + 1 = 0$ bisects the circumference of the circle

$$x^2 + y^2 - 2x + 8y - m = 0,$$

(i) passes through the centre of the second circle i.e.,

$(1, -4)$. Therefore, $6 - 56 + 1 + m = 0 \Rightarrow 1 + m = 50$

D) Common chord is given by $(x^2 + y^2 - 8) - ((x - a)^2 + y^2 - 8) = 0$

$$\Rightarrow 2ax - a^2 = 0 \Rightarrow \frac{2x}{a} = 1$$

$$\text{Now, } x^2 + y^2 - 8\left(\frac{2x}{a}\right) = 0$$

gives the combined equation of the straight lines joining the end points of this common chord and origin. If this common chord subtends a right angle at origin, then

$$1 - \frac{32}{a^2} + 1 = 0 \Rightarrow a^2 = 16$$

$$a = \pm 4$$

17. Match the statements of Column I with values of Column II

Column-I

A) If one of the diameters of the circle

$$x^2 + y^2 - 2x - 6y + 6 = 0 \text{ is a chord to the}$$

circle with centre at $(2, 1)$, then the radius of the circle is equal to

B) If the straight line $y = mx$ touches or lies

outside the circle $x^2 + y^2 - 20y + 90 = 0$, then $|m|$ can be equal to

C) AB is a chord of the circle $x^2 + y^2 = 25$

and the tangents at A and B intersect at C.

If $(2, 2\sqrt{3})$ be the mid-point of AB, then the area of the quadrilateral OACB is equal to (O being origin)

D) A circle C of radius unity touches both the coordinate axes and lies in the 1st quadrant.

If the circle C_1 , which also touches both the axes and lies in the 1st quadrant, intersects C orthogonally,

then the radius of C_1 can be equal to

Column - II

p) $2 - \sqrt{3}$

q) $2 + \sqrt{3}$

r) 2

s) 3

t) $\frac{75}{4}$

Key: A-s, B-prs, C-t, D-pq

Sol: A) $CD \approx \sqrt{5}$ and $AC=2$

$$\therefore AD = \sqrt{5+4} = 3$$

B) Distance from the centre $(0,10)$ of the line $y = mx = \frac{10}{\sqrt{1+m^2}}$ 2 radius $\Rightarrow |m| \leq 3$

$$C) OD = 4 \Rightarrow \sin \theta = \frac{4}{5}$$

So, $AC = OA \cot \theta = 5 \times \frac{3}{4} = \frac{15}{4}$ Area of quadrilateral

$$OACB = OA \times AC = \frac{75}{4}$$

D) The circle C is $x^2 + y^2 - 2x - 2y + 1 = 0$ and C_1 is $x^2 + y^2 - 2rx - 2ry + r^2 = 0$ They intersect orthogonally, if $2(r+r) = 1+r^2$

$$\Rightarrow r^2 - 4r + 1 = 0$$

$$\text{So, } r = 2 \pm \sqrt{3}$$

18. There are two circles in a parallelogram. One of them of radius 3, is inscribed in the parallelogram, and the other touches two sides of the parallelogram and the first circle. The distance between the points of tangency which lie on the same side of the parallelogram is equal to 3.

Column - I

A) The radius of the other circle is

B) Area of the parallelogram is equal to

C) Let d_1, d_2 denote the lengths of the diagonals of parallelogram, then the product d_1, d_2 is equal to

D) Let d_1, d_2 be the diagonals of the parallelogram

then the value of $d_1 + d_2$ is

Column - II

p) $\frac{75}{2}$

q) 75

r) $5\sqrt{31}$

s) $\frac{3}{4}$

Key: A-s, B-p, C-q, D-r

$$\text{Sol: } 2\sqrt{Rr} = 3 \Rightarrow r = \frac{9}{4 \times 3} = \frac{3}{4}$$

$$\frac{r}{R} = \frac{x}{x+3} = \frac{1}{4} \Rightarrow x = 1$$

$$\tan \frac{\theta}{2} = \frac{3}{4}$$

$$y = R \cot \left(\frac{\pi}{2} - \frac{\theta}{2} \right) = 3 \tan \frac{\theta}{2} = \frac{9}{4}$$

ABCD is a rhombus with side $= 4 + \frac{9}{4} = \frac{25}{4}$

$$\text{Area of ABCD} = \frac{25}{4} \times \frac{25}{4} \sin \theta = \frac{25}{4} \times \frac{25}{4} 2 \left(\frac{3}{4} \right) \frac{4}{5} = \frac{25 \times 25 \times 3}{8 \times 5} = \frac{156.25}{1}$$

$$A = \frac{75}{2} = \frac{1}{2}d_1d_2 \Rightarrow d_1d_2 = 75$$

$$\frac{d_1^2}{4} + \frac{d_2^2}{4} = \frac{625}{16} = \frac{(d_1 + d_2)^2 - 2d_1d_2}{16}$$

$$\Rightarrow d_1 + d_2 = \sqrt{775} = 5\sqrt{31}$$

19. A(-2,0) and B(2,0) are the two fixed points and P is a point such that $PA - PB = 2$. Let S be the circle $x^2 + y^2 = r^2$, then match the following

Column – I

A) If $r = 2$, then the number of points P

Satisfying $PA - PB = 2$ and lying on

$x^2 + y^2 = r^2$ is

B) If $r = 1$, then the number of points

satisfying $PA - PB = 2$ and lying on

$x^2 + y^2 = r^2$ is

C) For $r = 2$ the number of common tangents is

D) For $r = \frac{1}{2}$, the number of common tangents is

Column – II

(p) 2

(q) 4

(r) 0

(s) 1

Key: A $\rightarrow$ p, B $\rightarrow$ s, C $\rightarrow$ r, D $\rightarrow$ p

Sol. Locus of point P satisfying $PA - PB = 2$ is a branch of the hyperbola $x^2 - y^2 / 3 = 1$.

For $r = 2$ the circle and the branch of the hyperbola intersect at two points.

For $r = 1$ there is no point of intersection.

If m be the slope of the common tangent then $m^2 - 3 = r^2(1 + m^2)$

$$\Rightarrow m^2 = \frac{r^2 + 3}{1 - r^2}.$$

Hence no common tangents for $r > 1$ and two common tangents for $r < 1$.

20. If PQR is the triangle formed by the common tangents to the circles $x^2 + y^2 + 6x = 0$ and

$x^2 + y^2 - 2x = 0$ then

A) Centroid of ΔPQR is

B) In-centre of ΔPQR is

C) Circum-radius of ΔPQR is

D) In radius of ΔPQR is

p) (1,0)

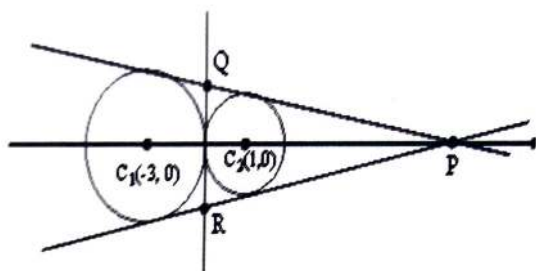
q) (0,1)

r) 1

s) 2

Key: A-p, B-p, C-s, D-r

Sol:



$$C_1 = (-3, 0) \quad r_1 = 3$$

$$C_2 = (1, 0) \quad r_2 = 1$$

ΔPQR is equilateral Δ^{le}

$$\therefore S = I = (1, 0)$$

In radius of $\Delta PQR = r_1 = 1$

Circum radius of $\Delta PQR = 2$

21. Column-I

A) If the circle $x^2 + y^2 - 4x - 6y + k = 0$ does not touch or intersect the axis and the points (2, 2) lies inside the circle, then k belong to

B) If the line $3x + ay - 20 = 0$ cuts the circle $x^2 + y^2 = 25$ at real, distinct or coincident points, then a belongs to the interval

C) If the point (2, k) lies outside the circle $x^2 + y^2 + x - 2y - 14 = 0$ and $x^2 + y^2 = 13$ then k belongs to the interval

D) If the point (k+1, k) lies inside the region bounded by the curve $x = \sqrt{25 - y^2}$, then k belongs to the interval

Column - II

(p) (-1, 3)

(q) $(-\infty, -3) \cup (4, \infty)$

(r) $(-\infty, \sqrt{7}] \cup [\sqrt{7}, \infty)$

(s) (9, 12)

Key: A-s, B-r, C-q, D-p

Sol: The centre of the given circle is (2, 3) and the radius $= \sqrt{4 + 9 - k}$, i.e., $\sqrt{13 - k}$. Since the given circle does not touch or intersect lies inside coordinate circle. $\therefore$ x-coordinate of centre $>$ radius i.e., $2 > \sqrt{13 - k}$, y-coordinate of centre $>$ radius i.e., $3 > \sqrt{13 - k}$ and, $4 + 4 - 8 - 12 + k < 0 \Rightarrow 4 > 13 - k, 9 > 13 - k$ and $-12 + k < 0$
 $\Rightarrow k > 9, k > 4$ and $k < 12 \Rightarrow 9 < k < 12$

B. (R) The length of the $\perp$ from the centre (0, 0) of the given circle to the line $3x + ay - 20 = 0$

$$= \frac{|3(0) + a(0) - 20|}{\sqrt{9 + a^2}} = \frac{20}{\sqrt{9 + a^2}}$$

Radius of the given circle = 5. Since the line cuts the circle at real, distinct or coincident

points. $\therefore \frac{20}{\sqrt{9 + a^2}} \leq 5 \Rightarrow a^2 + 9 \geq 16 \Rightarrow a^2 - 7 \geq 0 \Rightarrow (a - \sqrt{7})(a + \sqrt{7}) \geq 0$

$$\Rightarrow a \in (-\infty, -\sqrt{7}] \cup [\sqrt{7}, \infty)$$

C. (Q). Since the point (2, k) lies outside the circle $x^2 + y^2 + x - 2y - 14 = 0$

$$4 + k^2 + 2 - 2k - 14 > 0 \text{ or } k^2 - 2k - 8 > 0 \text{ or, } (k + 2)(k - 4) > 0 \text{ or, } k \in (-\infty, -2) \cup (4, \infty) \text{ (1)}$$

Also, the point (2, k) lies outside the circle

$$x^2 + y^2 = 13.$$

$$\therefore 4+k^2-13>0 \text{ or } k^2-9>0 \text{ or, } (k-3)(k+3)>0 \text{ or, } k \in (-\infty, -3) \cup (3, \infty)$$

The common solution of (1) and (2) given by $k \in (-\infty, -3) \cup (4, \infty)$

D. (P) Since the point $(k+1, k)$ lies inside the region bounded by $x = \sqrt{25-y^2}$ and y-axis, $\therefore (k+1)^2 + k^2 - 25 < 0$ And, $k+1 > 0 \Rightarrow 2k^2 + 2k - 24 < 0$ and $k > -1$

22. Circle S is touching the line $2x+3y+1=0$ at $(1, -1)$ and cutting orthogonally the circle S having line segment joining $(0, 3)$ and $(-2, -1)$ as diameter.

	Column I	Column II	Column III
	I) Radius of the circle S_1 is $\frac{1}{4}\sqrt{117}$	i) Normal to circle which goes through origin is $x-2y=0$	P) Length of tangent drawn from origin is $\sqrt{\frac{1}{2}}$
1.	II) Intercept of circle S_1 on x-axis $\sqrt{23}$	ii) Two tangents can be drawn from the point $(1, 1)$	Q) Chord $2x-5y-6=0$ through $(3, 0)$ is farthest from the centre
2.	III) Intercept of circle S_1 on y-axis $\sqrt{17}$	iii) Equation of Common chord of circles S_1 and S_2 is $14x+y-7=0$	R) The part of the circle does not lie in the third quadrant
3.	IV) Centre of the circle lies in the first quadrant	iv) The other end of diameter of circle through $(1, -1)$ is $(3, 7/2)$	S) Equation of tangent having slope $-2/3$ is $4x+6y-35=0$

Key:
Sol:

$$(x-1)^2 + (y+1)^2 + \lambda(2x+3y+1) = 0$$

[Here, $(x-1)^2 + (y+1)^2 = 0$ represents a point circle with centre at $(1, -1)$] or

$x^2 + y^2 + 2x(\lambda-1) + y(3\lambda+2) + (\lambda+2) = 0$ But given that this circle is orthogonal to the circle S_2 , the extremities of whose diameter are $(0, 3)$ and $(-2, -1)$. so, $x(x+2) + (y-3)(y+1) = 0$ or

$x^2 + y^2 + 2x - 2y - 3 = 0$ Applying the condition of orthogonality for (1) and (2) we get and

$$(2), \text{ we get } 2(\lambda-1) \cdot 1 + 2 \left(\frac{3\lambda+2}{2} \right) \cdot (-1) = \lambda+2 + (-3) \text{ [As } 2g_1g_2 + 2f_1f_2 = c_1 + c_2 \text{]}$$

Which of the following is the only CORRECT combination?

- a) (I) (i) (S)
c) (III) (i) (P)

Which of the following is the only INCORRECT combination?

- a) (I) (iii) (P)
c) (IV) (ii) (R)

Which of the following is the only IN-CORRECT combination?

- a) (II) (iii) (P)
c) (IV) (i) (R)

1-B, 2-C, 3-D

We are given that line $2x+3y+1=0$ touches a circle $S=0$ at $(1, -1)$ By drawing figure, we have So, equation of this circle can be given by

$$\Rightarrow 2\lambda - 2 - 3\lambda - 2 = \lambda - 1$$

$$\Rightarrow 2\lambda = -3$$

$$\Rightarrow \lambda = -3/2$$

Substituting this value of λ into equation (1)

we get the required circle $ax^2 + y^2 - 5x - \frac{5}{2}y + \frac{1}{2} = 0$ or $2x^2 + 2y^2 - 10x - 5y + 1 = 0$

Column-I

$$\text{Radius} = \sqrt{\frac{25}{2} + \frac{25}{16} - \frac{1}{2}} = \frac{1}{4}\sqrt{117} \quad x\text{-intercept} = 2\sqrt{g^2 - c} = 2\sqrt{\frac{25}{4} - \frac{1}{2}} = \sqrt{23} \quad y\text{-intercept}$$

$$= 2\sqrt{f^2 - c} = 2\sqrt{\frac{25}{16} - \frac{1}{2}} = \frac{1}{2}\sqrt{17}$$
 Also, centre of the circle lies in the first quadrant. Therefore,

from column I, options (I), (II) and (IV) are correct.

Column-II

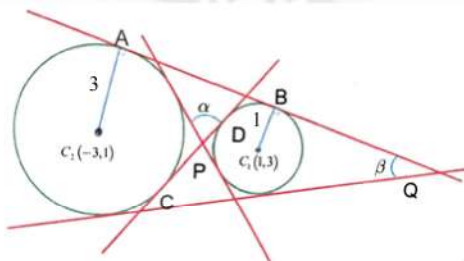
Normal to circle through origin is the line joining origin and centre of the circle which is $x - 2y = 0$. Point (1,1) lies inside the circle. Hence, no tangent can be drawn through this point. Common chord of the circles S_1 and S_2 is equation (1) - (2) . or $14x + y - 7 = 0$ If Q is the other end of diameter through the point $P(1, -1)$ then the centre $R(5/2, 5/4)$ is the mid-point of PQ So, coordinates of point Q are $(4, 7/2)$ Therefore, from column-II, options (i) and (iii)

Column-III.

$$\text{Length of tangent drawn from origin} = \sqrt{(0)^2 + (0)^2 - 5(0) - \frac{5}{2}(0) + \frac{1}{2}} = \sqrt{\frac{1}{2}} \quad \text{Chord through } T(3, 0)$$

is farthest from the centre is bisected at point T Slope of $RT = -5/2$ So, slope of chord $= 2/5$ Hence, required chord is $2x - 5y - 6 = 0$ Clearly, part of the circle does not lie in third quadrant $P(1, -1)$ has slope $-2/3$. So, another tangent having slope $-2/3$ must be at point $Q(4, 7/2)$. Hence, equation of tangent at point Q is $4x + 6y - 37 = 0$. Therefore, from column III, options (P), (Q) and (R) are

23. From the figure



Column I	Column II	Column III
I) Possible equation of AB is $y = 4$	i) $AB = 4$	P) $\alpha = 2 \tan^{-1} \frac{1}{2}$
II) Possible equation of CD is $y = 2$	ii) $CD = 2$	Q) $\beta = 2 \tan^{-1} 2$
III) locus of the centres of the circles	iii) Diameter of the smallest circle touching is 5. Both	R) Distance of P from origin
1. Which of the following is the only INCORRECT combination?	A) (I) (i) (P) B) (I) (i) (S) C) (IV) (i) (P) D) (IV) (i) (R)	
2. Which of the following is the only INCORRECT combination?	iv) Diameter of the circle just sufficient to contain is 5	S) Distance of Q from origin
3. Which of the following is the only INCORRECT combination?	A) (I) (iv) (P) B) (I) (iv) (S) C) (IV) (iv) (Q) D) (IV) (iv) (S)	

Key: 1-D, 2-D, 3-C

Sol: By drawing figure, we have Given circles are: $S_1 : (x-1)^2 + (y-3)^2 = 1$ or $x^2 + y^2 - 2x - 6y + 9 = 0$ and $S_2 : (x+3)^2 + (y-1)^2 = 9$ or $x^2 + y^2 + 6x - 2y + 1 = 0$ Point Q divides line C_1C_2 in the ratio $r_1 : r_2$

$$\text{i.e., } 1:3 \text{ externally. } \therefore Q \equiv \left(\frac{1 \times (-3) - 3 \times 1}{1-3}, \frac{1 \times 1 - 3 \times 3}{1-3} \right) \text{ or } Q = (3, 4) \text{ Thus,}$$

$$\text{equation of any line through point } Q(3, 4) \text{ is } y - 4 = m_1(x - 3) \text{ or } m_1x - y + 4 - 3m_1 = 0$$

If line (1) is tangent to first circle, then length of perpendicular from centre $C_1(1, 3)$ on (1) is

$$r_1 \text{ (radius)} \Rightarrow \frac{m_1 - 3 + 4 - 3m_1}{\sqrt{(m_1^2 + 1)}} = \pm 1 \Rightarrow (1 - 2m_1)^2 = m_1^2 + 1 \Rightarrow 3m_1^2 - 4m_1 = 0 \Rightarrow 3m_1^2 - 4m_1 = 0$$

$$\Rightarrow m_1 = 0, 4/3 \Rightarrow m_1 = 0, 4/3$$

Substituting $m_1 = 0$ and $4/3$ into (1), the equation of direct common tangents are $y = 4$ and $4x - 3y = 0$. point P divides the line $4x - 3y = 0$. point P divides the the ratio $r_1 : r_2$, i.e., 1:3

$$\text{internally. } \therefore P \equiv \left(\frac{1 \times (-3) + 3 \times 1}{1+3}, \frac{1 \times 1 + 3 \times 3}{1+3} \right) \text{ or } P = (0, 5/2) \text{ Thus, equation of any line through}$$

$$P(0, 5/2) \text{ is } y - 5/2 = m_2(x - 0) \text{ or } m_2x - y + 5/2 = 0 \text{ If line (2) is tangent to first circle, then}$$

$$\text{length of perpendicular from centre } C_1(1, 3) \text{ on (2) is } r_1 \text{ (radius). } \Rightarrow \frac{m_2 - 3 + 5/2}{\sqrt{m_2^2 + 1}} = \pm 1$$

$$\Rightarrow (2m_2 - 1)^2 = 4(m_2^2 + 1) \Rightarrow 0m_2^2 - 4m_2 - 3 = 0 \therefore m_2 = -3/4 \text{ and } \infty \text{ [as coeff. of } m_2^2 = 0 \text{]}$$

Substituting $m_2 = -3/4$ and ∞ into (2), the equations of transverse common $3x + 4y - 10 = 0$ and $x = 0$. Locus of the centres of the circle which intersect given circles orthogonally is

radical axis which is $S_1 - S_2 = 0$ or $2x + y - 2 = 0$. Equation of PQ is same as equation of C_1C_2

from the figure, $\sin\left(\frac{\pi}{2} - \frac{\alpha}{2}\right) = \frac{C_1D}{C_1P} = \frac{1}{\sqrt{5}/2} = \frac{2}{\sqrt{5}} \Rightarrow \cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}} \Rightarrow \tan \frac{\alpha}{2} = \frac{1}{2} \Rightarrow \alpha = 2 \tan^{-1} \frac{1}{2}$

Also, $\sin \frac{\beta}{2} = \frac{C_1B}{C_1Q} = \frac{1}{\sqrt{5}} \Rightarrow \tan \frac{\beta}{2} = \frac{1}{2} \Rightarrow \beta = 2 \tan^{-1} \frac{1}{2}$ Diameter of the smallest circle touching

both the circles $EF = C_1C_2C_1F - C_2E = 2\sqrt{5} - 1 - 3 = 2\sqrt{5} - 4$ Diameter of the circle just sufficient of Diameter of the circle contain both circles

$TR = TE + EF + FR = 6 + 2\sqrt{5} - 4 + 2 = 2\sqrt{5} + 4$ ΔMAB , $AB = \sqrt{MB^2 - AM^2} = \sqrt{C_1C_2^2 - AM^2} = \sqrt{20 - 4} = 4$ In triangle

In triangle ΔC_1NC_2 , $C_1N = \sqrt{C_1C_2^2 - C_2N^2} = \sqrt{20 - 16} = 2 = CD$ Thus, from column- I, (I) and (IV) are correct options. from column II, (i), (ii) and (iv) are correct options. from column III, (P) and (S) are correct options. Column I, II and III contain conics, equation of tangents to the conics and points of tangents to the conics and points contact, respectively.

24. If a chord of the circle $x^2 + y^2 - 4x - 2y - c = 0$ is trisected at the points $\left(\frac{1}{3}, \frac{1}{3}\right)$ and $\left(\frac{8}{3}, \frac{8}{3}\right)$

then

A) Length of the chord is

p) 5

B) $c =$

q) $4\sqrt{6}$

C) Length of intercept of the circle on x -axis

r) $7\sqrt{2}$

D) The radius of circle is

s) 20

Key: A-r, B-s, C-q, D-p

Sol: Chord equation is $y = x \dots (1)$ Distance $= \frac{7}{3}\sqrt{2} \Rightarrow$ chord length $= 7\sqrt{2}$ Mid point $= (3, 3)$

$$\therefore 2^2 + 1^2 + c = \left(\frac{7}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 \Rightarrow c = 20$$

25. Column I

A) Locus of centre of circles

Touching $x^2 + y^2 - 4x - 4y = 0$

Internally and

$x^2 + y^2 - 6x - 6y + 17 = 0$ is

Externally is,

B) The locus of the point

$(3h - 2, 3k)$ where (h, k)

lies on the circle

$x^2 + y^2 - 2x - 4y - 4 = 0$

C) Locus of centres of

the circles touching the

two circles $x^2 + y^2 + 2x = 0$ and

Column II

p) Straight line

q) Circle

r) Ellipse

$$x^2 + y^2 - 6x + 5 = 0$$

externally is

D) The extremities of a diagonal of a rectangle are (0,0) and (4,4).

s) Part of Hyperbola

The locus of the extremities of the diagonal is

Key: $a \rightarrow r, b \rightarrow q, c \rightarrow s, d \rightarrow q$

Sol: a) $sp + s'p = 2a$

$$b) \alpha = 32 - 2, \beta = 3k \Rightarrow \frac{\alpha + 2}{3} = h, \frac{\beta}{3} = k$$

$$c) |sp - s'p| = 2a$$

d) Locus is a circle with the given diagonal as diameter

26. Match the column

Column I

Column II

(A) The value of k for which

P) $2\sqrt{5}$

$$x^2 + y^2 = 4 \text{ and } x^2 + y^2 - 2x + 2ky = k$$

Cut Orthogonally

(B) The length of the chord cut by

q) -8

$$4y = 3x + 1 \text{ on the circle}$$

$$x^2 + y^2 - 6x = 0$$

(C) The value of k for which lines

r) 4.5

$$2x - ky + 3 = 0, 4x + y + 5 = 0$$

cut axes in concyclic point

(D) In any triangle, the maximum

s) -4

$$\text{Value of } \frac{r_1 + r_2 + r_3}{R}$$

Key: A-s, B-p, C-q, D-p

Sol: (A) $2gg' + 2ff' = c + c' \Rightarrow -(4+k) = 0 \Rightarrow k = -4$

(B) Distance of chord from centre (3,0) = 2, radius = 3

$$\text{Length of chord} = 2\sqrt{3^2 - 2^2} = 2\sqrt{5}$$

$$(C) a'' = bb' \Rightarrow 8 = -k \Rightarrow k = -8$$

(D) Since maximum is attained in an equilateral triangle, where

$$r_1 = r_2 = r_3 = \frac{3R}{2} \Rightarrow \frac{r_1 + r_2 + r_3}{R} = \frac{9}{2}$$

27. Column I

Column II

A) For how many values of p, the circle

p) 12

$$x^2 + y^2 + 2x + 4y - p = 0 \text{ and the coordinate axes have}$$

exactly three common points?

B) If $y = 2[2x - 1] - 1 = 3[2x - 2] + 1$ then the

q) 8

values of $[y + 5x]$ can be, []

denotes the G.I. F

C) The number of solution of

r) 7

equation $\cos x \sqrt{16 \sin^2 x} = 1$ in

$(-\pi, \pi)$ is

D) If one root of the equation

s) 2

$(a - 6)x^2 - (a + 6)x + 10 = 0$

is smaller than 1 and the

other root greater than 2 then

the value of a can be

t) 4

Key: A-s, B-qrs, C-t, D-pqr

Sol: (2) Case I: When $p = 0$ (i.e., circle passes through origin).

Here, equation of circle is $x^2 + y^2 + 2x + 4y = 0$.

Clearly, this circle meets coordinate axes at exactly three points.

Case II: When circle intersects x-axis at 2 distinct points and touches y-axis.

This occurs when $(g^2 - c) > 0$ and $f^2 - c = 0$

$\Rightarrow 1 - (-p) > 0$ and $4 - (-p) = 0$

$\Rightarrow p > -1$ and $p = -4$

Not possible.

Case III: When circle intersects y-axis at 2 distinct points and touches x-axis.

This occurs when $g^2 - c = 0$ and $(f^2 - c) > 0$

$\Rightarrow 1 - (-p) = 0$ and $4 - (-p) > 0$

$\Rightarrow p = -1$ and $\Rightarrow p > -4$

Here, $p = -1$ is possible.

Hence, possible values of p are 0 and -1.

(b) $y = 2[2x] - 3 = 3[2x] - 5 \Rightarrow [2x] = 2 \Rightarrow y = 1$

$2 \leq 2x < 3 \Rightarrow 5 \leq 5x \leq 7.5 \Rightarrow 6 \leq 5x + y \leq 8.5$

$[5x + y] = 6, 7, \text{ or } 8.$

(c) $4|\sin x| \cos x = 1$ solutions are possible if $\cos x > 0$

$\Rightarrow \frac{-\pi}{2} \leq x \leq \frac{\pi}{2} \Rightarrow -\pi \leq 2x \leq \pi$

if $x \in \left[-\frac{\pi}{2}, 0\right)$

$\sin 2x = \frac{-1}{2} \Rightarrow 2x$

$= \frac{-\pi}{6}, \frac{-5\pi}{6} \Rightarrow x = \frac{-\pi}{12}, \frac{-5\pi}{12}$

$$\text{if } x \in \left(0, \frac{\pi}{2}\right)$$

$$\sin 2x = \frac{1}{2} \Rightarrow 2x = \frac{\pi}{6}, \frac{5\pi}{6} \Rightarrow x = \frac{\pi}{12}, \frac{5\pi}{12}$$

$$d)(a-6)f(1) < 0 \Rightarrow a > 6$$

$$(a-6)f(2) < 0 \Rightarrow 6 < a < 13.$$

Exercise : V

(Assertion – Reason / Statement – I & II Type Questions Including PYQ's)

Circles & Analysis of two circles

1.Circles

1) Standard equation of a circle, Lengths of tangents and chords

- 1) A: The centre of the circle passing through the points (3,8) and (5,4) and having smallest radius is (4,6)

R : The centre of circle passing through two given points lies at mid point of the line joining the points

- a) A is true, R is true, R is a correct explanation for A
 b) A is true, R is true, R is not a correct explanation for A
 c) A is true, R is false
 d) A is false, R is true

Ans: (c)

Solution : The radius of the circle is smallest when the line joining the given points is a diameter of the circle and thus the centre lies at the mid point of the line joining (3,8) and (5,4) i.e. (4,6), so statement-1 is true but statement-2 is false as the line joining the given points may not be a diameter for each circle passing through these points.

- 2) A : Angles in the same segment of circle are equal

R : In cyclic quadrilateral opposite angles are supplementary

- a) A is true, R is true, R is a correct explanation for A
 b) A is true, R is true, R is not a correct explanation for A
 c) A is true, R is false
 d) A is false, R is true

Ans : (a)

A: We know that the angle subtended by an arc of a circle at the centre is double of the angle subtended by it at any point on the remaining part of the circle

$$\therefore \angle PAQ = \frac{1}{2} \angle POQ$$

$$\text{Also } \angle PBQ = \frac{1}{2} \angle POQ$$

$$\therefore \angle PAQ = \angle PBQ$$

$$\alpha = 2\angle C, \beta = 2\angle A$$

$$\alpha + \beta = 360^\circ$$

$$\angle C + \angle A = 180^\circ$$

$$\& \angle B + \angle D = 360^\circ - 180^\circ = 180^\circ$$

2) Intercepts made by a circle on axes , Equation of chord, Position of a point w.r.t circle

3) A : A square is inscribed in the circle $x^2 + y^2 - 2x + 4y + 3 = 0$ its sides are parallel to the coordinate axes then one vertex of the square is (2,-1)

R : A square is inscribed in the circle with centre (h,k) and radius r its sides are parallel to the co-ordinate axis then its vertices are $\left(h \pm \frac{r}{\sqrt{2}}, k \pm \frac{r}{\sqrt{2}}\right)$

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (a)

Solution : Centre (1,-2) radius $r = \sqrt{2}$

$$\text{Vertex} = \left[h \pm \frac{r}{\sqrt{2}}, k \pm \frac{r}{\sqrt{2}}\right] = (2, -1)$$

4) A : Let L_1 be a line passing through origin and L_2 be the line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 and L_2 are equal then L_1 is $x + 7y = 0$

R : If the chords are of equal length, then the distances of the centre from the lines are equal

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (a)

Solution : Let L_1 be $y - mx = 0$ centre $\left(\frac{1}{2}, -\frac{3}{2}\right)$

$$\frac{\left|-\frac{3}{2} - \frac{m}{2}\right|}{\sqrt{m^2 + 1}} = \frac{\left|\frac{1}{2} - \frac{3}{2} - 1\right|}{\sqrt{2}} \Rightarrow 7m^2 - 6m - 1 = 0 \Rightarrow m = 1, -\frac{1}{7}$$

5. A : If 'a' is constant and two chords of length $a^2 - 1$ and $3(a+1)$ of a circle bisect each other, the radius of the circle is $15/2$

R : Two chords of a circle bisect each other only when both are the diameters of the circle

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans: (a)

Sol : R is True because centre of the circle is the only point such that all chords passing through it are bisected by this point. Using this result in A, as both the chords are diameters $a^2 - 1 = 3(a + 1) \Rightarrow a^2 - 3a - 4 = 0 \Rightarrow a = 4$ or -1 .

But $a = -1$ is not possible as in that case $a + 1 = 0$ and the length of one of the chords

is zero. So $a = 4$ and the radius of the circle is $\frac{3(a+1)}{2} = \frac{15}{2}$

and thus A is also true

- 6) Assertion (A) : AB and CD are two parallel chords of a circle whose diameter is AC then $AB = CD$

Reason(R) : Perpendicular from the centre of a circle does not bisect the chord

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (c)

Solution : $\triangle AOB \cong \triangle COD$

$\angle CAB = \angle ACD$ (alternate angles)

$\angle COD = \angle AOB$ vertically opposite angles

$AO = OC$ (radius)

So, $AB = CD$ by corresponding points of congruent triangle

$\angle OPA = \angle OPB$ (both 90°)

$\Rightarrow OA = OB$ (both are radius)

$\Rightarrow OX = OX$ (common side)

$\triangle OAX \cong \triangle OBX$

$AX = BX$ (by property of congruent $\triangle^{les}$)

- 7) A : Perpendicular bisectors of two chords of circle intersect at its centre

R : A line drawn through the centre of circle to bisect a chord is perpendicular to the chord

Ans : (A)

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (A)

Solution : The perpendicular from centre of circle to the chord bisects the chord and perpendicular bisector of two . Chords of a circle intersects at the centre.

AB is closed

P is mid point of AB

PQ ⊥ AB

$$\angle APQ = 90^\circ, \angle OPA = 90^\circ$$

It is possible where PQ passing through 'O'

3) Tangent line equation

(Position of a line w.r.t a circle in circles, Condition for tangency in circles, Perpendicular distance from centre to the line is radius in circles, Equation of tangent in parametric form in circles, Point of contact of tangent in circles, Number of tangents from an external point in circles, Sum of slopes of tangents in circles, Equation of pair of tangents drawn from external point to circle in circles, Product of slopes of tangents in circles, Angle between pair of tangents drawn from external point to circle in circles, Director circle of a circle in circles)

8. A : The number of circles touching x-axis, y-axis and the line $3x + 3y - 12 = 0$ are 2

R : The circles touches both the coordinate axes with centre (h,k) and radius r then,

$$|h| = |k| = r$$

a) A is true, R is true, R is a correct explanation for A

b) A is true, R is true, R is not a correct explanation for A

c) A is true, R is false

d) A is false, R is true

Ans : (4)

Sol : Case (i) : Centre lies in 1st quadrant C(r,r)

Use $r = d \Rightarrow r = 1$ or 6

Case (ii) $C(-r,r) \Rightarrow r = 2$

Case (iii) $C(r,-r) \& r = d$

$$r = 3$$

Number of circles = 4

9. A : A ray of light incident at the point (-3,-1) gets reflected from the tangent at (0,-1) to the circle $x^2 + y^2 = 1$. If the reflected ray touches the circle, then equation of the reflected ray is $4y - 3x = 5$

R : The angle of incidence = angle of reflection . i.e. $\angle i = \angle r$

a) A is true, R is true, R is a correct explanation for A

b) A is true, R is true, R is not a correct explanation for A

c) A is true, R is false

d) A is false, R is true

Ans : (2)

Sol : Equation of tangent $y = mx \pm \sqrt{1+m^2}$

$$(-3, -1) \Rightarrow -1 = -3m \pm \sqrt{1+m^2}$$

$$m = 0 \text{ (or) } m = \frac{3}{4}$$

$$\text{Eq. of reflected ray is } y+1 = \frac{3}{4}(x+3) \Rightarrow 4y-3x=5$$

10. A : Tangents can not be drawn from the point $(1, \lambda)$ to the circle $x^2 + y^2 + 5x - 5y = 0$ if $2 < \lambda < 3$

$$R : \left(1 + \frac{5}{2}\right)^2 + \left(\lambda - \frac{5}{2}\right)^2 < \left(\frac{5}{2}\right)^2 + \left(\frac{5}{2}\right)^2$$

- a) A is true, R is true, R is a correct explanation for A
 b) A is true, R is true, R is not a correct explanation for A
 c) A is true, R is false
 d) A is false, R is true

Ans : (a)

$$\text{Sol : } P(1, \lambda) \quad C\left(-\frac{5}{2}, \frac{5}{2}\right)$$

$$r = \sqrt{\left(\frac{5}{2}\right)^2 + \left(\frac{5}{2}\right)^2}$$

Point $P(1, \lambda)$ must be inside the circle

$$\therefore CP < r$$

$$\Rightarrow CP^2 < r^2$$

$$\left(1 + \frac{5}{2}\right)^2 + \left(\lambda - \frac{5}{2}\right)^2 < \left(\frac{5}{2}\right)^2 + \left(\frac{5}{2}\right)^2$$

$$\Rightarrow 2 < \lambda < 3$$

11. A : If the perpendicular tangents to the circle C_1 meet at P, then the locus of P has a circle C_2 .

$$\text{Then } \frac{\text{radius of } C_1}{\text{radius of } C_2} = \frac{1}{\sqrt{2}}$$

R : Circles C_1 and C_2 are concentric

Ans : (2)

- a) A is true, R is true, R is a correct explanation for A
 b) A is true, R is true, R is not a correct explanation for A
 c) A is true, R is false
 d) A is false, R is true

Ans : (b)

Solution : C_2 is the director circle of C_1

If radius of C_1 is r_1 then radius of C_2 is $\sqrt{2}r$ and circles are concentric.

12. A: A circle can be inscribed in a quadrilateral where sides are

$$3x - 4y = 0, 3x - 4y = 5, 3x + 4y = 0 \text{ and } 3x + 4y = 7$$

R : A circle can be inscribed in a parallelogram if and only if it is a rhombus

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (d)

$$\text{Solution : Distance between parallel lines} = \frac{|C_1 - C_2|}{\sqrt{a^2 + b^2}} = \frac{5}{5} = 1$$

$$\text{Distance between parallel lines} = \frac{7}{5} \quad \therefore 1 + \frac{7}{5}$$

It is not rhombus

13. A : The locus of the point of intersection of the tangents to the circle $x = a \cos \theta, y = a \sin \theta$ at the points whose parametric angles differ by $\pi/2$ is $x^2 + y^2 = 2a^2$

R : Tangents at the extremities of a diameter of a circle are parallel.

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans. (b)

$$\text{Solution : } x \cos \theta + y \sin \theta = a,$$

$$x \cos(\pi/2 + \theta) + y \sin(\pi/2 + \theta) = a,$$

$$\Rightarrow (x \cos \theta + y \sin \theta)^2 + (-x \sin \theta + y \cos \theta)^2 = 2a^2$$

$$\Rightarrow x^2 + y^2 = 2a^2$$

$\Rightarrow$ statement-1 is true, statement-2 is also true but does not lead to statement-1

14. $C_1 : (x-3)^2 + (y-4)^2 = a^2$

A : C_1 touches the axis of x if $a = 4$

R : C_1 touches the line $y = x$ if $a = 3$

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (c)

Sol : A is true as the distance of the centre (3,4) from x-axis is equal to the radius. R is false, because $\frac{4-3}{\sqrt{2}} = a \Rightarrow a = 1/\sqrt{2}$

4) Normal to a circle

(Equation of normal line at point of contact in circles, Condition for normal to circle, Equation of normal at P (theta))

15) A : The lines $y = mx + 1 - m$ for all values of m is a normal to the circle $x^2 + y^2 - 2x - 2y = 0$

R : The line L passes through the centre of the circle

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (d)

Solution : Centre (1,1) satisfies the line $L = 0$

$\therefore L = 0$ is a normal to the circle

16) A : The line $x + 9y - 12 = 0$ is a chord of contact of a point P with respect to the circle

$$2x^2 + 2y^2 - 3x + 5y - 7 = 0$$

R : The line joining the points of contacts of the tangents drawn from a point P, outside a circle to the circle is the chord of contact of P with respect to the circle.

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (d)

Solution : Chord of contact of P(h,k) is $S_1 = 0$

$$\text{i.e. } 2xh + 2ky - \frac{3}{2}(x+h) + \frac{5}{2}(y+k) - 7 = 0 \text{ is same as } x + 9y - 12 = 0$$

$$\frac{4h-3}{1} = \frac{4k+5}{9} = \frac{3h-5k+14}{12} \Rightarrow h=1, k=1$$

$$\text{But } P(1,1) = 2 + 2 - 3 + 5 - 7 < 0$$

$\therefore P(1,1)$ lies inside the circle & given line is not a chord.

5) Chord of contact of a circle, Pole and polar of a circle

(Equation of chord of contact of a circle in circles, Equation of chord of a circle when midpoint is given in circles, Length of chord when midpoint is given in circles, Definition of pole and polar in circles, Polar of a point with respect to circle Pole of a polar with respect to a circle, Conjugate points with respect to given circle, Conjugate lines with respect to circle, Properties of pole and polar of a circle, Problems based on locus)

17) A : The area of the triangle formed by the tangents drawn to the circle $x^2 + y^2 = 9$ from the point (4,3) and the chord of contact is $7\frac{17}{25}$

R : The area of the Δ^{le} formed by the tangents drawn to the circle $S = 0$ from P(x,y) and chord of contact is $\frac{r(S_{11})^{3/2}}{S_{11} + r^2}$

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (a)

Sol : $x^2 + y^2 = 9 \Rightarrow r = 3$

$P(4,3) \Rightarrow S_{11} = 16 + 9 - 9 = 16$

Area = $\frac{r(S_{11})^{3/2}}{S_{11} + r^2} = \frac{3(16)^{3/2}}{16 + 9} = \frac{192}{25} = 7\frac{17}{25}$

18) A : If the line $2x + 3y = 3$ intersects the circle $x^2 + y^2 - 4 = 0$ at A and B and $M(\alpha, \beta)$ is point of intersection of the tangents at A and B then $\frac{\alpha}{\beta}$ is equal to $\frac{2}{3}$

R : The pole of the line $lx + my + n = 0$ with respect to the circle

$$x^2 + y^2 = r^2 \text{ is } \left[\frac{-r^2 l}{n}, \frac{-r^2 m}{n} \right]$$

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (a)

Solution: $2x + 3y - 3 = 0$ $x^2 + y^2 = 4$

Pole : $\left[\frac{-4.2}{-3}, \frac{-4.3}{-3} \right] = \left[\frac{8}{3}, \frac{12}{3} \right]$

19) A : Equation of the polar of P(-7,-9) with respect to the circle $x^2 + y^2 - 12x - 8y - 48 = 0$ is $13x + 13y - 30 = 0$

R : P is a point outside the circle C. A line passing through P meets the circle C at Q and R. The tangents at the extremities of the chord QR of the circle C intersect at S. The locus of S, for different chords of the circle C passing through P is called the Polar of P w.r.t. the circle C and its equation is same as the chord of contact of P with respect to the circle C.

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false

d) A is false, R is true

Ans : (a)

Solution : R is true. Using it in A, the required equation is

$$-7x - 9y - 6(x - 7) - 4(y - 9) - 48 = 0$$

$$\text{Or } 13x + 13y - 30 = 0$$

Showing that statement-1 is also true.

20. Assertion(A) : The inverse point of (1,-1) with respect to the circle $x^2 + y^2 = 4$ is (-2,2)

Reason(R) : (1,-1), (-2,2) lie on either side of the center

Solution : Inverse point of (1,-1) is the foot of the perpendicular from (1,-1) to the polar of (1,-1)

a) A is true, R is true, R is a correct explanation for A

b) A is true, R is true, R is not a correct explanation for A

c) A is true, R is false

d) A is false, R is true

Ans : (d)

Equation of the polar of (1,-1) w.r.t. $x^2 + y^2 = 4$ is $x - y - 4 = 0$

$$\frac{h-1}{1} = \frac{k+1}{-1} = \frac{-[1(1) - 1(-1) - 4]}{1^2 + 1^2} = 1$$

$$h = 2, k = -2$$

Inverse point = (2,-2)

R : Reason is correct

2. Analysis of two circles

6) Relative positions of two circles

(Angle between intersecting circles in circles, Condition for orthogonal circles, Non-intersecting circles in circles, Intersecting circles in circles, touching circles in circles, Circles touch internally in circles, Circles touch externally in circles, One circle lie inside the other in circles, Centres of similitude in circles, Different forms of common tangents to circles in circles, Direct common tangents in circles, Transverse common tangents in circles, Length of direct common tangent in circles, Length of transverse common tangent in circles, Equation of common tangent of two circles in circles)

21. A : Let C_1 and C_2 be two circles with lying inside C_1 , A circle C lying inside C_1 touches C_1 internally and C_2 externally. Then the locus of the centre of C is an ellipse.

R : If A and B are foci and P is any point on the ellipse, then $AP + BP = \text{constant}$

a) A is true, R is true, R is a correct explanation for A

b) A is true, R is true, R is not a correct explanation for A

c) A is true, R is false

d) A is false, R is true

Ans : (a)

$$\text{Sol : } OO_2 = r + r_2$$

$$OO_1 = r_1 - r$$

$$OO_1 + OO_2 = r_1 + r_2$$

Which is greater than O_1O_2 as $O_1O_2 < r_1 + r_2$

Locus of O is an ellipse with foci O_1 and O_2

22. A : Circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 8x + 7 = 0$ intersect each other two distinct points

R : Circles with centres C_1 and C_2 and radii r_1 and r_2 intersect at two distinct points, if

$$|C_1C_2| < r_1 + r_2$$

a) A is true, R is true, R is a correct explanation for A

b) A is true, R is true, R is not a correct explanation for A

c) A is true, R is false

d) A is false, R is true

Ans : (c)

$$\text{Sol : } C_1(0,0), r_1 = 2$$

$$C_2(4,0), r_2 = 3$$

$$C_1C_2 = 4 \quad |r_1 - r_2| = 1$$

$$|r_1 - r_2| < C_1C_2 < r_1 + r_2$$

23. A : Three circles touch are another externally, the tangents of their points of contact meet at a point whose distance from a point of contact is 4. Find the ratio of the product of the radii to the sum of the radii of the circle is 16.

R : If A,B,C are the centres of three circles touching mutually externally then the radical centre of the circles for $\triangle ABC$ is incentre

a) A is true, R is true, R is a correct explanation for A

b) A is true, R is true, R is not a correct explanation for A

c) A is true, R is false

d) A is false, R is true

Ans : (a)

$$\text{Solution: } r = r_1 + r_2 + r_3, a = r_2 + r_3, b = r_1 + r_3, c = r_1 + r_2$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{r_1 r_2 r_3 (r_1 + r_2 + r_3)}$$

$$\text{In radius, } r = 4 = \frac{\Delta}{s}$$

$$\frac{r_1 r_2 r_3}{r_1 + r_2 + r_3} = 16$$

24. A : The locus of the centres of the circles which touch the two circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = 4ax$ externally is $12x^2 - 4y^2 - 24ax + 9a^2 = 0$

R : Two circles neither intersect nor touch each other then locus of the centre of a variable circle which touches both the circles externally is a hyperbola

- a) A is true, R is true, R is a correct explanation for A
 b) A is true, R is true, R is not a correct explanation for A
 c) A is true, R is false
 d) A is false, R is true

Ans : (d)

Solution : Centre A (0,0) radius $r_1 = a$

B(2a,0) $r_2 = 2a$

$PA = r + r_1$ $PB = r + r_2$

$$PA - PB = r_1 - r_2 \Rightarrow PA - PB = a$$

$$PA = O + PB$$

$$S.O.B.S. \Rightarrow 12x^2 - 4y^2 - 24ax + 9a^2 = 0$$

25. A : The centre of a circle passing through the points (0,0),(1,0) and touching the circle $C : x^2 + y^2 = 9$ lies inside the circle C.

R : If a circle C_1 passes through the centre of circle C_2 and also touches the circle : The radius of the circle C_2 is twice the radius of the circle C_1

- a) A is true, R is true, R is a correct explanation for A
 b) A is true, R is true, R is not a correct explanation for A
 c) A is true, R is false
 d) A is false, R is true

Ans : (a)

Solution : Statement-2 is true as circle C_1 touches the circle C_2 internally and the diameter of C_1 is equal to the radius of the circle C_2 . Using it in statement-1, the centre of the required circle lies inside C.

7) System of circles, Radical axis

(Family of circles passing through P.O.I of circle and line in circles, Family of circles touching the circle at point (x_1, y_1) in circles, Family of circles passing through P.O.I of two circles in circles, Equation of radical axis, Properties of radical axis, Equation of common chord of two circles in circles, Co-axial system of circles, Radical centre in circles, Properties of radical centre, Limiting points of a co-axial system of circles in circles)

- 26) A: Circle $x^2 + y^2 - 6x - 4y + 9 = 0$ bisects the circumference of the circle $x^2 + y^2 - 8x - 6y + 23 = 0$

R : Centre of first circle lie on the second circle

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (b)

Solution : Eq. of common chord is $S - S^1 = 0$

$$\Rightarrow x + y - 7 = 0$$

Centre of second circle (4,3) lies on $x + y - 7 = 0$

Hence first circle bisects the circumference of the second circle.

27. A: The equations $S_1 : x^2 + y^2 + 7x - 9y + 5 = 0$, $S_2 : 2x^2 + 2y^2 + 5x + 7y + 11 = 0$ and $S_3 : 4x^2 + 4y^2 + x + 39y + 23 = 0$ belongs to a coaxial system of circles

R : $2S_1 - S_2 = 0$ and $2S_2 - S_3 = 0$ represents the same line.

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (a)

Sol : $2S_1 - S_2 = 0 \Rightarrow 9x - 25y - 1 = 0$

$$2S_2 - S_3 = 0 \Rightarrow 9x - 25y - 1 = 0$$

Hence, S_1, S_2 and S_3 belongs to a coaxial system of circles

28. A : From a point A(3,2) tangents are drawn to the circle $x^2 + y^2 - 2x + 4y - 4 = 0$, If BC is the chord of contact of tangents the line joining the midpoints of AB and AC is $4x + 8y - 17 = 0$

R : The radical axis of two circles bisects all common tangents

- a) A is true, R is true, R is a correct explanation for A
- b) A is true, R is true, R is not a correct explanation for A
- c) A is true, R is false
- d) A is false, R is true

Ans : (a)

Solution : $(x-3)^2 + (y-2)^2 = 0 \Rightarrow s = x^2 + y^2 - 6x - 4y + 13 = 0$

$$s' = x^2 + y^2 - 2x + 4y - 4 = 0$$

Radical axis is $S - S' = 0 \Rightarrow 4x + 8y - 17 = 0$

29. A : If locus of centre of circle which cuts the circles $x^2 + y^2 - 2x + 4y + 3 = 0$ and $x^2 + y^2 + 7x - 3y - 1 = 0$ orthogonally is $lx + my - 4 = 0$ then $l + m$ is equal to 2

R : The centre of a circle cutting two circles orthogonally lie on the radical axis of two circles

- a) A is true, R is true, R is a correct explanation for A

- b) A is true, R is true, R is not a correct explanation for A
 c) A is true, R is false
 d) A is false, R is true

Ans : (1)

Solution: Radical axis of $S = 0$ & $S^1 = 0$ is $S - S^1 = 0$
 $\Rightarrow 9x - 7y - 4 = 0$

Statement Questions

30. A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3}, 1)$. A straight line L perpendicular to PT is a tangent to the circle $(x - 3)^2 + y^2 = 1$

Statement-1 : Possible equation of L is $x - \sqrt{3}y = 1$ or $x - \sqrt{3}y = 5$

Statement-2 : The distance between these lines is $= 4$

- a) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for A
 b) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for A
 c) Statement-1 is true, Statement-2 is false
 d) Statement-2 is false, Statement-2 is true

Ans : (c)

Solution : Equation of tangent at $(\sqrt{3}, 1)$ to $x^2 + y^2 = 4$ is $\sqrt{3}x + y = 4 \dots (1)$

As L is perpendicular to (1)

$$\Rightarrow L = x - \sqrt{3}y = K \rightarrow (2)$$

(2) is tangent to $(x - 3)^2 + y^2 = 1$

$$\frac{|3 - 0 - k|}{\sqrt{3 + 1}} = 1 \Rightarrow |3 - k| = 2$$

$$3 - k = \pm 2$$

$$K = 1, 5$$

$$\text{Distance between parallel lines} = \frac{|C_1 - C_2|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|5 - 1|}{\sqrt{1 + 3}} = \frac{4}{2} = 2.$$

31. Tangents are drawn from the point $(17, 7)$ to the circle $x^2 + y^2 = 169$.

Statement-I : The tangents are mutually perpendicular

Statement-II : The locus of the points from which mutually perpendicular tangents can be drawn to the given circle is $x^2 + y^2 = 338$

- a) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for A

- b) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for A
 c) Statement-1 is true, Statement-2 is false
 d) Statement-2 is false, Statement-2 is true

Ans : (a)

Since, the tangents are perpendicular

$\Rightarrow$ Locus of perpendicular tangents to circle $x^2 + y^2 = 169$ is a director circle having equation

$$x^2 + y^2 = 338. \text{ Hence (a) is correct answer.}$$

32. Statement – I : Equation of a tangent the circle $x^2 + y^2 = 50$ at a point which has positive integral co-ordinates (α, β) ($\alpha \neq \beta$) is $x + 7y = 50$ or $7x + y = 50$

Statement-II : There are 12 points on the circle $x^2 + y^2 = 50$ with integral co-ordinates

- a) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for A
 b) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for A
 c) Statement-1 is true, Statement-2 is false
 d) Statement-2 is false, Statement-2 is true

Ans : (a)

Solution : Statement-2 is true, as the required points are $(\pm 1, \pm 7); (\pm 7, 1) (\pm 5, \pm 5)$ out of which points satisfying conditions in statement are (1,7) & (7,1) only and tangents at these points are respectively $x + 7y = 50$; $7x + y = 50$. So, statement-I is also true.

33. Statement-1 : The only circle having radius $\sqrt{10}$ and diameter along the line $x + 3y = 0$ is $x^2 + y^2 - 6x + y = 0$

Statement-2 : $x + 3y = 0$ is normal to circle $x^2 + y^2 - 6x + 2y = 0$

- a) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for A
 b) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for A
 c) Statement-1 is true, Statement-2 is false
 d) Statement-2 is false, Statement-2 is true

Ans : (a)

Solution :

$$\text{Circle } x^2 + y^2 - 6x + 2y = 0$$

$$\text{Centre} = (3, -1) \quad r = \sqrt{9+1} = \sqrt{10}$$

$$\text{Given line } x + 3y = 0$$

$$\therefore (3) + 3(-1) = 0$$

Hence, $C(3, -1)$ lies on line.

Therefore, line passes through centre.

Given line is normal to circle thus $st-2$ is true

But $st-1$ is not true as there are infinite circle according to the given conditions.

34. Statement-I : If the circles $x^2 + y^2 = a^2$ and $x^2 + y^2 - 6x - 8y + 9 = 0$ touch externally then $a = 1$

Statement-II : Two circles with centres C_1, C_2 and radii r_1, r_2 respectively touch externally if $C_1C_2 = r_1 + r_2$

- a) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for A
- b) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for A
- c) Statement-1 is true, Statement-2 is false
- d) Statement-2 is false, Statement-2 is true

Key : (1)

Solution:

Statement-I :

$$C_1(0,0)$$

$$C_2(3,4)$$

$$r_1 = a$$

$$r_2 = \sqrt{9+16-9} = 4$$

$$\text{Touch each other externally} \Rightarrow \sqrt{9+16} = a + 4 \Rightarrow a = 1$$

Statement I is true, Statement II is true

Statement I is true, Statement II is true, Statement II is the correct explanation for statement I.

35. Statement-I : For two non-intersecting circles, direct common tangents subtends a right angle at either of point of intersection of circles with line segment joining the centres of circles

Statement-II : If the distance between the centres is more than sum of radii, then circles are non intersecting

- a) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for A
- b) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for A
- c) Statement-1 is true, Statement-2 is false
- d) Statement-2 is false, Statement-2 is true

Ans : (d)

Solution : Draw circle C with direct common tangent as diameter then both the points of intersection of line joining the centres lie in interior of circle C. Therefore the angles are obtuse angles.

So, S-I is false and S-II is true. Hence, (d) is correct answer.

36. Statement – I : If $x^2 + y^2 - 2x + 3y + k = 0$, $x^2 + y^2 + 8x - 6y - 7 = 0$, cut each other orthogonally then $k = 10$

Statement- II : The circles $x^2 + y^2 + 2gx + 2fy + c = 0$,

$x^2 + y^2 + 2g'x + 2f'y + c' = 0$, cut each other orthogonally if $2gg' + 2ff' = c + c'$.

- a) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for A
 b) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for A
 c) Statement-1 is true, Statement-2 is false
 d) Statement-2 is false, Statement-2 is true

Ans : (4)

Solution :

I: $x^2 + y^2 - 2x + 3y + k = 0, x^2 + y^2 + 8x - 6y - 7 = 0$

Cut orthogonally $\Rightarrow 2gg' + 2ff' = c + c' \Rightarrow k = -10$

$\therefore$ I is false

II : II is true

Statement-I is false, Statement II is true

37. I : The length of the tangents from any point on the line $2x + 3y = 5$ to the circles

$x^2 + y^2 = 9$ and $x^2 + y^2 + 4x + 6y = 19$ are equal in length.

II : There is only one point such that the tangents from it to the three given circles are equal in length.

Then the correct statement is :

- 1) only I 2) only II 3) Both I & II 4) Neither I nor II

Key : (3)

I. $2x + 3y = 5$ is the Radical axis of given two circles

$\therefore$ length of the tangents from any point on Radical axis to given circles

II. There is only one point i.e. Radical centre such that the tangents from it to the three given circles are equal in length.

Hence both I & II are correct.

38. Statement – 1: The equation $x^2 + y^2 + 2\lambda x + 4 = 0$ represents a real circle for all $\lambda \in R$

Statement- 2: The radical axis of any two circles of the family represented by

$x^2 + y^2 + 2\lambda x + 4 = 0$ is the x-axis

- 1) A is true, R is true, R is a correct explanation for A
 2) A is true, R is true, R is not a correct explanation for A
 3) A is true, R is false
 4) A is false, R is true

Key : (4)

Circle $x^2 + y^2 + 2\lambda x + 4 = 0$

Radius = $\sqrt{\lambda^2 - 4}$ for real circle $\lambda^2 - 4 > 0 \Rightarrow |\lambda| > 2$

St-2

Given circle $x^2 + y^2 + 2\lambda x + 4 = 0$

$$x^2 + y^2 + 4 + 2\lambda x = 0$$

 $S_b + \lambda L = 0$ from this represents coaxial circle with radical axis $x = 0$; i.e. y-axis.

39. Statement – I : If two circles touch to each other $x^2 + y^2 + 2gx + 2fy = 0$ and $x^2 + y^2 + 2g'x + 2f'y = 0$ then $fg' = gf'$

Statement – II : Two circles touch to each other, if line joining there centres is perpendicular to all possible common tangents

- 1) A is true, R is true, R is a correct explanation for A
- 2) A is true, R is true, R is not a correct explanation for A
- 3) A is true, R is false
- 4) A is false, R is true

Ans : C

Solution: The statement II is false, since line joining centres may not be parallel to common tangents. The statement – I can be proved easily by using distance between centres = sum of radii.

Hence, (c) is correct answer.

40. Statement-I : The equation of the straight lines joining origin to the points of intersection of

$$x^2 + y^2 - 4x - 2y - 4 = 0 \text{ and } x^2 + y^2 - 2x - 4y - 4 = 0 \text{ is } (y - x)^2 = 0$$

Statement-II : $y + x = 0$ is a common chord of $x^2 + y^2 - 4x - 2y - 4 = 0$ and

$$x^2 + y^2 - 2x - 4y - 4 = 0$$

- 1) A is true, R is true, R is a correct explanation for A
- 2) A is true, R is true, R is not a correct explanation for A
- 3) A is true, R is false
- 4) A is false, R is true

Ans : C

Solution : Statement I is true because common chord itself passes through origin.

Statement II is false (common chord is $x - y = 0$)

Hence (C) is correct answer.